



What a "Strong and Rich Japan" could mean

July 24, 2026

Executive summary

- Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. Japan's new economic blueprint seeks to put the country on a path of investment-driven growth to meet the challenges of a new era.
- Japan is not alone in its bid to prioritize industry and rearm, but given a higher starting point for government debt, it will have to find room to spend while maintaining fiscal sustainability. This will inform Japan's approach in very significant ways.
- First, Japan is likely to remain focused on supporting nominal GDP growth, aiming to keep it above government funding costs to create deficit space. Second, Japan will look to actively mobilize domestic savings to fund investments and manage yield pressure. Each of these shifts could have implications for the currency.
- Japan's excess savings have arguably been kept in two places: in cash and abroad. The government is looking to get both moving. Households have around 50% of their \$15tn of savings in cash and deposits. Meanwhile, GPIF has 50% of its \$1.8tn portfolio abroad. If industry, energy, defence need to come home, so might the capital to fund them.
- Over the past few months, Japanese policy makers have been very focused on stabilizing the currency. However, if fiscal capacity is becoming the most important policy criteria, incentives could shift from FX to yield management.
- The ultimate impact on USD/JPY will depend on the levers the government chooses to pull to manage yields. If GPIF is mandated to bring money back into domestic assets, this could be very bullish for the JPY. However, if the BoJ is coopted to support bonds through renewed JGB purchases, this could be very negative for the JPY.
- Suppressing yield volatility could come alongside bigger moves in FX from here, depending on the approach the government takes. This favours higher JPY vol.

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Echoes of Meiji

Japan could be at the precipice of a very significant shift in fiscal and industrial policy, with echoes of the Meiji Restoration. In 1868, pre-modern Japan was staring at the threat of foreign involvement and colonization, having seen the Opium Wars unfold in China. Japan had signed unequal trade treaties that opened Japanese ports to free trade imperialism, prevented Japan from setting tariffs, and exempted foreigners from local laws. Japan's response changed history: under the Meiji Restoration, Japan returned rule to the emperor, strengthened the state, and pursued extremely rapid industrialization.

One could argue that Japan again faces existential challenges given fractures in globalization, intensifying great power competition, insecurity around energy and critical minerals, the need to participate in AI innovation, and external dependence for defence. Japan's answer under PM Takaichi: to return to its roots as an industrial nation driven by state and fiscal policy. The slogan of the 1868 Meiji Restoration was "Fukoku Kyohei" which translates into "Enrich the Country, Strengthen the Armed Forces" or more simply: rich country, strong army. It is perhaps telling that Takaichi's government is introducing an economic blueprint that calls for a "Strong and Rich Japan". ([Nikkei Asia](#)).

Invest, invest, invest

Japan's Cabinet has now passed the [Honebuto](#), or the government's new Basic Policy on Economic and Fiscal Management. The document calls for moving beyond "conventional thinking", "putting an end to the trend of underinvestment", and "responding to the era of intense global industrial policy competition". The cornerstone of Japan's proposed policy shift seems to be Y370trillion of public-private partnership investments with the government having identified 17 key strategic areas from AI and quantum, to defence, aviation, shipbuilding, critical minerals, and others. These investments are seen as crucial to boosting potential growth. Indeed, the decline in Japan's potential growth rate has not principally been about demographics as is commonly assumed, but about the collapse in the capital stock ([Figure 1](#)). Japan went through decades of very low investment after the 1990 asset bubble burst. It now wants to return to being an investment-driven economy.

Finding the funding: two big shifts

Japan is not alone in its bid to reindustrialize and rearm, but compared to peers, Japan arguably has less fiscal space. Japan's debt to GDP hovers above 200%, while Germany and Korea are under 50%, making it easier for them to pursue significant expansion in their fiscal deficits. While Germany is moving from austerity to expansion, Japan will have to find spending room while continuing to chart a fiscally sustainable path. This will hugely inform Japan's policy approach. There are two important shifts Japan appears to be making to enable fiscal spending.

First, changing the fiscal target. The Honebuto has confirmed a shift in Japan's core fiscal target away from the primary balance. The primary balance will now no longer be a mechanical target; it will be managed over many years, and be allowed to deteriorate temporarily as required. The new fiscal target is now the "stable reduction of the total debt-to-GDP ratio for national and local governments." The second notable and enabling shift is the pursuit of an "asset management nation" strategy. Japan is looking to more actively mobilize domestic savings to fund

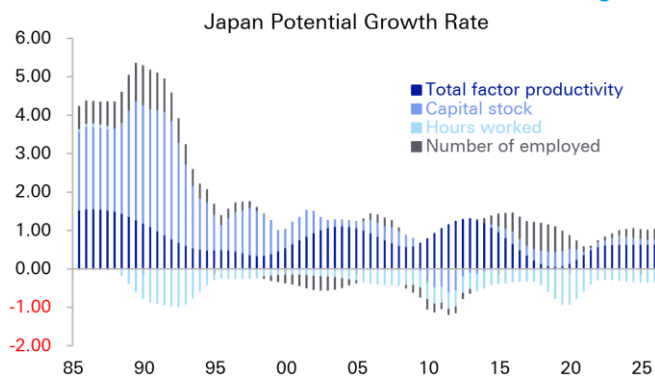


investments and help keep government financing costs manageable. Each of these shifts could have significant long-term implications for the currency.

The new fiscal approach means keeping r below g

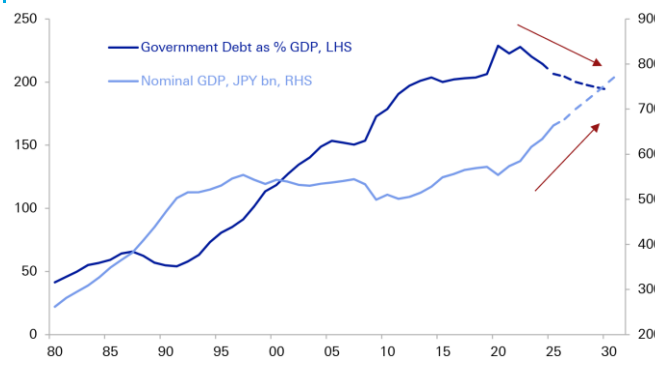
We think debt sustainability arithmetic has become key to understanding Japan's economic policy. Recall that: $\Delta d_t \approx (r-g)d_{t-1} + pd_t$ where d is the debt-to-GDP ratio, pd is the annual primary deficit, g is the nominal GDP growth rate, and r is the average interest rate paid on outstanding debt. The formula implies that Japan can run primary deficits and still achieve a declining debt-to-GDP ratio, as long as r is below g . In other words, Japan's cost of funding needs to be below nominal GDP growth for Japan to have fiscal deficit space. One might consider Abenomics' bid to return Japan to inflation as crucial to putting it on a path to fiscal health. As Figure 2 illustrates, the peaking in Japan's debt-GDP ratios has gone hand-in-hand with the revival in Japan's nominal GDP.

Figure 1: Japanese potential growth has declined on decades of underinvestment – this could now change



Source: Deutsche Bank Research, Bank of Japan

Figure 2: Rising nominal GDP has helped bring down debt-to-GDP



Source: Deutsche Bank Research, Haver Analytics

The extent of fiscal deficit space will crucially depend on the interplay between r and g with Figure 3 illustrating the potential space created by different $(r-g)$ scenarios. We note that the Honebuto outlined a 3% nominal GDP target (g) for Japan till 2040, made up of a 2% inflation target, and 1% real growth target, with aims of increasing it. This perhaps sheds light on why the government has become sensitive to the 3% level on 10Y yields. To be sure, Japan's overall cost of funding remains much lower for now at near 1% given just a fraction of debt is refinanced every year, but there are points of vulnerability. While the weighted average maturity of debt outstanding is around [9 years](#), BoJ holds close to 50% of outstanding JGBs against which they pay interest on excess reserves to banks. This means that almost half the consolidated liability base is in fact linked to front-end floating rates. Therefore, not only are long-run government bond yields above 3% not likely to be sustainable, front-end rate hikes also contribute to consolidated fiscal costs. Very roughly, every 100bps of rate hikes costs ¥5tn or around 0.7% of GDP.

The $(r-g)$ formula not only explains the importance of understanding the drivers of ' r ', but also explains the sensitivity to ' g '. It helps one appreciate why policymakers remain so worried about upsetting the balance on hard-won nominal GDP growth. Indeed, Japan tends to react very cautiously and dovishly



to any signs of growth risks (tariffs, Iran war) and has preferred to let inflation run hot and durable before tightening.

Japanese policy makers face a fine balancing act in optimizing this formula: dovish monetary policy may help to support g , but is also driving up r via inflation expectations reflected in long-run bond yields. Conversely, hawkish monetary policy could hurt ' g ' and also raise the cost of ' r ' via reserve payments. In light of this, the focus on trying to manage long-end yields makes more sense.

How might Japan do this? Changes to supply-side of debt management have been tried via reduced long-end issuance; the government may now be trying to more actively influence demand.

An Asset Management Nation

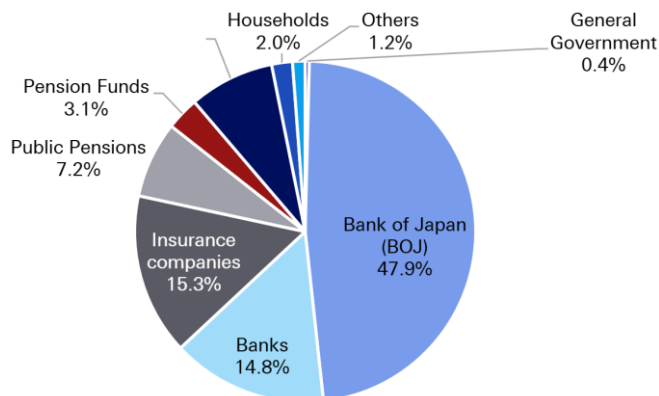
Very simply: Japan needs to find more buyers of JGBs (Figure 4). It is natural to look at where its own domestic capital has been invested. The answer points to two big places: in cash and overseas. Japan has close to USD5tn in portfolio assets held abroad across central bank, household, pension fund, insurance, and bank balance sheets, more than the size of GDP. Indeed, our research has long shown that one of the best metrics of fiscal space is a country's external position, and Japan may need to draw on it. Intuitively, the bid for strategic autonomy around the world should coincide with a growing "nationalism of capital." If industry, energy, defence all need to come home, so might the capital to fund it. We have explored this in earlier research [here](#). More of Japan's money could now be needed at home. Finance Minister Katayama has been explicit that Japan's ambitions to create a growth-oriented economy will coincide with a "strong stock market and positive interest rates" in which domestic citizens should reap benefits. The other perspective is they will help provide the money.

Figure 3: Keep g above r is key to Japan's fiscal space. Japan does not want to return to the lost decades

Scenario	Nominal GDP	Nominal Rates	Debt Stabilizing Primary Balance	Scenario Description
1	4.5%	1.75%	-5.5%	2025 Regime
2	3.0%	2.0%	-2.2%	Japan manages rise in yields to keep them below g
3	0.0%	1.5%	1.5%	Return to Lost Decades (1995-2010)

Source: Deutsche Bank Research, Bloomberg Finance LP

Figure 4: Japan will need to find more buyers of JGBs



Source: Deutsche Bank Research, Ministry of Finance

Where the money could come from

Pension fund and household assets appear to be the two big areas of focus for markets. The government pension fund, GPIF, currently has 50% of its \$1.8tn [portfolio](#) abroad. Meanwhile, households have around 50% of their \$15tn (!) of savings in cash and deposits (Figure 5).



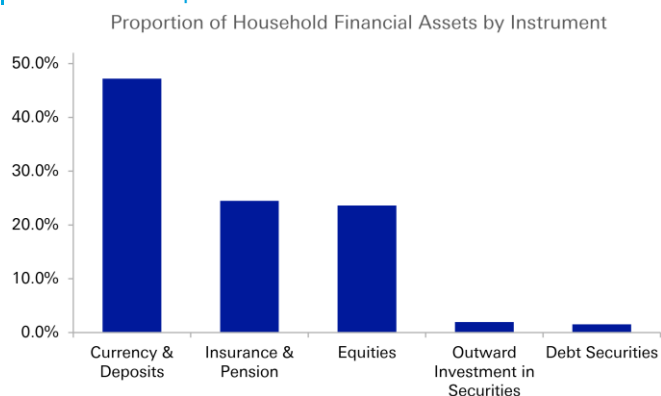
Movement of household deposits into JGBs could be encouraged through a few channels. JGBs may be brought under the NISA tax-exempt investment scheme; the retail JGB offering may be expanded or made more attractive. There were also references in the Honebuto to advancing on-chain finance, with the power of tokenization potentially being tapped to encourage financialization of savings. Mobilizing household savings could be material for yield management given their sheer size. For context, the 1000tn yen of household deposits are almost the size of the entire JGB market. It is not however likely to have any FX impact as it would mostly be an internal domestic flow.

In contrast, a shift from GPIF to invest more in domestic assets could be very material for FX, with PM Takaichi noting this is a key initiative. GPIF's [current policy mix](#) prescribes 25% investment in domestic equities and debt respectively, with +/-6% bands around it. If GPIF simply shifted to the top end of current permissible ranges from current levels (as of March 2026), this could bring around \$200bn back to Japan across domestic equities (\$130bn) and domestic bonds (\$75bn).

The bigger shift to play for however is an early policy review of GPIF's overall asset mix that revises up the domestic asset targets themselves. In [Figure 6](#) we illustrate scope for inflows at different levels of target allocation to domestic bonds alone. Hypothetically, if the target share of domestic bonds doubled to 50%, over USD400bn could flow into JGBs from GPIF. While this may seem extreme before Abenomics in 2012, GPIF had a 67% target allocation to domestic bonds in their policy mix. One could reasonably expect any flows to occur with a glide path, but the scale alone could be significant enough to impact the currency.

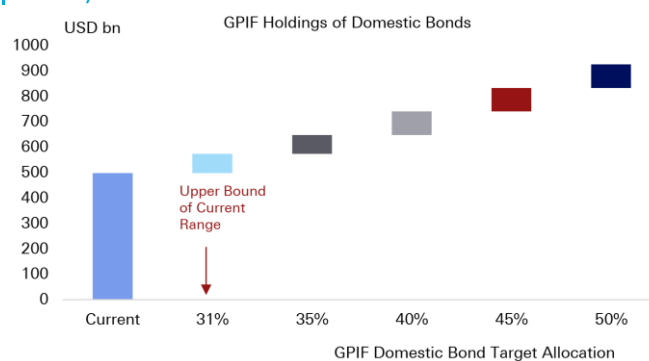
We note that GPIF has a current return objective of a 1.9% real return above nominal wage growth. The extent of any domestic reallocation will depend on how much Japan's domestic assets can deliver that return, or whether government pension returns are made subservient to financing costs for now. Another group that may be called upon to play a bond absorption role could be local banks. This is a source of financial repression has been used extensively around the world and could return where deficit financing demands it.

Figure 5: Japanese households hold half their savings in cash and deposits



Source: Deutsche Bank Research, Haver Analytics

Figure 6: GPIF could be encouraged to bring more money back into domestic bonds



Source: Deutsche Bank Research, GPIF



A shift to yield management could lead to bigger FX moves

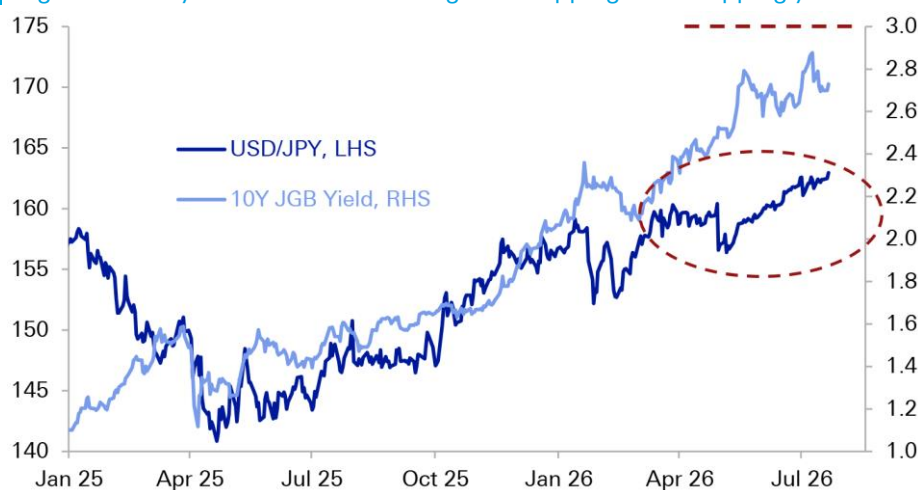
Over the past few months, Japanese policy makers have arguably been solving for stabilizing the currency. Household angst and imported inflation risks have likely encouraged this. Policymakers have engaged in verbal, rate check and actual intervention this year. And, Japan has in fact been successful in keeping USD/JPY near 160 amid the US-Iran war, a major global energy shock, and volatility in Fed pricing.

If, however, fiscal capacity is becoming the biggest and more existential policy criteria for Japan, then incentives may be shifting from FX management to yield management: from capping USD/JPY to capping 10Y yields and borrowing costs.

Suppressing interest rate volatility may come at the cost of amplifying it in FX, depending on the specific levers that the government chooses to pull. If GPIF is mandated to bring a larger share of their assets back into domestic assets, this would be very bullish for the JPY. We have previously identified GPIF as the biggest weapon in Japan's external arsenal. Equally, if BoJ is encouraged to accelerate tightening to manage inflation risk premium in long-end yields this too could be bullish JPY. If, however, the BoJ is [coopted](#) to support yield management through a more active role in the bond market, potentially resuming JGB purchases, this would be negative for the JPY. Similarly, if BoJ is encouraged to keep monetary policy dovish to prioritize (r-g) considerations this too would be bearish JPY.

Ultimately, the direction of travel for USD/JPY will depend on which measures the government favours. Clearer and more imminent signs of GPIF repatriation would favour the stronger yen tail, while pressure on BoJ to manage rates and yields again could open the weaker tail. Efforts to suppress yield volatility could come alongside bigger moves in FX from here. This favours higher JPY vols. We note USD/JPY vols are near multi-year lows with a 1Y USD/JPY straddle breaking even roughly below 150 or around 170.

Figure 7: Policy focus could be shifting from capping FX to capping yields



Source: Deutsche Bank Research, Bloomberg Finance LP



Appendix 1

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Time billionaires want more risk. Is AI to blame?

July 22, 2026

Executive summary

Are you a time billionaire? Some say “yes” if they expect to have more than 1bn seconds (or 31.5 years) in front of them. And as the number of younger investors increases, and their time horizons span 30, 40, or 50 years, they want two things: 1) More risk than they have today, 2) AI to manage their money and trade for them.

So, is AI convincing investors, particularly young people, to increase the current level of their investment risk? In this report, we use proprietary data from our dbDataInsights team to break down exactly who wants more investment risk and over what timeframe. We highlight that both Americans and Britons under the age of 55 are particularly keen to increase (relative to today) both their risk tolerance and their dependence on AI to manage their portfolio. The younger the age group, the more they say they will increase their risk over the coming 1, 3, and 5 years.

We then discuss various factors that are contributing to the increased desire for both risk and AI advice. Behavioural psychology feeds directly into the equation, particularly as ‘social’ investing has become a trend. ‘Aspiration inflation’ is also very tangible but we point out that young investors (and their risk appetite) are the product of the environment that has been created for them. In the end, there are pros and cons.

Several conflicts and conclusions come out of our work. The first is that it is intriguing that consumer sentiment is so low at the same time that the average person wants to take on more investment risk. A second is that this desire for risk at a household level comes as experts increasingly point to intensifying global systemic risks. A third is that the willingness to turn to AI for investment advice comes as the social winds against AI more broadly are growing in strength.

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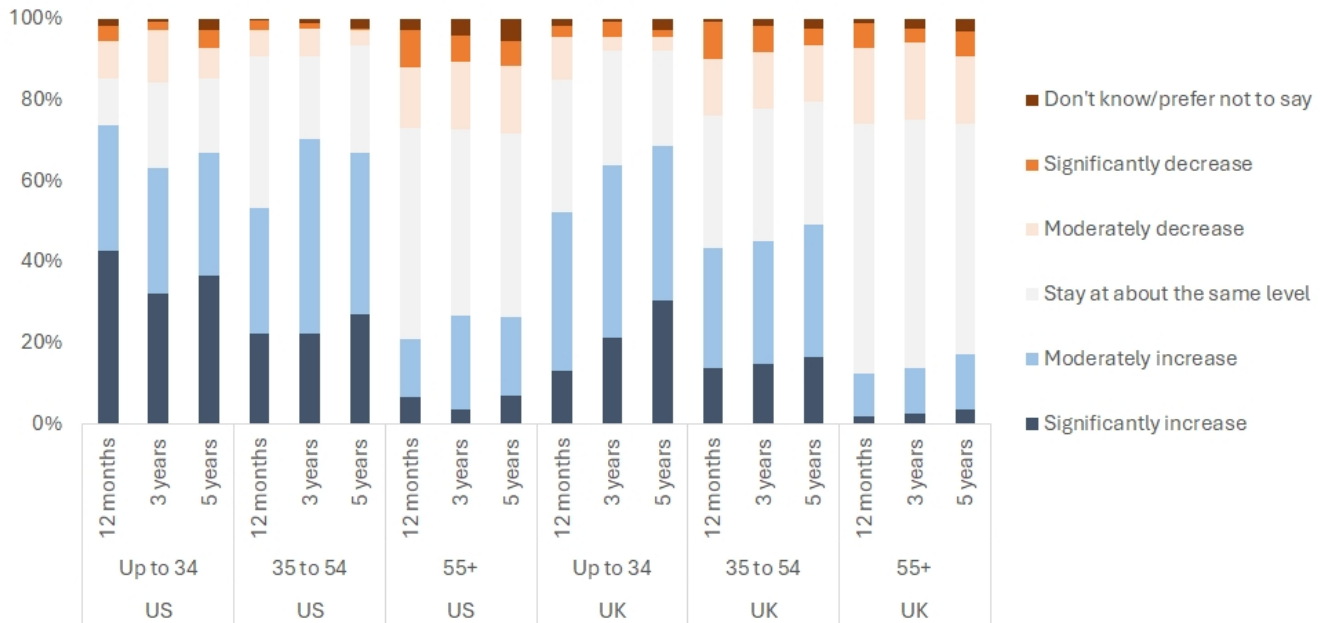
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An appetite for risk

At one level, it seems strange that six years into a decade that has been so volatile, people want to take more, not less, risk with their investments. Indeed, the following chart shows that all age groups (up to the 55+ group) want to substantially increase their investment risk over a 1-, 3-, and 5-year time frame.

Figure 1: Over each of the following timeframes, do you expect your level of risk-taking in relation to your main investment portfolio to...



Source : dbDataInsights

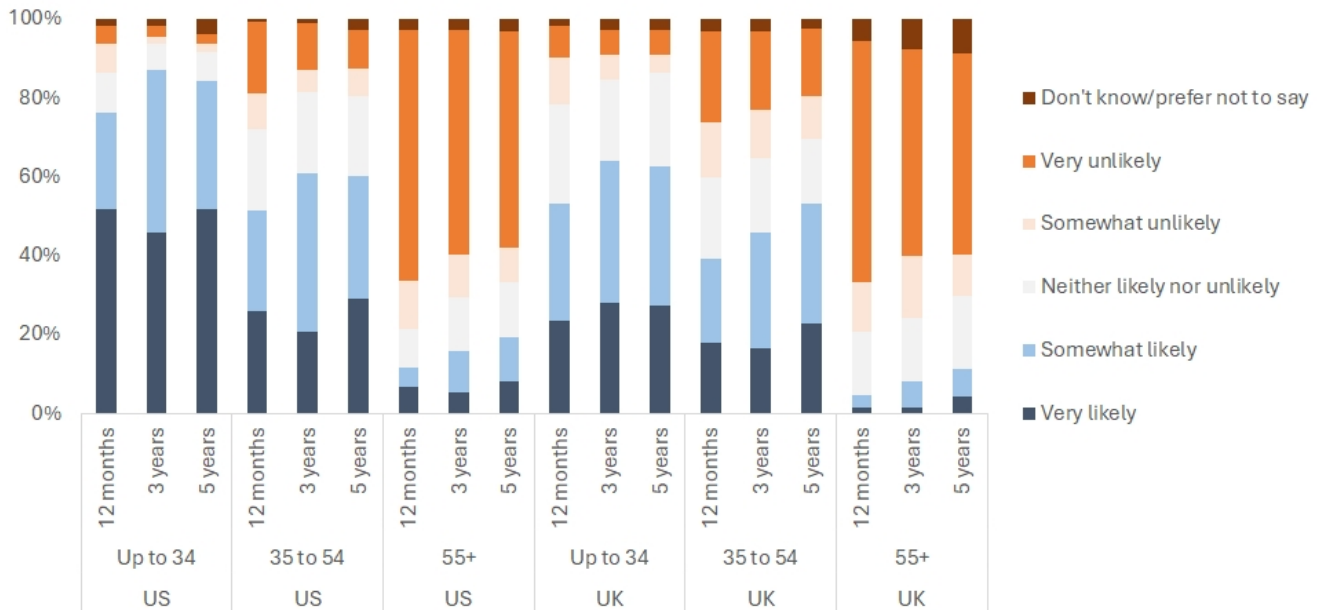
A stark observation is the age gradient. In short, the younger you are, the more likely you are to want to take even more investment risk in the future. The effect is most pronounced in the US but the trend is the same in the UK.

Is AI encouraging risk?

People are not only using AI to find investment advice, but also their shift towards it is accelerating. As the following chart shows, people under the age of 55 expect to show a strong desire to move to an AI managed investment method over the next 1, 3, and 5 years. This phenomenon is particularly aggressive in the US but still pronounced in the UK.



Figure 2: Within the following timeframes, what is the likelihood you will switch from your current investment management method (self-managed or human advisor) to a Generative AI-managed solution?



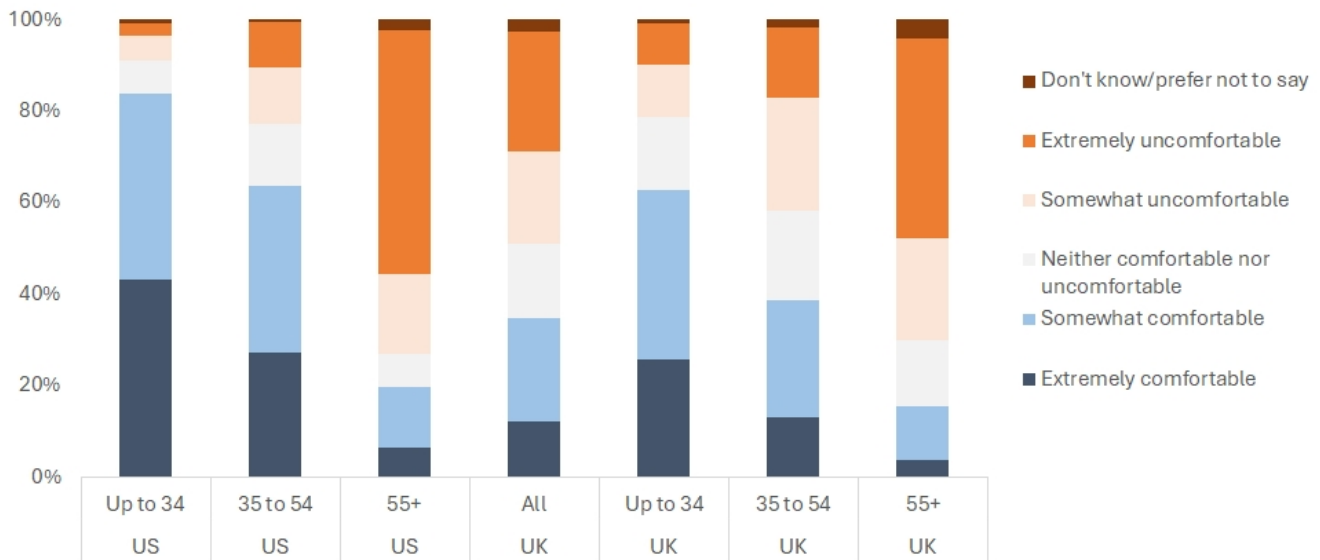
Source : dbDataInsights

Will AI trade for you?

Regardless of how AI is or is not encouraging risk, people are certainly increasingly comfortable with AI automatically making trades in their investment account. For example, Americans up to 34 years old are happy giving AI responsibility for automatically executing trades on their account by a ratio of 10:1. In the UK, the same age group is happy with AI by a ratio of 3:1. The ratio falls as people age.



Figure 3: How comfortable would you be with a generative AI automatically executing investment trades on your behalf, if you had set the parameters (e.g., buy/sell limits, asset allocation)?



Source : dbDataInsights

Does AI give confidence to time billionaires?

AI may amplify risk appetite, particularly for young people, because it reduces the friction around decision-making. Everyday investors can now ask an AI tool to explain asset allocation, compare funds, summarise macro risks or model hypothetical scenarios. Even if the output is imperfect, the interaction can create a sense of fluency. That matters because people are more likely to take action when a complex problem appears legible.

Some research supports this direction of travel. The World Economic Forum has previously highlighted that Gen-Z and millennial investors are embracing AI and digital platforms for financial advice and education, and that younger investors place greater importance on intuitive, technology-enabled wealth-management systems. The WEF has noted that 45% of Gen-Z and millennials began investing in early adulthood, compared with 15% of Gen X and baby boomers.

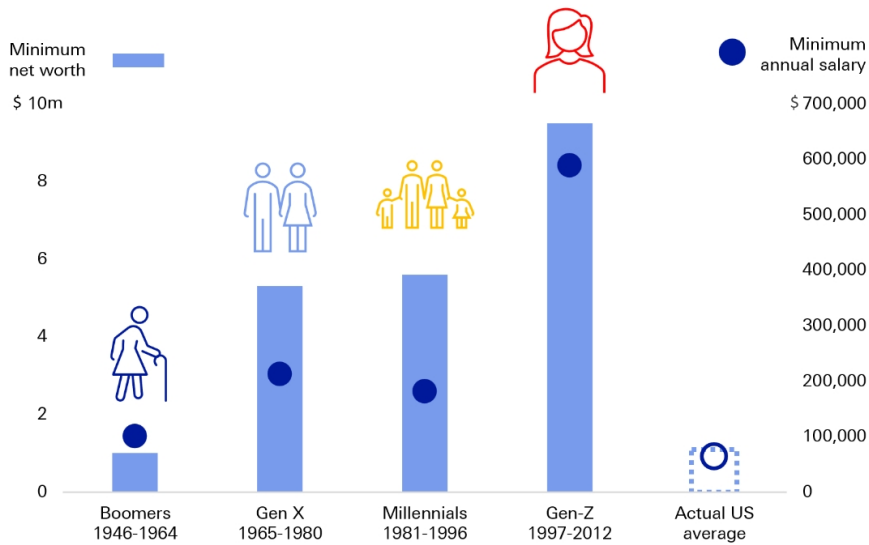
What else is driving greater risk taking?

Aspiration inflation

Some theories suggest the ease of 'aspirational connections' and 'dream scrolling' on social media has fuelled sky-high expectations for Gen-Z. In fact, one survey shows that Gen-Z thinks they need twice as much wealth and triple the annual salary as the previous generation to achieve "financial success".



Figure 4: Survey: "What does financial success look like?"



Source : Empower "Secret to Success" study: [Secret to Success](#) Empower, November 2004, Deutsche Bank. (Note: Average salary as of 2024, average net worth as of 2022)

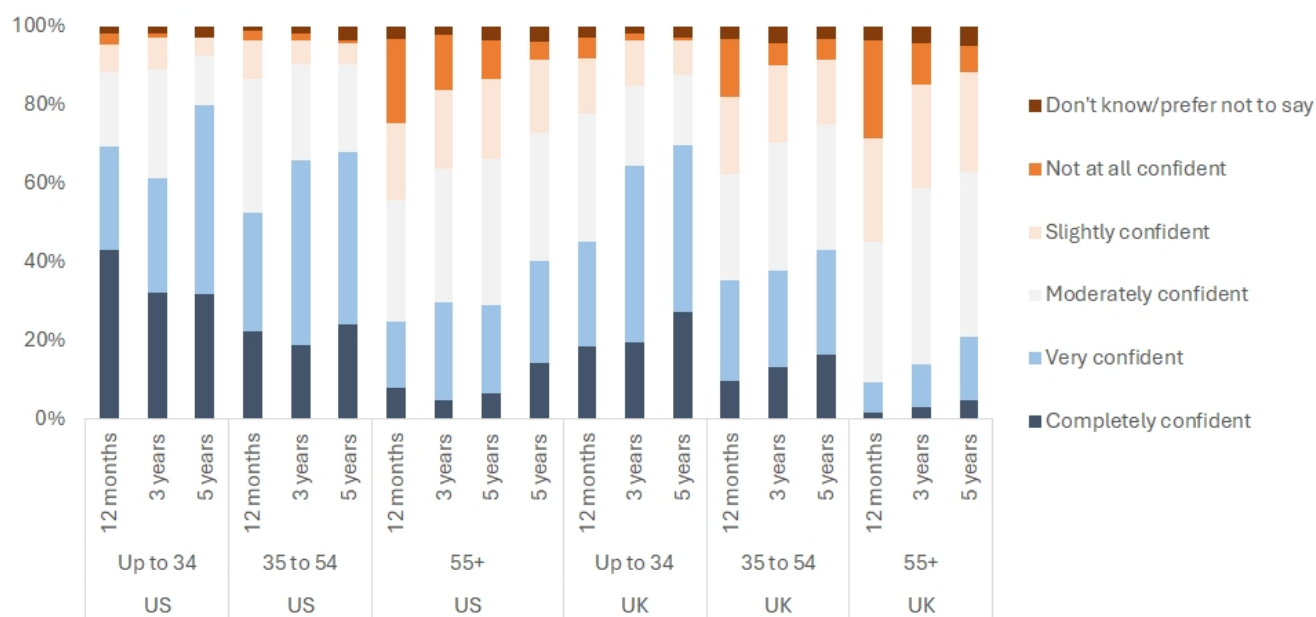
Of course, the flip side of the same coin is that younger people are increasingly worried about housing affordability, education costs, uncertain retirement prospects and perceptions of stagnant real income growth. As a result, this may encourage them to see traditional saving as insufficient. If higher-risk investments can appear less like speculation and more like necessity then risk-taking is not a rejection of prudence, it is an attempt to close an opportunity gap.

Bullish return expectations

The chart below shows that people in all age groups are confident that their investment returns will meet or exceed their expectations. Again, age groups matter. Younger people are overwhelmingly confident of strong returns across 1, 3, and 5 years.



Figure 5: Over the following time frames, how confident are you that your main investment portfolio will meet or exceed your expectations?



Source : dbDataInsights

Rallying against the risk of doing nothing

Portfolio risk appetite is not inversely related to global risk perception. In a risky world, households may feel that the cost of remaining conservative is also rising. If inflation, technological disruption and asset-price dispersion are perceived as threats, risk-taking can become a hedge against being left behind.

Young investors are a product of their environment

Many younger investors have been socialised as they came of age in a market culture shaped by mobile trading, social media, crypto assets, zero-commission brokerage, online financial influencers and rapid cycles of boom-and-bust narratives. Investment decisions are discussed, compared, gamified and sometimes celebrated in digital communities. This can make risk-taking feel less isolated and less exceptional. In turn, this amplifies psychological issues of social proof, herding, and fear of missing out. In the end, the emotional cost of failure is lower.

Anthropological work on risk argues that societies do not merely discover risks; they classify, narrate and prioritise them. Mary Douglas's cultural theory of risk argued that communities highlight different hazards depending on their social organisation, values and forms of trust. What counts as a salient risk is, in part, culturally produced.

Studies also emphasise that risk-taking is often mediated by rituals, expertise and tools that make uncertainty feel manageable. In modern finance, models, apps, dashboards and AI systems can serve a comparable psychological function. They do not eliminate uncertainty, but they make uncertainty appear structured and therefore more actionable.



The increased ease at which people can access risky products raises the idea of 'prospect theory'. This highlights that people become risk-seeking when they perceive themselves to be in the domain of losses. They then can be more likely to gamble to return to their reference point rather than accepting a smaller certain loss. An investor who is underwater relative to a recent peak, or who feels they have 'missed out' relative to peers who bought earlier, is, under this framework, more likely to reach for a higher-variance trade to close that gap.

Other psychologists have highlighted how sensation seeking can correlate with risk-taking in gambling, extreme sports, and financial speculation. It tends to be higher in younger populations — a pattern consistent with our survey findings.

Do consequences feel lower than in the past?

Risk-taking is not simply a function of appetite; it is also a function of perceived downside protection. Behavioural researchers analyse how humans recalibrate their risk-taking in response to changes — real or perceived — in the safety of their environment.

Consider the Peltzman effect. This describes how people can take greater risks if they know they are protected from downside risk. Evidence for this has been argued by researchers in various domains:

- Some drivers were found to be involved in more accidents when their cars had greater safety features
- The introduction of HANS safety devices in Nascar was argued by some to be related to an increase in more aggressive racing
- Skydiving/wingsuit users readily admit that the increase of the quality of equipment has made them push the limits of the sports

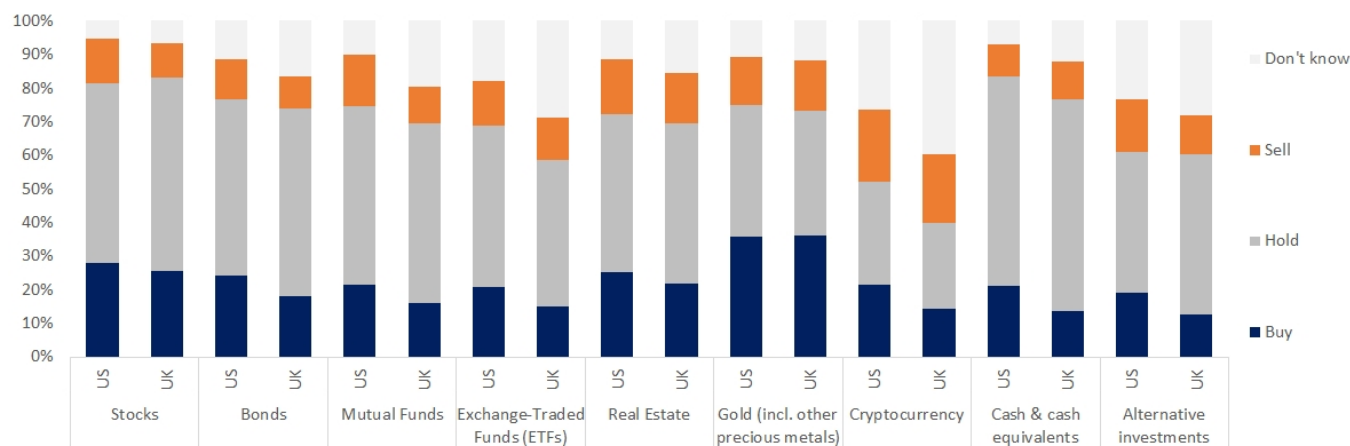
Application to finance and investing

Post-2008, investors have become used to policymakers increasing their response to shock events. At the same time, financial innovation has created the perception in some that risk is either lower or can be managed better. Consider that every major US market decline since the 2008 financial crisis has been followed by a relatively timely recovery to new highs. A generation of investors has now only ever experienced dips as buying opportunities — a very different experience from the multi-year drawdowns and stagnant markets of the 1970s or the 2000s.

Hence, the number of people who say they treat uncertainty as an opportunity to take on traditional risk assets. The following chart shows that people think that going into traditional 'risk' assets is the best strategy during periods of uncertainty. In fact, the average American wants to buy, rather than sell, equities during uncertain periods by a ratio of 2:1. In the UK that ratio rises to 2.5:1.



Figure 6: In any period of economic uncertainty, what do you think is the best strategy for each of the following investment asset categories?



Source : dbDataInsights

Of course, investors may agree that the world is risky while also believing that AI, technology stocks, private markets, crypto assets or thematic funds offer the best path to future security.

In addition to general faith in policymakers, several other factors may be encouraging risk:

- Deposit insurance, broker-dealer protections, and post-2008 regulatory reforms have reduced certain tail risks that were live concerns a generation ago, such as a brokerage or bank outright failing with a customer's assets inside it.
- Diversified, low-cost index products have made it structurally harder to lose everything through a single-stock mistake compared with earlier eras of concentrated, high-fee retail investing.
- Information is more abundant and more immediate, which may create a (not necessarily accurate) sense that fewer unknowns remain and that outcomes are more predictable.
- The practical consequences of experimentation have been reduced by fractional shares, low-cost funds, commission-free trading and the ability to start with small sums. A young investor can now test higher-risk assets without the same minimum account sizes or advice costs that once limited participation.

Implications for wealth managers and policymakers

Younger investors are unlikely to be persuaded by generic warnings about risk. As such, AI should be framed as a support mechanism rather than a substitute for suitability, behavioural coaching and long-term planning.

For policymakers, the issue is not whether AI-driven investment engagement is good or bad. It is how to ensure that increased access is matched by adequate disclosure, financial literacy and safeguards against misleading personalisation. If AI makes investing easier, it should also make risk more understandable.



If the current environment of ever greater risk taking continues, there are several key concerns that may exacerbate the downside risks:

- Leverage positioning: US margin debt has pushed to over 4.5% of GDP. That is well above prior peaks in 2021, 2007 and 2000. Retail investors can easily access leverage through brokers, ETFs, or other specialised products.
- Speculation has broadened: New avenues for risk taking include the ease of options trading (particularly same-day and binary options), leveraged ETFs, and meme assets, prediction markets and sports gambling.
- Safety buffers are shrinking: Household savings rates are falling in the US and other key countries



Conclusion

The common thread across not only financial decisions, but also auto safety, extreme sports, medicine, and gambling is that humans do not hold risk-taking constant when their environment changes. They recalibrate around a target level of risk, and that target appears to rise when protective factors (real or perceived) are introduced. Applied to investing, this suggests the relevant question is not just, 'Why are people taking more risk?' but, 'What changed in their perceived environment that permitted it?'

AI may be one driver of this desire for risk but there are others, too. Yet, it is concerning that perceived consequences may have declined faster than the actual potential consequences. AI can widen this gap if it gives users the impression that risks are fully understood or controllable. Our data points to pros and cons: AI may democratise investment engagement, but it may also compress the distance between curiosity and action.

The younger cohorts in our survey may have greater time horizons, but they also face housing, employment and debt pressures that can constrain their ability to absorb volatility. The next generation of advice must therefore be both more digital and more behavioural.



Appendix 1

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El Nino: The next supply shock?

July 20, 2026

Current forecasts point to one of the strongest El Nino events on record in Q4. That matters because El Nino events can act as a multi-dimensional supply shock, with a causal link to higher food and energy prices, alongside broader supply-chain disruption. This would be problematic at any point in time, but it's a particular issue today because of existing supply-chain stresses and the backdrop of above-target inflation. So a very strong El Nino event would be another negative supply shock, at a point when the global economy has limited room to absorb one.

This paper explores what an El Nino is, and why a very strong one later this year would be so concerning. It also takes a look at what happened on previous occasions, and the channels through which it affects the broader economy and financial markets.

We also consider the potential consequences this time round. Much will be weather dependent, but the precedents from the 1970s show how an El Nino event can coalesce with other shocks to keep inflation persistently above target, just as we've seen in recent years. Moreover, this is particularly damaging for many EM economies, where food can make up as much as a third of consumer expenditure.

Authors

Henry Allen

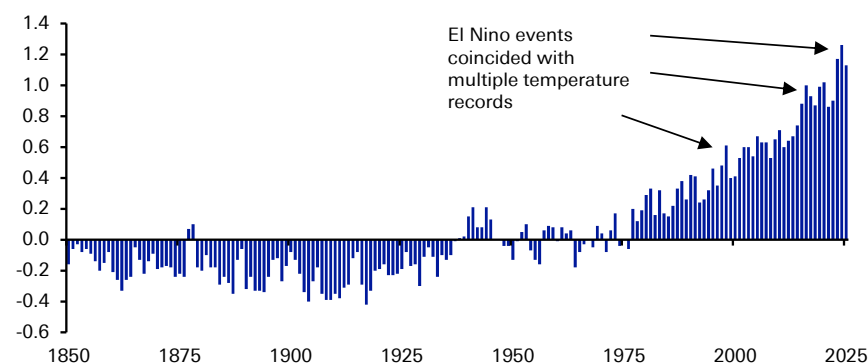
Macro Strategist



What is an El Nino?

- An El Nino event is the warm phase of the El Nino-Southern Oscillation (ENSO), which is a variation in winds and sea surface temperatures in the Pacific Ocean. This oscillates between warming phases (called El Nino) and cooling phases (called La Nina).
- The warming El Nino phase occurs when there are unusually warm sea surface temperatures in the eastern Pacific. These warmer waters mean that the Pacific jet stream moves south relative to normal, which creates changes in weather patterns and ecosystems.
- We can see the effect on global temperatures too. Even as temperatures have been rising anyway over recent decades, the record years have often coincided with El Nino events, including 1998, 2016, and 2024.
- Unfortunately, El Nino events are correlated with a higher frequency of natural disasters, with more people affected by disasters in El Nino years relative to neutral years. Flooding is one that sees a particular increase, but it is also associated with temperature changes too.
- The effects of an El Nino vary considerably by region. In the US, it normally coincides with wetter conditions in the southern US along the Gulf Coast. In stronger episodes, this can extend to California and the Southwest too. But in other regions it is more likely to cause drought, like Australia, whilst India sees weaker monsoon rains.
- In practical terms the main effect of an El Nino is disruption to food production. But there are other effects too. El Ninos have been correlated with the spread of climate-sensitive diseases, and one study found that the El Nino in 2015 contributed to the spread of the Zika virus since it created conditions beneficial for mosquito-borne transmission.¹ Furthermore, natural disasters like flooding increase the frequency of poor hygiene conditions or undernutrition, so there is an impact there too, and El Ninos have been linked to many famines and even conflicts.
- For markets, an El Nino is capable of hitting agricultural production, hydropower, shipping, and even has effects on public health and political stability. In other words, it operates as a multi-channel supply shock.

Figure 1: Global land and ocean average temperature departure with respect to 1901-2000 average (degrees celsius)



Source : National Centers for Environmental Information, Deutsche Bank

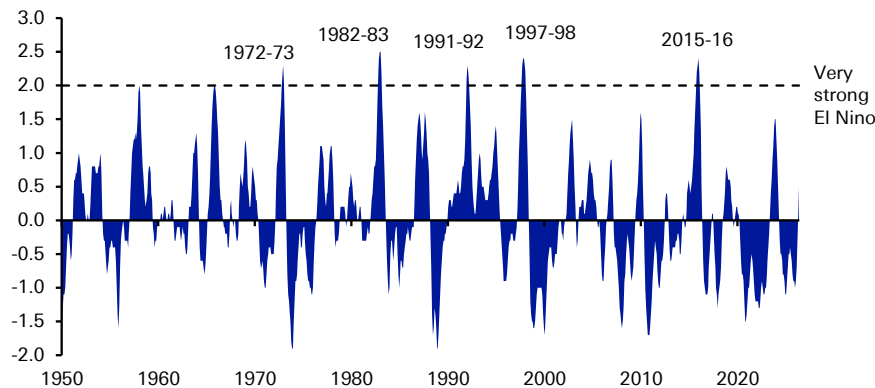
1 Caminade C, et al. Global risk model for vector-borne transmission of Zika virus reveals the role of El Niño 2015. Proc Natl Acad Sci U S A. 2017 Jan 3;114(1):119-124.



What's particularly bad about this one?

El Nino cycles typically repeat every few years with irregular frequency. However, today's episode is particularly concerning because forecasts suggest it will be a very strong one, ranking among the most serious in recent history.

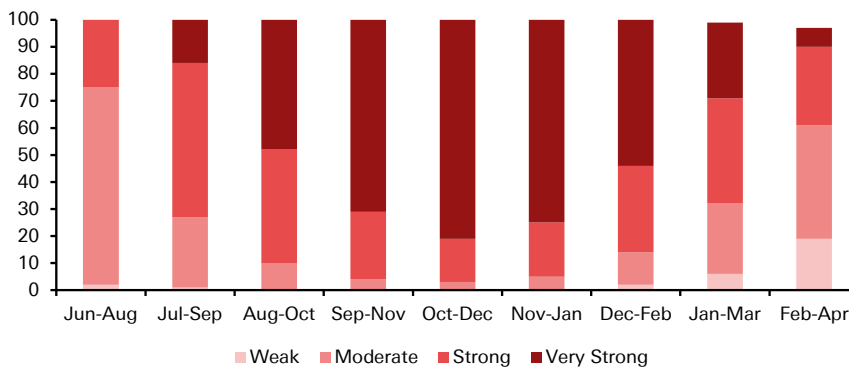
Figure 2: Relative Oceanic Niño Index



Source : Climate Prediction Center, Deutsche Bank

Among others, the US Climate Prediction Center have said this could be one of the strongest in over 75 years, with an 81% chance of it reaching the "very strong" threshold over Oct-Dec. It even has a 73% chance of reaching the "strong" threshold of 1.5C as soon as the July-September period.

Figure 3: US Climate Prediction Center forecasts (%) for El Nino



Source : Climate Prediction Center, Deutsche Bank

A very strong El Nino means it's more likely to shift rainfall and temperature patterns across multiple regions simultaneously. And from a macro perspective, the risks are non-linear. For instance, a modest food shock can be absorbed via inventories, imports or substitution. But a large shock propagating across multiple countries can't be absorbed as easily, with second-round inflation effects more likely to result.

The timing is particularly bad right now, because the global economy is already facing significant supply-chain stress. Most notably, the closure of the Strait of Hormuz has led to a spike in oil and gas prices this year. And even if the El Nino is temporary, the resulting inflation shock will push inflation higher at a time when it's



already above target in many countries.

These risks are already being considered by policymakers in decision making, particularly in emerging markets. Indeed, this year has seen central banks in India, Peru and Colombia all reference El Nino in official publications.

What happened on previous occasions?

2023-24: Supply-chain disruption and record temperatures occurred with the last El Nino.

- The most recent El Nino offered a clear example of what can happen, and this didn't even hit the "very strong" threshold.
- 2024 was the warmest year on record, and the episode saw severe droughts in parts of the world, including parts of South America, along with severe flooding in East Africa and South America, consistent with the typical redistribution of rainfall during El Nino.
- Meanwhile, the Panama Canal had its 3rd driest year on record in 2023, meaning that authorities had to cut shipping passing through. The disruption to commercial shipping demonstrated how El Nino is more than just a food price story.

2014-16: The last "very strong" El Nino had broader food security and public health implications

- Just as in 2024, the El Nino of 2016 also saw the biggest upward global temperature anomaly to date.
- Ethiopia suffered its worst drought in decades, leaving millions of people in need of food assistance. Moreover, the Pacific hurricane seasons saw 16 hurricanes in each of 2014 and 2015, the joint-highest number of record.
- One study found that El Nino was a contributing factor to the spread of the Zika virus, since it created conditions beneficial for mosquito-borne transmission.

1997-98: The "very strong" El Nino also had major climactic and public health implications

- 1998 was another record year for global temperatures, and included many extremes around the world. For instance, San Francisco saw its wettest rainfall season since 1889-90, whilst Peru saw 16 times its average rainfall.
- Meanwhile, after major flooding in east Africa, there was an outbreak of Rift Valley fever in Kenya, Somalia and Tanzania. The outbreak also led to major losses of livestock too.

1982-83: The El Nino occurs when high inflation is still lingering

- By 1982-83, inflation had fallen substantially from its earlier heights, aided by monetary tightening like that seen under Paul Volcker at the Fed. However, the El Nino didn't help matters, as it caused major climactic disruption once again, including flooding in the United States, and droughts in Indonesia and Australia.

1972-73: The El Nino played a role in raising inflation before the first oil shock



- Although the oil shock of 1973 was the primary factor that drove the stagflation of the 1970s, several other shocks including the El Nino had already occurred beforehand. This was significant because collectively, they served to push up inflation and inflation expectations. For instance, US fiscal spending had already risen in the late-1960s, thanks to the Johnson Administration's Great Society programmes and the Vietnam War. Then in 1971, President Nixon ended the dollar's link to gold, ending the system of fixed exchange rates that had prevailed after WWII, and in February 1973 there was a dollar devaluation. Amidst all these inflationary pressures, the El Nino event further drove up food prices.

Even further back

- Back in the mid-17th century, there was a period when El Nino events happened about twice as often on average. This was an unusually turbulent period of global history that's been termed the "General Crisis". Some historians have proposed that climactic changes such as El Nino played a role in fomenting instability.

How does an El Nino affect the economy and global markets: The main transmission channels

A useful way to think about El Nino is a set of different supply shocks via multiple channels: food prices; energy prices; supply chains; EM balance sheets; and growing risk premia resulting from the heightened risk of social/political unrest. For individual economies, the impact depends on their exposure to food in the consumer basket, and the extent of intervention from policymakers, be that via food subsidies or tighter monetary policy to counteract inflation.

1. Food: The clearest inflation channel

- **Direct supply hit:** El Nino shifts rainfall patterns, raising the odds of drought in parts of Asia and Australia, and flooding in parts of the Americas. This causes damage to crops, kills livestock, and leads to shipping disruption.
- **Emerging market exposure:** The impact of a food shock varies considerably across different economies. In the US for instance, food-at-home makes up just 8% of the consumer basket for CPI inflation. By contrast, in several EM economies, over a third of household expenditure is on food, and an IMF paper² found that the inflation response to El Nino was biggest in economies where food made up a higher share of the consumer basket. So emerging markets like India and Indonesia are among those more severely affected. Moreover, food inflation is a problematic category because it's a necessity, and consumers can't easily substitute away from it, as they can with luxury goods.
- **Second-round effects:** Beyond the immediate impact, El Nino also has second round effects on inflation. For instance, governments may cushion the impact of higher food prices with subsidies. That might reduce the domestic hit temporarily, but by further boosting demand and raising fiscal deficits, it pushes the adjustment elsewhere and risks creating further inflation overall.

2. Energy and Hydropower: The less-obvious inflation channel

2 IMF Working Paper: Fair Weather or Foul? The Macroeconomic Effects of El Niño (2015), Cashin, Mohaddes and Raissi.

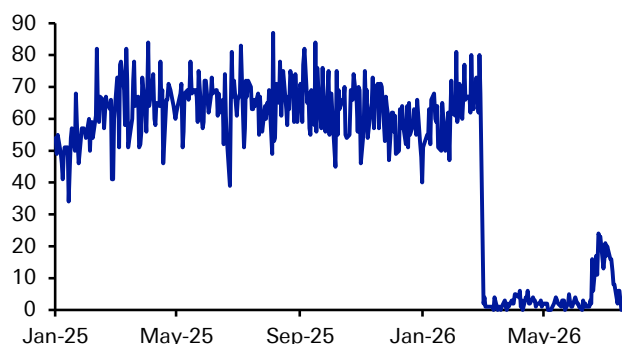


- **Hydropower shortfall:** Lower rainfall in some regions means there's less electricity generated from hydro-electric dams. Moreover, the droughts mean this cut to electricity generation is happening just as heat is raising demand for cooling, and more water is required for irrigation. So there's less supply of energy, just as demand goes up.
- **Fuel-switching channel:** With hydropower output down, this then forces a substitution into other forms of energy, such as gas, coal or oil-fired generation, again raising demand for those other fuels and lifting power prices.

3. Shipping/supply chains

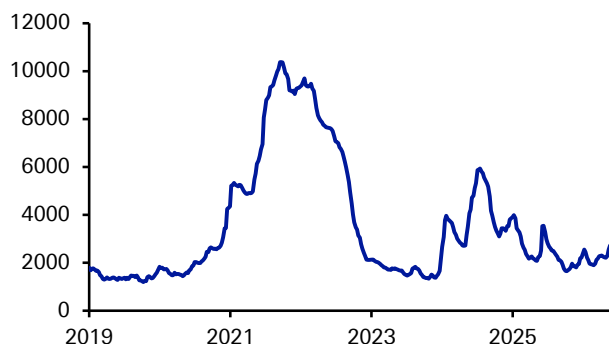
- **Climate effects on supply chains:** The last El Nino in 2023-24 showed how it can impact supply chains, quite aside from a direct impact on the food supply. For instance, Panama had its third-driest year on record in 2023, with rainfall around 30% below average, forcing authorities to restrict the number of ships passing through. This is a point covering around 5% of global maritime trade.
- **Interaction with other chokepoints:** Any supply-chain issues are particularly troublesome today, because the Strait of Hormuz has seen traffic almost completely stop thanks to the US-Iran conflict. So the more transit points that are impacted, the more difficult it is for other routes to absorb any disruption. Indeed, some measures of shipping rates have already doubled since the start of the year
- **Macro Transmission:** If key routes are restricted or closed, then that raises shipping rates further, increases delivery times, and means firms want to hold more inventories against future disruption. So it's another negative supply shock, on top of that seen from higher food and energy prices.

Figure 4: Strait of Hormuz tanker vessel crossings



Source : Bloomberg Finance LP, Deutsche Bank

Figure 5: World Container Index Composite Container Freight Benchmark Rate (USD/40 Foot Box) Drewry



Source : Bloomberg Finance LP, Deutsche Bank

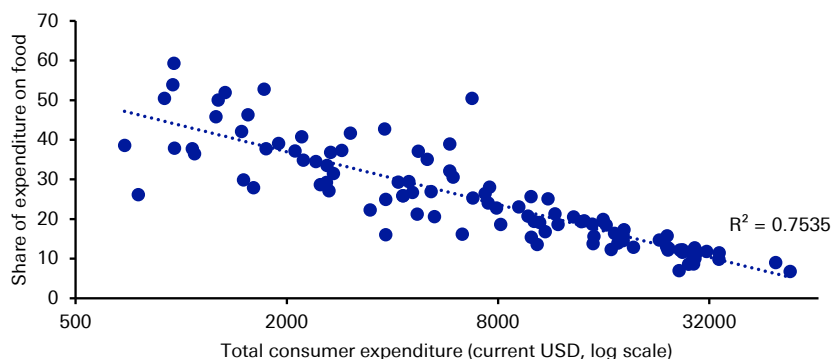
4. EM vulnerability

- **Climate impact:** By virtue of their geographic position, emerging markets are more likely to face the impact of flooding or drought associated with El Nino.
- **Household exposure:** Emerging market economies have more exposure to food prices, so any price shock hits them harder, with food making up over a third of consumer expenditure in several countries.



- **Policy challenges:** This vulnerability makes the policy response to an El Nino event particularly acute. Central banks might raise rates to keep inflation expectations in check and defend the currency, but that doesn't solve the actual supply shock itself. Moreover, if fiscal policy is used to cushion the impact, the shock is then shifted onto public balance sheets, and sovereign risk can become an issue.

Figure 6: Total consumer expenditure vs share of expenditure spent on food, 2023



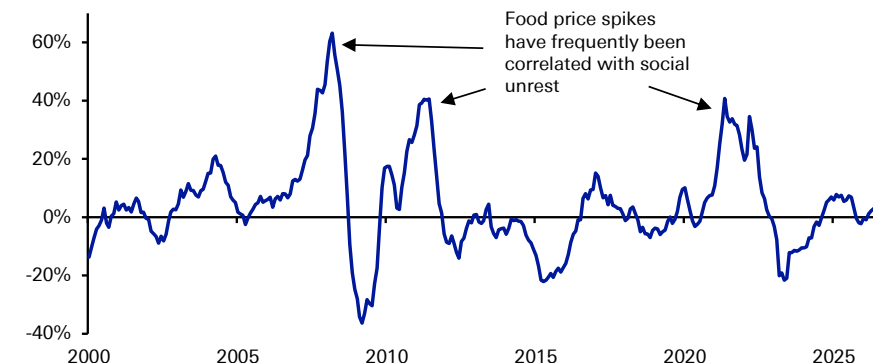
Source : USDA Economic Research Service, Our World in Data, Deutsche Bank

5. Social/political unrest

- **Higher food prices raise the risk of unrest:** Throughout history, higher food prices have often been correlated with political unrest. Clearly it takes many factors to spark unrest, but the squeeze on disposable incomes means that any existing dissatisfaction is magnified. If an El Nino leads to higher food prices, that in turn makes unrest more likely.
- **Recent precedents:** In 2010-11, higher food prices were regarded as a contributing factor behind the Arab Spring. A few years earlier in 2007-08, there was a wave of instability across many EM countries, including Tunisia, Egypt, Cameroon, Ivory Coast, and Haiti. And more recently, the 2022 price shock was a contributor to protests in Ecuador, Sierra Leone, Haiti and Panama.
- **Historical precedents:** Looking further back, several of history's most important revolutions took place against the backdrop of higher food prices. The French Revolution of 1789 came after a succession of harvest failures. The 1848 revolutions in Europe happened at the end of the so-called "Hungry Forties", when there was a severe famine after the failure of potato crops. And in Russia in 1917, food shortages played a key role in fomenting discontent, with supplies hampered in part by the reallocation of farm workers into the military for WWI.



Figure 7: UN Food and Agriculture World Food Price (year-on-year % change)



Source : Bloomberg Finance LP, Deutsche Bank

What does it mean for economies and markets?

A very strong El Nino event would be another negative supply shock when the global economy has limited room to absorb one. The combination of higher food and energy prices will mechanically raise inflation, and broader supply-chain disruption risks creating further price pressures when the Strait of Hormuz has already been blocked.

This is particularly problematic, because experience shows that a prolonged period of above-target inflation raises the risk of inflation expectations becoming unanchored. That in turn can create a self-fulfilling wage-price spiral, as if firms and individuals expect high inflation, then they will set their prices and bargain for wages accordingly.

In several respects, this is reminiscent of the experience in the 1970s. Back then, a series of seemingly unconnected supply shocks (including an El Nino) occurred that collectively kept inflation high. Even though each might have been temporary on paper, implying policy should look through and not over-react, the reality was that the shocks began to feel permanent because they kept on happening. So inflation expectations rose and policy eventually had to react.

Emerging markets are particularly vulnerable as well, with the risk greatest in places where food is a larger part of the consumer basket, and where fiscal space is more limited to accommodate any shock. Moreover, we know from recent history that higher food prices are also linked to higher instances of social unrest, creating further economic uncertainty.

Finally, there is a clear human cost to severe El Nino events. The frequency of natural disasters goes up and there is a greater risk of famine. Moreover, public health issues are often correlated with El Nino events, with several channels in operation, including the heightened risk of cholera after flooding, or the added risk of malaria from conditions more favourable for mosquitos.

Given the forecasts for the strength of this El Nino, this will be an important event to look out for over the months ahead.



Appendix 1

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Open-source AI 101: the battle for the future of AI

July 21, 2026

Open-source versus proprietary AI models in a nutshell

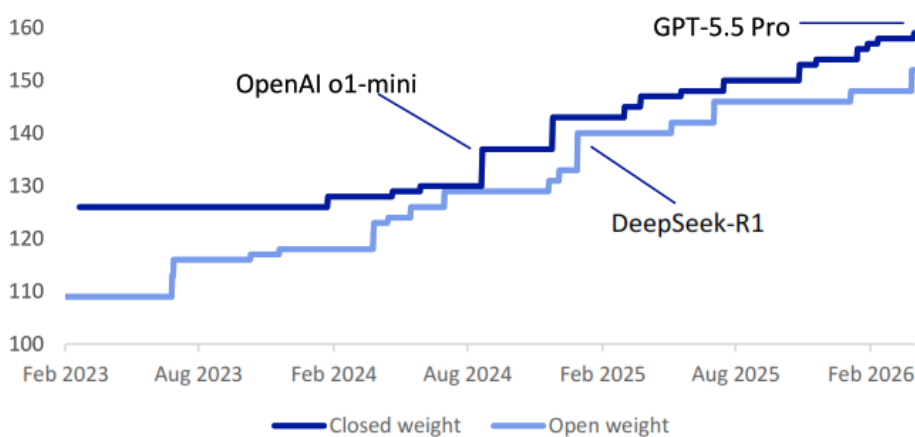
AI-related stocks tumbled at the end of last week on a second DeepSeek moment, as the latest Chinese open-weight AI model reminded investors that cheap, light alternatives are snapping at the heels of the state-of-the-art proprietary models.

In this brief report for generalists, we explain this latest technology format war:

1. Open-what? A new twist on age-old format wars
2. What are open AI models?
3. Why are they disrupting the markets?
4. Why would a company give its model away?
5. What comes next?

The emergence of DeepSeek's breakthrough R1 model in January last year wiped \$600bn off the market value of Nvidia in one day on concerns about whether vast hyperscaler capital expenditure was justified in the light of a "good enough" model developed in China using older chips and trained for a fraction of the cost.

Figure 1: Epoch Capabilities Index score: open-weight models narrow the gap with closed-weight proprietary models (data to end April 2026)



Source : Epoch AI, Deutsche Bank Research

Authors

Adrian Cox
Research Analyst



A similar release on Friday revived those concerns, with the Magnificent Seven (mostly hyperscaler) stocks shedding 1.8 percent and the Philadelphia Stock Exchange Semiconductor Index turning down 1.6 percent, bringing its loss for the week to more than 10 percent. Alphabet's Google, Microsoft, Amazon's AWS, Meta and Oracle are set to spend about \$700bn building AI capacity this year, up by about 70 percent from last year.

Even so, while powerful open models raise questions about the business models of proprietary model makers and semiconductor demand, they may be bullish for AI adoption, applications and enterprise customisation. Greater competition, more efficient models and lower prices may supercharge a new phase of growth.

1. Open-what? A new twist on age-old format wars

Open-source software is nothing new.

- A [study](#) by Harvard Business School in 2024 found that about 96 percent of commercial programs include some code created or distributed for free by public-facing tech forums. Without it, enterprises would pay 3.5 times more, or roughly \$8.8 trillion, to build their software and platforms, it found.

But open AI models are contentious.

- Some critics saying that once released they may be less secure and more likely to be misused. (We mainly use the term "open models" for readability and explain the distinction between open-source and open-weight in the definition section.)
- Some open models have also been [accused](#) of being trained through "distillation", using the outputs of more capable proprietary models to teach a new model, which may breach providers' terms and magnify errors, biases or safety weaknesses in the original models. Critics also wonder out loud about whether some open models have been geared more towards scoring highly on benchmarks – so-called [benchmaxxing](#) – than for solving real-world problems.

Meanwhile, concerns have been rising about whether proprietary models can be [depended on](#).

- Anthropic's most powerful Claude models were recently de facto banned for several weeks on security concerns. Critics also say calls by proprietary model makers for safety regulation would in fact protect incumbents.

The debate is reminiscent of past format battles.

- Every great technology needs a rival. In the "war of the currents" of the late 1880s, Thomas Edison defended his direct-current (DC) empire by portraying the alternating-current (AC) system promoted by George Westinghouse and Nikola Tesla as lethally dangerous—even supporting demonstrations in which animals were electrocuted with AC.
- And remember VHS versus Betamax video cassette players in the late 1970s and early 1980s? VHS did not win by being the better technology: JVC licensed it broadly, manufacturers produced cheaper machines, recording times were longer and a self-reinforcing ecosystem developed around it. Open-source AI could follow the same path. It does not need to beat every proprietary model on every benchmark. It needs to be good enough, inexpensive and ubiquitous.



However, while the contest between open and proprietary AI is often presented as a winner-takes-all format war, the most likely outcome is a combination of the two.

- At the consumer level, it may resemble Apple and Google's Android, a controlled premium ecosystem coexisting with a larger, more customisable ecosystem. And at the industry level, it may resemble the IBM personal computer, where widespread compatibility commoditised the foundational technology while shifting significant value into chips, software, services and applications built around it. Many organisations will use both.

2. What are open AI models?

There are three distribution models. For users, it boils down to control and – to coin a phrase – with greater control comes greater responsibility.

Proprietary, or closed, models are generally the plug and play option, accessed through the developer's application, an application programming interface (API), or a managed cloud services.

- For a rough analogy, think of Apple's iOS operating system or Microsoft's Windows, where the company controls the underlying technology and the terms on which others can use it.
- With proprietary AI models, the developer keeps the model weights, training methods and most details of the training process private. It controls pricing, updates and safety restrictions, and can alter or withdraw access. Prominent examples include OpenAI's flagship GPT models and Anthropic's Claude models.

Open-weight models are open-source-lite.

- Their trained parameters – the billions or trillions of numerical values that encode much of the model's learnt behaviour – are typically released for users to download, subject to an applicable licence. You can run the model on your own infrastructure, adapt or fine-tune it, and technically alter or remove its safeguards.
- But the release may not include the original training data, the complete training code or enough information for you to reproduce the model from scratch. Many of the models referred to casually as open-source – including DeepSeek's R1 and Meta's Llama models, as well as many released by France's Mistral – are in fact more accurately described as open-weight.

Genuinely open-source AI systems are much rarer.

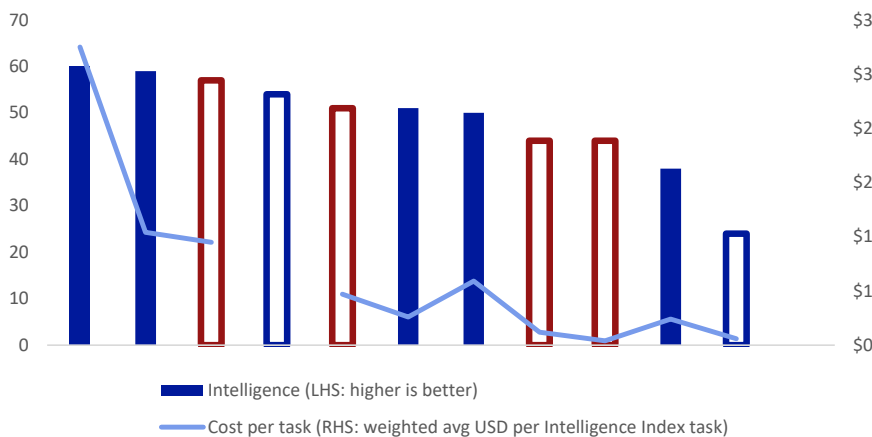
- Under the [Open Source Initiative](#), they should give users the freedom to use, study, modify and redistribute the system, supported by access to its parameters, training and inference code and detailed information about the training data. The grand-daddy of open-source is Linux, a free operating system created in 1991 and found in nearly all cloud infrastructure and the majority of the world's servers and mobile devices.



Consequently, open-source models often offer:

1. **Lower costs, particularly at scale or for specialised tasks.** If you run the model yourself, you can avoid the developer's per-token API charges and can select smaller or compressed models requiring fewer calculations. **The downside?** Downloadable does not mean free: you still have to pay for compute, electricity, engineering, monitoring and maintenance. Open models may also not be as universally capable as the leading proprietary systems and may lack their polished integration with wider products and services.
2. **Greater customisation and control.** You can fine-tune a model for your industry, language or task, rather than relying on a one-size-fits-all service. **The downside?** Although open models are cheaper over the long-run, you need to invest a lot upfront in infrastructure and expertise. You have to take more responsibility for testing, guardrails, cybersecurity, updates and reliability—though managed cloud providers can fill some of the gaps.
3. **Local deployment.** You can install some models on your own servers, or even on "edge devices" such as cars and phones, rather than relying on a connection to the cloud network run by the provider. That makes them potentially faster (less "latency") and more secure as you don't need to share your data. Once you download it, it's yours, and API access cannot be suspended without warning. That is key for some regulated organisations where data can't leave the premises. **The downside?** Not every model can run economically on local hardware, and responsibility for maintaining performance and protecting the system falls on your shoulders.

Figure 2: Intelligence index of selected AI models vs cost per tasks: US (blue) and proprietary (solid colour) models are no longer far ahead of Chinese (red) and open-weight (white with border) models



Source : Artificial Analysis Intelligence Index, Deutsche Bank Research. Data as of July 20, 2026



3. Why are they disrupting the markets?

Open models challenge four assumptions about proprietary models embedded in AI valuations.

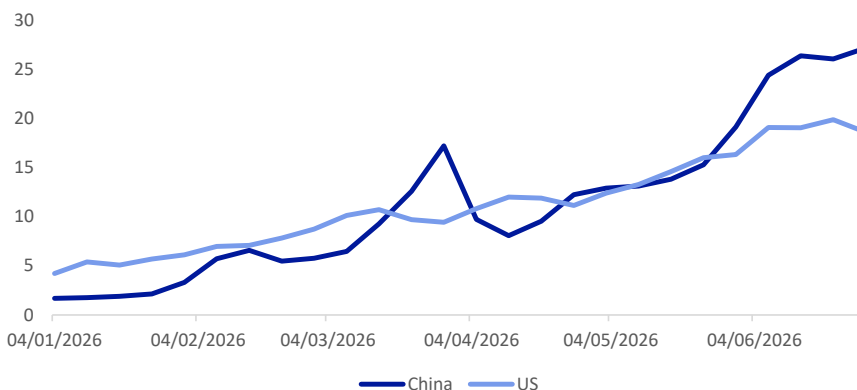
First, scarcity. If capable models can be copied, modified and hosted by many providers, AI models might begin to look like a commoditised software infrastructure rather than a uniquely monetisable product.

Secondly, pricing power. AI agents and “tokenmaxxing” employees are guzzling up vast numbers of tokens, the small units of data that AI models process. As pilots scale up and AI subsidies come to an end, the leading makers of comprehensive models are starting to charge for each token rather than a flat rate. For cost-conscious companies, open models that are good enough for most purposes are looking increasingly attractive.

Thirdly, capital investment. If you can get something like the same performance out of an open model with lighter training and inference (running) horsepower, can the projected spending on chips, networking equipment, electricity and data centres be justified?

Fourth, US technological moat. Chinese open models weaken the perceived durability of US supremacy. The Trump administration says it is in “a [race](#) to achieve global dominance in AI” and has driven policies to strengthen domestic production and adoption and stifle Chinese access to cutting-edge chips and other technology. Chinese models have overtaken US models this year in terms of the number of tokens processed on each by the OpenRouter developer platform.

Figure 3: OpenRouter traffic: trillion tokens going through model platform this year, broken down by country of model



Source : OpenRouter, Deutsche Bank Research

For more, see our recent report “[AI IPOs 101: why they’ll change the boom forever](#)”.



4. Why would a company give its model away?

Open models may be commercially viable because producers do not need to monetise the weights themselves. It's about losing the battle to win the war.

1. Monetising complementary products

- Developers can charge for convenient hosted APIs, premium subscriptions, faster inference, enterprise support, fine-tuning, security and guaranteed service levels. The equivalent of in-app purchases for a free app gives less tech-engaged customers the best of both worlds.

2. Building an ecosystem and standard

- A widely adopted model attracts developers, tools, optimisations, fine-tuned variants and research. All of that can improve the model, lower the cost of acquiring new customers, and make its interface an industry standard.

3. Commoditising competitors' profit pool

- This is a classic case of [commoditising the complement](#): giving away something for free (or very little) to stop your rival from earning excess returns, while increasing demand for (and monetising) your own surrounding infrastructure. So cloud providers that produce or encourage open models can benefit from selling compute to host open models; consumer platforms can benefit from better recommendations and advertising; and application companies can benefit from lower input costs.

4. Boosting reputation, talent and capital

- In a world where you could fit all of the elite AI researchers (about [5,000](#)) into a mid-sized concert hall, and where companies are racing to attract capital to fund exponentially increasing compute costs, a leading open model demonstrates technical competence, attracts researchers and polishes the case for fundraising or an initial public offering.

5. Increasing geopolitical influence

- Open models can spread a country's technical standards, software ecosystem and cultural standards internationally. For China, capable open models reduce its dependence on US APIs and present it as the provider of choice to emerging markets. France has championed Mistral's open models as a European alternative. The US is not sitting idle either. OpenAI itself released smaller, open-weight gpt-oss models as a defensive move last August, and Thinking Machines launched Inkling, its own open model, last week.



5. What comes next?

History shows that formats can coexist where each offers something different.

- The contest between doesn't have to be an either/or. With so much at stake, OpenAI's honeymoon of a near-monopoly on generative AI in the 18 months or so after the launch of ChatGPT in November 2022 was never going to last.
- This is in line with the oft-quoted Jevons Paradox, where more efficient steam engines increased, not decreased, demand for steam power – and hence coal. Similarly, demand for compute will likely continue to surge as the market bifurcates into a handful of cutting-edge proprietary models on the one hand and a myriad of smaller open-source models on the other – just like trains and cars both found their market and continue to live side by side. No one ever said: "That's enough intelligence for now".

But both will exert a positive gravitational force on each other.

- Open models will force proprietary models to keep a lid on prices and remain flexible; proprietary models will set the benchmark for industrial-grade quality, safety and fully integrated products. The competitive moat will increasingly move away from raw weights and towards the balance of proprietary data, product design, memory, security and efficiency.

The greatest unknown in this uneasy coexistence is the direction of AI regulation.

- A growing backlash against AI in the US and the West significantly raises the chance of dramatic policy that could significantly affect its trajectory. While a handful of key US model makers presents a large target for legislators, they are also able to speak with one voice in a way that is not possible for fragmented, often foreign, open-source providers.
- For example, Anthropic released an [Advanced AI Framework](#) last month that called for obligations on frontier AI developers to test their models for catastrophic risks, engage external evaluators and disclose incidents on an ongoing basis. Their message echoes warnings from scientists and economists ranging from job losses to extinction.
- But critics say onerous regulations could favour incumbents with deep enough pockets to cover the costs and would stifle innovation, putting national security at risk. Former White House AI tsar David Sacks posted over the weekend on social media that there was "no reason to limit American models on tasks that Chinese models handle without issue. We're only making ourselves less competitive".



Recent relevant reports

[AI IPOs 101: why they'll change the boom forever](#) (July 1, 2026)

[Assessing AI in finance with Evident Co-CEO Alexandra Mousavizadeh](#) (June 9, 2026)

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[AI Q&A: Why have market vibes suddenly shifted on AI?](#) (Feb 16, 2026)

[Three AI themes for 2026: the honeymoon is over](#) (Jan 20, 2026)



Appendix 1

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Deutsche Bank
Research

The Rise of the RMB Financing Ecosystem

From Panda and Dim Sum Bonds to Deeper Capital Markets

July 2026

Perry Kojodjojo | Chen Kan

Deutsche Bank Research Institute

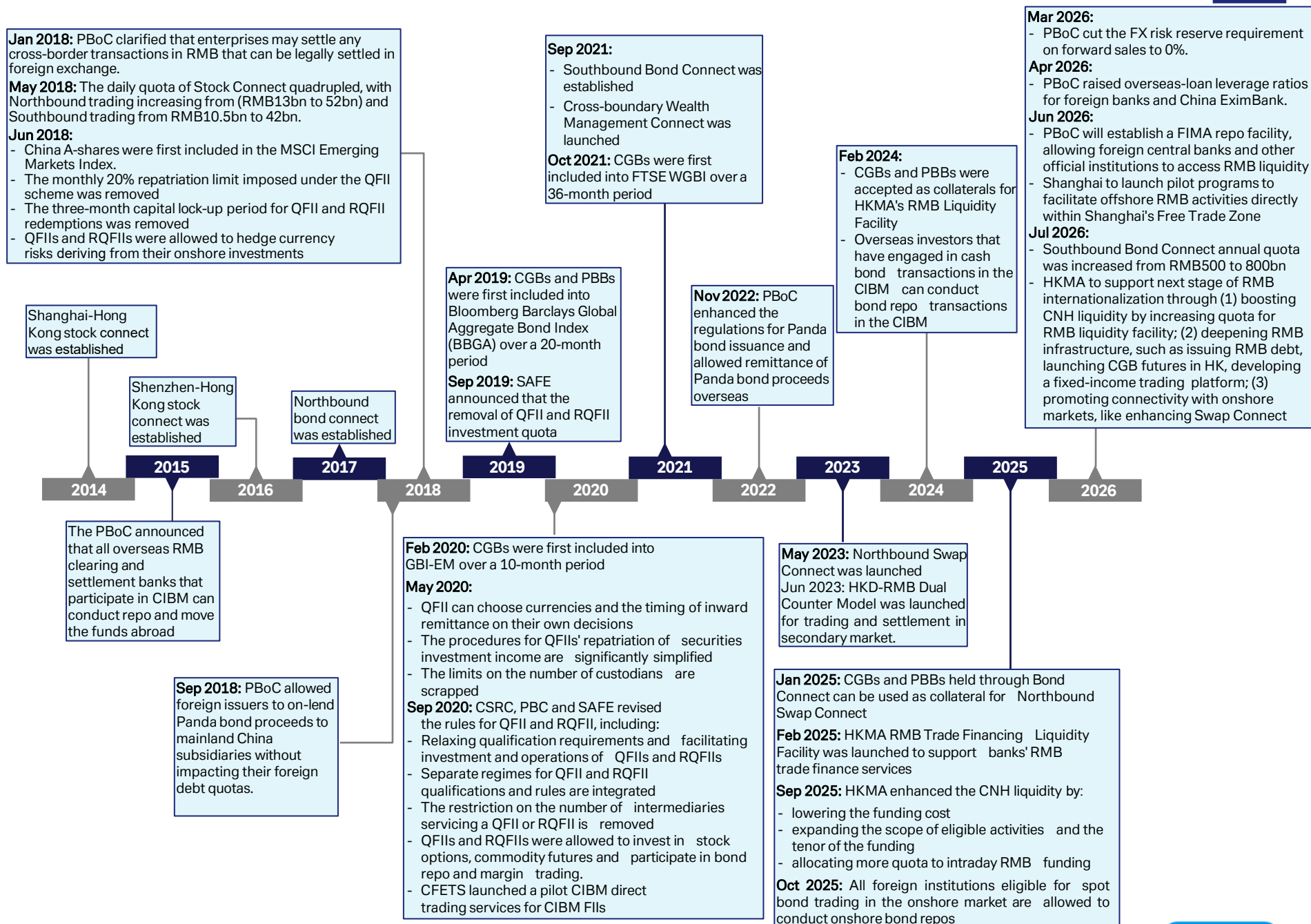
IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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- **RMB's global ambition.** China is actively shifting RMB from one used for trade to a global investment and financing currency. The Panda (CNY bonds issued by foreign entities within China) and Dim Sum (CNH bonds issued in Hong Kong) bond markets are the core pillars of this strategy, creating the necessary ecosystem for the RMB's international expansion.
- **Panda bonds - Unlocking onshore China for global use.** The onshore Panda bond market has matured into a strategic funding tool for foreign issuers. Thanks to regulatory easing on fund usage, over half of the proceeds are now used offshore, making it a flexible funding instrument and cementing the role of foreign participants who now account for over 30% of issuance.
- **Dim Sum bonds - A deepening offshore hub.** The offshore Dim Sum market is experiencing a surge in sophistication and diversity, signaling its evolution from a policy-driven market into a mature offshore funding hub. It is now driven by new issuers like tech firms, is offering longer-term bonds, and is set to receive a massive demand boost from new regulations allowing mainland insurers to invest in the market.
- **Next frontier - a focus on building a complete financial ecosystem.** While both markets are growing, challenges remain, including broadening the issuer base beyond financial institutions. The key to addressing these challenges is building a complete RMB financing ecosystem, supported by deep secondary-market liquidity, reliable benchmark yield curves for price discovery, and comprehensive risk-management tools. Ultimately, this is a gradual but foundational journey to establish the RMB as a more widely funded, invested, and held international currency.
- **Expanding European participation in RMB funding markets.** While European entities dominate foreign issuance in both Panda and Dim Sum markets, this participation is highly concentrated among a few financial sector repeat borrowers, with corporates largely absent. Barriers include smaller deal sizes, a narrower investor base, and lack of market familiarity. As RMB internationalization progresses, we anticipate broader international issuer participation, particularly from European corporates given strengthening EU-China ties.

RMB internationalization timeline



RMB Internationalization: A New Chapter Driven by Diverging Bond Markets



China's policy stance on RMB internationalization has shifted from cautious to explicit, with the 15th Five-Year Plan (2026-30) elevating it to a strategic priority. This shift is supported by a series of concrete actions in early 2026 amidst a global de-dollarization trend. However, progress remains uneven. While there is strong official commitment, evidenced by record transaction volumes on China's Cross-Border Interbank Payment System (CIPS) and a surge in RMB goods trade settlement to a projected 30% by 2025, the currency's broader global footprint is still modest. The RMB holds only a 1-4% share across global payments, FX reserves, and debt securities, which is disproportionately low compared to China's approximately 19% share of global GDP.

A fundamental change has occurred in the nature of cross-border RMB usage. The center of gravity has moved from trade to investment flows, with capital and financial account transactions now constituting around 75% of total RMB settlements. **This means the next phase of internationalization requires building a deeper and more accessible financial ecosystem for raising and deploying RMB capital globally.** This includes creating more robust channels for funding, such as the onshore Panda and offshore Dim Sum bond markets, alongside more sophisticated hedging tools and a broader array of attractive RMB-denominated assets.

Since 2025, these two key bond markets have entered a new, more dynamic chapter, accelerating on different tracks and developing distinct characteristics. **The onshore Panda bond market is experiencing a record year, with issuance in the first half of 2026 exceeding RMB160bn.** This boom is driven by fundamental improvements to its architecture, including streamlined issuance procedures and more flexible rules that allow issuers to remit proceeds overseas. This has transformed Panda bonds from a ring-fenced currency pot into a source of globally usable RMB funding.

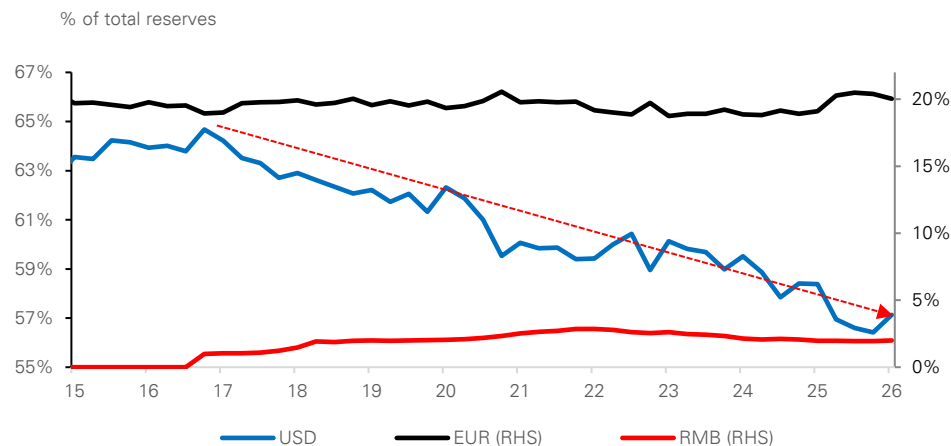
In contrast, **the offshore Dim Sum market's strong growth is driven by demand diversification.** It is poised for a significant boost as mainland insurers gain access via the Southbound Bond Connect, whose quota has been expanded to RMB 800bn, creating a new source of demand for an increasingly foreign-led issuer base.

The markets are also evolving with unique "DNA." **The Panda market's issuer base remains dominated by traditional sectors like financials, while the Dim Sum market is diversifying and embracing tech.** Issuance from IT giants has risen sharply in the Dim Sum market as they seek cheap, long-tenor funding for their AI and cloud capex cycles. This reflects a difference in their duration profiles as well. The Panda market remains overwhelmingly short-term, with nearly all issuance at three years or less due to bank-dominated demand and a flat yield curve. Conversely, the Dim Sum market is showing clear maturity with a growing share of bonds in the 5-year and longer tenors, which better aligns with the investment horizons of the tech sector.

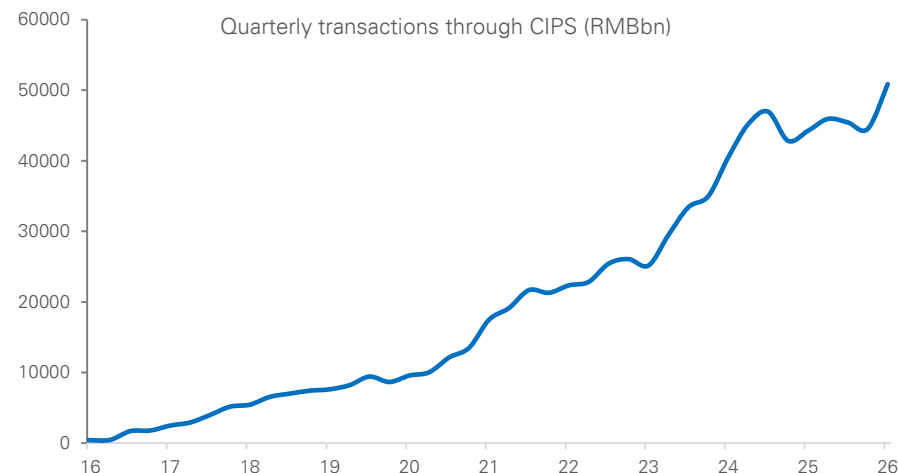
A strategic push amid ongoing de-dollarisation theme



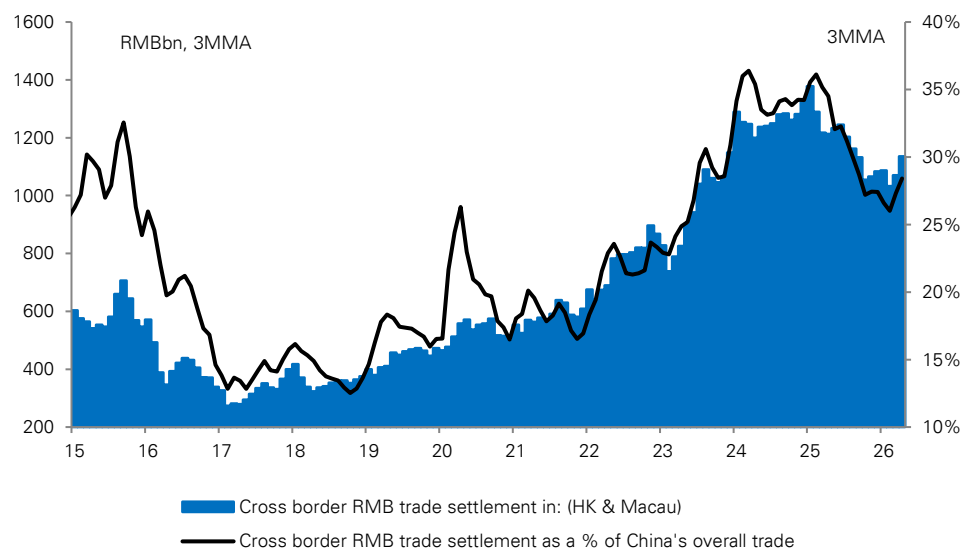
USD's declining share of reserves points to ongoing de-dollarization



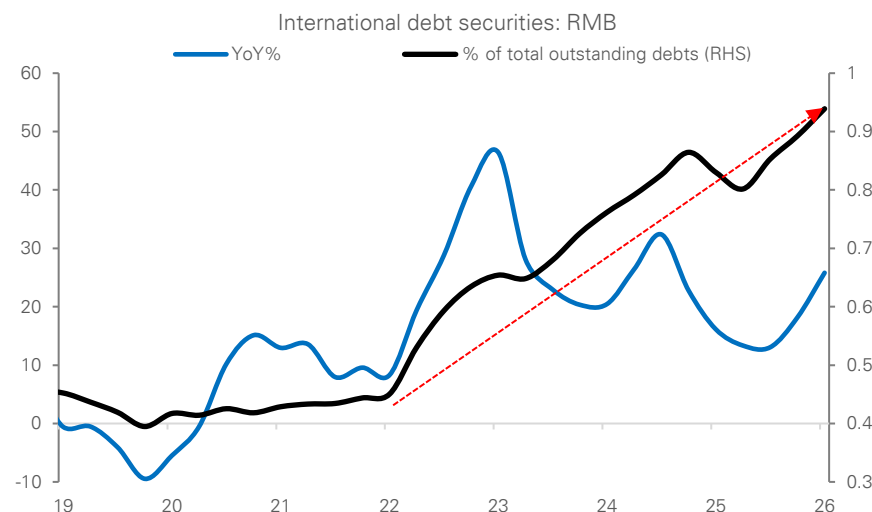
RMB transactions through CIPS hit a record high



A more active use of RMB in trade settlement



RMB-denominated debt issuance has risen notably particularly as China pushes to develop the Panda and Dim Sum Bond markets

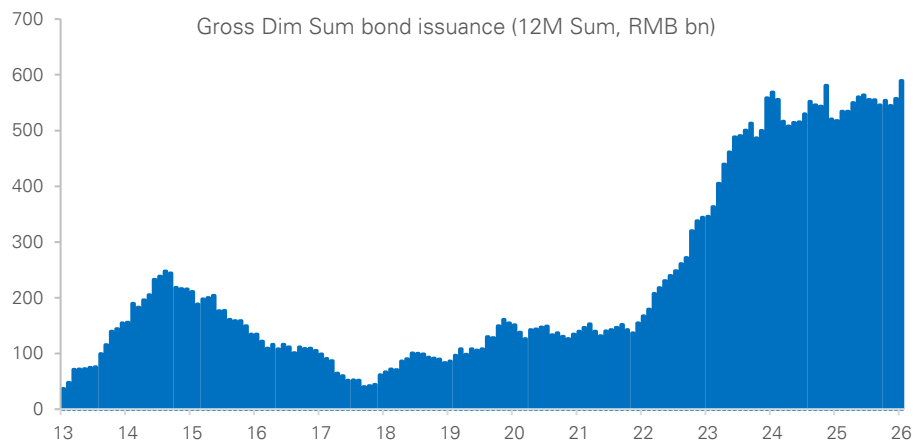


Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

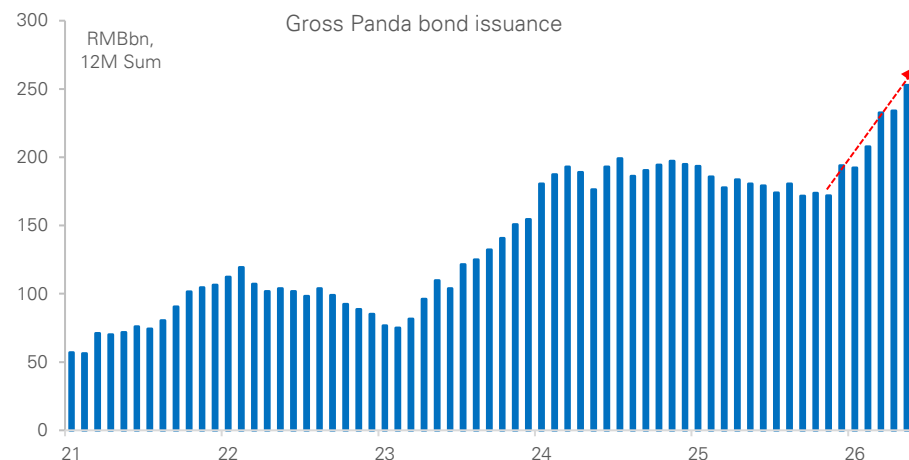
The divergence in focus - key trends to note



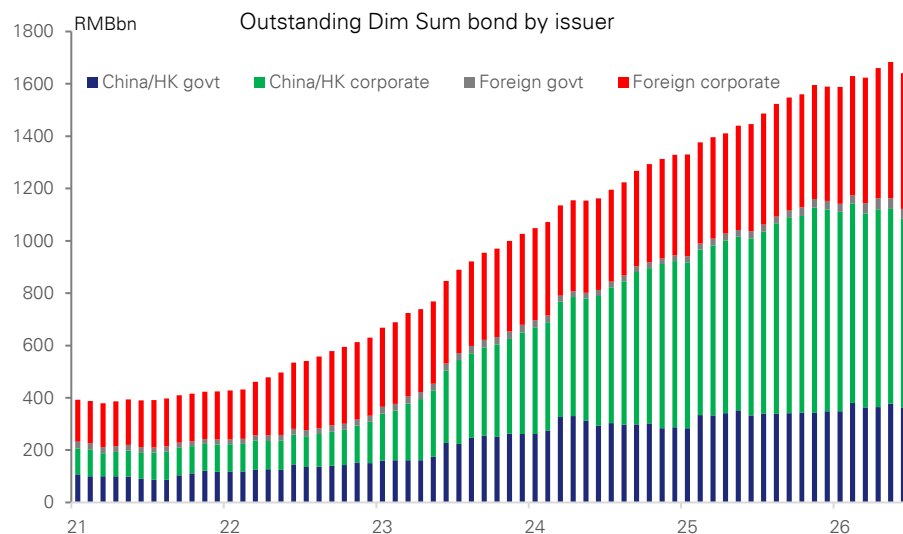
Steady growth in the Dim Sum bond market poised to deepen



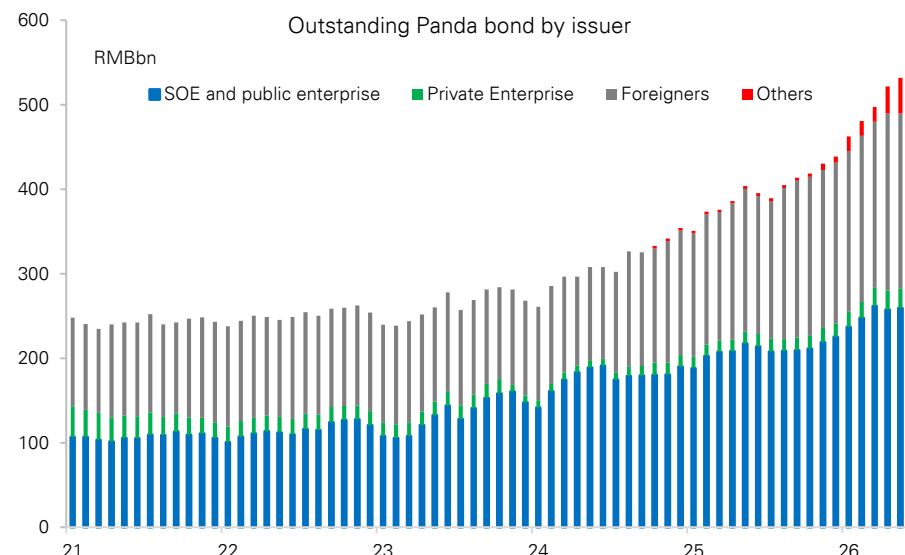
Policy easing unlocks new growth phase in the Panda bond market



Becoming the Go-To offshore RMB hub not just for government entities but also corporates



A hybrid market set to attract more foreign issuers



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

Panda Bonds - Maturing into a funding tool



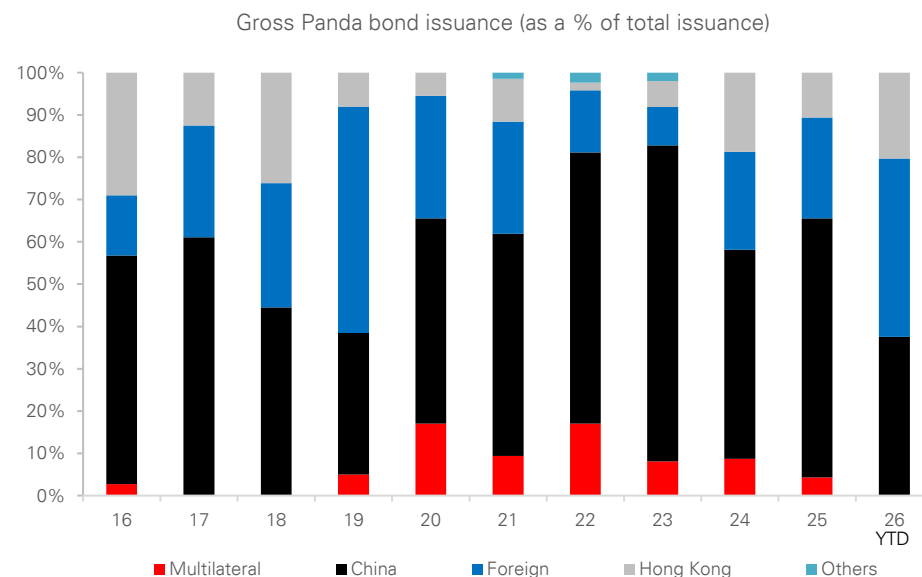
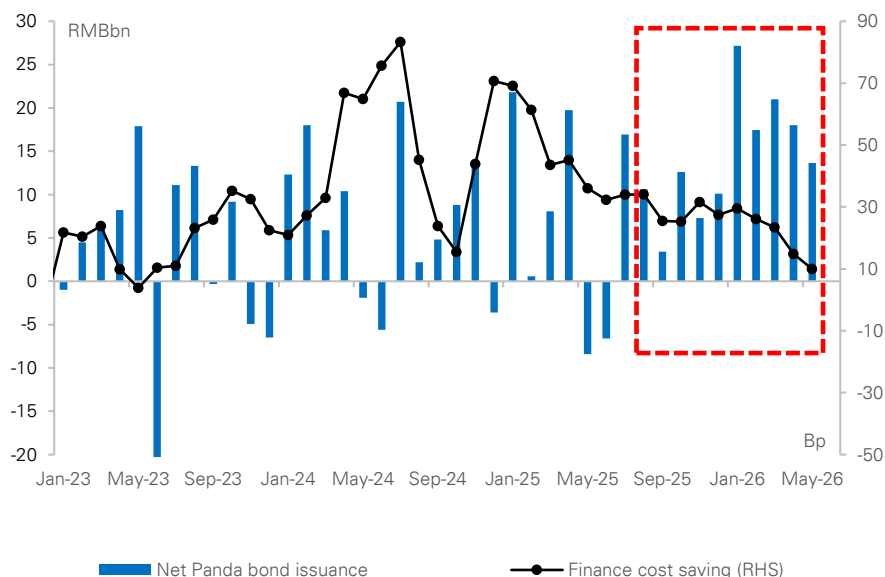
The Panda bond market's record issuance year, even without a clear cost advantage, confirms its shift from a tactical choice to a strategic funding tool. This maturation is driven by two key structural changes:

- **From onshore tool to global funding.** Regulatory easing has been a game-changer. Over half of Panda bond proceeds (53%) are now used offshore, making it a flexible, globally-integrated RMB funding instrument.
- **From niche to mainstay.** Foreign issuers are now a cornerstone of the market, consistently accounting for over 30% of issuance and showing early signs of diversification beyond just financial institutions.

While the market remains concentrated in short-term debt (<3 years) and dominated by domestic bank investors, the modest return of foreign buyers is a positive signal for future market depth.

Issuance remains resilient as market moves beyond cost saving

Foreign participation becomes a cornerstone of the market

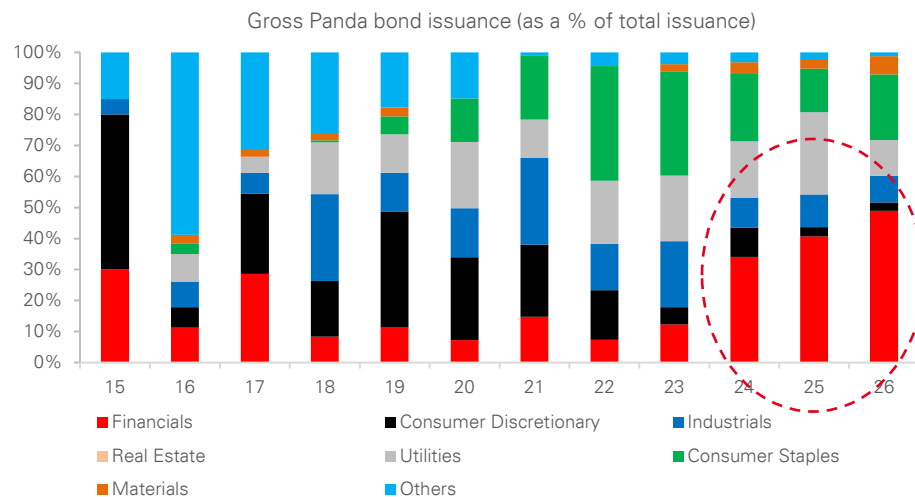


Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

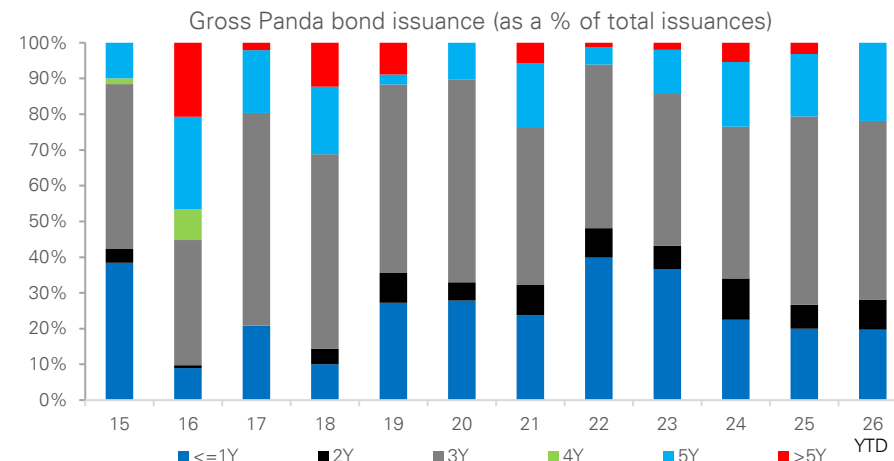
Key trends shaping the Panda market



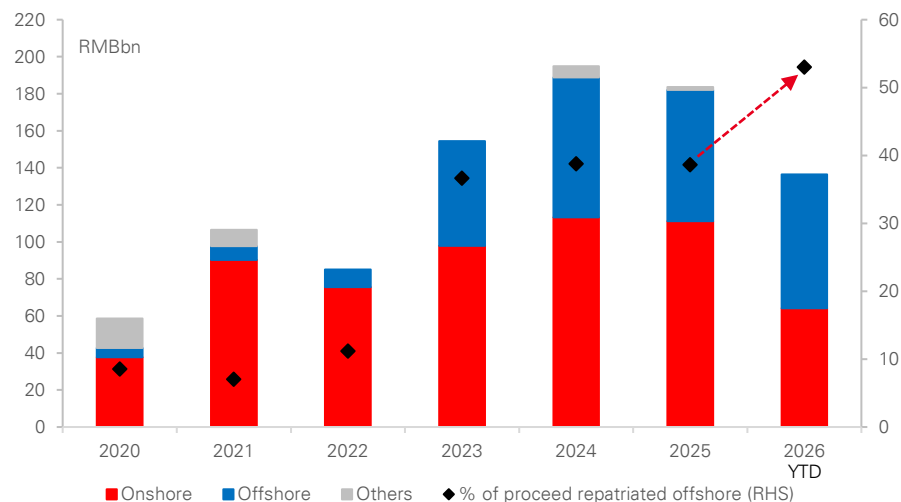
Financials continue to pave the way for broader issuer growth



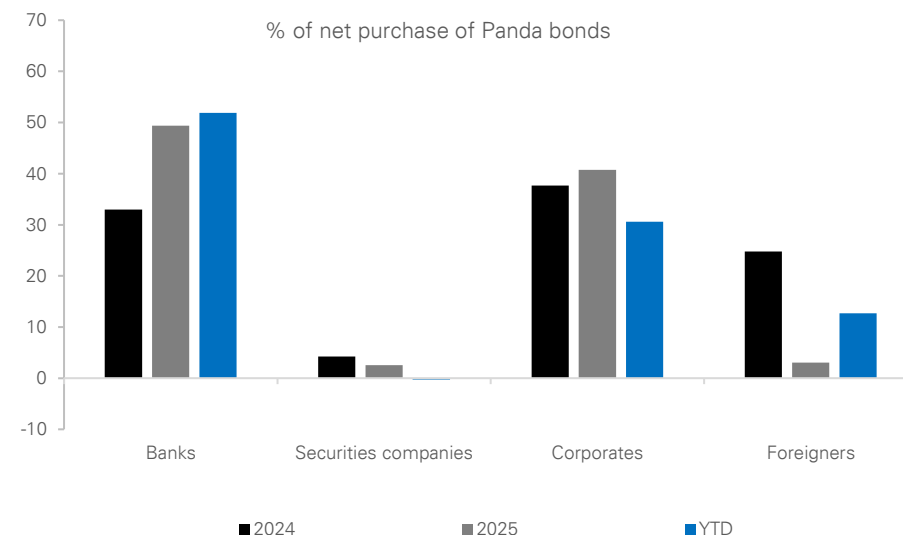
Short tenors still dominate, highlighting room for market deepening



Flexibility increases as offshore use becomes the default



Foreign investor demand returns, signalling renewed confidence



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

Dim Sum market is becoming deeper and more diversified

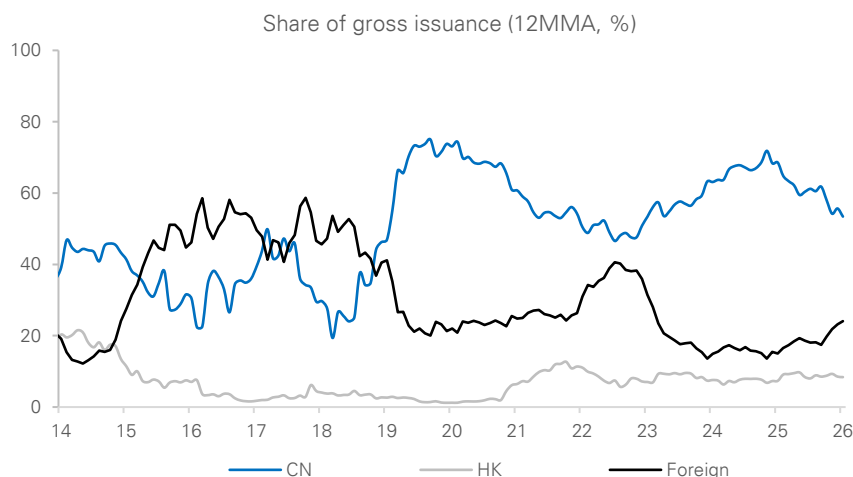


The offshore Dim Sum market is undergoing a structural transformation, becoming deeper and more sophisticated. Three key trends define its new trajectory.

- **Issuer base revolution.** The market is now driven by new participants. Foreign entities have become a cornerstone, accounting for over 25% of issuance. Domestically, high-growth Tech/IT companies have decisively replaced real estate as the primary issuers, reflecting growing demand for long-duration RMB funding to support AI, cloud and digital infrastructure investment.
- **Market maturation.** Issuance tenors are extending significantly. The growing share of bonds with 5+ year maturities, and the emergence of a nascent 15Y+ market, signal rising investor confidence and more sophisticated funding strategies.
- **Demand catalysts unlocked.** Recent policy changes are a game-changer for demand. Allowing mainland insurers to buy Dim Sum bonds via Southbound Bond Connect, alongside an increased quota (to RMB800bn) and wider product scope, is set to unlock a massive and sustainable new pool of capital.

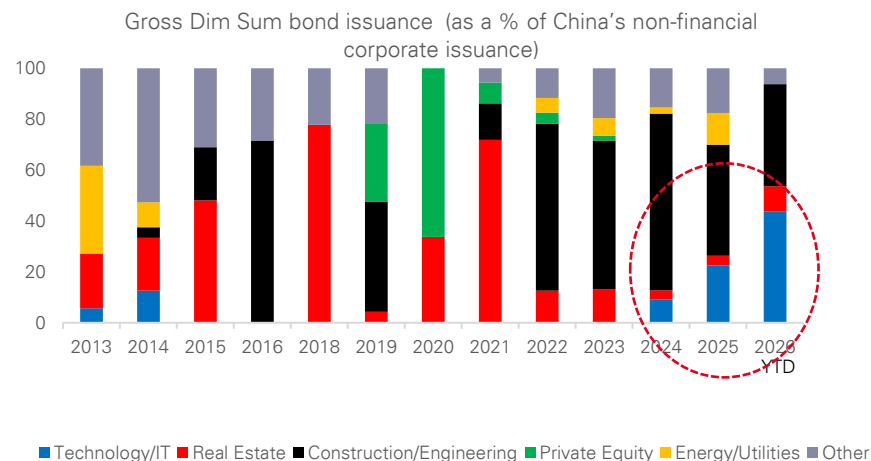
In short, the Dim Sum market is evolving from a niche offshore CNH pool into a more mature, diversified, and institutionally supported market with a powerful new demand base on the horizon. Together, these developments are helping to establish the foundations of a deeper and more liquid offshore RMB capital market.

Foreign participation becomes a cornerstone of the market



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind

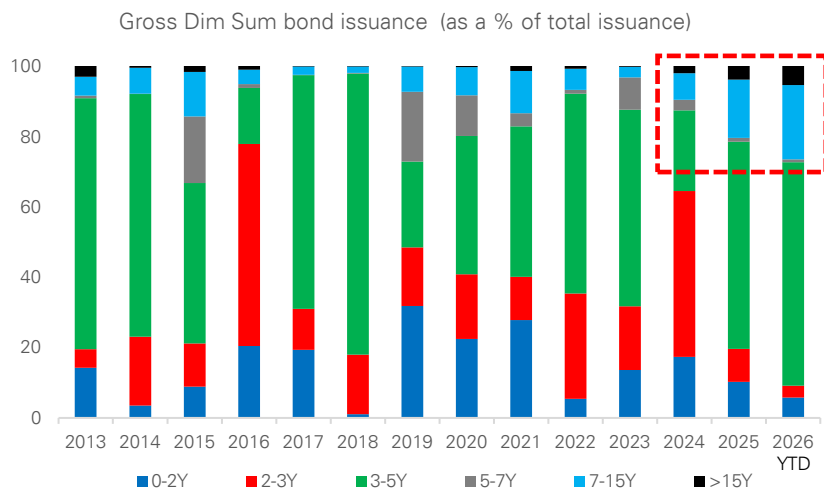
Tech firms replace property-linked sectors among domestic issuers



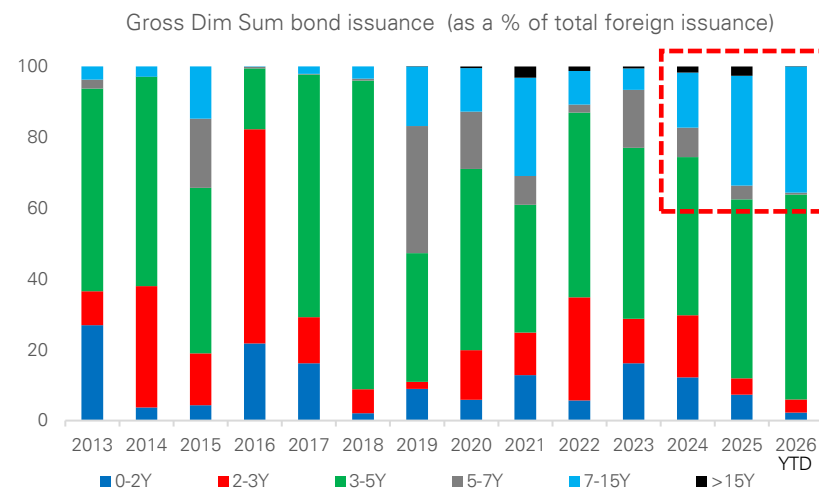
Market depth and demand continue to improve for Dim Sum bonds



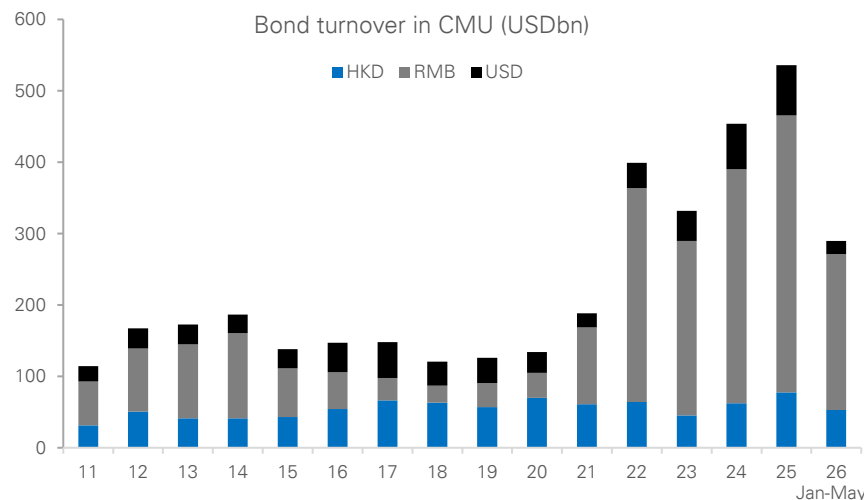
Extended issuance tenor for Dim Sum bonds...



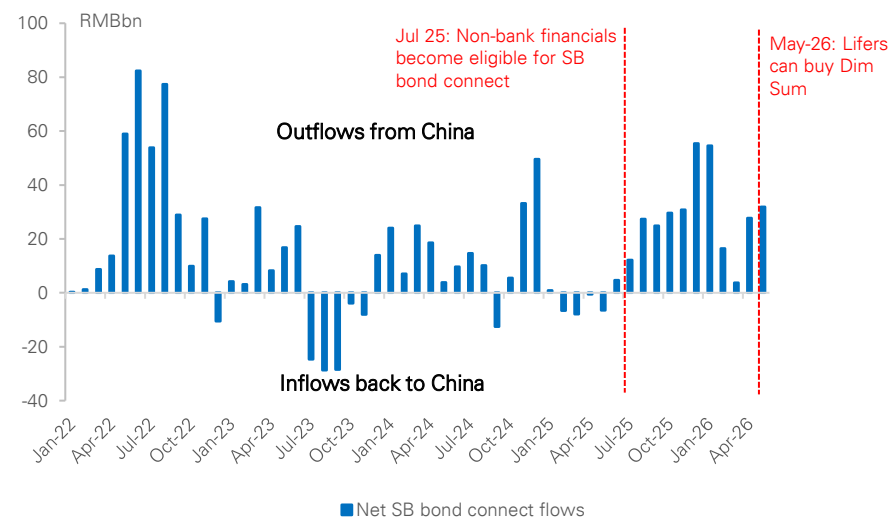
...led by foreign issuers



RMB bonds dominate Hong Kong bond market



Supportive demand-side regulation should underpin Dim Sum demand



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind



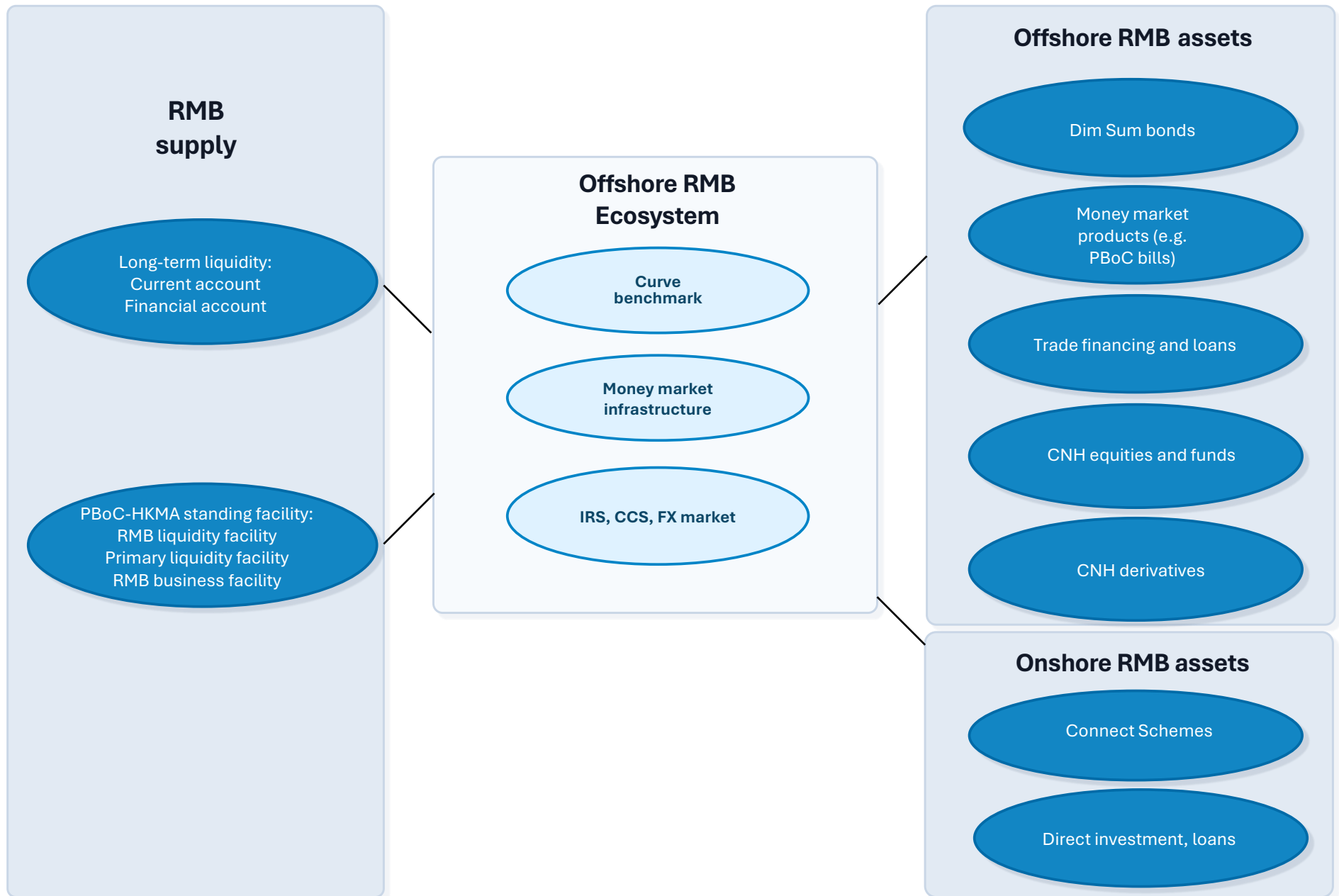
Over the next five years, **Panda and Dim Sum bonds are set to be the two core pillars driving the RMB's evolution from a trade-settlement currency into a global investment and financing currency.** Together, these markets are creating a comprehensive ecosystem that provides the essential assets for international players to fund, invest, and manage RMB holdings.

The Panda bond market, where foreign entities issue RMB-denominated debt inside mainland China, is successfully transitioning from a policy initiative to a viable funding channel. Regulatory reforms have improved access for a diverse range of international issuers, leading to growing market confidence and projected annual volume growth of 20-30%. However, significant hurdles remain, including a narrow investor base, a lack of robust tools for hedging interest rate risk, and complex issuance rules that need further standardization.

Concurrently, **the Dim Sum bond market, consisting of RMB bonds issued outside mainland China (primarily in Hong Kong), is solidifying its position as the primary offshore RMB hub.** The focus here is on building a reliable benchmark yield curve through regular government issuance and upgrading market infrastructure with enhancements to Bond Connect and innovations like digital bonds. This market's growth, however, remains highly sensitive to offshore RMB liquidity, currency expectations, and overall investor demand. Its long-term success depends on creating a deep and sustainable ecosystem rather than just growing issuance volume.

Ultimately, **the development of these bond markets is a gradual but foundational journey.** Much like the "Samurai" market was for the internationalization of the Japanese Yen, Panda and Dim Sum bonds are creating the necessary building blocks for the RMB's global future. While this evolution is subject to broader factors such as capital account policies, market reforms, and the geopolitical landscape, the strategic goal is clear: to establish a robust foundation for the RMB to be more widely funded, invested, and held across the world.

Offshore RMB Ecosystem: from RMB supply to RMB asset formation



Source: Deutsche Bank Research

Expanding European participation in RMB funding markets



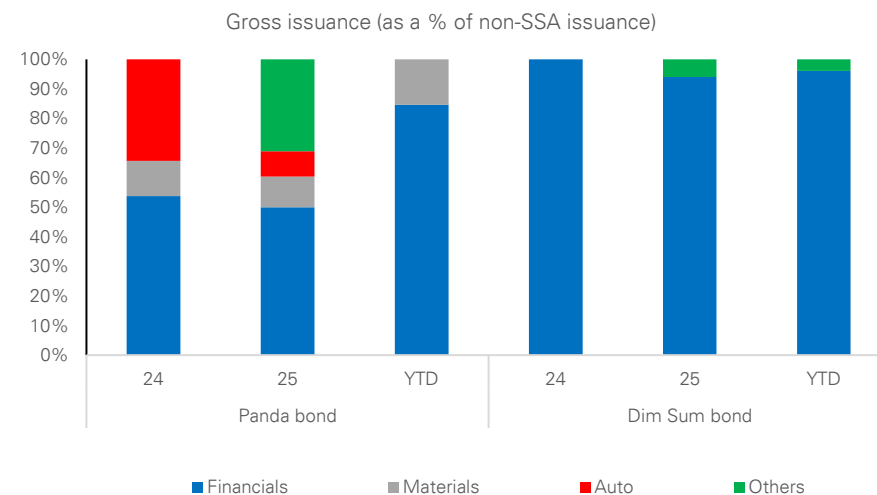
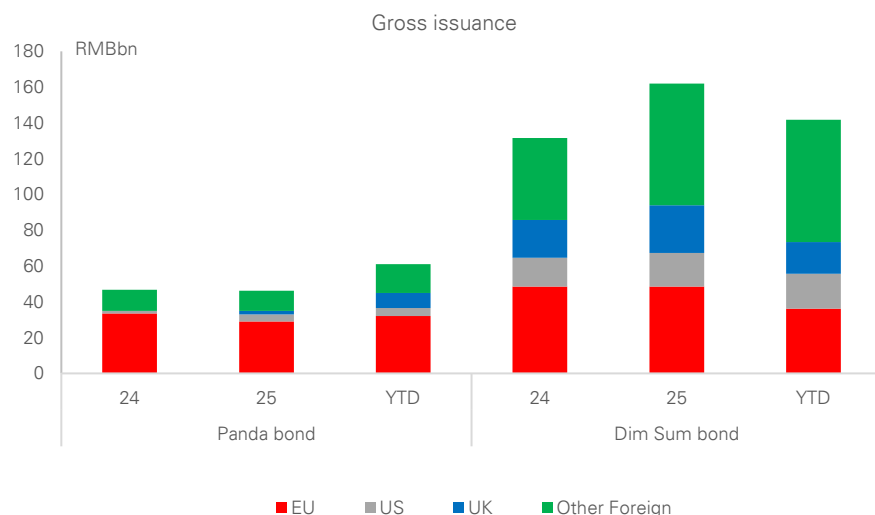
European entities are major players in the RMB bond market, representing over half of foreign Panda and a third of Dim Sum issuance. However, the issuer base is highly concentrated, with financial institutions dominating (85-95%) and repeat borrowers accounting for over 90% of activity. The key challenge lies in attracting new issuers, not retaining existing ones.

Primary hurdles include smaller deal sizes compared to major markets, a narrower investor base, and lack of market familiarity. These factors often make the upfront investment in RMB funding infrastructure, documentation, and investor relationships economically unfeasible.

Looking ahead, we anticipate these barriers will ease with RMB internationalization. A deeper offshore RMB ecosystem will support larger benchmark issuance, broaden the investor base, and improve liquidity. Further development of CNH derivatives markets will simplify currency risk hedging and RMB proceeds swaps. Additionally, China's low interest rates and growing China-related business exposure could enhance RMB funding appeal for international corporates. As the cost-benefit equation improves, we expect future market development to be driven by a broader range of international issuers, particularly European corporates, rather than larger issuances from existing borrowers.

EU issuers account for a large share of foreign RMB bond issuance...

...but the issuances are overwhelmingly concentrated in financial institutions

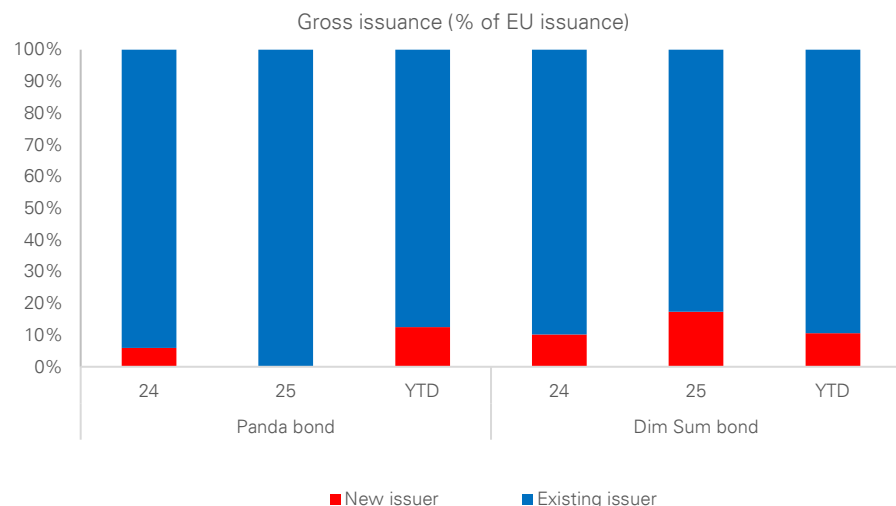


Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind, BIS

Structural tailwinds point to a broader EU issuer universe



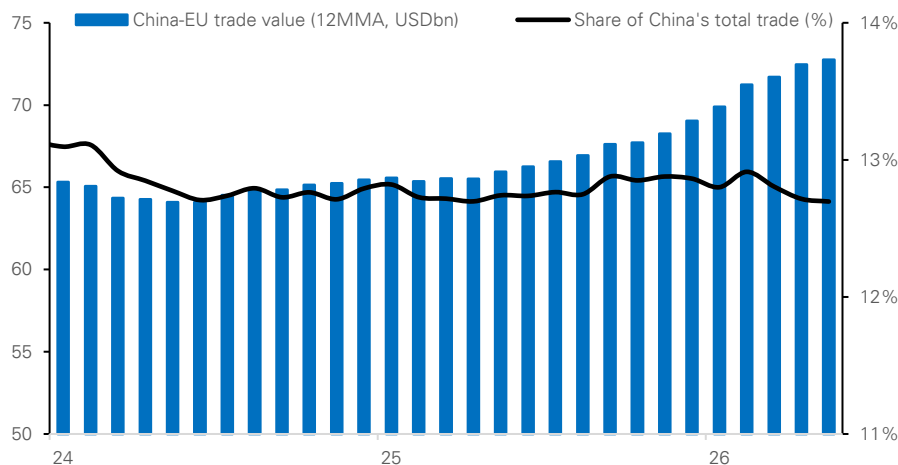
EU RMB bond issuance remains dominated by repeat borrowers



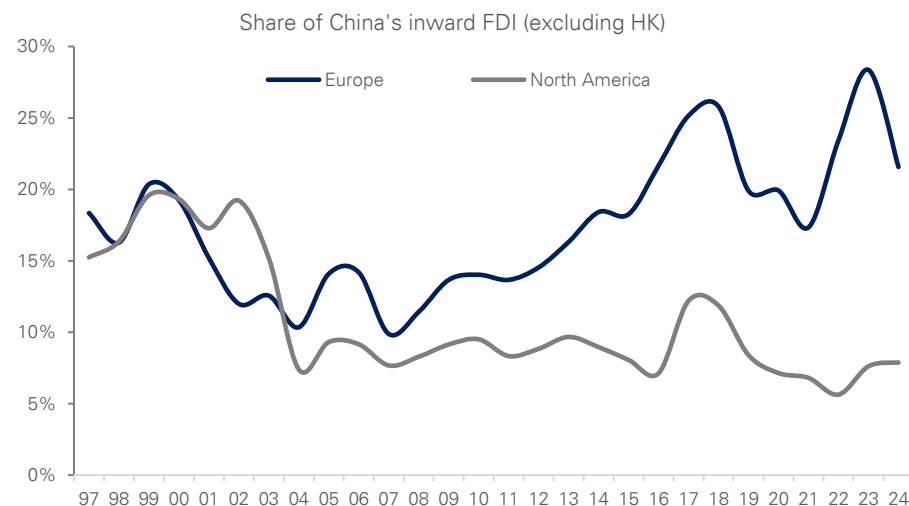
Global debt markets are gradually diversifying away from USD



EU is China's biggest trading partner...



...the most important source of FDI



Source: Deutsche Bank Research, Bloomberg Finance LP, CEIC, Haver, Wind, BIS



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Food For Thought: Sowing The Seeds of Food Inflation

July 15, 2026

Ongoing tensions in the Middle East have raised the prospect of indirect effects on inflation - particularly in relation to food prices. Indeed, much has been said about the rise in energy prices and its spill over into other commodities such as fertiliser, CO₂, and other key ingredients that sit within the agricultural pipeline. To be sure, both the size and duration of the shock matter for food prices.

In this note, we do four things.

- **First**, we highlight what's happened to key commodity prices in the wake of the Iran shock.
- **Second**, we set out a framework, estimating the potential impact of commodity price moves on food inflation for both the euro area and the UK.
- **Third**, we stress test our estimates by looking at the duration of the shock and whether pass-through of cost increases change when the shock is deemed 'transitory' or 'persistent'.
- **Finally**, we highlight some of the broader implications for inflation and where risks lie around our estimates.

Our analysis points to renewed upside risks to food prices stemming from recent oil and fertiliser market developments. The effects appear larger in the UK than in the Euro area.

Importantly, we find that pass-through isn't immediate, and in fact, can build over a long period of time given the mapping of commodity prices into food retail prices. What also matters is whether shocks prove persistent or transitory. Notwithstanding the MoU agreed to by the US and Iran, we estimate that the net effects from oil and urea prices will lead to modest inflationary pressure in the quarters ahead – something we will be watching for closely. That said, considerable uncertainty surrounds both the scale and timing of these effects, with a range of other supply-side factors also likely to influence outcomes. We will be monitoring market developments closely and will calibrate our projections accordingly.

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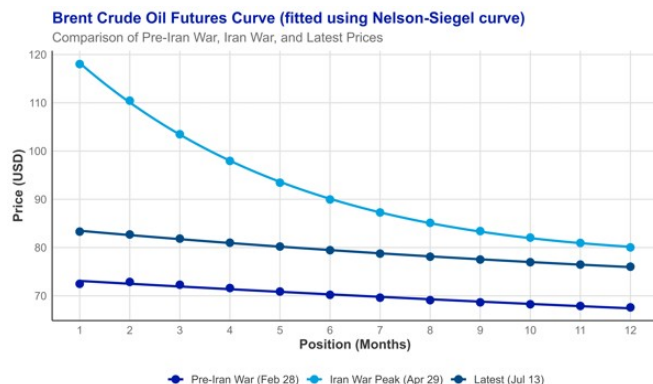


A recap on recent price movements

Recent geopolitical developments have been associated with large swings in fertiliser and oil prices. Oil prices initially rose as markets priced supply-disruption concerns. Subsequently, however, price rises retraced meaningfully as these concerns eased with continuing peace deal negotiations. To add context to these movements, we compare spot and futures prices before the episode, after the initial Iran shock, and using the latest available market data. For the futures curves, we fit a Nielsen-Siegel curve to trace out the shape of the curve across contract positions.

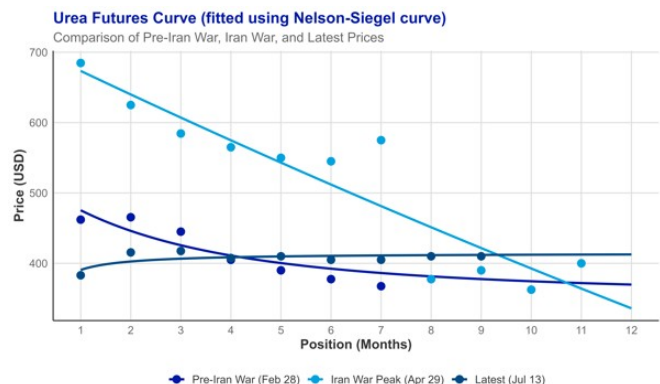
For fertiliser prices, we use the World Bank urea price series as a proxy for global nitrogen fertiliser costs. Urea is one of the most widely traded nitrogen fertilisers and is closely linked to the production process for agricultural inputs. We note, however, that this captures only one important fertiliser-cost channel rather than the complete range of farm input costs.

Figure 1: Brent futures pricing downward sloping...



Source : Deutsche Bank, Bloomberg Finance LP

Figure 2: ...same as urea futures pricing



Source : Deutsche Bank, Bloomberg Finance LP

Before the Iran War, the spot price for Brent crude oil was trading at about USD 72.50/barrel. This then spiked to USD 118/barrel during the peak of the Iran/US conflict, before settling at around USD 83/barrel at current market prices ([figure 1](#)). Interestingly, the futures curve is currently flatter than the steeper curve observed at the start of the war, which could imply a more gradual decline in prices in the next few months. The urea spot price also increased sharply at the beginning of the Iran war. In contrast, these prices have settled at a much lower level ([figure 2](#)).

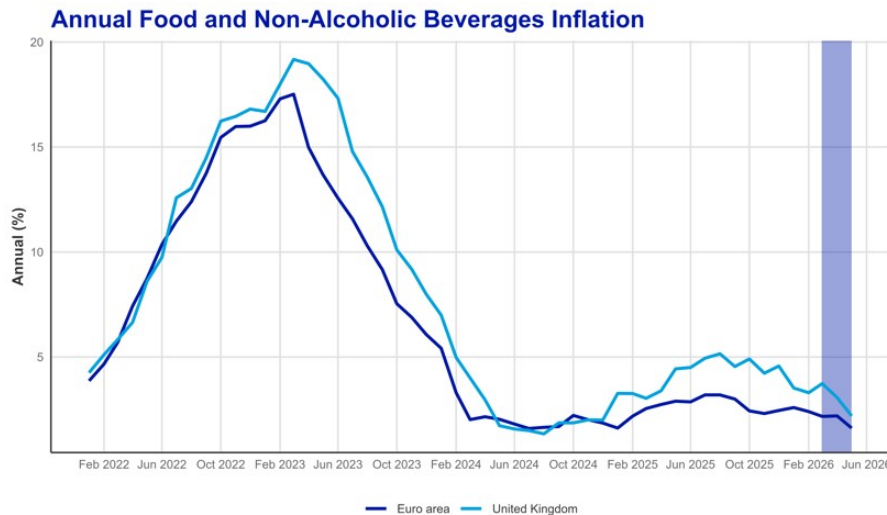
Perhaps unsurprisingly, in terms of recent food price developments, we have yet to see any tangible impact on retail prices.

Euro area food inflation has softened more than expected this year, declining from 2.5% y-o-y in Dec-25 to 1.6% in Jun-26 based on the flash estimate. This marks the lowest reading for food HICP since mid-2021, with the bulk of the adjustment occurring over the past two months. Focusing on food stuff, the largest disinflationary impulses have come from dairy products, sugar and sweets, and coffee. In 2025, these categories were among the main drivers of food inflation, accounting for roughly 35% of annual food inflation despite representing only 18% of the HICP food basket. In recent months, their contribution has fallen to just over 20%. Meat prices have also eased somewhat from their 2025 peak, although they remain the largest contributor to food inflation.



In the UK, we see a similar story. Heavy discounting has led to a drop-off in food prices. Food and non-alcoholic beverages CPI slowed to 2.2% y-o-y – reaching its lowest rate in 17 months. The slowdown in food prices can be seen across both processed and unprocessed food. The latest PPI data for food points to further disinflation in the pipeline, with producer prices across food, beverages and tobacco slowing to 1.1% y-o-y in May-26.

Figure 3: Food (and non-alcoholic beverages) inflation in the EA and the UK has been on a downtrend this year



Source : Deutsche Bank, Macrobond

How does energy and fertiliser costs pass-through into food inflation? There are two channels in which oil and fertiliser prices work their way into CPI.

Channel 1: the direct energy cost. Oil enters food at pretty much every node in the food chain. For example, manufacturers need diesel for tractors and irrigation pumps, marine freight and trucking, processing heat, packaging (plastics) and refrigeration. Energy-related inputs account for around 5–10% of farm production costs, although the share can be significantly higher in energy-intensive sectors such as horticulture and poultry.

Channel 2: fertiliser cost. Urea is essentially manufactured from natural gas. The higher gas prices go, the higher fertiliser prices go too. The response can effectively be summarised into two segments: an immediate margin squeeze, then a supply response as farmers cut application rates or switch acreage to lower Nitrogen crops (soybeans over maize), which lifts prices with a lag of a full crop cycle.

The transmission from input costs to retail prices, as a result, is heavily staggered. The upstream leg is fast: ammonia and urea reprice off gas within weeks. Put simply, an energy shock is visible in fertiliser markets almost immediately. The middle leg is slow and seasonal — application decisions are made pre-season, so a nitrogen cut driven by a spring price spike doesn't show up until (as early) as the following harvest. Ultimately, the final leg is both lagged and heavily damped.



Estimating pass-through to food prices

There are several ways to estimate the effect of oil and fertilizer price shocks to food inflation. In this case, we opt for a simple approach that allows the "data to speak": the local projection method ([Jorda, 2005](#)). Essentially what we do is trace out the impact of a shock over time, in this case oil and fertilizer price shocks, by running a separate regression for each forecast horizon (1-month ahead, 2-months ahead and so on...).

A key challenge is identifying the commodity-price shock itself. Oil and fertiliser prices move for many different reasons, including global demand, exchange rates, trade conditions, inflationary pressures and supply disruptions. To reduce the risk of conflating commodity-specific surprises with broader macroeconomic developments, we use a Romer and Romer-style approach ([Romer & Romer, 2004](#)).

The above approach involves regressing the shocks of interest on a set of potentially confounding variables, including the USD effective exchange rate, monthly dummy variables to capture seasonality, lags of the shocks of interest to account for persistence, global trade volumes and global CPI inflation. We then extract the residuals from this regression. These residuals could then be interpreted as movements in oil and urea prices that are not explained by our macroeconomic controls.

The shocks of interest that we use are the month-on-month percentage changes in Brent crude oil spot prices and the World Bank urea commodity price series.

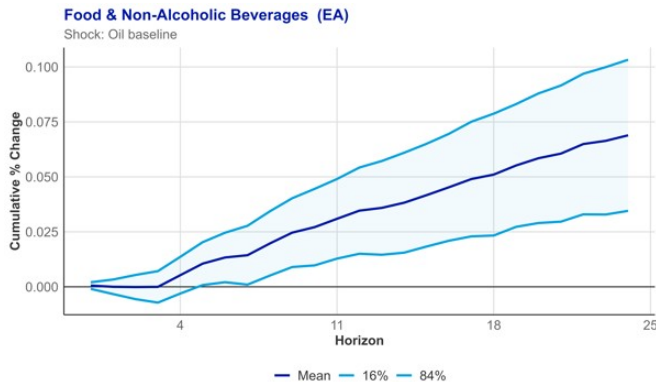
For the local projections, we use the HICP Food & Non-alcoholic beverages series for the Euro area and the UK Consumer Price Index Food and Non-alcoholic beverages. The regressions include the shock of interest (oil & urea), and three lags of the price variables. We then estimate the cumulative percent impact on the price level from a percentage point surprise in both oil and urea costs.

Baseline results

The baseline estimates suggest that oil and fertiliser price surprises have a measurable effect on food prices. The estimated response varies across the two economies and commodities, with the UK showing larger pass-through than the euro area in our specifications. Our estimates, however, become relatively more uncertain the further out we go as shown by the widening error bands.

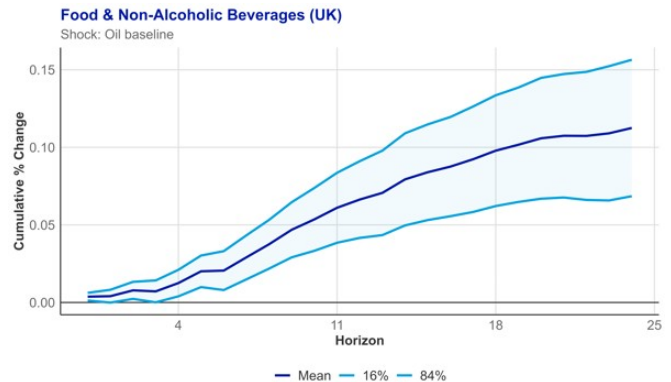


Figure 4: A 1pp shock in oil price leads to nearly a 7bps increase in food HICP (EA)...



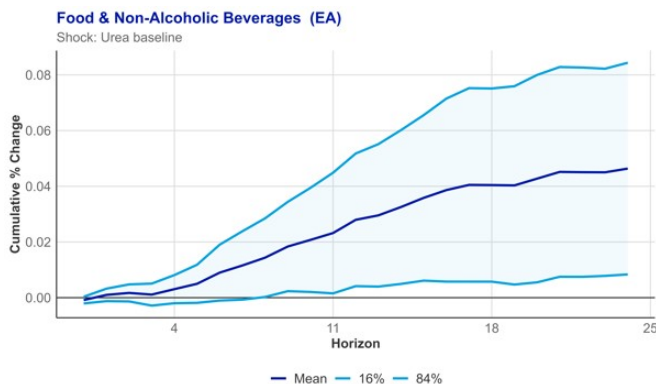
Source : Deutsche Bank

Figure 5:...and just over 10bps in the UK (CPI)



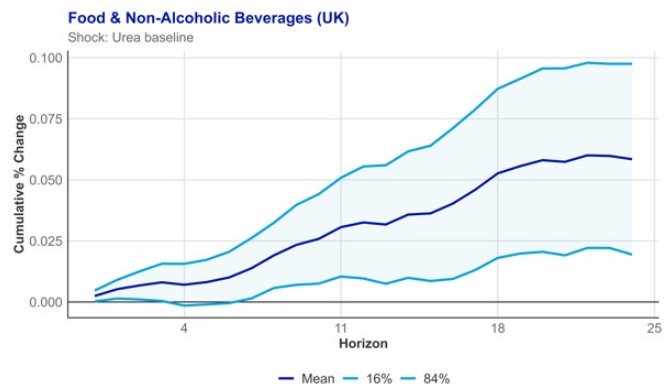
Source : Deutsche Bank

Figure 6: A 1pp shock to urea prices results in just over 4bps to EA HICP (food & non-alcoholic beverages)...



Source : Deutsche Bank

Figure 7:...and close to 6bps for UK CPI (food and non-alcoholic beverages)



Source : Deutsche Bank

In further detail, [figures 4-7](#) present the impact of a 1 percentage point increase in monthly oil and urea price growth on the cumulative food & non-alcoholic beverages price level. Looking first at fertiliser price shocks, we find that the peak impact on UK food prices occurs after around 20 months with a 0.07% cumulative impact on the food price level ([figure 7](#)). In the Euro area, the peak impact is slightly lower at around 0.05% and also occurs after roughly 20 months ([figure 6](#)). A broadly similar pattern is visible for oil price shocks. The UK experiences a larger cumulative increase of 0.1% in food prices while the Euro area sees an impact of about 0.07%.

There are a few possible explanations for the slightly larger estimated effects in the UK relative to the Euro area. One possible explanation that may be driving the difference in pass-through between the Euro area and the UK is the higher imported share in the UK food basket. This would make importers and producers more exposed to externally driven input cost pressures.

A second explanation may come from sterling weakness during global energy shocks.

A third explanation may be in fact that many UK farming sectors operate on narrow margins. When energy, fertiliser or feed costs rise sharply, farmers have less ability

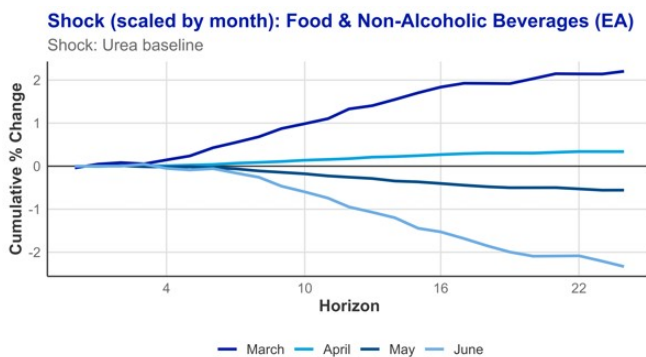


to absorb the shock and are more likely to pass it on through the supply chain. Defra's Farm Business Survey consistently shows that profitability is highly sensitive to movements in input costs. UK agriculture is also relatively intensive, particularly in cereals, oilseeds and grassland systems.

The impact of the Iran war

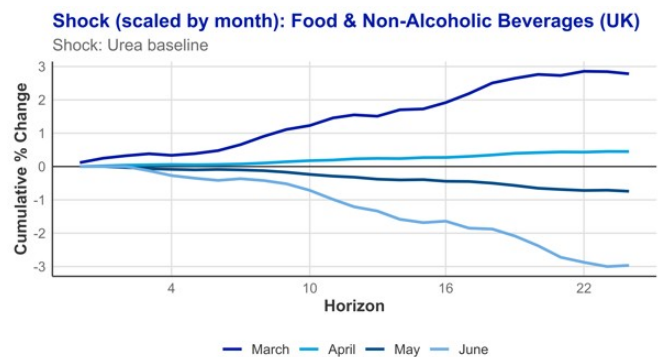
So far, the results have shown a 1 percentage point increase in fertiliser and oil price inflation. However, events such as the Iran war have led to big swings in both urea and oil prices – and in both directions. In order to better gauge these impacts, we scale the shocks in the baseline results to the actual movements observed from March to June. The figures below show the estimated cumulative food price pass-through implied by each monthly commodity price move.

Figure 8: Estimated urea shock pass-through for each month since the conflict began in the EA...



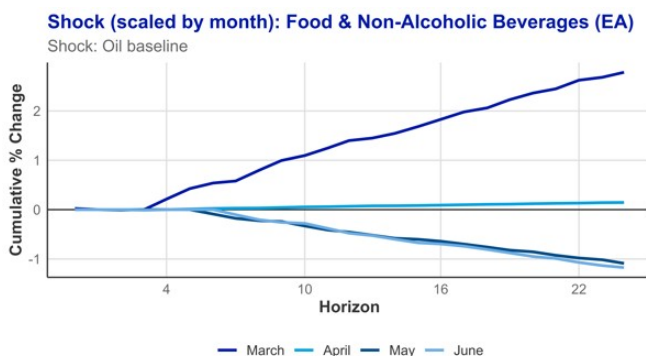
Source : Deutsche Bank

Figure 9:...and the UK



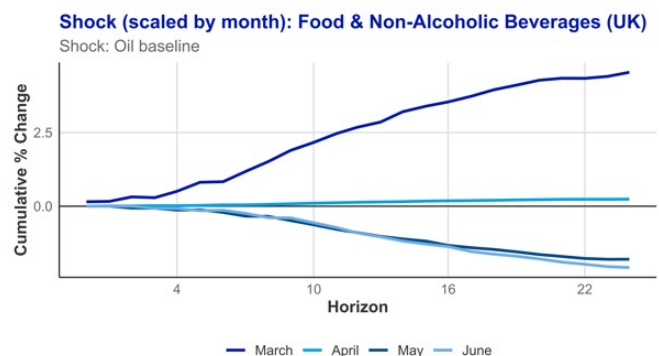
Source : Deutsche Bank

Figure 10: Scaled oil shock between March and June for the EA...



Source : Deutsche Bank

Figure 11: ...and the UK



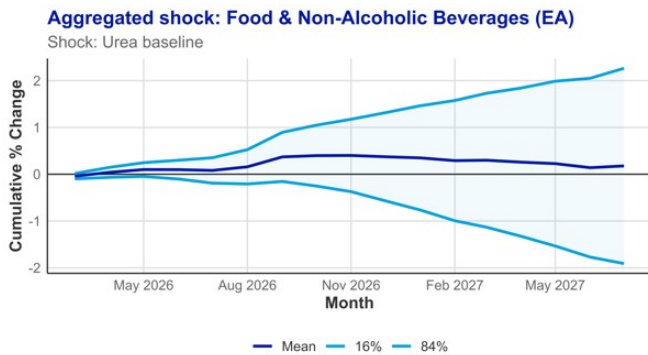
Source : Deutsche Bank

To be sure, the March shock impact alone would have had a big impact on food prices in the Euro area and the UK. For both oil and urea, it was an upside surprise of over 40 percentage points (over 3 standard deviations). However, over the course of the past few months, there have been significant reversals in both oil and urea prices. We attempt to capture these offsetting shocks and obtain the overall impact since March. How do we do this? We do this by aggregating the impacts of the



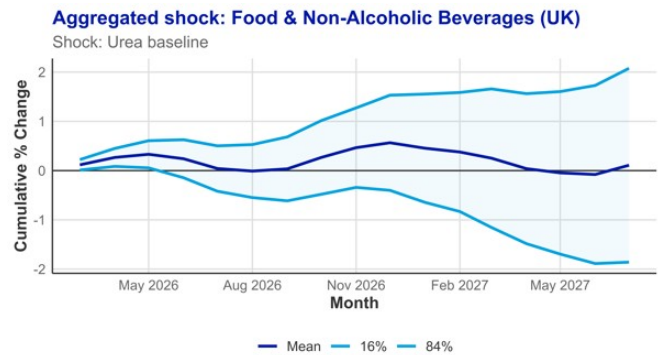
monthly urea and oil price shocks – i.e. we aggregate the shocks from oil and urea via superposition of the estimated local projections.

Figure 12: Estimated aggregate urea shock on food HICP in the EA...



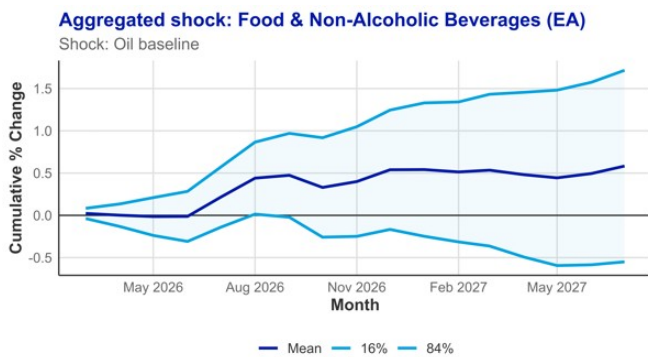
Source : Deutsche Bank

Figure 13: ...and on UK food CPI



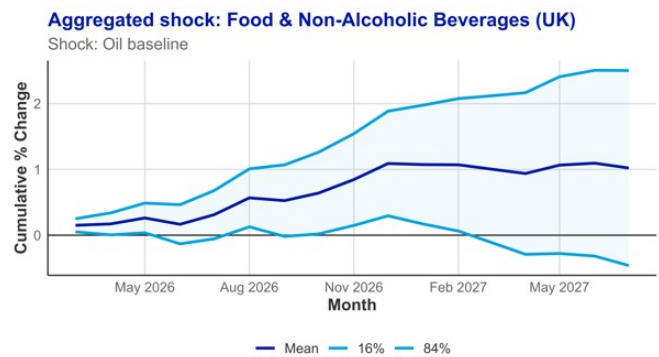
Source : Deutsche Bank

Figure 14: Estimated aggregated oil shock on food HICP in the EA...



Source : Deutsche Bank

Figure 15: ...and on UK food CPI



Source : Deutsche Bank

We trace out these shocks and find some interesting implications for the overall impact. For the UK, we see a cumulative impact of around 1% on food prices from the oil shocks a year from March and 0.3% from a urea shock peaking also in a year's time from March, i.e. totalling nearly 130bps (though the effect starts to unwind four quarters out). For the Euro area, the oil shocks lead to about a 0.5% cumulative increase in a year from March and about a 0.3% impact from urea shocks in a year from March. Overall, we observe similar patterns where the UK experiences a slightly stronger impact from the recent shocks compared to the Euro area.

Stress testing our results: persistent vs transitory shocks

A simple scaling of the shocks is unlikely to fully reflect firms' pricing behaviour. In practice, firms may respond differently to the size and duration of the shock. A large but temporary increase in oil or fertiliser prices may generate less pass-through than a smaller but more sustained rise.

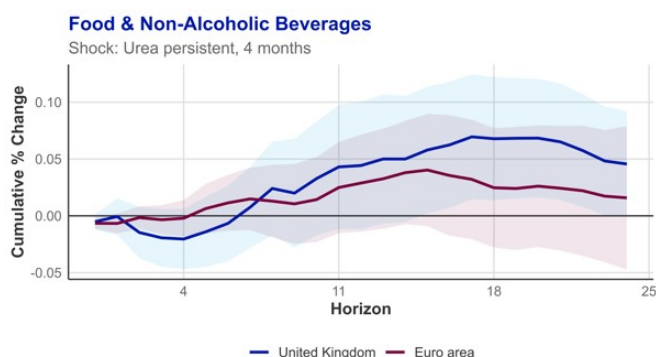
To capture this distinction, we split our oil and fertiliser shocks between persistent



and transitory. We classify a shock as persistent if the price remains at least half as large after four months. If the price falls by more than half over that period, we treat it as transitory. We only apply this classification to sizeable shocks, defined as moves larger than one standard deviation in the relevant shock series. This gives us a simple way to separate temporary price spikes from more sustained input-cost pressures.

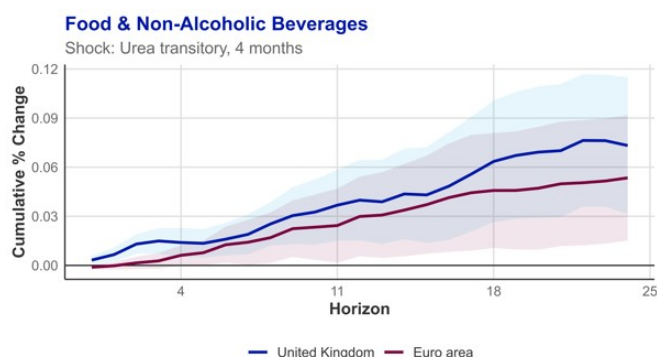
We then run local projections again on the persistent and short-lived shocks and we obtain the following results:

Figure 16: Persistent urea price shock...



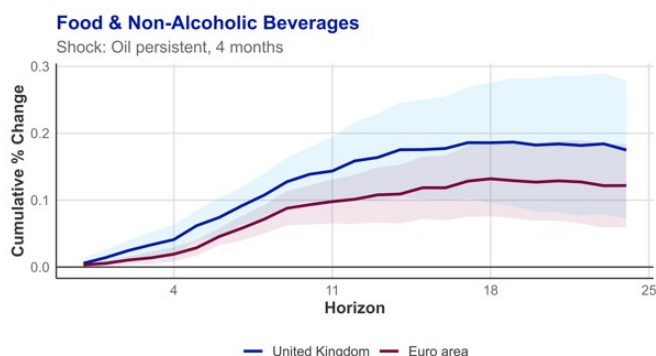
Source : Deutsche Bank

Figure 17: ...vs transitory urea price shock



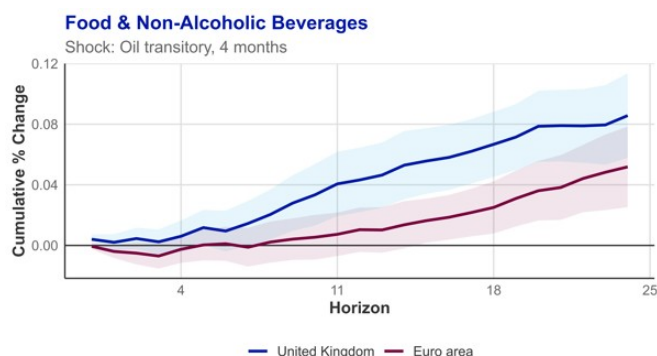
Source : Deutsche Bank

Figure 18: Persistent oil price shock...



Source : Deutsche Bank

Figure 19: ...vs transitory oil price shock



Source : Deutsche Bank

The results suggest that persistence matters, particularly for oil shocks ([figures 18 and 19](#)). When the shock is persistent, we find a stronger peak impact for both the Euro area and the United Kingdom for a given 1 percentage point surprise compared to a short-lived shock. For urea, the pattern is clearer in the UK, where persistent shocks are associated with a larger food price pass-through. In the Euro area, however, the difference between persistent and short-lived shocks is less clear. Overall, the evidence points to a stronger role for shock persistence in the UK, while the Euro area results are more mixed.

While it is too early to definitely classify the latest energy price shocks, the above results highlight an important distinction. If recent recent rises in oil and fertiliser prices prove short-lived and unwind over the coming months, then pass-through to



food prices may be more limited. However, if prices remain elevated, history suggests we could expect a larger and more sustained impact on food inflation.

Based on the shocks observed so far, our classification leans towards the recent developments being transitory in nature. That said, the classification is far from certain and will depend on how oil and fertiliser prices evolve through July and the months ahead.

Conclusions

What to make from the above analyses? We highlight several key findings.

First, the impact of a fertiliser and oil price shock and its pass-through to food price levels tends to be stronger in the United Kingdom than in the Euro area.

Second, after scaling and aggregating the fertiliser and oil price shocks from March up to June, we find that in a year's time from March, the cumulative impact on the UK food price level could be about 1.3% while for the Euro area it would be around 0.8% instead – implying a 0.1-0.15pp increase in headline HICP/CPI.

Third, we also recognise the limitations of simply scaling these shocks. The size and duration of the shock can result in very different implications for food price pass-through. The persistence of a shock can have a much larger impact, especially in the UK. It's too early to say where the current energy shock will fall into. Recent price action suggests a further wave of energy pressures on the back of escalating Middle East tensions. We will be monitoring this closely. The longer commodity prices remain elevated, the more they feed through into food prices.

Lastly, the confidence bands around our estimates suggest wide variation in pass-through. Given the swings in price rises across oil and urea, the staggered pass-through of cost rises into the food supply chain may be more varied. While our estimates provide a measure of the average conditional pass-through, other factors will also impact pass-through. For example, warmer global temperatures, less rainfall, and changing labour costs alongside ongoing geopolitical tensions may also shape pricing strategies.

To be sure, despite the pull back in oil and urea prices from their peak, the net effects still point to a modest jump in food (and non-alcoholic beverages) prices. Furthermore, the recent jump back in price momentum will do little to assuage concerns of indirect effects brewing within inflation. We will reflect some of our findings over the coming week(s) as we update our inflation projections.



Appendix 1

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Which decade was the most uncertain?

July 13, 2026

Executive summary

The cliché of our time is “uncertainty”. And for good reason. So far this decade, we have lived through a pandemic, Europe’s biggest ground war since WWII, a historic rate-hiking cycle, a tariff shock, and the closure of the Strait of Hormuz – all within roughly six years.

So, ask people whether today’s investment environment is more or less certain than a prior decade, and the intuitive answer is that today is more uncertain than ever.

Yet, when the dbDataInsights team put this question to a representative sample of the US and UK populations, we received a surprising response. By a considerable margin, people tend to think there is more certainty today than in any other post-war decade. That is despite many other factors indicating great disaffection and very low economic confidence.

This report sets out to examine why. In short, our conclusions rest on three pillars, which relate to the way in which the six megatrends that impact economies and markets are changing over time.

- A government backstop that did not previously exist at such scale or speed is available today. This has been highly visible to households and investors and we believe markets are pricing this in.
- The frequency of shocks has risen this decade but each individual disruption has, on average, been relatively contained rather than compounding to the extent that shocks have in prior decades.
- Technology has been a tremendous buffer that people have experienced personally in the 2020s.

We do not think today’s certainty is unconditional or permanent. In fact, it has a great cost that we expect will come due soon. Our megatrend model shows how this cost is building and how other trends are compounding it. We close this report with a brief discussion on the implications we see for portfolio construction.

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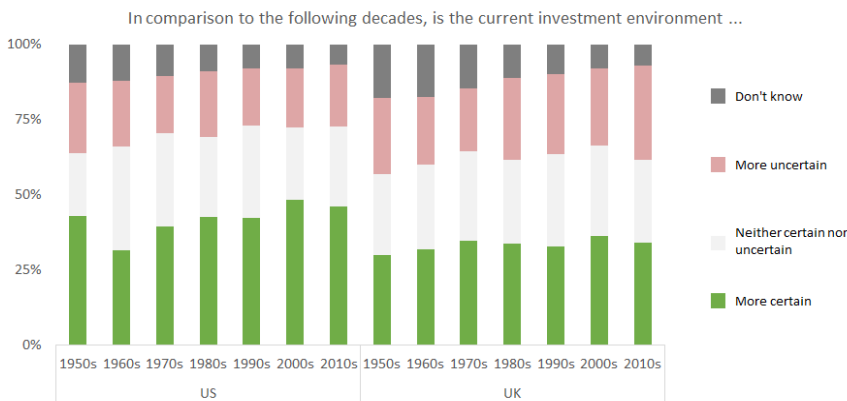


An unexpected finding

A common theme we hear from investors is that they are struggling to deal with bracing uncertainty in global politics, economics, and markets. Indeed, the geopolitical upheaval of the last few years is compounding other concerns such as an energy crisis, sovereign debt levels, high equity valuations, private capital exposure, tech volatility, rising populism and more.

Yet, when we actually asked people how they feel about today's investment environment, their answers were unexpected. Our dbDataInsights survey of a representative sample of people in the US and UK showed that people overwhelmingly believe that today's investment environment is more certain than it was any other post-war decade.

Figure 1: People think today is "more certain" than any prior post-war decade



Source : dbDataInsights, Deutsche Bank Research

Rearranging the numbers into a table, we can rank each decade from the most uncertain to least uncertain.

Figure 2: Americans and Britons have very different ideas about which decade was the most uncertain

The decades since WWII ranked **from most uncertain to least uncertain**
(Net % of people ranking today as being more certain each prior decade)

US		UK	
2000s	29%	1970s	14%
2010s	26%	2000s	11%
1990s	23%	1960s	9%
1980s	21%	1980s	6%
1970s	21%	1990s	6%
1950s	20%	1950s	4%
1960s	10%	2010s	3%

Source : dbDataInsights, Deutsche Bank Research

The result seems a bit paradoxical: investors claim today is more certain than prior decades, however, many consumer-sentiment surveys and polls of views on the state of the world show multi-decade lows.



Yet, we believe people can experience relative investment certainty at the same time they are worried about the state of the world. This belief reflects two structural changes in recent years:

1. a demonstrated government capacity to backstop shocks quickly and at scale;
2. a series of shocks in the 2020s that, so far, have been contained rather than having compounded as did prior crises the 1970s, 2008 and more; and
3. the impact that the non-AI technology evolution of the 2020s has had on people's work lives.

The three pillars are certainly a positive for the world, however, they come with a price tag largely in the form of sovereign debt. Our megatrend model rates this as one of the most persistently negative forces in our six-trend framework. Today's certainty may be earned, but it is also financed.

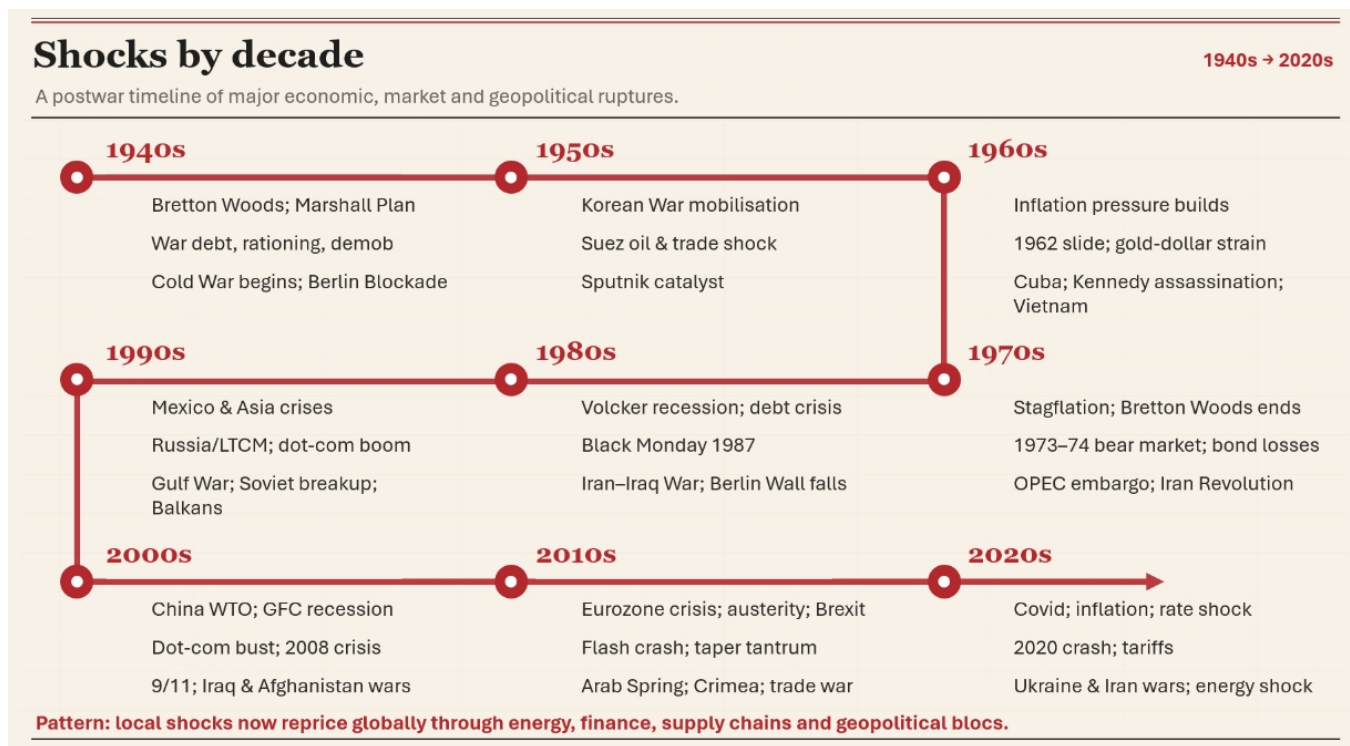
Shocks in context

The world has absorbed a remarkable run of shocks this decade: the covid pandemic, the fastest rate-hiking cycle in a generation, the Russia-Ukraine conflict, a tariff shock, and the war with Iran. These touched almost all global markets, yet we have emerged feeling more, not less, settled about its financial future.

To understand today's perceived certainty, we must first appreciate the magnitude of past uncertainties. The post-war era was defined by a series of profound shocks that fundamentally threatened the global economic and geopolitical order and brought us to the brink of nuclear war. As the decades wore on, the shape of those shocks changed from geopolitical to financial. Crucially, they all occurred at a time when fiscal and monetary stimulus was not used in the way it is today.



Figure 3: Are the shocks of the 2020s worse than those in prior decades? Investors think not.



Source : Deutsche Bank

Digging into the certainty paradox: narrative versus lived experience

The frequency of headline events is not the same thing as depth of realised economic damage, and it is not the same thing as an investor's confidence in the system's ability to respond. Our survey suggests people are, implicitly, drawing that distinction — separating “how often does something happen” from “how worried am I about my own financial position as a result.”

The following are three key points to help explain the certainty paradox.

Argument 1: The government backstop

Perhaps the single biggest structural change in global government of the past two decades is not a technology or a demographic shift — it is the demonstrated willingness to respond to shocks with fiscal and monetary firepower, quickly and at scale. The response to the shocks of the 2020s has been orders of magnitude greater than that for the crises of the 1970s or even 2008.

Consider the sequence of the 2020s alone. The Covid pandemic shock was met within weeks by the largest peacetime fiscal package in US history, combined with emergency central bank balance sheet expansion. The 2025 tariff shock and the 2026 Iran-linked energy shock both triggered rapid policy and market-stabilisation responses — some of which were reflexive as policy changed to placate markets.

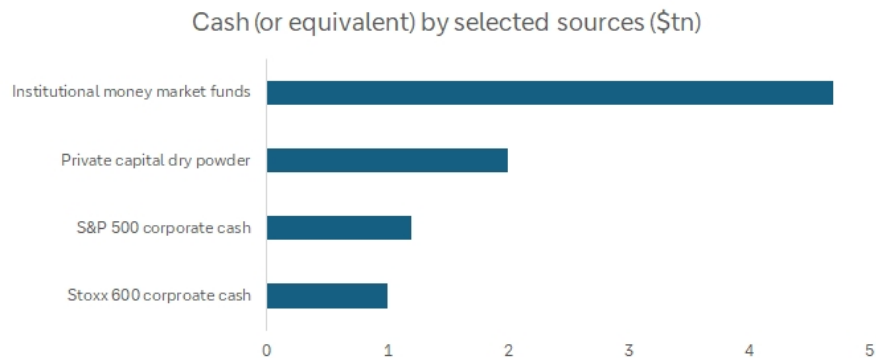
Each of these episodes was, in isolation, alarming. But each was met with a visible, fast institutional reflex — and our megatrend model (see [here](#)) shows that reflex increasingly priced into how megatrends interact with markets.



The other backstop – plenty of cash

That institutional capacity is not just a central-bank story. It also shows up in the sheer volume of private capital sitting ready to be deployed at the first sign of dislocation — itself a backstop of sorts, because a market that expects buyers to show up on any large drawdown behaves differently to one that does not.

Figure 4: The shock absorption capacity sitting on the sidelines



Source : Bloomberg Finance LP, Haver Analytics, Deutsche Bank. Note: corporate cash excludes financials, real estate and utilities.

None of this capital is a government backstop but its existence is itself downstream of a decade and a half of policy conditions (first ultra-low rates, now a smoother path through the 2022 hiking cycle) that left the corporate and institutional sector unusually flush. A system with this much capital behaves more like one with a backstop than one without.

Argument 2: Are today's shocks less shocking?

In spite of the greater frequency of shocks in the 2020s, each of them individually have, from a certain perspective, done less to shake the confidence of the average person compared with the defining shocks of prior decades.

Each shock has, so far, been contained rather than compounding

Consider some prior shocks:

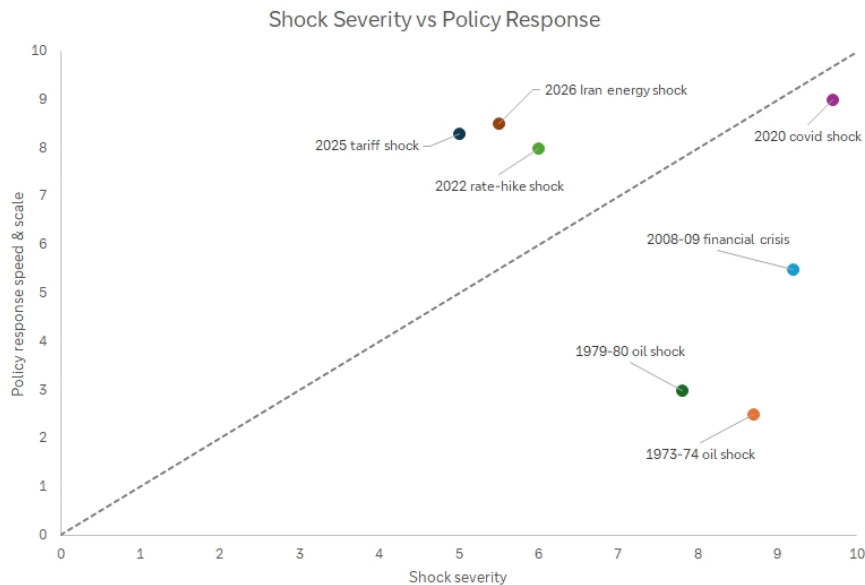
- The 1970s oil shocks were compounded by unresolved geopolitical tension, a technology cycle that had not yet turned, and worsening sovereign finances. These multiple negative megatrends reinforced each other while offsetting fiscal or monetary policy was unavailable or insufficient.
- The 2008 financial crisis was amplified in the same way. It was compounded by negative domestic politics, deteriorating demographics, and a geopolitics/globalisation backdrop that was also turning negative.

In contrast, the 2020s has been a decade in which shocks have been relatively isolated (despite their proximity). In addition, our megatrend model shows that technology and, more recently, the energy transition, have been consistently positive throughout the decade, partially offsetting the negative effects.

To illustrate the different shock profile, we have produced the figure below based on our AI-driven estimations. It is not a precise empirical index but it illustrates the underlying logic: more recent shocks have been met with a more aggressive response.



Figure 5: An illustrative guide that estimates recent shocks have had more aggressive policy responses



Source : Deutsche Bank

On this reading, the 2020s have delivered more shocks per year than prior periods but each individual shock has, so far, been met quickly enough. As a result, none of the shocks have been allowed to compound in the way that they did in, say, the 1970s or in 2008.

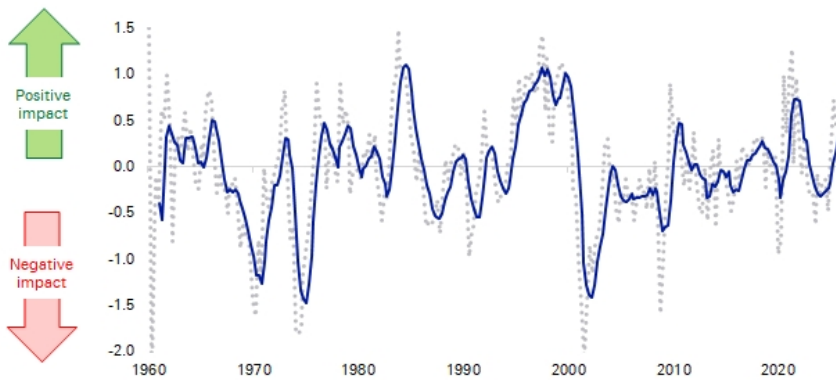
Argument 3: Technology as a buffer

The acceleration in technology in the 2020s has allowed large parts of the economy to keep functioning when physical movement, office attendance, and cross-border coordination were suddenly constrained. The clearest example is the ability to work remotely during the pandemic. But the advances extended to other areas, such as global supply chains which have weathered trade clashes, sanctions, tariffs, and pandemic bottlenecks via better data analytics, digital logistics platforms, real-time inventory systems, supply-chain 'control towers', AI-assisted forecasting and more.

The following chart shows our technology megatrend indicator for technology. During the early 2020s, this accelerated rapidly in terms of its impact on markets and economies, putting the advancement on par with the jumps during the 1980s computer rollout and the 1990s tech boom.



Figure 6: The impact of the technology megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank

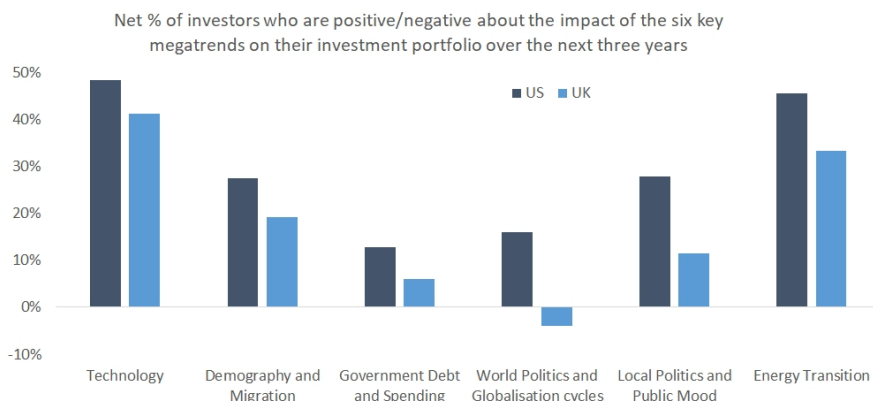
Technology has changed the way shocks actually propagate around the economy and has prevented shocks from compounding as severely as they might have done in earlier eras.

Technology-as-a-buffer is arguably more important for people's perceptions of certainty than is the potential for productivity gains that may stem from evolving technology – certainly reinforced by the very tangible and personal ways that people experienced technology during the pandemic.

Megatrends helping portfolios

Our survey also shows that investors feel today's megatrends will have a positive impact on their portfolios. The following chart shows that more Americans think that every megatrend will be positive for their portfolio than negative. In the UK, the trend is the same, albeit less pronounced, and with one small exception in geopolitics.

Figure 7: People generally feel positive about how megatrends will impact their portfolios



Source : dbDataInsights, Deutsche Bank Research



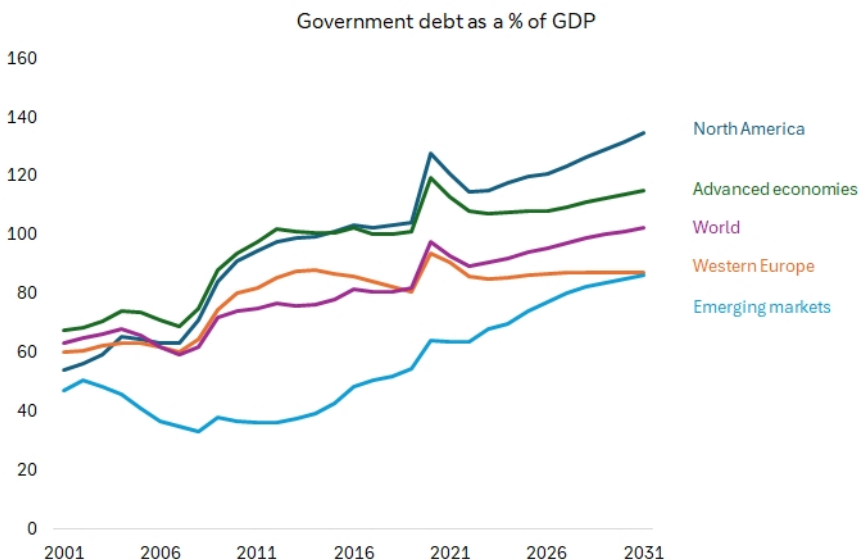
The main thing is that people's expectations of technology and energy are particularly strong. This result should perhaps be no surprise given the upheaval of AI, and the multiple energy shocks of the 2020s. Both have driven the type of disruption that has resulted in substantial investment into these sectors.

The bill for today's certainty will come due

Today's certainty is not costless or permanent. Indeed, the backstop that provides certainty is itself a source of risk. Specifically, we worry that investor faith in a rapid policy response to shocks may mean they simultaneously underestimate the negative megatrends that are impacting the world's economies and markets.

Most importantly, every fiscal intervention adds to the sovereign balance sheet, and our megatrend model indicator for sovereign deficits is sharply negative. That increases the risk that future shocks will not only have their own impact but also may contribute to a market risk event for sovereign debt.

Figure 8: The expected acceleration in sovereign debt raises the potential severity of a market shock when it does eventually occur



Source : IMF, Deutsche Bank

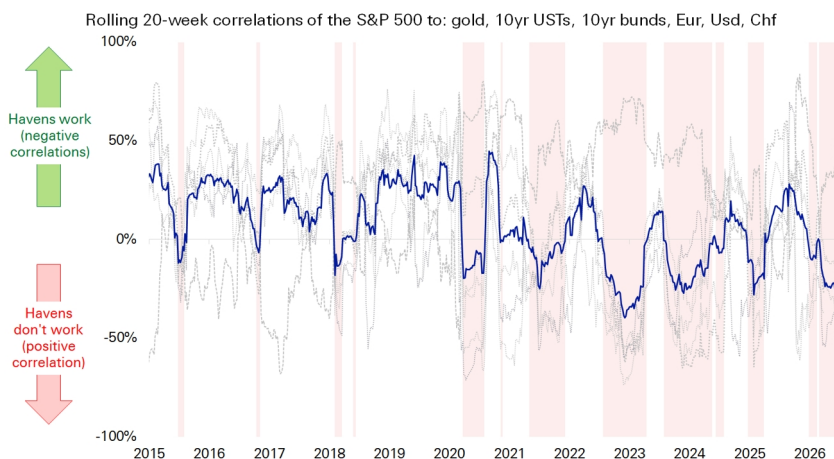
None of this contradicts the finding that today feels, and arguably is, more certain than decades past. But it does indicate that certainty is conditional on both the continued fiscal capacity to backstop shocks and the ability of technology to deliver meaningful productivity upgrades that will outweigh the drag from sovereign debt, demographics, and domestic politics.

Traditional havens are not the hedge most investors expect

When shocks do occur they are not being hedged as people expect. Our haven indicator below shows that six key hedging assets (gold, the US dollar, the Swiss franc, the Japanese yen, and long-dated government bonds) have become less reliable as hedges this decade.



Figure 9: Our haven indicator shows that, during the 2020s, the usual havens have been ineffective, especially during shocks



Source : Bloomberg Finance L.P., Deutsche Bank

This matters because it adds to the potential for future shocks to be compounded by problems with portfolios that are not able to sustain the losses investors currently expect. Leverage would compound the problem further.

Implications for investors

If the certainty investors feel today is real but conditional, we believe three practical implications follow.

- Do not assume traditional havens will do their job in the next shock. With gold, the US dollar, the Swiss franc, the yen, and core government bonds all showing an unreliable hedge relationship with equities since 2020, portfolio construction may need to lean more on genuinely uncorrelated assets and private-market diversification.
- Treat policy-response speed as a variable to monitor, not a constant to assume. The backstop that explains today's calm depends on fiscal and political capacity that is itself being eroded by the sovereign-deficits megatrend. A shock that arrives when that capacity is more constrained — rather than during a period of ample dry powder — would test the very mechanism this report describes.
- Watch the technology megatrend as the single most important swing factor. So much of today's certainty rests on technology offsetting other negative megatrends. A stalling or disappointing AI adoption curve would remove the offset that has, so far, helped keep shocks from compounding the way they did in the 1970s and 2008.

In short: the certainty expressed in our survey results is not a mirage, but neither is it a permanent state of affairs. It is the observable output of a strong policy backstop and a positive technology cycle. Both are doing more work than they have typically had to do in the past. Investors who understand why today feels calmer are likely to be better placed to notice if either force starts to shift.



Conclusion

Shocks in the second half of the 20th century were existential, system-breaking crises: the threat of nuclear war, the collapse of the monetary system, the oil embargoes etc. These have been replaced by challenges that, while significant, are more manageable within the current economic framework.

Yet, the bill for this certainty will come due and when it does it will likely be at a very inconvenient time. Indeed, our megatrend analysis suggests the defining feature of the next five years will be volatility amidst a greater frequency of shocks. The risk is that, at the same time that a shock occurs, markets fear the fiscal response and sovereign debt markets sell-off, thereby limiting the ability to mitigate the shock.

So, what of people's perception today of relative investment certainty? From one point of view it is logical they feel this way - they have become used to the policy response. However, that attitude only increases the potential negative impact when a shock occurs and policymakers find they cannot simply throw money at the problem.



Appendix 1

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Mapping the World's Prices - 2026

July 13, 2026



Which are the world's cheapest and most expensive cities?

Mapping the World's Prices, now in its tenth edition, is the definitive guide to quality of life, rental costs and prices for phones, coffee, taxis and more in 69 cities from Abu Dhabi to Zurich.

Authors

Jim Reid

Global Head of Macro and
Thematic Research

Galina Pozdnyakova

Research Analyst



Mapping the World's Prices – 2026

Welcome to the tenth edition of Mapping the World's Prices. When this series launched in 2012, the eurozone was in crisis, the yen was strong, and a cappuccino in Zurich already cost more than a full meal in Mumbai. Nearly 15 years on, the global cost-of-living map has been redrawn – by a pandemic, an energy crisis, the strongest dollar cycle in a generation, and now by tariffs and geopolitical conflict.

This year's edition covers 69 cities across 6 continents, with prices converted to US dollars and sourced primarily from Numbeo's crowdsourced database - augmented by our own cleaning and cross-checks. As always, crowdsourced data carries inherent noise: we flag potential outliers where they appear and urge readers to focus on broad trends rather than any single data point.

In this edition, as well as looking at year-on-year changes, we compare 2026 prices against 2016 levels - a full decade - as well as against 2019 (a pre-Covid baseline) and also against 2012 when we first published. 2012 was a pivotal year as it marked peak Yen and a historically cheap USD. So it's a fascinating benchmark, especially as the exact reverse conditions are now in place. Reading this in 2012 would have probably led you to increase your exposure to the US. Will you want to do the same with Japan after reading 2026's edition?

These longer-term perspectives reveal structural shifts that annual data alone cannot capture: which cities have genuinely become more expensive, where living standards have improved, and where an entire region's pricing has been fundamentally reset.

If nothing else, it will give you a good idea of how expensive your summer holiday will be over the next few weeks and where you may want to go next year for the best value break!

Please delve into the full report where we have synthesized tens of thousands of global price data points in user friendly summaries, graphs and tables.

As always, feel free to quote and reference the report. If so we would be grateful if you could attribute to the Deutsche Bank Research Institute. Feel free to direct people to this report and many others at <https://www.dbresearch.com/PROD/IE-PROD/HOME.alias>



At a Glance: The Headlines:

- **Zurich and Geneva** continue to battle it out to be **the most expensive cities in the world** followed now by Tel Aviv, New York and San Francisco. Of the largest developed market cities, Tokyo stands out as being very “cheap”.
- **Luxembourg** holds the **#1 quality-of-life spot** for a second year. Copenhagen, Amsterdam, Vienna and Munich are others in the top 5 with Germany also seeing Frankfurt in 6th. Global financial hubs like Paris (43rd), New York (46th), London (47th), and Hong Kong (55th), score lower on liveability, hampered by expensive housing, long commutes, and high pollution levels. However, this is a highly subjective measure.
- **Zurich** tops the **global salary table**, followed by San Francisco, Geneva, Boston and New York. Over a decade, Budapest salaries have risen +161%, Prague +121%, and Warsaw +120%.
- Cities that make **the top 10 for both quality of life and disposable income** are Luxembourg, Copenhagen, Frankfurt, Geneva and Zurich - **all European**. No US city makes both lists.
- **Hong Kong** retains **the most expensive property**, even if prices are down - 10% from pre-Covid levels, but Seoul (3rd) is closing in as tech fever envelops a supply constrained market. Zurich, Singapore and Geneva round out the top 5. Budapest property prices have tripled (+209%) in 10 years - the largest gain in our sample.
- **Japan** has completed a **remarkable structural repricing**: Tokyo's PPP-implied price level has fallen from 173 in the mid-1990s to just 60 today.
- Renting a 3-bed apartment in **Tokyo** is only around a quarter of that in New York. A meal for two in Tokyo now costs less than in Warsaw or Prague and is a third of the cost of Zurich and New York. A McDonald's costs a quarter of that in Tel Aviv and Japan is the cheapest place in the world to buy an iPhone.
- **An iPhone in Türkiye** costs 2.2x the US price, the most expensive across 41 economies we track. South Korea and Japan are the only places where the iPhone is cheaper than in the US.
- **US tariffs**, by contrast, have left a surprisingly **light consumer-level footprint** - so far. US grocery and car prices are broadly flat YoY.
- **Tel Aviv** has experienced a decade of climbing the global prices table and now sits at the top of the most expensive city in the world to buy a McDonald's meal. The city is in top 5 for prices of a few items including a meal for two, a summer dress and a pair of jeans.
- Back to **3-bed rentals**, and **New York** retains its top spot, followed by Zurich, San Francisco, Boston and Singapore. London (8th), Paris (21st), Frankfurt (30th) and Tokyo (40th) are interesting big financial city datapoints. Seoul (44th) is much cheaper to rent in than to buy.
- A **monthly public transport pass** remains the most expensive in **London** (by a big margin) and remains free in Luxembourg. Sydney, New York, Melbourne and Auckland are the other most expensive. **Gasoline prices** are almost 9 times as expensive in Hong Kong than Cairo, although a Zurich **taxi** costs 23 times that of a Cairo one.
- **German and Central European cities** heavily populate the top 10 in terms of **highest utility bills**, a lasting legacy of the energy crisis and the shift away from Russian gas and nuclear power.
- Our **“Cheap Date” index** sees Geneva, Zurich, Tel Aviv, Oslo and Copenhagen edge London out of the top 5 as the most expensive place to be a romantic. A date in Geneva costs nearly 5 times that of Delhi.



- If you love **wine and coffee** head to **Rome**. It has the cheapest bottle of wine in the world and the 4th cheapest cappuccino. Lisbon is next cheapest for wine while Singapore is the most expensive. Other cheap places for wine include Buenos Aires, Budapest, Johannesburg and Cape Town, and even Frankfurt.
- Don't spill the cheap wine on your **summer dress** in **Buenos Aires** as it's the most expensive place to buy a replacement.
- Access to the **internet** is the most expensive in the **Middle East** (Dubai top) and India, Russia and China have cheap connectivity.

Methodology: *We've continued to improve consistency, standardising goods and services across geographies and converting all figures into USD. Crowdsourced data remains our backbone and thanks to Numbeo for their wealth of data here. Numbeo offer data on many more global cities beyond those we selected on their website. For our sample, we augment the data by rigorous cleaning where necessary across thousands of datapoints. There will always be a few puzzling outliers—particularly in the historical numbers—but we're confident this edition offers the most comprehensive snapshot for cities relevant to the financial world.*

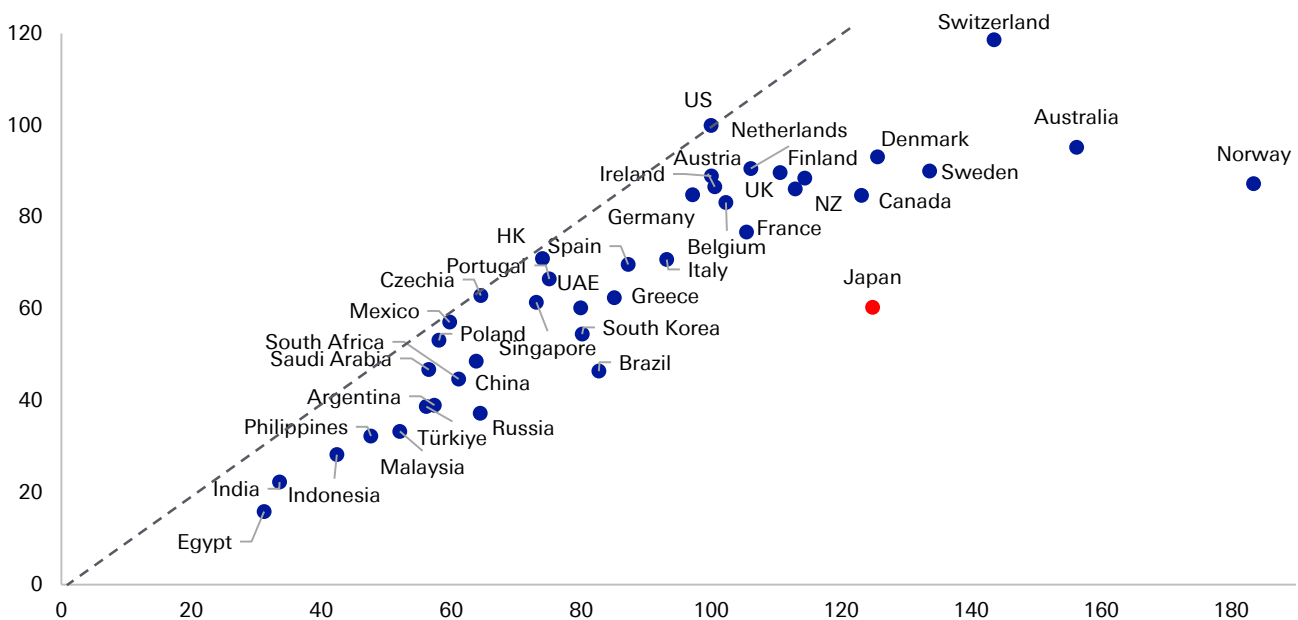


The Macro Backdrop — Currencies, Inflation, and Purchasing Power

Since this report was first published in 2012 we have seen big swings in inflation, growth, prices and exchange rates but one of the main enduring trends has been that the US has become notably more expensive and Japan has travelled from expensive to very cheap.

Over this period, on a PPP basis, the US has got more expensive relative to every single one of the 42 economies covered in this report. Australia and Norway have cheapened up considerably but still rank highly on a PPP basis. The real outlier is Japan.

Figure 1: Relative Price Levels as Implied by PPP (US = 100), 2012 (x-axis) and 2025 (y-axis)



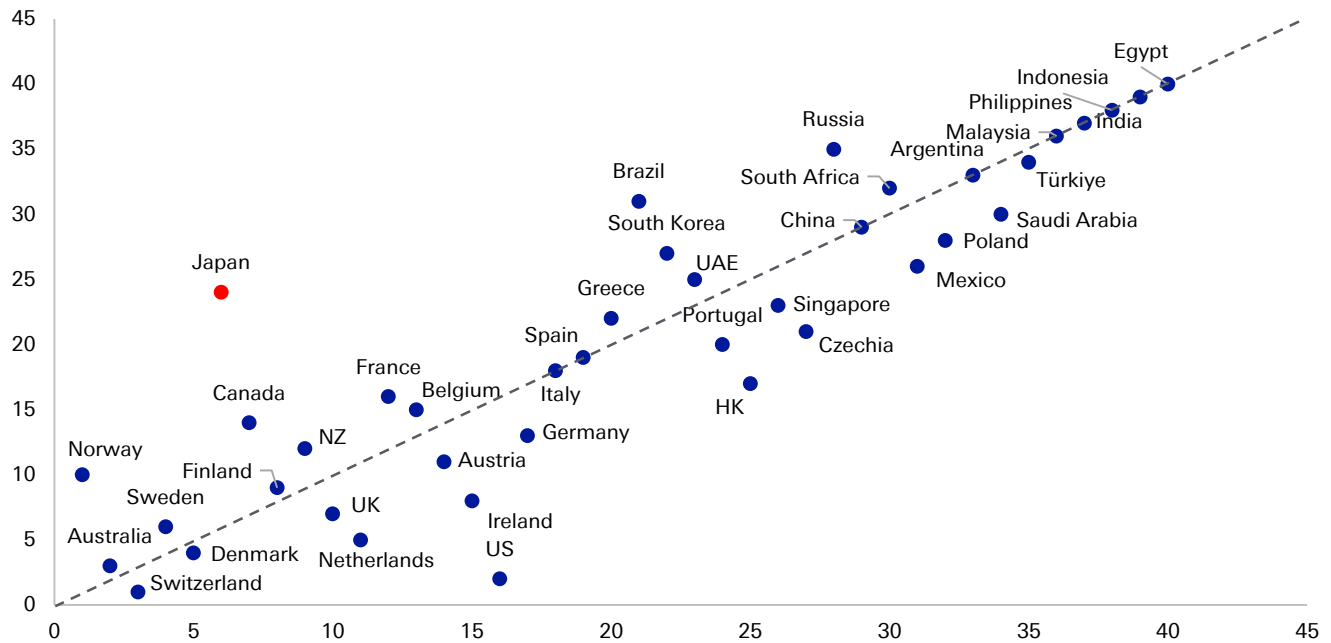
Source: IMF, WEO, Haver Analytics, Bloomberg Finance LP, Deutsche Bank calculations.

Note: The PPP conversion rates as published by the IMF have been adjusted with actual foreign exchange rates (as on 31 December each year) to derive the implied price levels. Also note that some of the above numbers may not match those published last year exactly due to data revisions.

Japan's PPP-implied price level has fallen from 173 in the mid-1990s, 125 in 2012, to just 60 today. A meal for two in Tokyo now costs less than in Warsaw or Prague and has gone from being around three-quarters of the cost of the meal in New York in 2012 to only around 30% today in USD terms.



Figure 2: Relative Price Levels as Implied by PPP (US = 100), ranked by economy between 2012 (x-axis) and 2025 (y-axis)

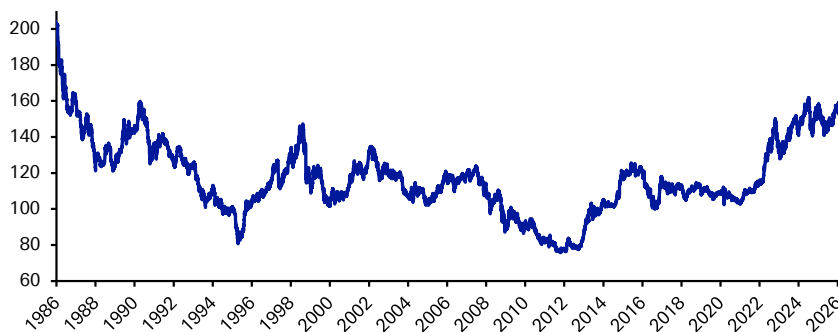


Source : IMF, WEO, Haver Analytics, Bloomberg Finance LP, Deutsche Bank calculations.

Note: The PPP conversion rates as published by the IMF have been adjusted with actual foreign exchange rates (as on 31 December each year) to derive the implied price levels. Also note that some of the above numbers may not match those published last year exactly due to data revisions.

2012 marked the point of maximum Yen strength when after years of funding carry trades due to ultra-low rates, money flowed back to Japan as other global central banks also slashed rates close to zero (like Japan had previously for years). The Yen was also seen as a safe haven in a world of European sovereign stress and uncertainty about the US post the GFC. Nearly a decade and a half later and Japan has spent the last few years defying the global round of interest rate normalising and the Yen has become the premier funding currency of the world again.

Figure 3: USDJPY



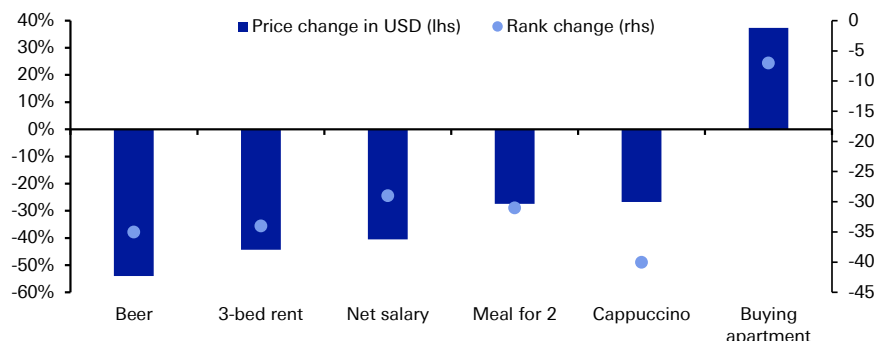
Source : Bloomberg Finance LP, Deutsche Bank.

Given Japanese inflation has been relatively low globally, while its currency has slumped, Japan now stands out amongst its developed market peers in this report as being extremely cheap when converting to USDs. The yen is down by -51% since 2012 and inflation is 'only' up by +20%. The result is a Tokyo that is now genuinely cheap by developed-market standards:



- If you account for taxes, Japan is the cheapest place in the world to buy an iPhone.
- You can have nearly two dates in Tokyo for the price of one in London according to our “Cheap Date” index.
- A meal out for two in Tokyo is ranked 57th out of 69 and is actually a third of the cost of the same in Zurich and New York. It’s cheaper than in Johannesburg, Sao Paulo, Istanbul, Warsaw and Prague amongst dozens of other global cities on this measure.
- A McDonalds meal now costs four times as much in Tel Aviv (the most expensive city on this measure) than in does in Tokyo (63rd) with a beer more almost two times as expensive in London. A cappuccino in Tokyo is cheaper than in 44 of our 69 focus cities, including EM economies.
- Tokyo property and rental prices, the most expensive in the world around 35 years ago, are now closer to the middle section of our global tables.
- Renting a 3-bedroom apartment in Tokyo (40th) is almost a quarter of the cost of doing so in New York. In USD terms, it’s down -23 percent since 2016.
- Buying an apartment in Tokyo (25th) is nearly a third of the cost of Asian rivals Hong Kong and Seoul, and half that of London. Similar to rent costs, the price per square meter is down by almost a quarter over the last 10 years.
- Salaries have also lagged with Zurich (1st) seeing wages 3.5 times that in Tokyo (39th), which sits next to Madrid, Prague and Warsaw.

Figure 4: Tokyo, price and rank change since 2012 (USD)



Source : www.numbeo.com, Deutsche Bank.

So on virtually all measures, including those in this study, Tokyo and Japan are very cheap internationally. Note that when we launched this series in 2012, the US was very cheap versus its peers. For example, New York has moved up +32 places to the 8th most expensive city for a date using our Cheap Date index that includes prices of a meal for two, coffees, alcohol, taxi and clothes. It is also now ranked 4th for a fitness membership, jumping +11 ranks since 2012. It is a similar picture for other US cities that we track.

While economies can stay cheap or expensive for a prolonged period of time, the AI boom does potentially provide Japan with a potential catalyst for mean reversion over the medium-term. The Nikkei is up over +70% over the last 12 months and has finally shaken off the peak level seen at the end of 1989.



Figure 5: Nikkei 225 index level



Source : Bloomberg Finance LP, Deutsche Bank.

While the US leads in frontier model creation, Japan's opportunity lies in becoming a leading real-world implementer of AI, especially in industries where it already has deep manufacturing and robotics expertise.

1. **Demographics as a catalyst.** In many Western economies, AI adoption is often framed through the lens of job displacement. In Japan, the starting point is different: a shrinking and ageing population has created persistent labour shortages across sectors, making productivity-enhancing technologies more economically and politically attractive. This does not eliminate displacement risk, particularly in white-collar roles, but it means AI is more likely to be viewed as a tool to ease capacity constraints rather than simply as a substitute for labour.
2. **A more adoption-oriented policy stance.** Japan's AI framework is also relatively supportive of deployment. Unlike the EU AI Act, which establishes a binding risk-based regulatory regime, Japan's 2025 AI Act and subsequent AI Basic Plan are more promotion-oriented, focused on accelerating research, development and utilisation while addressing risks through governance, transparency and guidance. The government's stated ambition to become "the most AI-friendly country in the world" underscores this pro-adoption bias.
3. **A potential edge in physical AI.** Japan's industrial base gives it a potential advantage as AI moves beyond chatbots and software into robotics, factories, logistics, healthcare and other physical-world applications. The country has longstanding strengths in industrial robotics, precision manufacturing and process engineering. So Japan's opportunity lies in combining its hardware and manufacturing expertise with faster progress in AI-enabled systems.

Ultimately, Japan's back is against the wall demographically. This makes aggressive AI adoption not just a competitive advantage, but a strict economic necessity for the country's survival over the next two decades.

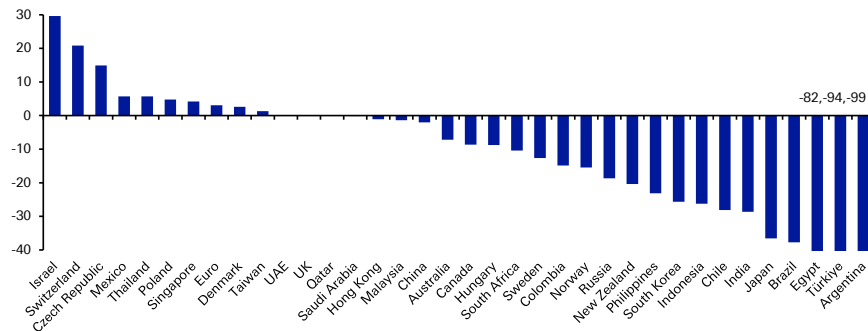
So Japan stands out in this study but it's not the only economy where there have been big price and FX moves over the last decade or so.



Other Movers in The Last Decade

Exchange rate moves are the single largest driver of year-to-year changes in our USD-denominated data. The past decade has produced a dramatic divergence:

Figure 6: Cumulative changes in currencies of our sample economies vs the USD (%) over the past 10 years



Source : Bloomberg Finance LP, Deutsche Bank. Note: calculated from 30 June 2016 to 30 June 2026.

The Israeli shekel's +30% gain is one of the strongest in our sample and helps explain why Tel Aviv has climbed so many price rankings. The Czech koruna (+15%) and Polish zloty (+5%), combined with strong domestic wage growth, and higher inflation than in Western Europe, have transformed Prague and Warsaw from bargain cities into mid-price destinations.

At the other extreme, the Argentine peso's near-total collapse (-99%) means Buenos Aires remains near the bottom of most price tables in USD terms despite domestic hyperinflation. As already discussed, the yen's -37% decline has turned Tokyo from one of the world's most expensive cities into a mid-table one.

Israel: Climbing to The Top

You will see in the report that Tel Aviv has seen a big climb up the tables as the economy gets more expensive in dollar terms partly due to the Shekel being one of the best performing currencies over the last 12 months (+13%), which keeps it firmly at the top of the charts over the last decade of major currencies. Strong domestic tech and defence industries help, as do regional supply disruptions.

Here are some Tel Aviv facts from our study:

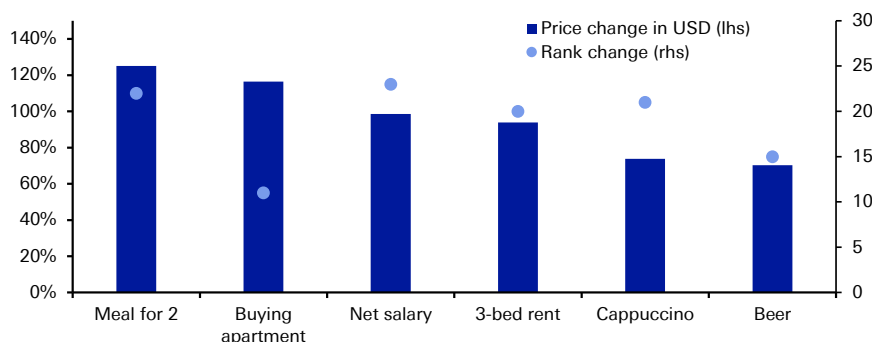
- It's now the most expensive city in the world to buy a McDonald's meal. The city is in top 5 for prices of a summer dress and a pair of jeans.
- It also has the second most expensive utility prices in the world, ranking in between Berlin and Oslo.
- It's the second most expensive for gasoline, climbing +6 places this year and +13 over the last decade and second most expensive to buy a car.
- Moves up to 4th most expensive place for a "meal for two" behind only Zurich, New York and Geneva. Climbing +3 places this year and +22 over the last decade.
- Moves up to the third most expensive place for a "cheap date" behind only Geneva and Zurich. Climbing +2 places this year and +12 over the last decade.



- Apartment prices are 6th globally with a 3-bed rent 16th.
- Salaries are 12th but up +10 places from last year.

Tel Aviv is no longer a "mid-price" Mediterranean city; it now sits firmly among the world's most expensive.

Figure 7: Tel Aviv, price and rank change since 2016 (USD)



Source : www.numbeo.com, Deutsche Bank.

Central Europe: The Great Convergence

Alongside Israel, Central Europe represents another example of structural ascent over the last decade. The numbers are striking:

Figure 8: Price changes for selected items in Central and Eastern Europe (USD)

Item	Prague				Warsaw				Budapest			
	2016	2026	USD price % change	Rank change	2016	2026	USD price % change	Rank change	2016	2026	USD price % change	Rank change
Net salary	1,063	2,348	121%	+9	1,009	2,221	120%	+9	678	1,770	161%	+16
3-bed rent	1,093	2,349	115%	+16	1,040	2,175	109%	+12	874	1,771	103%	+5
Buying apartment	3,868	9,744	152%	+18	2,796	6,350	127%	+11	1,980	6,116	209%	+14
Meal for 2	28	54	98%	+9	30	64	117%	+11	31	70	122%	+17
Cappuccino	1.91	3.80	99%	+15	2.29	4.55	99%	+23	1.42	3.56	151%	+23
Cheap date	170	250	47%	+13	168	263	56%	+21	166	276	66%	+28
Beer	0.7	1.2	65%	+2	0.8	1.2	49%	+4	0.9	1.3	51%	+7

Source : www.numbeo.com, Deutsche Bank.

Five forces have driven this convergence:

1. EU structural funds boosting investment and productivity
2. Tight labour markets driving 5–10% annual local-currency wage growth
3. Currency appreciation (koruna +15%, zloty +5% vs USD over the decade)
4. Post-Covid inflation (Hungary +57%, Poland +51%, Czech Rep +43% cumulative since end of 2019) that has not fully reversed
5. Capital inflows into property

The PPP data confirms the trend: Czech Republic's relative price level has risen from 48 to 63 (vs US = 100), Hungary's from 45 to 57 and Poland's from 42 to 53 over the past decade. These cities can no longer be considered "cheap" European destinations. A weekend in Prague or Budapest can now cost more than in many



Western European cities - a reversal of the dynamic that prevailed for years after the fall of communism.

Peak US - an Update

Last year we asked whether 2025 might represent "peak US" in pricing terms. One year on, the evidence is mixed.

Signs the US remains at the top:

- New York retains the #1 spot for rents (1-bed and 3-bed)
- US cities still dominate the top 5 for salaries
- US grocery prices remain among the world's highest
- No visible tariff-driven price acceleration, yet

Signs of potential moderation:

- Several US salary figures were broadly flat
- US car prices broadly flat despite tariffs, suggesting the strong dollar may be masking underlying pressures
- San Francisco property prices have fallen -15% over the past decade

The stabilisation of US prices, combined with sharp increases in European and other cities, suggests the gap may be starting to narrow.

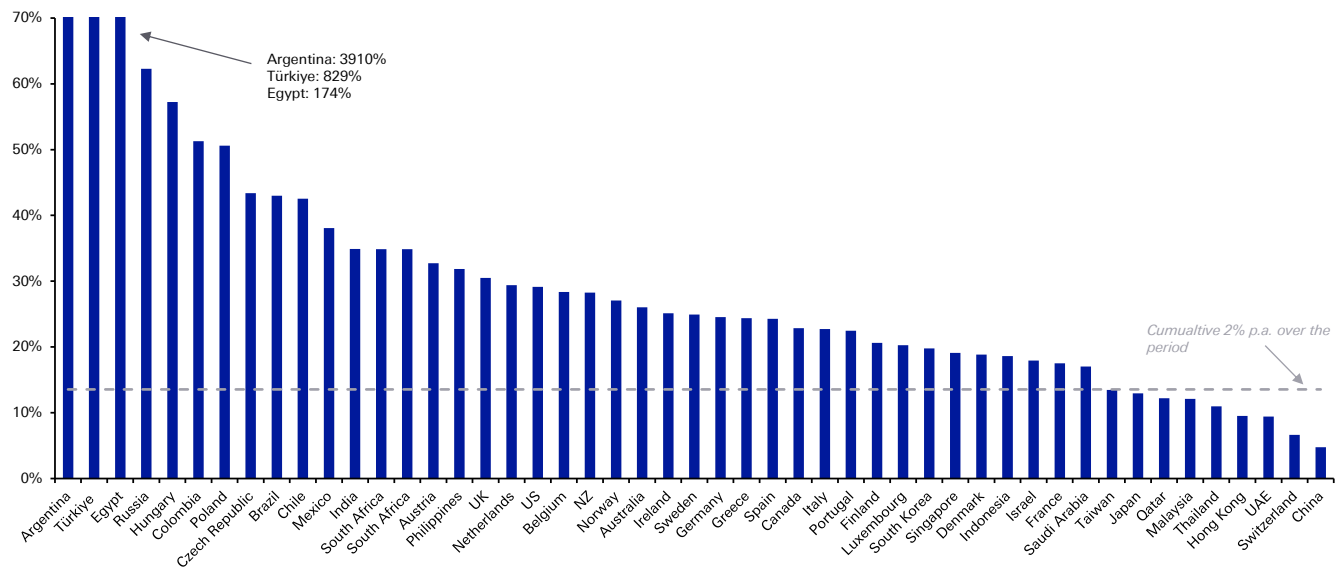
Contrast within the US emerges when tracking disposable income which adjusts for rents. New York is up four places to 39th place this year (-12 places since 2012 as rents have soared), and although Boston (5th) has slipped a place in 2026, it is up +25 places since 2012. San Francisco is 3rd in 2026, up 35 places in a decade. So despite the overall price increases and big affordability issues in New York in particular, a few other big US cities have been doing well as salaries have kept up with housing affordability.

The Post-Covid Inflation Scorecard

Inflation was always with us in various parts of the world in the 2010s when this report first started but for many parts, especially developed markets it was largely subdued. However since Covid global inflation is well and truly back and is shaping relative prices and currencies everywhere. Cumulative CPI inflation since 2019 reveals where the cost of living has risen most in local currency terms — stripping out FX effects:



Figure 9: Cumulative inflation since end of 2019 and just before Covid changed the landscape



Source : Haver Analytics, Deutsche Bank.



Quality of Life

The quality-of-life index combines purchasing power, safety, healthcare, cost of living, property-price-to-income ratio, commute time, pollution, and climate into a single score.

Figure 10: Quality of Life Top 10 (2026) vs 2025 edition

Current rank	City	Rank change from 2025
1.	Luxembourg	+0
2.	Copenhagen	+0
3.	Amsterdam	+0
4.	Vienna	-1
4.	Munich	+8
6.	Frankfurt	+1
7.	Helsinki	-2
8.	Geneva	-2
9.	Zurich	-1
10.	Edinburgh	+0

Source : www.numbeo.com, Deutsche Bank.

Luxembourg, Copenhagen, Amsterdam and Vienna keep their places at the top from last year. This is despite affordability pressures as both Vienna and Copenhagen have seen their property to income ratio ranks deteriorate by -13 and -9 positions from last year's edition, respectively. Still, low traffic commute times and pollution help keep those cities at the top.

Frankfurt now ranks just outside the top 5. Germany's financial capital has improved its cost-of-living rank (from 50th to 42nd) while maintaining very short commute times (1st globally) and low pollution. It now ranks above both Geneva and Zurich where cost of living is bottom of our 69-city pile. Another German city to enter top 10 this year is Munich (+8 ranks) amidst stronger purchasing power score. It now ties with Vienna on the fourth place for the overall index.

Hong Kong has fallen outside our top-50 list and now ranks 55th, down from 48th place in last year's edition. It is up just +2 places since 2019. Extremely high costs, property prices ranked 68th worst, pollution and long commutes all weigh on its score.

London continues to hover near the bottom of our top-50 table but has moved up +3 places since last year, with Birmingham also up by +7 to now sit in the middle of the table at 25th place.

Notably, no US city makes the top 10 for quality of life but there is much dispersion. New York at 46th and Los Angeles at 49th are towards the lower-end of the table, while Boston (17th) and San Francisco is (21st) are comfortably in the top half. The combination of high costs, long commutes, and safety concerns keeps American cities out of contention for the top ten though - despite their salary dominance.

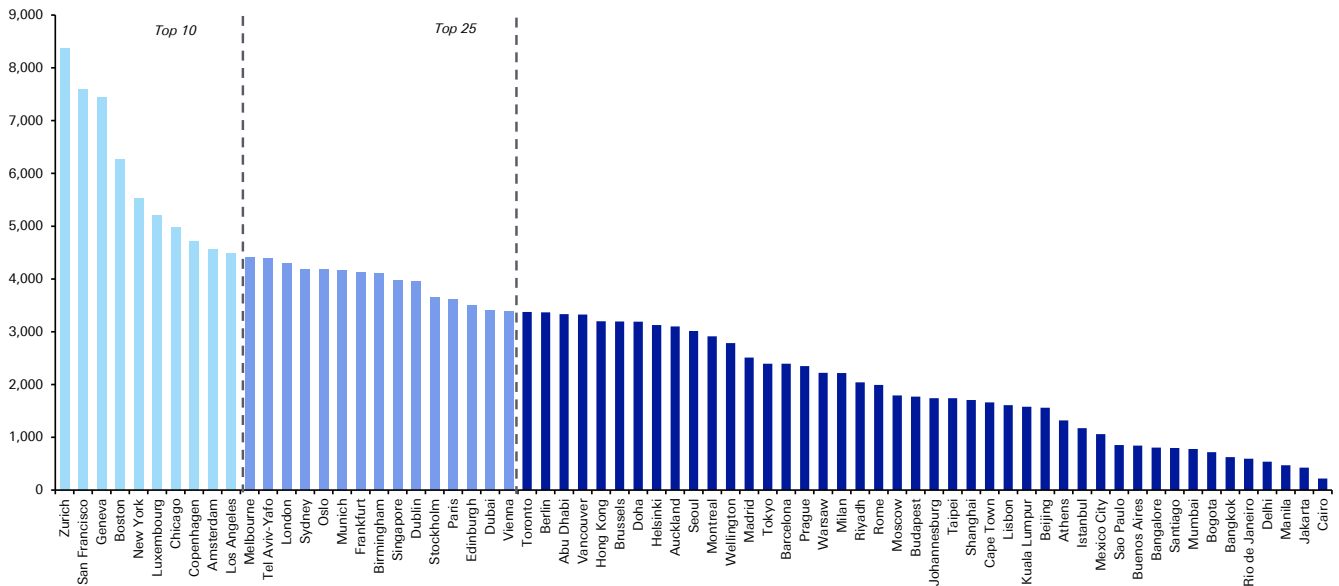


Salaries and Disposable Income

Net Salaries

Zurich leads globally, followed by San Francisco, Geneva, Boston, and New York for the top 5. The US continues to dominate the top tier, with five cities in the top ten.

Figure 11: Monthly Salary (Net of Taxes, USD)



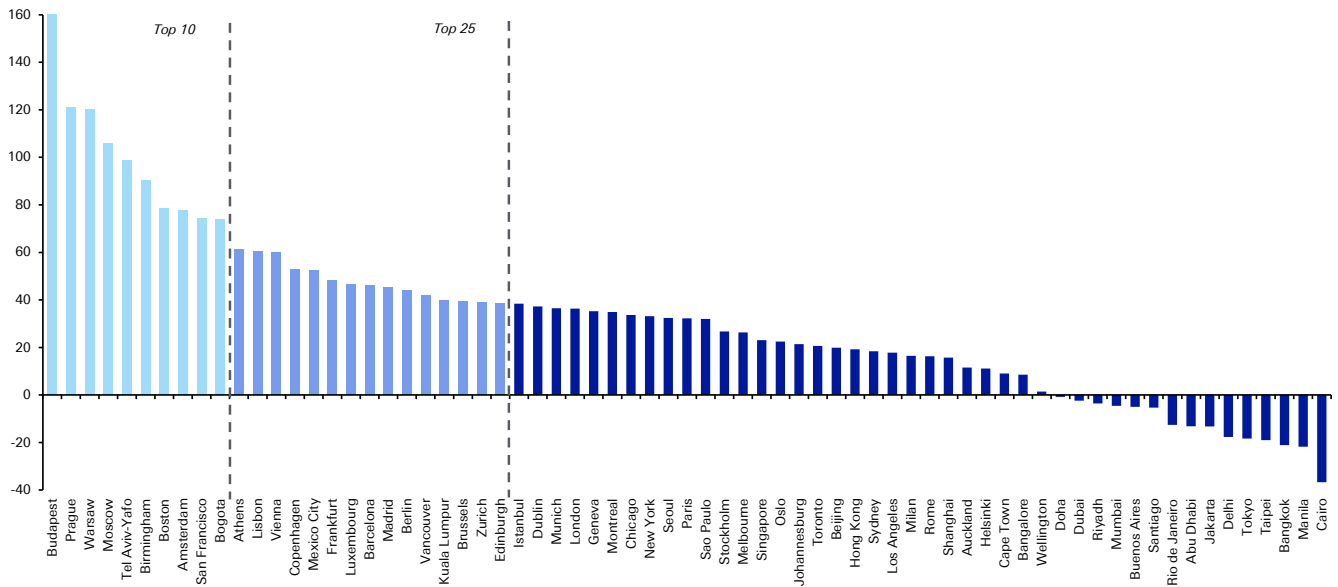
Source : www.numbeo.com, Deutsche Bank.

As mentioned above, over the last decade or so, several Central European cities have enjoyed strong wage growth and the chart below shows that salaries in Budapest, Prague and Warsaw have doubled in 10 years, outpacing every major DM city we track. Tel Aviv rounds up the top 5 growth over the period. Lisbon, Athens and Bogota have all outpaced salary growth of Copenhagen, Frankfurt and London.

On the other side of the chart the notable net salary decliner is Tokyo with the 5th smallest salary growth at -18%.



Figure 12: Monthly Salary (Net of Taxes, USD), cumulative change since 2016 (%)



Source : www.numbeo.com, Deutsche Bank.

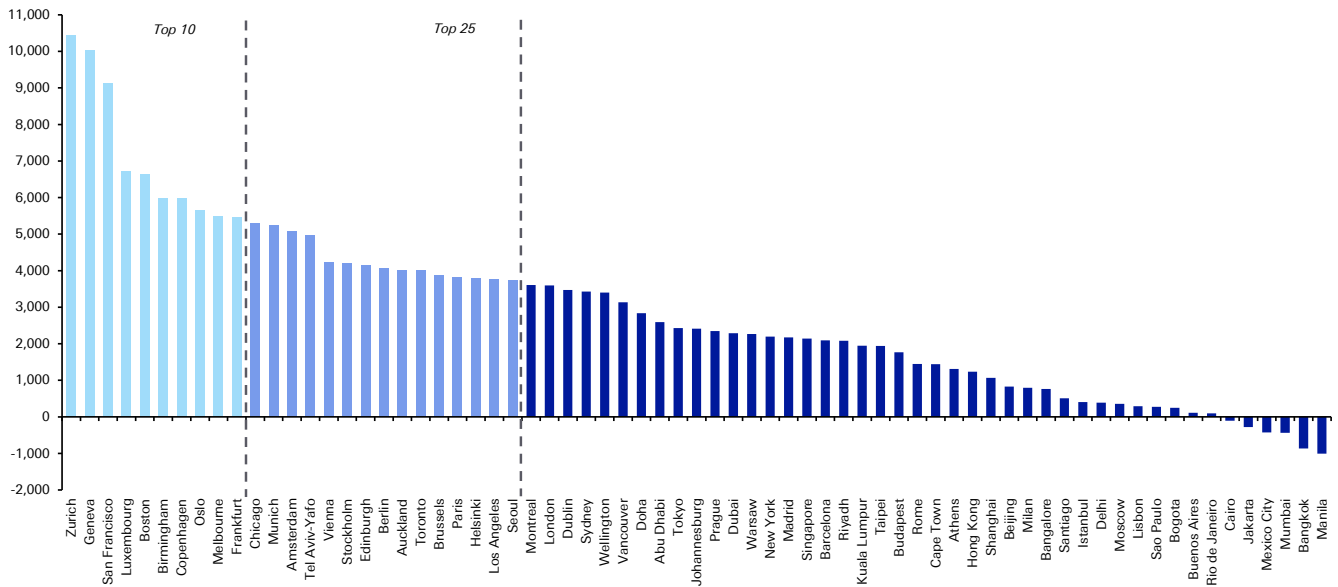
US salary growth has been dispersed, with San Francisco (+74%) and Boston (+79%) in the top 10 but New York and Los Angeles ranked 33rd and 46th, respectively. Salaries in Zurich and Geneva have grown by roughly 40% over the decade, somewhat above the average of 32% across our sample.

Disposable Income After Rent

Zurich, Geneva and San Francisco remain the standout cities even after salaries are adjusted for rent. Luxembourg and Boston close out top 5. The newcomers to top 10 this year are Birmingham (6th, +8 ranks) and Melbourne (9th, +3 ranks), edging out Chicago and Munich.



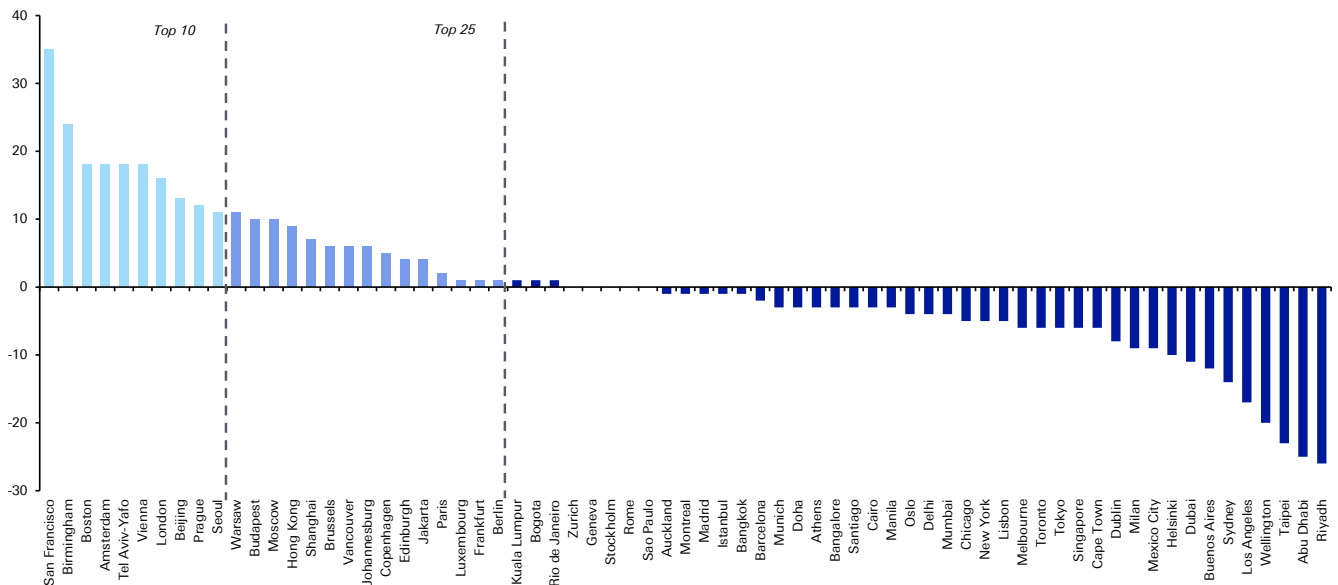
Figure 13: Disposable Income Index after rents (2 persons working and renting 3 bedroom apartment, USD)



Source : www.numbeo.com, Deutsche Bank.

On a 10-year lookback, San Francisco (+35) and Birmingham (+24) are the biggest gainers. Boston, Amsterdam, Tel Aviv, Vienna and London have also climbed by more than +15 places. Beijing, Prague and Seoul close out top 10 on this measure, with Warsaw and Budapest right after.

Figure 14: Disposable Income Index after rents (2 persons working and renting 3 bedroom apartment, USD), rank changes from 2016



Source : www.numbeo.com, Deutsche Bank.

On the other side of this the biggest losers in disposable income terms over the decade have been Riyadh, Abu Dhabi, Taipei, Wellington and Los Angeles. New York (-5 ranks) and Tokyo (-6 ranks) are also down. Amongst cities that have roughly

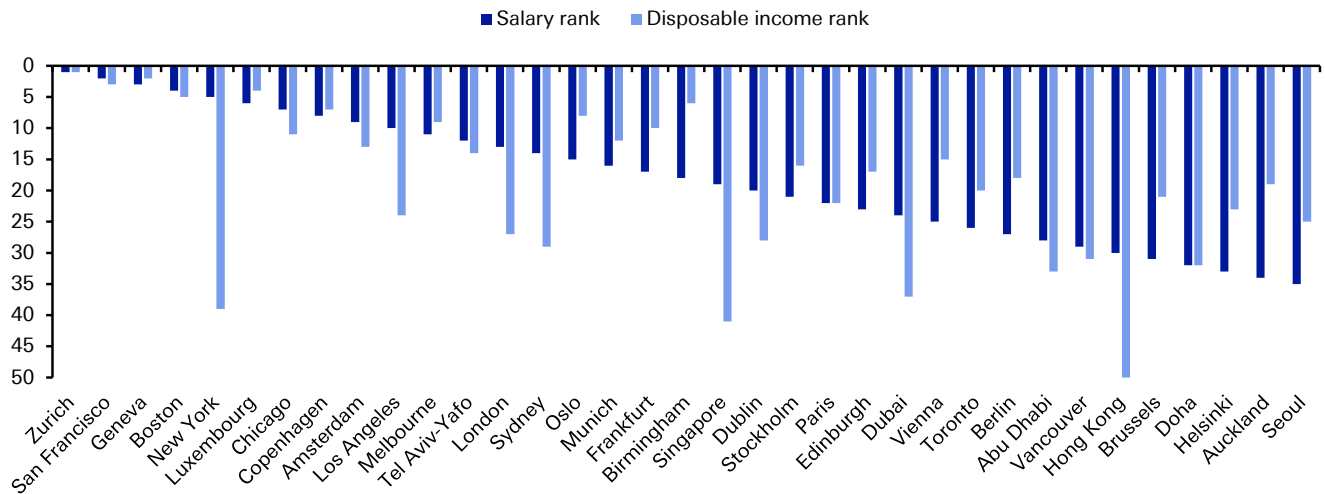


held onto their positions are Stockholm, Berlin (+1 rank) and Rome, with Geneva and Zurich understandably staying comfortably on top.

As the chart below shows, adjusting for rent reshuffles the rankings for the top cities. New York plummets -34 ranks to 39th as its very high rents consume a huge share of average income, lagging -28 places behind Chicago (11th) despite similar gross salaries. The US property market remains the most bifurcated in the world.

Other top salary cities with similar affordability issues include Hong Kong (-20 positions) and Singapore (-22), with Sydney (-15), Los Angeles (-14), London (-14) and Dubai (-13) also close in terms of affordability drops.

Figure 15: 2026 ranks for top 35 cities by salary and disposable income.



Source : www.numbeo.com, Deutsche Bank.

On the other hand, Birmingham (+12) and Auckland (+15) see their ranks improve significantly when income is adjusted for rents. Seoul, Helsinki and Vienna all climb +10 ranks.

Cities that make the top 10 for both quality of life and disposable income include Luxembourg, Copenhagen, Frankfurt, Geneva and Zurich - all European. No US city makes both lists.

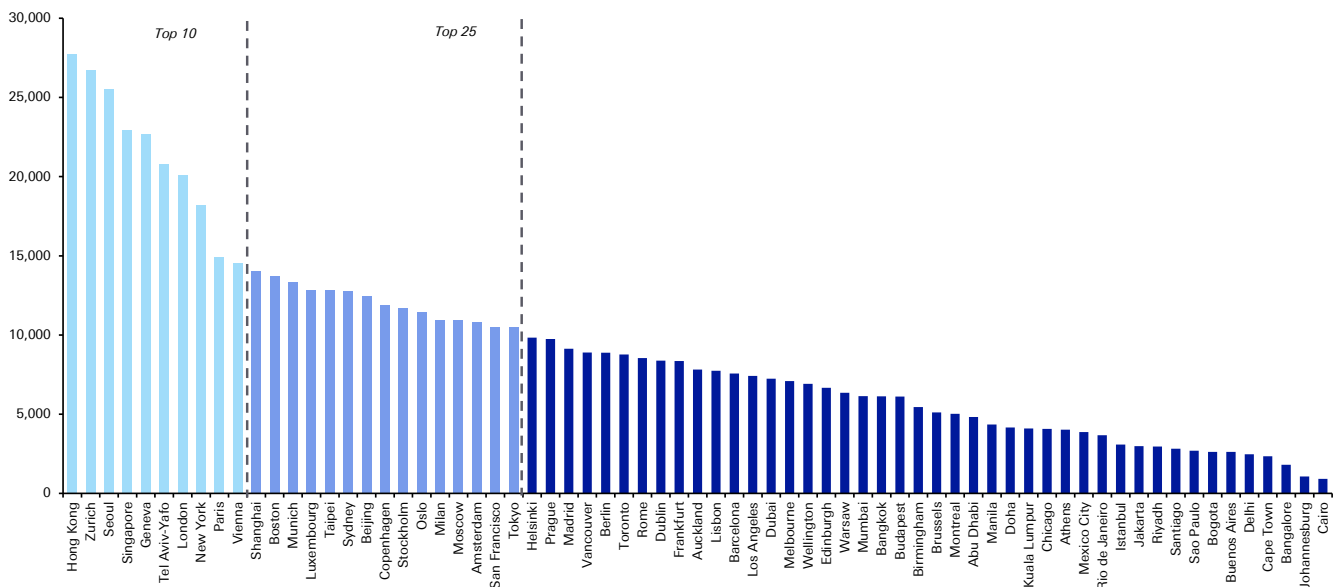


Real Estate

Most Expensive to Buy (per m²)

Hong Kong and Zurich are the top two most expensive places, followed by Seoul, Singapore and Geneva. Seoul's 3rd place - with prices more than doubling over the past decade (+8 ranks) - reflects South Korea's persistent housing affordability crisis, and tech leadership, despite the won's -26% decline against the dollar. In local currency terms, the increase is even more extreme. The top 10 is completed by Tel-Aviv, London, New York, Paris and Vienna.

Figure 16: Price per square meter to buy apartment in city centre (USD)

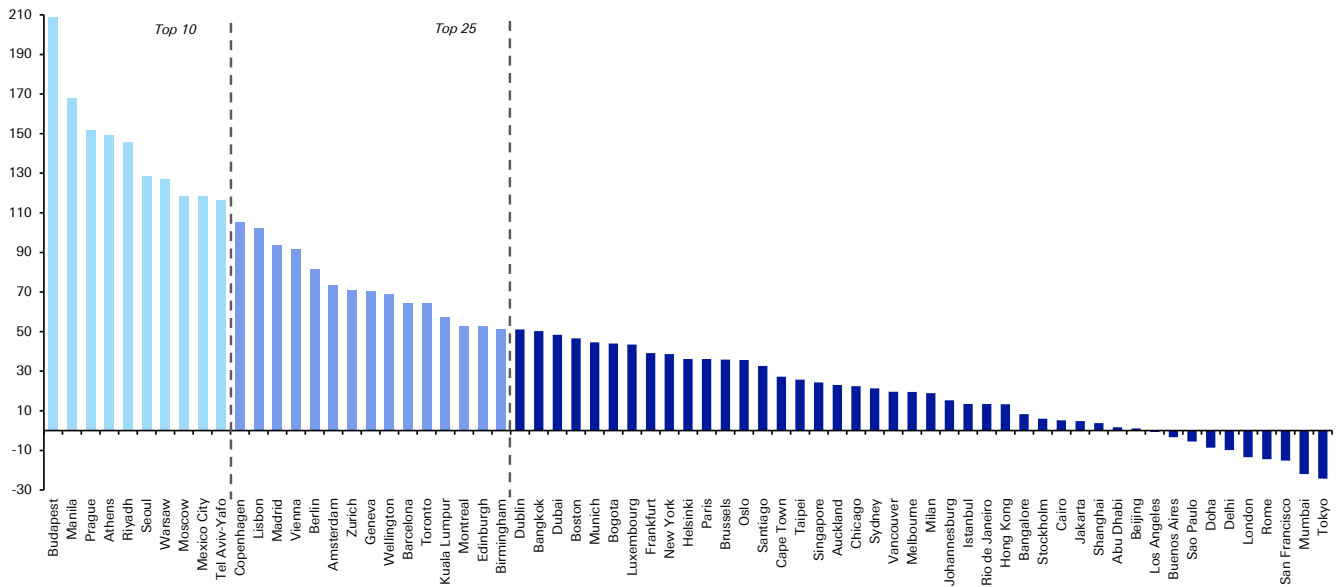


Source : www.numbeo.com, Deutsche Bank.

Looking back a decade, however, reveals stark changes beyond the most expensive places. Budapest, Manila and Prague have seen property prices rise by more than 150% over the period. While EM cities lead the top 10 growth-wise, some notable DM ones are amongst those that have seen the largest declines. This includes Tokyo (the largest at -24%), San Francisco, Rome and London (5th largest decline). The growth is below the cumulative US CPI change over the period for Chicago and Los Angeles and just in line for New York.



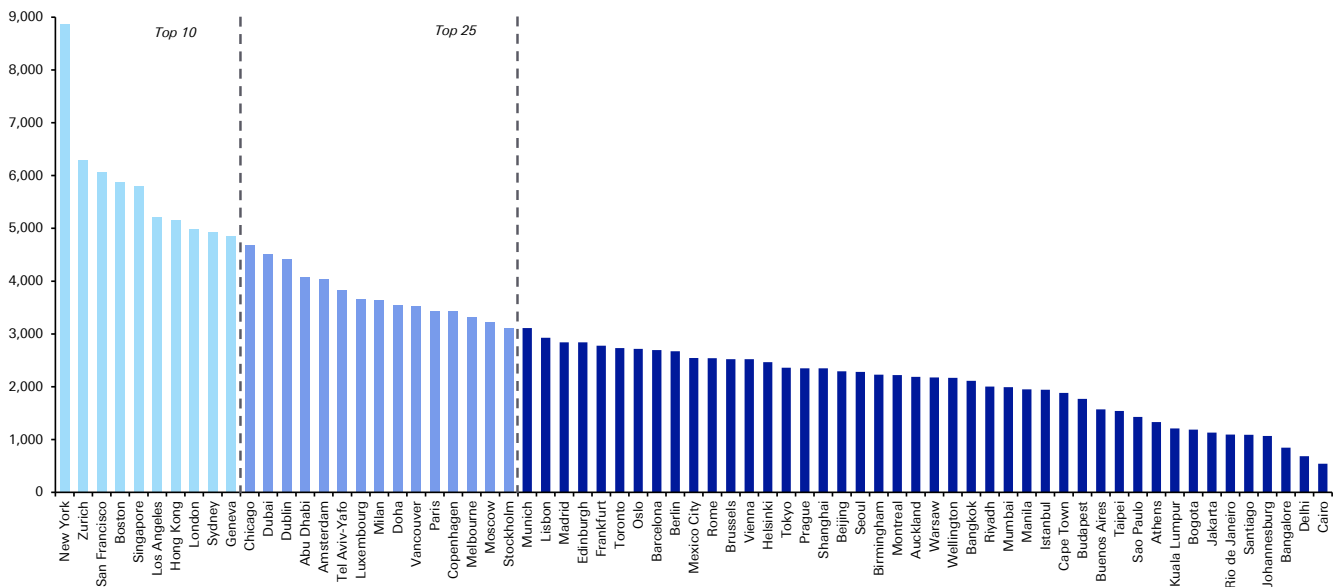
Figure 17: Price per square meter to buy apartment in city centre (USD), cumulative change since 2016 (%)



Source : www.numbeo.com, Deutsche Bank.

Most Expensive to Rent (3-bed apartment)

Figure 18: Rent per month for a 3-bedroom apartment in city centre (USD)

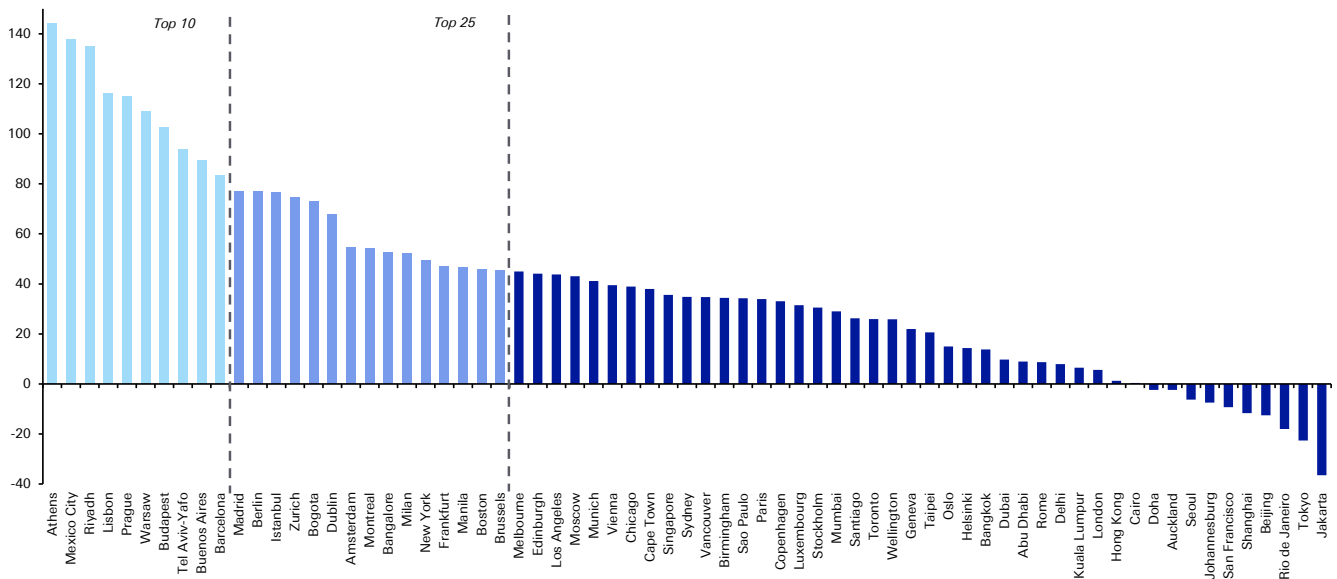


Source : www.numbeo.com, Deutsche Bank.

New York remains in the league of its own for the 3-bed apartment rent. Zurich, San Francisco, Boston and Singapore close out top 5 with a little changed top 10 list.



Figure 19: Rent per month for a 3-bedroom apartment in city centre (USD), cumulative change since 2016 (%)



Source : www.numbeo.com, Deutsche Bank.

As with property prices, the big growth has been happening in EM cities over the last decade. Seven of the top 10 cities on this measure have seen rent prices more than double. A few European cities, including Lisbon, Amsterdam and Dublin, have seen strong rent growth, driven by supply constraints, the return of tourism and remote-work migration.

Some of the highlights on the other end of the spectrum are Tokyo, Beijing, Shanghai, San Francisco and Seoul, which have all seen declines over the decade. Dubai, London and Hong Kong have seen some of the smallest positive growth rates in our sample.



The Cost of Living

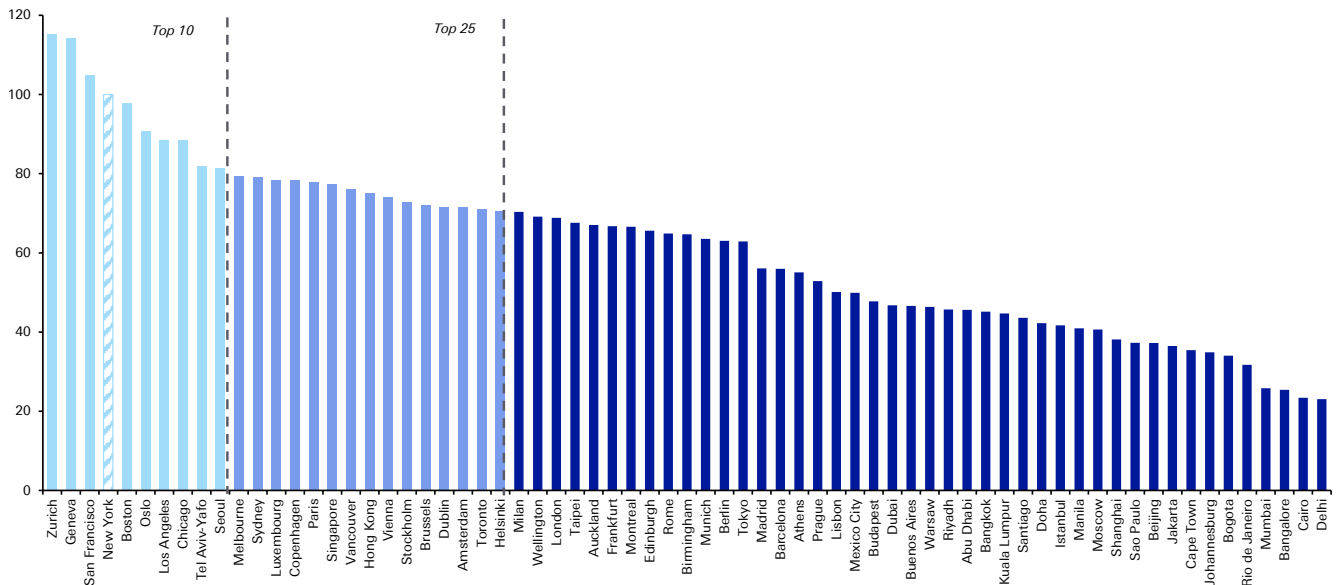
Utilities

The highlight is Tel Aviv's surge to 2nd place, up +10 ranks from 2025, amidst energy markets disruptions in the region. German cities continue to dominate the top ranks, with Berlin 1st, Munich 6th and Frankfurt 7th, a lasting legacy of the energy crisis and the shift away from Russian gas and nuclear power. Oslo, London and Edinburgh complete the top 5. Cheapest utilities remain in Egypt, India and China.

Groceries

The Numbeo groceries index reflects the cost of groceries in a city relative to New York (=100). Only Zurich, Geneva and San Francisco are more expensive on this measure than New York across our sample cities. The remaining US cities are all in top 10 on this measure, with Boston 5th, Los Angeles 7th and Chicago 8th. Oslo, Tel Aviv (+8 ranks from last year) and Seoul complete the top 10.

Figure 20: The Numbeo groceries index - cost of groceries in a city relative to New York (=100, shaded)



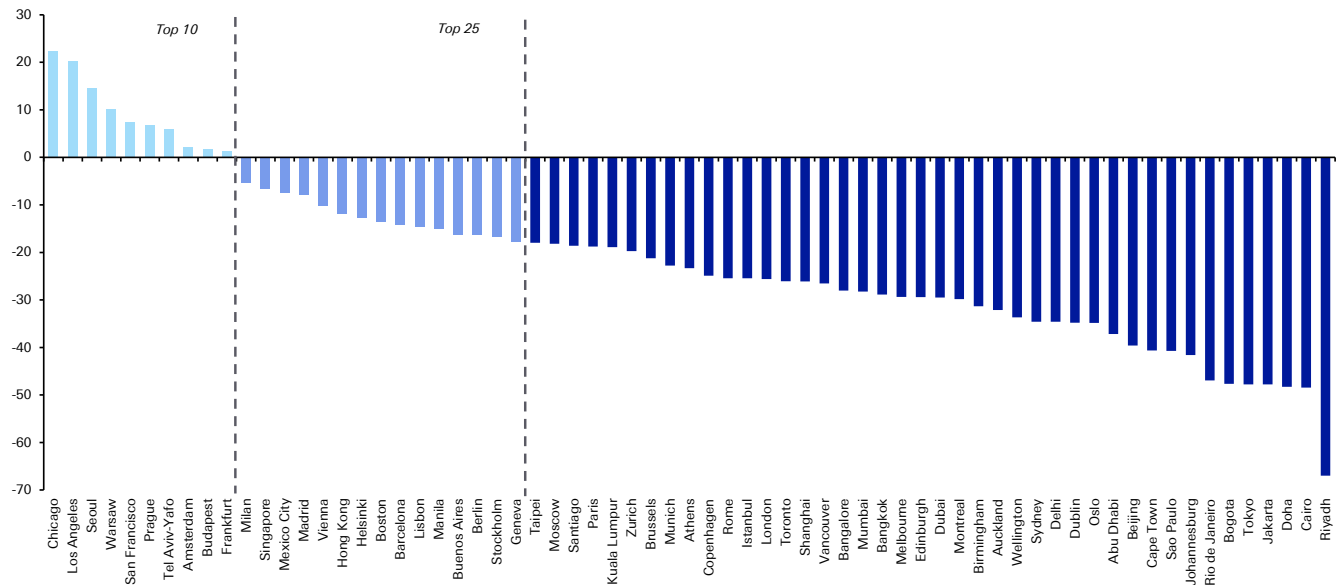
Source : www.numbeo.com, Deutsche Bank.

Since the index we use is calculated relative to New York, its changes are useful as a guide of how expensive our sample cities have become relative to US. Over the last decade, Warsaw, Prague and Budapest have stood out on this measure but many other European cities have also become more expensive. Meanwhile, many Asian cities, beyond Tokyo which has been a spotlight throughout this report, have cheapened on a relative basis.

Since 2012, when this report was first published, the change is more dramatic. The US was a much cheaper place and since then only 10 of our sample cities have seen their groceries price indices increase relative to New York, three of them are in the US.



Figure 21: Cumulative change in the Groceries index relative to New York (=100) since 2012 (index points)

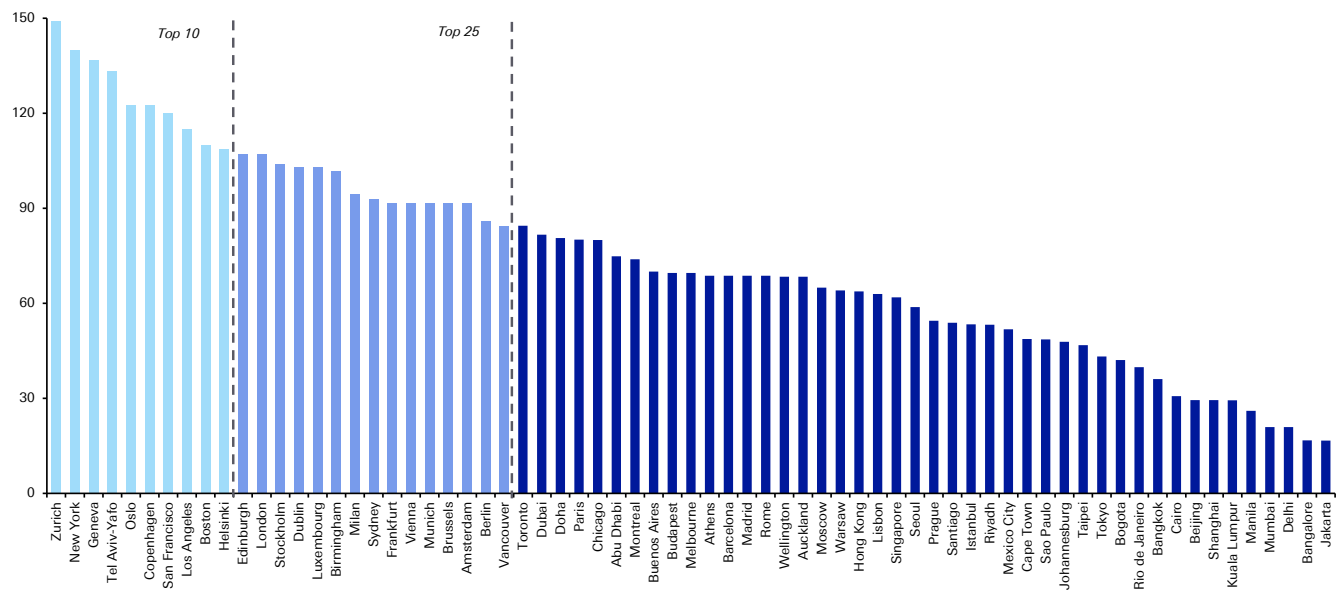


Source : www.numbeo.com, Deutsche Bank. * Note: Groceries index assesses the cost of grocery items in a city relative to New York City. It is calculated using the weighted prices of various food items commonly found in the "Markets" section of Numbeo's database.

Restaurants

Zurich continues to be the world's most expensive dining city, followed by New York and Geneva. Tel Aviv has moved up to 4th place, overtaking Oslo (5th). Four of the US cities are in top 10, including San Francisco, Los Angeles and Boston. Copenhagen and Helsinki are the other cities in top 10. London has dropped out to 11th place relative to last year, tied with Edinburgh.

Figure 22: Three-course meal for 2 people at a mid-range restaurant (USD)



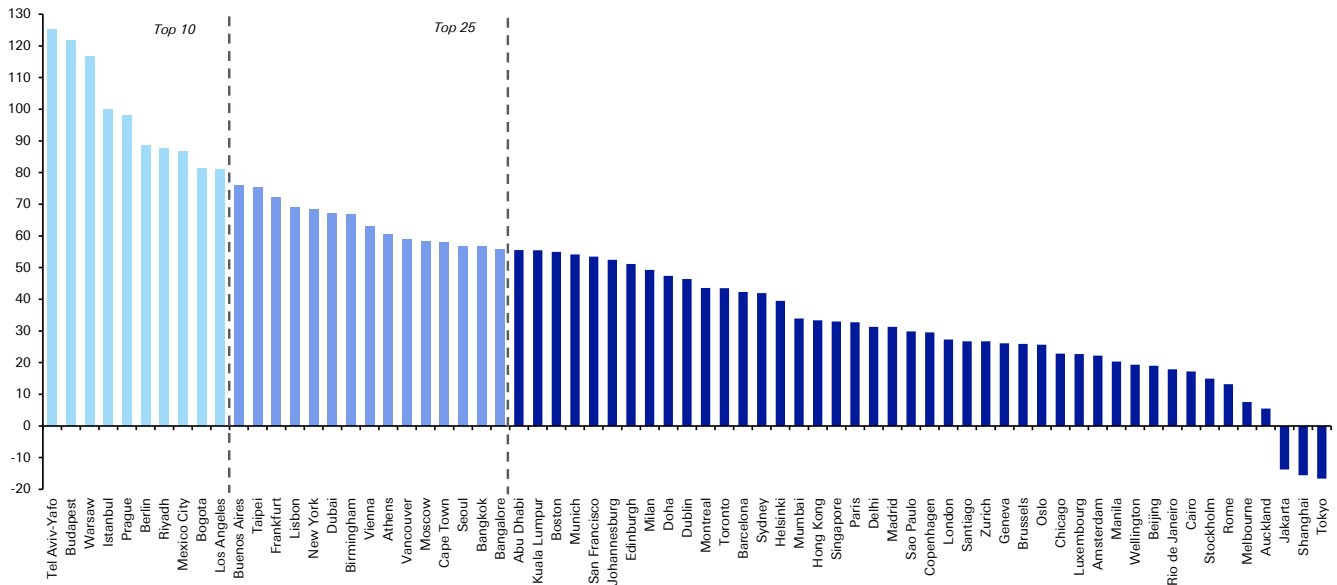
Source : www.numbeo.com, Deutsche Bank.

Remarkably, dining out has become more expensive over the last decade in each of



our sample cities bar Tokyo, Shanghai and Jakarta. Our spotlight cities are leading again here, with Tel Aviv, Budapest and Warsaw seeing prices more than double and showing further evidence of the Central European repricing.

Figure 23: Three-course meal for 2 people at a mid-range restaurant (USD), cumulative change since 2016 (%)



Source : www.numbeo.com, Deutsche Bank.

The McDonald's combo in Tel Aviv has surged past Zurich to become the world's most expensive place for a Big Mac meal, encapsulating the city's across-the-board cost surge.



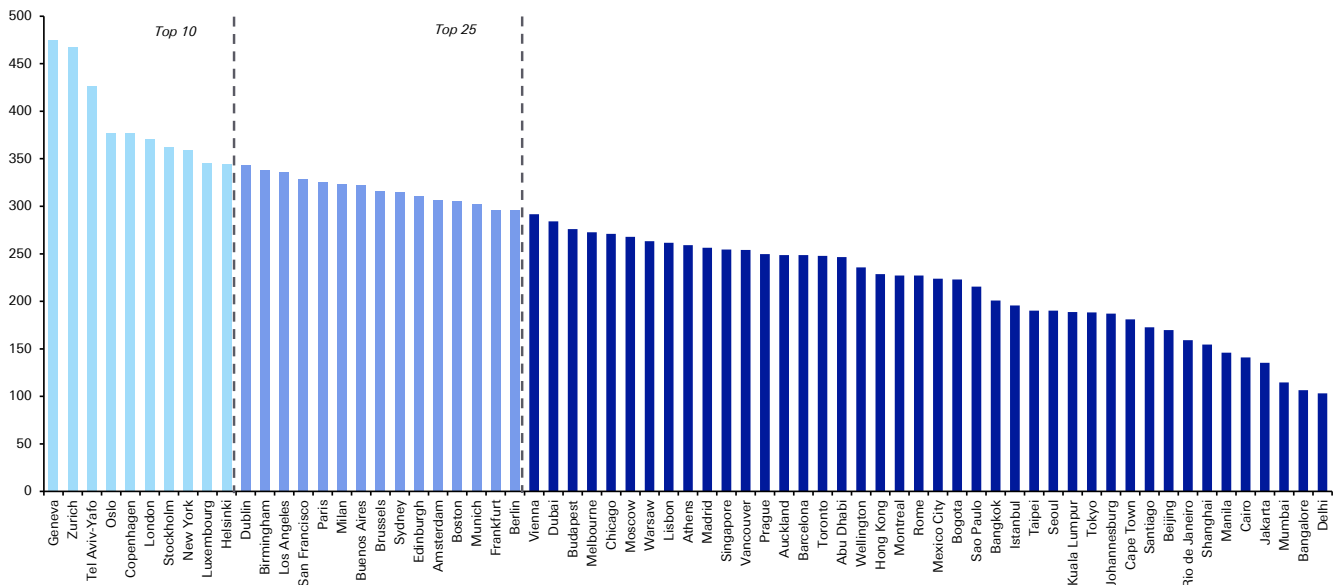
Lifestyle Indices

"Cheap Date" Index

Our composite index covers a cab ride, public transport, dinner, two cinema tickets, clothes and a couple of drinks.

Tel Aviv's increase from 5th to 3rd is driven by simultaneous increases across most of the index components. Geneva and Zurich continue to be at the top, with Oslo and Copenhagen rounding out top 5.

Figure 24: Our Cheap Date index (USD)



Source : www.numbeo.com, Deutsche Bank.

Note: We have defined a cheap date as follows: 1 bottle of wine, 1 pair of jeans, 1 dress, 2 coffees, 1 meal for 2 people in mid range restaurant, 2 cinema tickets, 2 transport tickets one way, 5km taxi trip

Over the last decade, dating has become more expensive not only in Emerging markets with Budapest and Mexico City at the top, but also in the US, with Los Angeles, New York and Boston in the top 10 based on the cost increase. The cheapening of Asia is clear in this index too, with 8 out of top 10 cities by decade-long declines located in Asia with Shanghai the most improved on cost.

If you're looking for romance on a budget, India remains the absolute cheapest answer though: Delhi (\$103), Bangalore (\$106), and Mumbai (\$115) occupy the bottom three spots and have also seen dating costs decline over the last decade.

The Oasis Index (Cigarettes & Beer)

Melbourne (\$111.2) is the world's most expensive city for cigarettes and beer, continuing Australia's relentless application of sin taxes, with Sydney 2nd. Over a decade, Melbourne's Oasis index has risen +89%. Auckland and Wellington rank 3rd and 4th respectively.

The UK also ranks near the top - London (5th, \$63.9), Birmingham (6th, \$62.8), and Edinburgh (8th, \$60.5) - reflecting high tobacco duties. Dublin, Oslo and New York close out the top 10. Cheapest: Shanghai (\$10.4), Rio (\$11.7) and Cairo (\$12.1).

Wine and Coffee



Wine: Singapore (\$23.2) remains the most expensive place to buy a bottle of mid-range wine, followed by Jakarta (\$21.1). The cheapest? Rome (\$4.6) and Lisbon (\$5.7).

Cappuccino: Zurich (\$7.13) reclaims the top spot, followed by Copenhagen (\$6.96) and Geneva (\$6.41). Italian cities remain reassuringly cheap: a cappuccino in Milan costs just \$2.18 - less than a third of the Zurich price. Three of US cities - New York, Los Angeles and San Francisco - are in top 10. Dubai, Tel Aviv and Doha are also in the list.

Similar to eating out, having a coffee has become more expensive almost everywhere except for a few big Asian cities, with only 6 seeing 10-year declines. Central European cities are ahead once again.



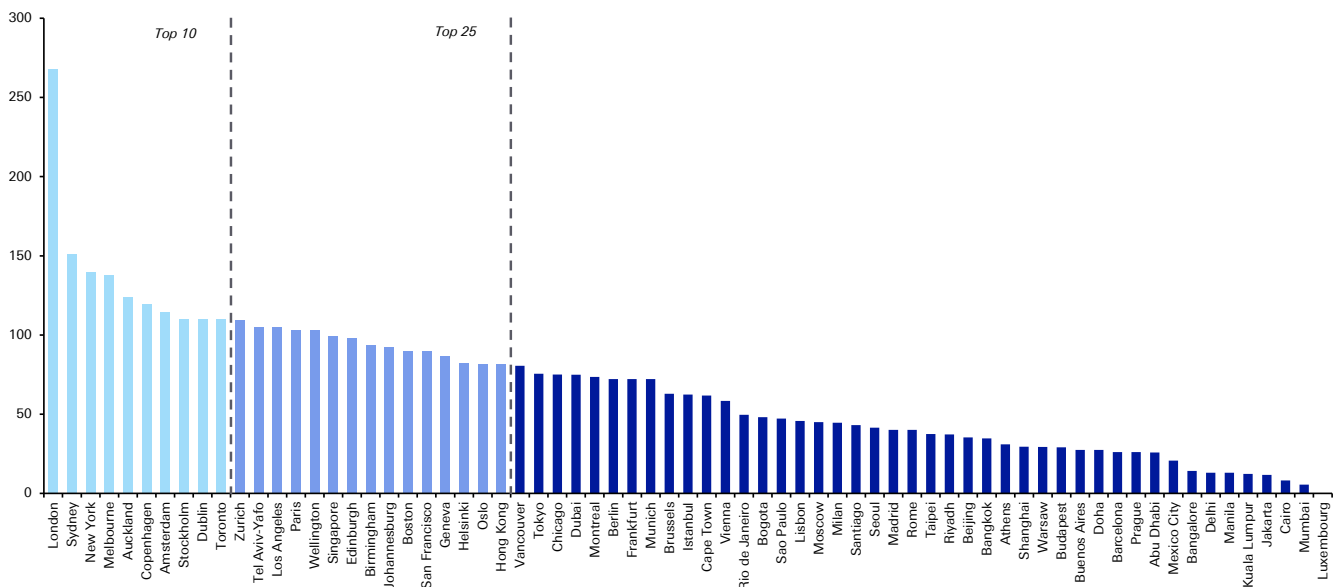
Transport

Public Transport

London extends its lead as the world's most expensive monthly pass — now +35% above its 2016 level. Sydney, New York, Melbourne and Auckland complete the top five.

Luxembourg remains the cheapest - most public transport continues to be free, a policy that has helped anchor its 1st quality-of-life ranking.

Figure 25: Public transport monthly pass (USD)



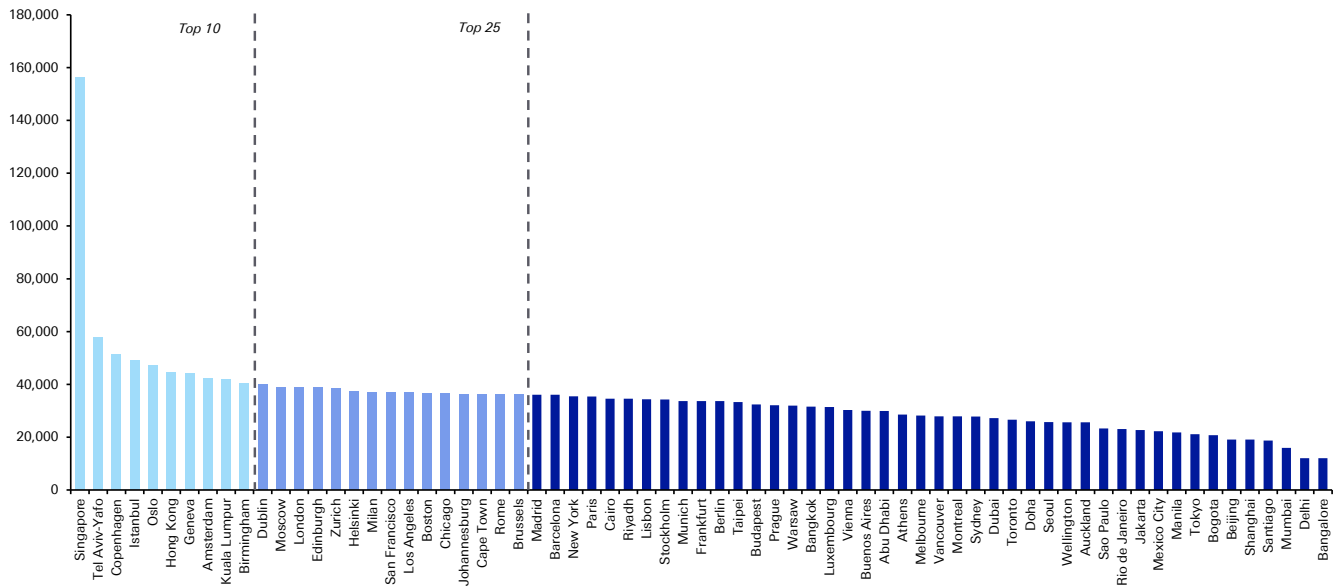
Source : www.numbeo.com, Deutsche Bank.

Cars

Singapore (\$156,223 for a VW Golf) remains in a league of its own - 4.4x the New York price - reflecting the city-state's Certificate of Entitlement system. Tel Aviv has moved to 2nd (\$57,859), with Copenhagen completing the top 3.



Figure 26: Prices of Volkswagen Golf 1.5 (or equivalent new car, USD)



Source : www.numbeo.com, Deutsche Bank.

Gasoline

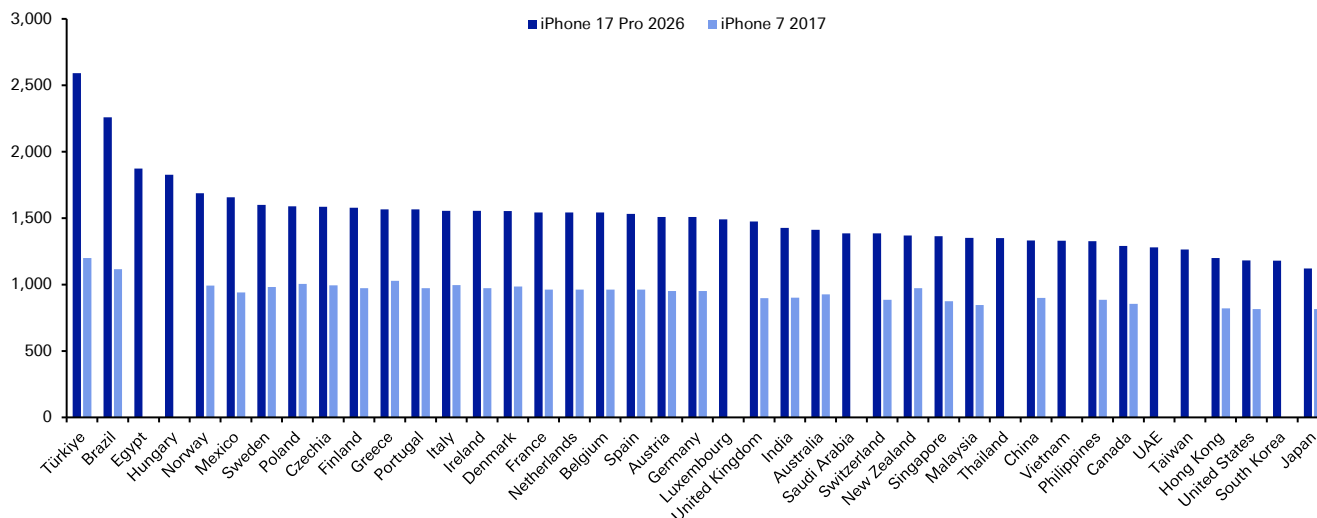
Hong Kong (\$3.88/litre) remains the most expensive - 8.6x the price in Cairo (\$0.45) and 6.3x that in Riyadh (\$0.62). Tel Aviv (\$2.08) has jumped +27% YoY, the sharpest increase among major cities, potentially reflecting the Iran conflict's impact on regional energy supply and logistics.



The iPhone Index

The iPhone has become one of our most reliable cross-economy price comparisons. Apple sets prices centrally and the product is identical worldwide — so price differences reflect taxes, tariffs, distribution costs, and currency effects, making it an excellent barometer of trade and fiscal policy.

Figure 27: Prices of iPhone 17 Pro (256GB) and iPhone 7 from the 2017 Mapping the World's Prices edition, full available list (USD)



Source : Official Apple Website, Online Retailers, Deutsche Bank.

Note: Prices are for iPhone 17 Pro (256GB). The prices are sourced from official Apple store websites. In the absence of official prices via websites, we have used prices from online retailers. As of mid-June 2026.

Key observations:

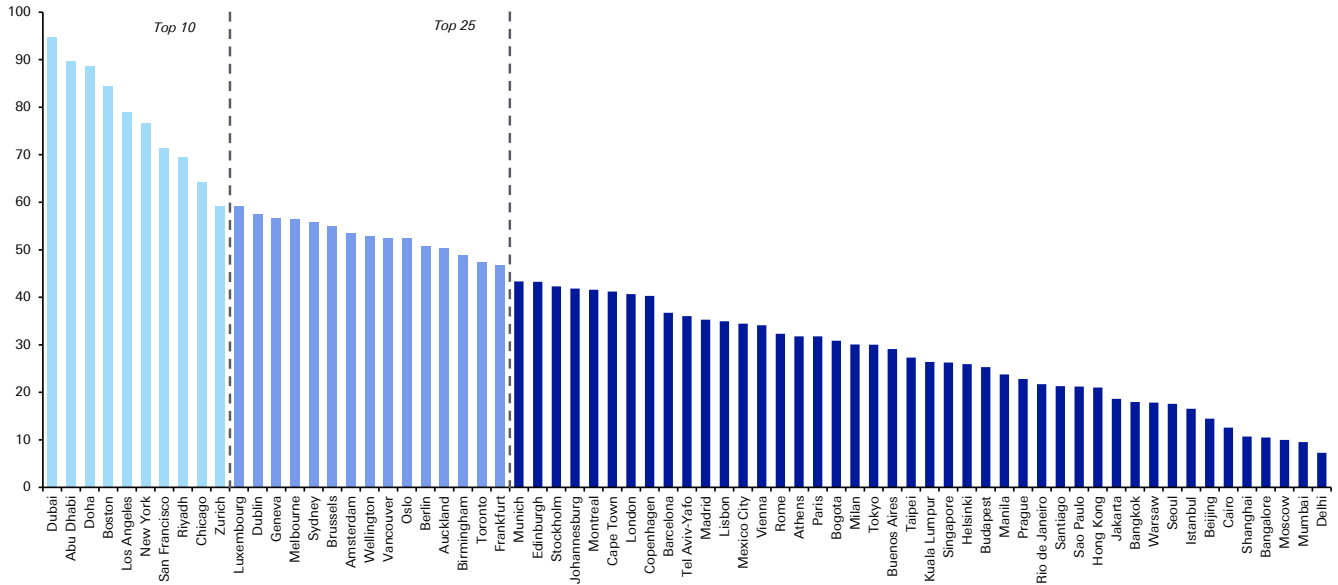
- Japan (\$1,121, -5% vs US) offers the cheapest iPhone amidst yen weakness.
- South Korea (2nd cheapest) remains among the cheapest globally, amidst domestic competition from Samsung.
- Türkiye at 2.4x the US price reflects high taxes and the lira's -92% collapse over the decade.
- Brazil (2.1x) also shows the impact of taxes designed to encourage local assembly.
- Taxes also raise the relative price of an iPhone in Hungary (27% VAT).
- Switzerland (\$1,385) is the cheapest European economy in our sample despite being the world's most expensive economy for nearly everything else - thanks to its low 8.1% VAT.
- Hong Kong (0% VAT, +9% premium) serves as a useful benchmark for Apple's base distribution margin outside the US.



Other Highlights

Internet: Most expensive in the Middle East (Dubai \$95, Abu Dhabi \$90, Doha \$89) and the US (Boston \$85, Los Angeles \$79, New York \$77). Cheapest: Russia (\$10) and India (Delhi \$7, Mumbai \$10).

Figure 28: Monthly internet cost (60 Mbps or higher, unlimited data) (USD)



Source : www.numbeo.com, Deutsche Bank.

Jeans: Most expensive in Geneva (\$141) and Zurich (\$135). Cheapest: India, Brazil and Egypt.



Conclusion

If there is one overarching lesson from this tenth edition of Mapping the World's Prices, it is that the global cost-of-living map is being redrawn less by one single shock than by the cumulative force of currencies, inflation, housing constraints, geopolitics and long-running structural change. In dollar terms, the world now looks very different from 2012, from 2016, and even from the pre-Covid benchmark of 2019. The old assumptions about which places are expensive, which are cheap, and where wages compensate for living costs no longer always hold.

The most dramatic structural story remains Japan. Tokyo, once synonymous with extreme expense, now sits in the middle or lower half of many developed-market price tables once converted into dollars. A weak yen, decades of low inflation and subdued wage growth have turned Japan into one of the clearest examples in this report of how relative prices can move over very long cycles. The irony is that this cheapness may now be part of Japan's opportunity set. If AI adoption, automation and the country's demographic urgency combine to lift productivity, today's apparent undervaluation could become one of the more interesting mean-reversion stories of the next decade.

At the other end of the spectrum, Tel Aviv has moved from being an expensive city to being one of the defining outliers of this year's survey. The strength of the shekel, the resilience of Israel's tech and defence sectors, regional shortages and the impact of conflict have combined to push it up a striking number of tables. It is now among the most expensive cities in the world for everything from meals out and utilities to cars, clothing, gym memberships and even McDonald's. Tel Aviv's repricing has been unusually broad-based.

Central Europe provides the other major convergence story. Prague, Warsaw and Budapest are no longer the obvious bargains they once were. Strong wage growth, EU-related investment, tighter labour markets, higher post-Covid inflation and capital inflows into property have all pushed these cities closer to Western European price levels. In some areas, particularly housing and leisure, the convergence has been startling. That is good news in the sense that it reflects genuine economic progress, but less welcome for anyone still hoping for a cheap weekend in Prague.

The US story is more nuanced. American cities still dominate several income and rent tables, and New York remains the global rental benchmark by some distance. Yet the evidence for a new tariff-driven consumer price shock is limited so far. Groceries and car prices in the US are not yet flashing red, even if lag effects, retailer margin absorption or dollar strength may be masking pressures that could appear later.

Housing remains the thread that runs through almost every liveability discussion. High salaries alone do not guarantee a comfortable life if rent consumes the gains. This is why cities such as Luxembourg, Copenhagen, Zurich, Geneva and Frankfurt stand out: they combine high quality-of-life scores with strong disposable income after housing costs. By contrast, New York, London, Paris and Hong Kong remain global magnets for talent and capital, but their liveability scores are weighed down by expensive housing, long commutes, pollution or affordability pressures. The lesson is that the richest cities are not always the easiest places to live.

The more playful indices tell the same story in miniature. A cheap date is no longer



especially cheap in much of Europe. London remains the world leader in expensive public transport. Rome is still cheap for wine and coffee. Singapore remains in a category of its own for cars. Türkiye and Brazil show how taxes, tariffs and currency weakness can distort the price of an iPhone. And, as ever, Switzerland somehow manages to be expensive in nearly everything while still preserving exceptional purchasing power.

Looking ahead, the key question is not simply which city tops which table in 2026, but which of these shifts proves cyclical and which proves structural. The strong dollar, the weak yen, the expensive shekel, the post-Covid inflation legacy, the European housing squeeze, the tariff shock and the AI investment cycle will not all move in the same direction forever. If history is any guide, the next few years will contain as many reversals as continuations.

For now, this year's report shows a world where Switzerland remains expensive, Japan has become very cheap, Tel Aviv has surged, Central Europe has converged, the US remains powerful but perhaps no longer accelerating, and quality of life increasingly depends on whether income can survive the rent bill. As always, these are highlights from a much larger dataset. Readers should focus less on any single rank and more on the direction of travel — because it is the direction, not the precise decimal point, that tells the real story.



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Note: Relative to last year's edition, we have included different lookback periods, including a 10-year change since 2016 and a 7-year change since 2019 that is meant as a post-Covid analysis of price changes.



Figure 29: Top 50 out of our 69 financial market relevant cities in terms of Quality of Life. Lower ranks are the most positive across the subcomponents.

Quality of life index	vs 2012	vs 2019	Economy	City	Purchasing Power Index	Safety Index	Health Care Index	Cost of Living Index	Property Price to Income Ratio	Traffic Commute Time Index	Pollution Index	Climate Index
1.	N/A	+3	Luxembourg	Luxembourg	7	19	22	57	24	7	4	40
2.	+6	+1	Denmark	Copenhagen	10	17	15	60	25	6	8	36
3.	N/A	+12	Netherlands	Amsterdam	14	23	14	58	21	1	12	28
4.	N/A	+3	Germany	Munich	11	5	9	53	37	9	6	47
4.	N/A	+1	Austria	Vienna	19	21	4	48	47	3	2	42
6.	N/A	+10	Germany	Frankfurt	5	39	12	42	12	1	17	32
7.	N/A	+2	Finland	Helsinki	28	12	3	46	29	4	1	56
8.	N/A	+0	Switzerland	Geneva	3	22	35	68	37	15	10	40
9.	-7	-8	Switzerland	Zurich	2	8	39	69	36	30	11	43
10.	N/A	-4	United Kingdom	Edinburgh	20	24	13	45	15	5	13	34
11.	+5	-1	Australia	Melbourne	9	37	25	52	10	44	14	13
12.	N/A	-10	New Zealand	Wellington	34	27	56	38	20	34	7	6
13.	N/A	+10	Norway	Oslo	21	30	7	64	27	7	5	58
14.	-10	-2	Australia	Sydney	29	29	17	55	33	49	15	8
14.	-3	+5	Canada	Vancouver	31	34	27	40	31	24	9	19
16.	N/A	+2	New Zealand	Auckland	17	44	36	36	23	32	16	1
17.	N/A	-6	United States	Boston	8	33	37	62	14	53	18	49
18.	-3	+15	Spain	Madrid	42	20	5	32	40	9	25	30
19.	-16	+4	Sweden	Stockholm	22	40	49	56	33	19	3	51
19.	+1	+8	Czech Republic	Prague	40	11	16	30	54	14	20	44
21.	-12	-8	United States	San Francisco	1	58	53	65	7	55	33	7
22.	-12	-3	Japan	Tokyo	30	8	6	24	48	46	28	31
23.	N/A	-2	Qatar	Doha	13	2	20	21	3	11	44	69
24.	N/A	-7	United Arab Emirates	Abu Dhabi	41	1	24	26	9	12	29	68
25.	N/A	+3	United Kingdom	Birmingham	4	60	28	41	5	16	37	27
26.	-20	-12	United Arab Emirates	Dubai	18	3	31	35	13	26	35	66
27.	-22	-2	Canada	Montreal	16	26	59	31	10	35	21	65
28.	-27	-6	Germany	Berlin	23	38	45	44	28	16	24	39
29.	-22	+2	Canada	Toronto	15	35	18	39	29	50	23	54
30.	N/A	+0	Israel	Tel Aviv-Yafo	37	16	18	37	67	28	30	14
31.	N/A	+8	South Korea	Seoul	12	14	2	32	62	36	31	52
32.	N/A	-6	United States	Chicago	6	63	55	49	2	42	36	53
33.	-16	+3	Portugal	Lisbon	56	25	22	27	56	20	22	3
34.	N/A	+4	South Africa	Cape Town	47	67	38	13	4	45	26	5
35.	N/A	+7	Taiwan	Taipei	50	4	1	24	66	20	34	33
36.	-23	+1	Ireland	Dublin	32	51	67	51	19	36	27	29
37.	-12	+3	Poland	Warsaw	43	13	63	23	31	22	45	48
38.	-17	-6	Belgium	Brussels	26	54	21	43	8	27	47	35
39.	N/A	-4	Saudi Arabia	Riyadh	39	10	44	19	5	18	50	67
40.	-22	-11	Singapore	Singapore	45	7	26	63	60	39	18	62
41.	-2	+14	Italy	Rome	52	41	53	34	46	23	32	15
42.	N/A	-1	Spain	Barcelona	48	46	11	33	42	12	48	10
43.	-20	+7	France	Paris	24	55	10	54	53	40	49	23
44.	-18	+4	Hungary	Budapest	51	28	66	28	45	33	39	46
45.	-23	-1	South Africa	Johannesburg	33	69	61	16	1	38	46	18
46.	-32	-3	United States	New York	35	45	57	66	26	48	43	45
47.	-20	+2	United Kingdom	London	36	53	33	61	50	52	42	24
48.	-20	+6	Malaysia	Kuala Lumpur	25	57	33	10	18	41	51	63
49.	-37	-16	United States	Los Angeles	26	49	59	59	16	68	57	11
50.	-10	-3	Greece	Athens	57	52	63	29	44	31	41	12

Source : www.numbeo.com, Deutsche Bank.

Note: Index ranks are based on indices published periodically on source website. Rank changes are affected by many cities missing 2012 data.

Note on methodology: The Numbeo Quality-of-Life index is an empirical measure of the living standards across various cities. The index is calculated by aggregating various sub-indices that measure certain broad factors that affect overall quality of life in a given location. These underlying sub-indices measure (1) consumer purchasing power (purchasing power index); (2) regional crime and safety (safety index); (3) overall quality of health care availability (healthcare index); (4) general cost of consumer goods (cost of living index); (5) housing affordability (property price to income ratio); (6) traffic congestion and commute times (traffic commute time index); (7) overall pollution (pollution index); (8) regional climate likeability (climate index). Further details on methodology around data collection and aggregation can be found at the source website [here](http://www.numbeo.com).



Figure 30: Monthly Salary (Net of Taxes, USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+0	+1	+0	Switzerland	Zurich	5,237	6,026	6,266	7,989	8,363	151%	38.8	33.5
2.	+4	+1	-1	+1	United States	San Francisco	4,484	4,364	7,397	7,415	7,601	137%	74.1	2.7
3.	-2	-1	+0	-1	Switzerland	Geneva	6,093	5,505	5,538	7,416	7,444	135%	35.2	34.4
4.	+13	+6	+1	+0	United States	Boston	3,487	3,504	5,022	6,558	6,258	113%	78.6	24.6
5.	+2	-1	-1	+1	United States	New York	4,170	4,155	5,132	5,328	5,534	100%	33.2	7.8
6.	-4	+2	+2	+1	Luxembourg	Luxembourg	6,035	3,548	4,057	5,048	5,196	94%	46.4	28.1
7.	+11	+0	+0	-2	United States	Chicago	3,257	3,729	4,232	5,461	4,984	90%	33.6	17.7
8.	+11	+9	+7	+0	Denmark	Copenhagen	3,256	3,077	3,166	4,714	4,703	85%	52.6	48.6
9.	+21	+20	+9	+0	Netherlands	Amsterdam	2,815	2,560	3,035	4,679	4,554	82%	77.9	50.0
10.	+6	-4	-4	+0	United States	Los Angeles	3,488	3,803	4,514	4,621	4,480	81%	17.8	-0.8
11.	-3	+0	+3	+4	Australia	Melbourne	4,089	3,490	3,184	3,996	4,409	80%	26.3	38.5
12.	+31	+23	+21	+10	Israel	Tel Aviv-Yafo	1,859	2,216	2,498	3,715	4,399	79%	98.5	76.1
13.	+7	+3	+6	-1	United Kingdom	London	3,253	3,146	3,018	4,396	4,289	78%	36.4	42.1
14.	-10	-5	-5	+4	Australia	Sydney	4,762	3,532	3,435	3,925	4,181	76%	18.4	21.7
15.	-10	-2	-5	+5	Norway	Oslo	4,562	3,411	3,396	3,798	4,177	75%	22.5	23.0
16.	-6	+2	+5	-2	Germany	Munich	3,061	3,053	2,999	4,191	4,169	75%	36.5	39.0
17.	-5	+8	+3	-4	Germany	Frankfurt	3,578	2,780	3,012	4,371	4,124	75%	48.3	36.9
18.	+3	+19	+18	+13	United Kingdom	Birmingham	3,066	2,155	2,337	3,308	4,105	74%	90.5	75.7
19.	+15	-5	-7	-8	Singapore	Singapore	2,497	2,231	3,272	4,451	3,975	72%	23.0	21.5
20.	+9	+1	+3	-4	Ireland	Dublin	2,941	2,875	2,868	3,986	3,947	71%	37.3	37.6
21.	+2	-1	+4	+5	Sweden	Stockholm	3,045	2,883	2,813	3,446	3,654	66%	26.8	29.9
22.	-8	+5	+9	-1	France	Paris	3,505	2,738	2,585	3,740	3,622	65%	32.3	40.1
23.	+8	+7	+12	+0	United Kingdom	Edinburgh	2,794	2,525	2,435	3,540	3,497	63%	38.5	43.6
24.	-11	-12	-7	-7	United Arab Emirates	Dubai	3,507	3,486	3,044	3,968	3,402	61%	-2.4	11.8
25.	+13	+15	+13	+2	Austria	Vienna	2,160	2,114	2,174	3,423	3,380	61%	59.8	55.5
26.	-11	-3	+3	+2	Canada	Toronto	3,501	2,796	2,636	3,417	3,373	61%	20.6	28.0
27.	+9	+5	+1	+2	Germany	Berlin	2,284	2,340	2,642	3,377	3,366	61%	43.9	27.4
28.	-17	-23	-4	-9	United Arab Emirates	Abu Dhabi	3,776	3,840	2,832	3,873	3,334	60%	-13.2	17.7
29.	-1	+2	-7	-5	Canada	Vancouver	2,967	2,346	2,883	3,519	3,327	60%	41.8	15.4
30.	-3	-2	+0	+2	Hong Kong	Hong Kong	2,986	2,682	2,622	3,243	3,199	58%	19.2	22.0
31.	-5	+2	+1	-6	Belgium	Brussels	2,999	2,291	2,576	3,467	3,194	58%	39.4	24.0
32.	-23	-17	-19	+2	Qatar	Doha	4,059	3,218	3,241	3,193	3,190	58%	-0.9	-1.6
33.	+0	-11	-7	+2	Finland	Helsinki	2,723	2,814	2,717	3,149	3,127	57%	11.1	15.1
34.	-2	-10	-7	-1	New Zealand	Auckland	2,790	2,781	2,652	3,216	3,102	56%	11.5	16.9
35.	+0	-1	+2	-5	South Korea	Seoul	2,292	2,276	2,331	3,313	3,015	54%	32.4	29.3
36.	-11	+0	-2	+0	Canada	Montreal	3,019	2,161	2,482	2,983	2,915	53%	34.9	17.4
37.	-13	-11	-26	+1	New Zealand	Wellington	3,020	2,747	3,380	2,619	2,786	50%	1.4	-17.6
38.	+6	+4	+1	+1	Spain	Madrid	1,670	1,727	1,860	2,528	2,509	45%	45.3	34.9
39.	-29	-20	-23	-2	Japan	Tokyo	4,023	2,932	3,054	2,722	2,394	43%	-18.4	-21.6
40.	+5	+4	+2	+1	Spain	Barcelona	1,622	1,638	1,657	2,286	2,394	43%	46.2	44.5
41.	+13	+9	+4	+2	Czech Republic	Prague	1,081	1,063	1,301	2,164	2,348	42%	121.0	80.4
42.	+16	+9	+8	+3	Poland	Warsaw	962	1,009	1,126	2,052	2,221	40%	120.2	97.3
43.	-6	-2	-3	-1	Italy	Milan	2,179	1,903	1,832	2,285	2,216	40%	16.4	21.0
44.	-4	-5	-1	-4	Saudi Arabia	Riyadh	1,982	2,117	1,498	2,381	2,041	37%	-3.6	36.3
45.	-6	-2	-4	-1	Italy	Rome	2,045	1,714	1,660	2,143	1,993	36%	16.3	20.1
46.	+1	+8	+7	+1	Russia	Moscow	1,411	871	1,033	1,695	1,794	32%	106.0	73.7
47.	+14	+16	+8	+6	Hungary	Budapest	753	678	876	1,464	1,770	32%	161.1	102.2
48.	-6	-1	+1	+0	South Africa	Johannesburg	1,911	1,434	1,229	1,569	1,740	31%	21.4	41.6
49.	-3	-11	-5	-3	Taiwan	Taipei	1,440	2,150	1,467	1,709	1,740	31%	-19.1	18.6
50.	+6	-4	-4	-1	China	Shanghai	1,018	1,476	1,298	1,519	1,708	31%	15.7	31.6
51.	-10	-6	-3	-1	South Africa	Cape Town	1,935	1,525	1,285	1,500	1,662	30%	9.0	29.4
52.	+1	+0	-1	+0	Portugal	Lisbon	1,099	1,003	1,042	1,469	1,608	29%	60.3	54.3
53.	-3	-4	-1	+1	Malaysia	Kuala Lumpur	1,222	1,129	1,042	1,356	1,577	29%	39.7	51.4
54.	+5	-6	-7	-3	China	Beijing	892	1,302	1,288	1,490	1,561	28%	19.9	21.2
55.	+2	+2	-1	+0	Greece	Athens	991	819	895	1,217	1,321	24%	61.3	47.6
56.	-4	-1	+9	+1	Türkiye	Istanbul	1,129	847	502	1,059	1,173	21%	38.4	133.9
57.	+3	+4	+2	+1	Mexico	Mexico City	768	695	651	950	1,059	19%	52.4	62.7
58.	-10	+7	+3	+5	Brazil	Sao Paulo	1,229	646	606	734	853	15%	32.0	40.7
59.	-10	-6	+1	+0	Argentina	Buenos Aires	1,223	886	624	809	842	15%	-5.0	34.9
60.	+5	+0	-3	-4	India	Bangalore	633	741	750	1,085	804	15%	8.5	7.2
61.	-6	-5	-5	+1	Chile	Santiago	1,060	844	768	741	798	14%	-5.4	3.9
62.	+5	-4	+0	-2	India	Mumbai	537	814	573	781	777	14%	-4.6	35.6
63.	+0	+5	+5	+2	Colombia	Bogota	641	412	353	531	718	13%	74.0	103.0
64.	-2	-5	-6	-3	Thailand	Bangkok	656	790	748	764	623	11%	-21.1	-16.6
65.	-14	-3	-2	+1	Brazil	Rio de Janeiro	1,175	680	558	511	594	11%	-12.6	6.6
66.	-2	-2	-2	-2	India	Delhi	633	653	556	607	538	10%	-17.7	-3.3
67.	+1	-1	-1	+0	Philippines	Manila	476	602	487	491	471	9%	-21.8	-3.4
68.	-2	-1	-1	+0	Indonesia	Jakarta	574	492	463	456	426	8%	-13.3	-7.9
69.	+0	+0	+0	+0	Egypt	Cairo	339	347	237	184	220	4%	-36.7	-7.2

Source : www.numbeo.com, Deutsche Bank.



Figure 31: Price per square meter to buy apartment in city centre (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NV	10-yr cum change (%)	7-yr cum change (%)
1.	+1	+0	+0	+0	Hong Kong	Hong Kong	15,407	24,501	30,968	26,749	27,753	153%	13.3	-10.4
2.	-1	+2	+10	+0	Switzerland	Zurich	25,773	15,619	12,690	24,480	26,727	147%	71.1	110.6
3.	+12	+8	+1	+0	South Korea	Seoul	8,487	11,178	15,710	24,220	25,545	140%	128.5	62.6
4.	-1	-1	-2	+1	Singapore	Singapore	14,498	18,444	20,009	23,595	22,333	126%	24.3	14.6
5.	-1	+2	+3	-1	Switzerland	Geneva	13,746	13,317	13,435	23,700	22,877	125%	70.3	68.8
6.	+6	+11	+8	-2	Israel	Tel Aviv-Yafo	8,797	9,604	11,789	17,803	20,786	114%	116.4	76.3
7.	-1	-5	-4	-1	United Kingdom	London	12,308	23,230	16,285	19,379	20,113	111%	-13.4	23.5
8.	+0	+0	-2	-1	United States	New York	10,764	13,120	14,732	19,000	18,192	100%	38.7	23.5
9.	-2	+4	+2	+1	France	Paris	11,451	10,943	12,682	14,480	14,899	82%	36.2	15.7
10.	+15	+14	+17	-1	Austria	Vienna	6,584	7,592	7,633	14,676	14,563	80%	91.8	90.8
11.	+6	-5	-4	+1	China	Shanghai	8,320	13,508	13,758	13,852	14,015	77%	3.8	1.9
12.	+15	+6	+5	+2	United States	Boston	6,563	9,354	9,979	13,454	13,710	75%	46.6	37.4
13.	+6	+6	+0	+2	Germany	Munich	7,572	9,231	11,861	12,814	13,340	73%	44.5	12.5
14.	+11	+7	+1	+2	Luxembourg	Luxembourg	6,584	8,954	11,034	12,706	12,845	71%	43.5	16.4
15.	-5	+0	-5	-4	Taiwan	Taipei	10,057	10,211	12,944	14,109	12,841	71%	25.8	-0.8
16.	-7	-2	+6	-3	Australia	Sydney	10,259	10,551	8,819	13,820	12,795	70%	21.3	45.1
17.	+6	-7	-12	+0	China	Beijing	6,715	12,316	15,258	12,424	12,447	68%	1.1	-18.4
18.	+16	+14	+15	+7	Denmark	Copenhagen	5,116	5,787	6,973	10,450	11,889	65%	105.4	70.5
19.	-6	-7	-1	-1	Sweden	Stockholm	8,718	11,047	9,859	11,928	11,709	64%	6.0	18.8
20.	-4	+2	+0	+0	Norway	Oslo	8,327	8,416	9,474	10,977	11,415	63%	35.6	20.5
21.	-10	-1	+0	+1	Italy	Milan	9,336	9,198	9,357	10,578	10,940	60%	18.9	16.9
22.	-2	+13	+16	-3	Russia	Moscow	7,471	5,012	5,561	11,139	10,940	60%	118.3	96.7
23.	+6	+6	+0	+0	Netherlands	Amsterdam	6,379	6,243	8,768	10,512	10,818	59%	73.3	23.4
24.	+4	-15	-15	+2	United States	San Francisco	6,458	12,386	13,265	10,191	10,506	58%	-15.2	-20.8
25.	-7	-20	-9	-4	Japan	Tokyo	7,639	13,856	10,830	10,707	10,492	58%	-24.3	-3.1
26.	-2	+1	-2	-2	Finland	Helsinki	6,650	7,224	8,354	10,456	9,836	54%	36.2	17.7
27.	+13	+18	+10	+0	Czech Republic	Prague	4,087	3,868	5,598	9,724	9,744	54%	151.9	74.1
28.	+4	+11	+8	+1	Spain	Madrid	5,489	4,714	6,049	9,324	9,136	50%	93.8	51.0
29.	-8	-3	-10	-1	Canada	Vancouver	6,911	7,439	9,539	9,531	8,898	49%	19.6	-6.7
30.	+14	+6	+4	+2	Germany	Berlin	3,519	4,894	6,932	8,796	8,881	49%	81.5	28.1
31.	+0	+3	-6	-1	Canada	Toronto	5,607	5,326	8,046	9,240	8,765	48%	64.6	8.9
32.	-27	-16	-4	-1	Italy	Rome	12,875	9,555	7,576	8,828	8,549	47%	-14.5	12.9
33.	+16	+0	+2	+2	Ireland	Dublin	3,045	5,548	6,265	8,113	8,382	46%	51.1	33.8
34.	+1	-4	-5	-1	Germany	Frankfurt	4,938	6,004	7,207	8,410	8,355	46%	39.2	15.9
35.	+12	-7	-5	-1	New Zealand	Auckland	3,310	6,361	7,185	8,144	7,823	43%	23.0	8.9
36.	+15	+10	+7	+1	Portugal	Lisbon	2,737	3,831	4,861	7,583	7,741	43%	102.0	59.2
37.	+0	+3	+5	+1	Spain	Barcelona	4,864	4,595	4,931	7,363	7,565	42%	64.6	53.4
38.	-16	-13	-12	+1	United States	Los Angeles	6,745	7,476	7,996	7,240	7,420	41%	-0.8	-7.2
39.	+7	-2	+11	+1	United Arab Emirates	Dubai	3,358	4,883	3,544	7,227	7,244	40%	48.3	104.4
40.	-26	-9	-8	+5	Australia	Melbourne	8,544	5,932	7,019	6,440	7,087	39%	19.5	1.0
41.	-5	+2	+0	-5	New Zealand	Wellington	4,887	4,095	5,080	7,620	6,910	38%	68.7	36.0
42.	-3	+0	-2	+0	United Kingdom	Edinburgh	4,112	4,364	5,169	6,760	6,659	37%	52.6	28.8
43.	+5	+11	+8	+1	Poland	Warsaw	3,078	2,796	3,492	6,497	6,350	35%	127.1	81.9
44.	-11	-21	-13	-3	India	Mumbai	5,442	7,860	7,132	6,850	6,134	34%	-22.0	-14.0
45.	-3	-1	-6	-2	Thailand	Bangkok	3,834	4,078	5,477	6,514	6,131	34%	50.3	11.9
46.	+19	+14	+6	+2	Hungary	Budapest	1,692	1,980	3,258	5,020	6,116	34%	209.0	87.7
47.	+3	+1	-1	+0	United Kingdom	Birmingham	2,769	3,606	4,233	5,488	5,449	30%	51.1	28.7
48.	-5	-1	+0	+2	Belgium	Brussels	3,621	3,762	3,760	4,969	5,110	28%	35.8	35.9
49.	-4	+1	-2	+0	Canada	Montreal	3,363	3,290	4,127	4,992	5,025	28%	52.8	21.8
50.	+2	-12	-1	+1	United Arab Emirates	Abu Dhabi	2,725	4,748	3,577	4,682	4,826	27%	1.6	34.9
51.	+11	+14	+5	-5	Philippines	Manila	2,016	1,622	2,829	5,655	4,351	24%	168.2	53.8
52.	-14	-11	-8	+1	Qatar	Doha	4,339	4,556	4,547	4,251	4,164	23%	-8.6	-8.4
53.	+0	+5	+2	+1	Malaysia	Kuala Lumpur	2,601	2,604	2,909	4,323	4,100	23%	57.5	41.0
54.	+6	-5	-9	+1	United States	Chicago	2,153	3,325	4,351	3,958	4,089	22%	22.4	-6.5
55.	+2	+11	+7	-1	Greece	Athens	2,355	1,611	2,203	3,984	4,016	22%	149.3	82.3
56.	+8	+7	+4	+2	Mexico	Mexico City	1,812	1,775	2,395	3,318	3,874	21%	118.2	61.8
57.	-27	-6	-3	-1	Brazil	Rio de Janeiro	6,200	3,240	2,913	3,541	3,674	20%	13.4	26.1
58.	-2	-2	+8	+1	Turkiye	Istanbul	2,379	2,721	1,495	3,208	3,087	17%	13.4	106.5
59.	-1	-6	+0	-2	Indonesia	Jakarta	2,235	2,842	2,649	3,375	2,978	16%	4.8	12.4
60.	-5	+7	+7	+0	Saudi Arabia	Riyadh	2,451	1,205	1,155	2,966	2,961	16%	145.8	156.3
61.	+5	-2	+0	+2	Chile	Santiago	1,609	2,128	2,287	2,445	2,822	16%	32.6	23.4
62.	-21	-10	-5	-1	Brazil	Sao Paulo	3,872	2,855	2,801	2,888	2,886	15%	-5.5	-3.7
63.	+0	-1	+1	+4	Colombia	Bogota	1,957	1,822	1,905	2,068	2,822	14%	43.9	37.6
64.	-10	-7	-11	-2	Argentina	Buenos Aires	2,561	2,712	3,172	2,608	2,621	14%	-3.4	-17.4
65.	-6	-10	-2	-1	India	Delhi	2,160	2,733	2,130	2,280	2,465	14%	-9.8	15.7
66.	-5	-5	-8	+0	South Africa	Cape Town	2,086	1,841	2,756	2,077	2,342	13%	27.2	-15.0
67.	+1	-3	-2	-2	India	Bangalore	1,369	1,665	1,893	2,088	1,803	10%	8.3	-4.7
68.	-1	+0	+0	+0	South Africa	Johannesburg	1,496	924	1,084	904	1,065	6%	15.3	-1.7
69.	+0	+0	+0	+0	Egypt	Cairo	964	870	765	814	915	5%	5.2	19.6

Source : www.numbeo.com, Deutsche Bank.



Figure 32: Monthly rent for 3 bedroom apartment in city centre (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+1	+1	+0	United States	New York	4,938	5,938	6,241	8,747	8,873	100%	49.4	42.2
2.	+12	+11	+8	+4	Switzerland	Zurich	3,428	3,601	3,373	5,250	6,285	71%	74.5	86.4
3.	+2	-2	-2	+1	United States	San Francisco	4,250	6,690	6,403	5,910	6,071	68%	-9.3	-5.2
4.	+7	+3	+1	-2	United States	Boston	3,700	4,028	4,143	6,161	5,873	66%	45.8	41.8
5.	-3	+0	+2	-2	Singapore	Singapore	5,034	4,283	3,914	6,033	5,807	65%	35.6	48.4
6.	+15	+6	+0	+3	United States	Los Angeles	2,750	3,619	4,119	4,684	5,203	59%	43.8	26.3
7.	-6	-4	-4	-2	Hong Kong	Hong Kong	5,839	5,100	5,024	5,335	5,164	58%	1.3	2.8
8.	-4	-4	-4	-1	United Kingdom	London	4,374	4,719	4,158	5,089	4,983	56%	5.6	19.8
9.	-1	+1	+5	+2	Australia	Sydney	3,999	3,660	3,103	4,622	4,934	56%	34.8	59.0
10.	+0	-2	-1	-2	Switzerland	Geneva	3,913	3,980	3,534	4,705	4,854	55%	22.0	37.3
11.	+6	+3	-3	+1	United States	Chicago	3,100	3,363	3,688	4,622	4,673	53%	39.0	26.7
12.	+3	-6	+1	-2	United Arab Emirates	Dubai	3,240	4,114	3,143	4,671	4,513	51%	9.7	43.6
13.	+21	+5	-2	+1	Ireland	Dublin	2,126	2,634	3,361	4,118	4,424	50%	68.0	31.6
14.	-7	-5	+11	+2	United Arab Emirates	Abu Dhabi	4,054	3,740	2,439	3,663	4,076	46%	9.0	67.1
15.	+20	+5	+0	-2	Netherlands	Amsterdam	2,037	2,616	3,082	4,429	4,046	46%	54.7	31.3
16.	+23	+20	+11	+7	Israel	Tel Aviv-Yafo	1,878	1,970	2,370	3,212	3,821	43%	93.9	61.2
17.	+8	-1	-5	+0	Luxembourg	Luxembourg	2,593	2,786	3,177	3,647	3,663	41%	31.5	15.3
18.	+9	+7	+2	+3	Italy	Milan	2,407	2,389	2,639	3,444	3,637	41%	52.2	37.8
19.	-10	-8	-2	-1	Qatar	Doha	3,980	3,628	2,919	3,580	3,542	40%	-2.4	21.4
20.	+0	+1	-2	-5	Canada	Vancouver	2,757	2,612	2,832	3,707	3,520	40%	34.8	24.3
21.	-5	+2	-5	-2	France	Paris	3,148	2,563	2,978	3,506	3,433	39%	34.0	15.3
22.	+4	+0	+2	-2	Denmark	Copenhagen	2,425	2,575	2,493	3,463	3,427	39%	33.1	37.5
23.	-10	+6	+5	+2	Australia	Melbourne	3,470	2,292	2,363	3,091	3,322	37%	45.0	40.6
24.	-6	+6	+15	+0	Russia	Moscow	3,022	2,258	1,957	3,207	3,230	36%	43.1	65.1
25.	-1	+1	+7	+11	Sweden	Stockholm	2,613	2,384	2,182	2,479	3,112	35%	30.5	42.6
26.	+10	+6	-3	-4	Germany	Munich	2,029	2,204	2,513	3,360	3,111	35%	41.1	23.8
27.	+28	+23	+14	-1	Portugal	Lisbon	1,276	1,354	1,919	2,902	2,926	33%	116.1	52.5
28.	+16	+16	+14	+1	Spain	Madrid	1,707	1,606	1,870	2,780	2,843	32%	77.0	52.0
29.	+17	+6	+15	-1	United Kingdom	Edinburgh	1,662	1,971	1,833	2,802	2,840	32%	44.1	54.9
30.	+12	+7	+3	+1	Germany	Frankfurt	1,790	1,893	2,181	2,695	2,779	31%	46.8	27.4
31.	-8	+2	-10	-4	Canada	Toronto	2,663	2,170	2,627	2,812	2,733	31%	26.0	4.0
32.	-13	-5	-6	+1	Norway	Oslo	2,837	2,363	2,393	2,540	2,716	31%	14.9	13.5
33.	+18	+14	+12	+5	Spain	Barcelona	1,385	1,469	1,743	2,462	2,695	30%	83.4	54.6
34.	+22	+12	+6	-4	Germany	Berlin	1,226	1,510	1,922	2,728	2,672	30%	76.9	39.0
35.	+17	+23	+18	+8	Mexico	Mexico City	1,365	1,071	1,215	2,271	2,545	29%	137.7	109.4
36.	-24	-8	+1	-4	Italy	Rome	3,537	2,337	2,074	2,655	2,541	29%	8.7	22.5
37.	+8	+4	+12	+0	Belgium	Brussels	1,687	1,736	1,621	2,469	2,523	28%	45.3	55.7
38.	+0	+1	+5	+2	Austria	Vienna	1,950	1,809	1,868	2,392	2,522	28%	39.5	35.0
39.	-11	-5	-3	-4	Finland	Helsinki	2,258	2,155	2,085	2,493	2,465	28%	14.4	18.2
40.	-34	-25	-21	-1	Japan	Tokyo	4,244	3,053	2,725	2,408	2,361	27%	-22.7	-13.3
41.	+16	+16	+10	+1	Czech Republic	Prague	1,188	1,093	1,463	2,354	2,349	26%	115.0	60.6
42.	-12	-25	-20	+6	China	Shanghai	2,216	2,659	2,554	2,122	2,348	26%	-11.7	-8.1
43.	-6	-24	-14	+2	China	Beijing	1,982	2,620	2,263	2,197	2,292	26%	-12.5	1.3
44.	-22	-20	-9	+0	South Korea	Seoul	2,688	2,434	2,094	2,225	2,282	26%	-6.3	9.0
45.	-14	-2	+1	+4	United Kingdom	Birmingham	2,205	1,657	1,680	2,108	2,227	25%	34.4	32.6
46.	+1	+2	+4	+4	Canada	Montreal	1,553	1,440	1,613	2,079	2,221	25%	54.3	37.7
47.	-15	-16	-17	-13	New Zealand	Auckland	2,186	2,242	2,238	2,534	2,188	25%	-2.4	-2.2
48.	+11	+12	+6	+3	Poland	Warsaw	1,079	1,040	1,202	2,038	2,175	25%	109.0	81.0
49.	-16	-7	-18	-8	New Zealand	Wellington	2,139	1,723	2,226	2,371	2,168	24%	25.9	-2.6
50.	-9	-12	-12	-4	Thailand	Bangkok	1,796	1,857	2,021	2,176	2,113	24%	13.8	4.6
51.	+11	+12	+11	+1	Saudi Arabia	Riyadh	967	853	785	1,924	2,003	23%	134.8	155.3
52.	-4	-7	-5	+1	India	Mumbai	1,457	1,545	1,639	1,902	1,993	22%	29.0	21.6
53.	-4	-1	-19	-6	Philippines	Manila	1,422	1,330	2,161	2,135	1,949	22%	46.6	-9.8
54.	+6	+2	+11	+0	Türkiye	Istanbul	1,016	1,100	639	1,830	1,943	22%	76.6	204.3
55.	-5	-6	-7	+1	South Africa	Cape Town	1,400	1,364	1,632	1,531	1,883	21%	38.0	15.4
56.	+13	+5	+1	+2	Hungary	Budapest	519	874	1,026	1,403	1,771	20%	102.7	72.7
57.	+4	+7	+11	+0	Argentina	Buenos Aires	989	831	579	1,416	1,572	18%	89.2	171.8
58.	-15	-5	-3	-3	Taiwan	Taipei	1,722	1,278	1,196	1,737	1,541	17%	20.6	28.8
59.	-19	+0	-3	+0	Brazil	Sao Paulo	1,802	1,065	1,058	1,308	1,430	16%	34.3	35.2
60.	+5	+8	+3	+1	Greece	Athens	715	545	752	1,242	1,330	15%	144.2	76.9
61.	-8	-6	-3	+1	Malaysia	Kuala Lumpur	1,311	1,137	973	1,140	1,210	14%	6.5	24.4
62.	+2	+3	+2	+2	Colombia	Bogota	776	686	665	1,098	1,186	13%	73.0	78.2
63.	-9	-23	-11	-3	Indonesia	Jakarta	1,308	1,781	1,240	1,250	1,131	13%	-36.5	-8.8
64.	-35	-13	-4	+2	Brazil	Rio de Janeiro	2,220	1,333	878	968	1,093	12%	-18.1	24.5
65.	-2	-3	-4	-2	Chile	Santiago	893	863	873	1,131	1,090	12%	26.2	24.9
66.	-8	-12	-7	-1	South Africa	Johannesburg	1,132	1,155	938	986	1,069	12%	-7.4	13.9
67.	+1	+0	-1	+0	India	Bangalore	556	554	630	885	846	10%	52.7	34.4
68.	-1	-2	-1	+0	India	Delhi	625	635	588	648	685	8%	7.9	16.5
69.	-3	+0	+0	+0	Egypt	Cairo	714	540	396	448	543	6%	0.4	37.0

Source : www.numbeo.com, Deutsche Bank.



Figure 33: Disposable Income Index after rents (2 persons working and renting 3 bedroom apartment)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+0	+0	+0	Switzerland	Zurich	7,045	8,451	9,159	10,728	10,442	476%	23.6	14.0
2.	+0	+0	+1	+0	Switzerland	Geneva	8,272	7,031	7,542	10,127	10,034	457%	42.7	33.0
3.	+4	+35	-1	+0	United States	San Francisco	4,717	2,038	8,391	8,921	9,130	416%	347.9	8.8
4.	-3	+1	+1	+1	Luxembourg	Luxembourg	9,477	4,311	4,936	6,449	6,729	307%	56.1	36.3
5.	+25	+18	-1	-1	United States	Boston	3,274	2,981	5,901	6,935	6,643	303%	122.9	12.6
6.	+10	+24	+20	+8	United Kingdom	Birmingham	3,928	2,653	2,993	4,508	5,984	273%	125.5	99.9
7.	+8	+5	+6	+1	Denmark	Copenhagen	4,088	3,579	3,838	5,965	5,978	272%	67.0	55.8
8.	-4	-4	+1	+1	Norway	Oslo	6,286	4,459	4,400	5,055	5,639	257%	26.5	28.2
9.	-1	-6	+2	+3	Australia	Melbourne	4,709	4,689	4,005	4,801	5,496	250%	17.2	37.2
10.	-4	+1	+2	-3	Germany	Frankfurt	5,366	3,667	3,843	6,047	5,468	249%	49.1	42.3
11.	+15	-5	-4	-5	United States	Chicago	3,413	4,095	4,777	6,300	5,294	241%	29.3	10.8
12.	+2	-3	+5	-2	Germany	Munich	4,092	3,903	3,485	5,023	5,227	238%	33.9	50.0
13.	+10	+18	+14	-2	Netherlands	Amsterdam	3,583	2,505	2,989	4,928	5,062	231%	102.1	69.4
14.	+27	+18	+18	+6	Israel	Tel Aviv-Yafo	1,839	2,462	2,627	4,217	4,977	227%	102.2	89.4
15.	+21	+18	+19	+1	Austria	Vienna	2,371	2,420	2,480	4,453	4,237	193%	75.1	70.9
16.	+9	+0	+2	+1	Sweden	Stockholm	3,477	3,382	3,443	4,413	4,197	191%	24.1	21.9
17.	+0	+4	+8	+2	United Kingdom	Edinburgh	3,927	3,079	3,037	4,279	4,154	189%	34.9	36.8
18.	+11	+1	+2	+4	Germany	Berlin	3,342	3,169	3,362	4,026	4,060	185%	28.1	20.8
19.	+9	-1	+5	+6	New Zealand	Auckland	3,393	3,320	3,066	3,897	4,015	183%	20.9	30.9
20.	-10	-6	+10	+3	Canada	Toronto	4,340	3,423	2,644	4,022	4,014	183%	17.3	51.8
21.	-10	+6	-5	-6	Belgium	Brussels	4,311	2,846	3,532	4,465	3,866	176%	35.8	9.5
22.	-3	+2	+15	+2	France	Paris	3,863	2,914	2,191	3,975	3,811	174%	30.8	74.0
23.	+8	-10	-1	+5	Finland	Helsinki	3,188	3,473	3,349	3,804	3,790	173%	9.1	13.2
24.	-12	-17	-18	-11	United States	Los Angeles	4,226	3,988	4,908	4,557	3,756	171%	-5.8	-23.5
25.	+14	+11	+8	+7	South Korea	Seoul	1,895	2,118	2,568	4,402	3,748	171%	76.9	45.9
26.	-17	-1	-5	+0	Canada	Montreal	4,485	2,882	3,351	3,887	3,608	164%	25.2	7.7
27.	+10	+16	+11	+2	United Kingdom	London	2,132	1,572	1,877	3,703	3,596	164%	128.7	91.5
28.	-6	-8	+7	-1	Ireland	Dublin	3,756	3,117	2,375	3,855	3,471	158%	11.4	46.1
29.	-24	-14	-15	+3	Australia	Sydney	5,524	3,404	3,767	3,227	3,428	156%	0.7	-9.0
30.	-12	-20	-22	+5	New Zealand	Wellington	3,901	3,772	4,535	2,867	3,403	155%	-9.8	-25.0
31.	+1	+6	-2	-1	Canada	Vancouver	3,176	2,080	2,935	3,331	3,133	143%	50.6	6.8
32.	-19	-3	-17	+5	Qatar	Doha	4,137	2,807	3,563	2,807	2,838	129%	1.1	-20.4
33.	-9	-25	-10	-12	United Arab Emirates	Abu Dhabi	3,497	3,941	3,224	4,084	2,592	118%	-34.2	-19.6
34.	-14	-6	-15	-1	Japan	Tokyo	3,803	2,812	3,394	3,035	2,427	111%	-13.7	-28.3
35.	-1	+6	+7	+4	South Africa	Johannesburg	2,691	1,713	1,521	2,151	2,412	110%	40.8	58.6
36.	+14	+12	+8	+6	Czech Republic	Prague	973	1,033	1,140	1,974	2,347	107%	127.3	105.9
37.	-16	-11	-9	-6	United Arab Emirates	Dubai	3,773	2,868	2,944	3,264	2,291	104%	-19.8	-22.2
38.	+14	+11	+8	+3	Poland	Warsaw	845	978	1,050	2,065	2,268	103%	132.0	115.9
39.	-12	-5	-29	+4	United States	New York	3,403	2,372	4,024	1,908	2,194	100%	-7.5	-45.5
40.	+2	-1	-1	-2	Spain	Madrid	1,633	1,848	1,850	2,277	2,176	99%	17.7	17.6
41.	+21	-6	-10	-7	Singapore	Singapore	-40	2,179	2,631	2,869	2,143	98%	-1.6	-18.6
42.	-2	-2	-1	-2	Spain	Barcelona	1,858	1,806	1,571	2,111	2,093	95%	15.9	33.2
43.	-10	-26	-7	-7	Saudi Arabia	Riyadh	2,997	3,380	2,211	2,838	2,080	95%	-38.5	-5.9
44.	+4	+1	+1	+2	Malaysia	Kuala Lumpur	1,134	1,122	1,111	1,572	1,945	89%	73.3	75.0
45.	+2	-23	-5	-1	Taiwan	Taipei	1,158	3,022	1,739	1,682	1,939	88%	-35.8	11.5
46.	+3	+10	+5	+1	Hungary	Budapest	987	482	725	1,526	1,769	81%	267.1	143.9
47.	+9	+0	-4	-2	Italy	Rome	552	1,091	1,246	1,630	1,445	66%	32.5	16.0
48.	-13	-6	+1	+0	South Africa	Cape Town	2,470	1,685	937	1,470	1,441	66%	-14.5	53.9
49.	-5	-3	-2	+1	Greece	Athens	1,268	1,093	1,038	1,193	1,312	60%	20.0	26.4
50.	+9	+9	+8	+1	Hong Kong	Hong Kong	133	265	220	1,152	1,233	56%	364.6	460.8
51.	+13	+7	+13	+2	China	Shanghai	-180	292	42	915	1,067	49%	265.4	2,454.2
52.	+13	+13	+4	+2	China	Beijing	-199	-15	313	783	830	38%	5,598.8	164.9
53.	-15	-9	-5	-1	Italy	Milan	1,951	1,418	1,024	1,127	796	36%	-43.8	-22.3
54.	-1	-3	-4	-5	India	Bangalore	710	928	870	1,286	762	35%	-17.9	-12.4
55.	-9	-3	-2	+1	Chile	Santiago	1,226	824	664	352	507	23%	-38.5	-23.6
56.	-11	-1	-1	+1	Türkiye	Istanbul	1,241	594	365	288	403	18%	-32.1	10.6
57.	-2	-4	-3	-2	India	Delhi	642	672	524	565	390	18%	-42.0	-25.6
58.	+8	+10	+3	+1	Russia	Moscow	-201	-515	109	182	358	16%	169.4	229.5
59.	-8	-5	+0	+3	Portugal	Lisbon	922	652	165	37	289	13%	-55.7	75.3
60.	-6	+0	+0	+0	Brazil	Sao Paulo	657	227	155	159	276	13%	21.4	78.5
61.	-4	+1	+4	+2	Colombia	Bogota	506	139	42	-35	249	11%	79.0	500.5
62.	-19	-12	-10	-4	Argentina	Buenos Aires	1,456	941	670	201	112	5%	-88.1	-83.3
63.	-3	+1	-6	-2	Brazil	Rio de Janeiro	129	27	237	54	96	4%	252.8	-59.6
64.	-3	-3	-1	+0	Egypt	Cairo	-36	155	78	-79	-103	-5%	-166.5	-232.4
65.	-2	+4	+1	+0	Indonesia	Jakarta	-160	-798	-315	-338	-278	-13%	65.2	11.7
66.	-8	-9	-4	+1	Mexico	Mexico City	171	319	87	-371	-427	-19%	-234.1	-592.6
67.	+0	-4	+0	-1	India	Mumbai	-383	84	-493	-340	-439	-20%	-622.9	10.9
68.	+1	-1	+0	+0	Thailand	Bangkok	-483	-276	-526	-647	-867	-40%	-213.6	-64.9
69.	-1	-3	+0	+0	Philippines	Manila	-470	-126	-1,187	-1,154	-1,008	-46%	-698.2	15.1

Source : www.numbeo.com, Deutsche Bank.



Figure 34: Monthly rent for 1 bedroom apartment in city centre (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+0	+1	+1	+0	United States	New York	2,763	2,916	3,195	4,136	4,285	100%	47.0	34.1
2.	+1	-1	-1	+0	United States	San Francisco	2,279	3,414	3,504	3,376	3,774	88%	10.5	7.7
3.	+5	+1	+0	+0	United States	Boston	1,972	2,386	2,580	3,373	3,401	79%	42.5	31.8
4.	+2	+5	+7	+5	Switzerland	Zurich	2,092	1,942	1,870	2,546	3,021	71%	55.5	61.6
5.	-1	-2	+1	-1	United Kingdom	London	2,237	2,480	2,225	3,152	3,004	70%	21.2	35.1
6.	-4	+0	+2	+1	Singapore	Singapore	2,670	2,185	2,058	3,068	2,958	69%	35.4	43.7
7.	+3	+6	+0	-1	Switzerland	Geneva	1,921	1,791	2,064	2,820	2,791	65%	55.9	35.3
8.	+12	+9	+5	+0	Netherlands	Amsterdam	1,427	1,487	1,793	2,562	2,635	61%	77.2	46.9
9.	+12	+1	-4	-2	United States	Los Angeles	1,400	1,910	2,260	2,572	2,563	60%	34.2	13.4
10.	+22	+6	+0	+3	Ireland	Dublin	1,084	1,491	1,893	2,348	2,516	59%	68.7	32.9
11.	+6	+3	-2	+0	United States	Chicago	1,518	1,776	1,934	2,388	2,443	57%	37.6	26.3
12.	-7	-4	+0	+2	Australia	Sydney	2,189	1,954	1,793	2,295	2,395	56%	22.5	33.5
13.	+2	-6	+2	-3	United Arab Emirates	Dubai	1,547	2,054	1,805	2,475	2,290	53%	11.4	42.7
14.	+5	+1	+0	+1	Luxembourg	Luxembourg	1,471	1,512	1,731	2,189	2,289	53%	51.4	32.3
15.	-8	-10	-11	-3	Hong Kong	Hong Kong	2,023	2,256	2,307	2,387	2,253	53%	-0.1	-2.3
16.	+20	+12	+4	+2	Israel	Tel Aviv-Yafo	1,000	1,163	1,426	1,891	2,210	52%	90.1	55.0
17.	+7	+2	+4	+1	Denmark	Copenhagen	1,299	1,396	1,407	2,065	1,978	46%	41.7	40.6
18.	-7	-6	+6	+5	United Arab Emirates	Abu Dhabi	1,800	1,861	1,318	1,691	1,961	46%	5.3	48.8
19.	-3	-1	+0	+1	Norway	Oslo	1,543	1,417	1,439	1,748	1,934	45%	36.5	34.4
20.	-11	-9	-3	-1	Qatar	Doha	1,942	1,894	1,560	1,838	1,933	45%	2.0	23.9
21.	+4	+0	-3	-4	Canada	Vancouver	1,274	1,302	1,522	1,962	1,831	43%	40.6	20.3
22.	+9	-2	+3	+6	Sweden	Stockholm	1,087	1,382	1,315	1,553	1,774	41%	28.4	34.9
23.	+5	+7	+4	+1	Italy	Milan	1,145	1,114	1,266	1,670	1,721	40%	54.6	36.0
24.	-10	-2	+2	+1	Australia	Melbourne	1,630	1,282	1,308	1,598	1,708	40%	33.2	30.6
25.	+5	+2	-3	-3	Germany	Munich	1,101	1,176	1,379	1,728	1,669	39%	41.9	21.1
26.	+18	+16	+10	+5	Spain	Barcelona	856	856	1,085	1,528	1,669	39%	94.9	53.9
27.	-5	-4	-4	+0	France	Paris	1,385	1,252	1,345	1,574	1,640	38%	31.0	21.9
28.	+22	+18	+12	-2	Portugal	Lisbon	728	700	1,001	1,562	1,639	38%	134.1	63.8
29.	+13	+5	+10	+0	United Kingdom	Edinburgh	862	1,011	1,008	1,544	1,630	38%	61.2	61.7
30.	-7	-6	-14	-9	Canada	Toronto	1,365	1,245	1,586	1,737	1,604	37%	28.8	1.1
31.	+7	+10	+6	-1	Spain	Madrid	953	873	1,057	1,532	1,596	37%	82.7	51.0
32.	-14	+6	+13	+3	Russia	Moscow	1,493	880	924	1,390	1,561	36%	77.4	68.9
33.	+13	+11	+5	+0	Germany	Berlin	824	800	1,014	1,418	1,493	35%	86.5	47.3
34.	+3	+1	+9	+2	United Kingdom	Birmingham	990	988	941	1,379	1,407	33%	42.4	49.6
35.	+8	+1	-4	-1	Germany	Frankfurt	858	981	1,155	1,391	1,375	32%	40.2	19.1
36.	+3	+3	+6	-2	Belgium	Brussels	938	878	945	1,288	1,304	30%	48.5	37.9
37.	-24	-12	-7	+3	Japan	Tokyo	1,662	1,222	1,210	1,263	1,295	30%	5.9	7.0
38.	+3	+5	+3	+1	Austria	Vienna	872	841	971	1,270	1,280	30%	52.2	31.8
39.	-27	-10	-6	-7	Italy	Rome	1,690	1,146	1,114	1,456	1,257	29%	9.7	12.8
40.	-14	-8	-8	+3	Finland	Helsinki	1,248	1,066	1,127	1,229	1,246	29%	16.8	10.5
41.	-1	+4	+3	+1	Canada	Montreal	898	767	924	1,240	1,243	29%	62.1	34.5
42.	-15	-5	-14	-1	New Zealand	Wellington	1,188	919	1,259	1,248	1,232	29%	34.1	-2.1
43.	+15	+11	+7	+2	Poland	Warsaw	555	572	673	1,147	1,228	29%	114.6	82.5
44.	-11	-18	-15	-7	New Zealand	Auckland	1,084	1,213	1,247	1,297	1,223	29%	0.9	-2.0
45.	+10	+3	+1	-1	Czech Republic	Prague	578	628	857	1,168	1,193	28%	90.0	39.2
46.	+17	+13	+6	+1	Mexico	Mexico City	390	506	595	1,046	1,136	27%	124.6	91.0
47.	+10	+13	+15	-1	Saudi Arabia	Riyadh	563	500	446	1,053	1,052	25%	110.2	135.9
48.	+3	+1	+0	+4	South Africa	Cape Town	721	615	778	859	1,030	24%	67.4	32.4
49.	-4	-18	-14	+0	China	Beijing	833	1,094	1,087	944	1,021	24%	-6.7	-6.1
50.	+6	+2	+15	-2	Türkiye	Istanbul	570	595	329	1,042	968	23%	62.6	193.9
51.	-16	-18	-17	+0	China	Shanghai	1,021	1,035	1,101	887	960	22%	-7.2	-12.8
52.	+15	+9	+2	+2	Hungary	Budapest	291	462	548	757	902	21%	95.3	64.4
53.	-19	-6	+5	+11	Brazil	Rio de Janeiro	1,065	645	478	565	820	19%	27.2	71.5
54.	-5	+3	+3	-1	Taiwan	Taipei	772	552	538	817	788	18%	42.9	46.5
55.	-7	-15	-8	-5	South Korea	Seoul	785	875	799	906	776	18%	-11.4	-3.0
56.	+8	+10	+7	+2	Greece	Athens	384	311	422	664	737	17%	136.9	74.4
57.	+2	+5	+9	-2	Argentina	Buenos Aires	553	444	328	728	736	17%	65.7	124.6
58.	-11	-3	-2	+1	Brazil	Sao Paulo	792	567	540	658	688	16%	21.3	27.3
59.	-7	-6	-10	-2	Thailand	Bangkok	638	590	679	667	682	16%	15.6	0.5
60.	+0	+4	-9	-4	India	Mumbai	508	554	599	679	653	15%	18.0	9.0
61.	+0	+4	+3	+2	Colombia	Bogota	461	406	371	571	644	15%	58.5	73.4
62.	-9	-12	-7	-1	Malaysia	Kuala Lumpur	633	611	541	618	637	15%	4.2	17.7
63.	-1	+1	-3	-1	Chile	Santiago	448	434	456	605	595	14%	37.3	30.5
64.	+1	-6	-11	-4	Philippines	Manila	375	535	586	636	541	13%	1.2	-7.7
65.	-11	-2	-6	+0	South Africa	Johannesburg	612	439	471	488	532	12%	21.0	12.8
66.	-37	-15	-5	+0	Indonesia	Jakarta	1,119	605	450	373	357	8%	-41.1	-20.8
67.	+2	+2	+0	+0	India	Bangalore	205	220	254	328	327	8%	48.9	29.0
68.	+0	+0	+0	+0	India	Delhi	217	257	238	244	241	6%	-6.5	1.0
69.	-3	-2	+0	+0	Egypt	Cairo	320	272	218	218	238	6%	-12.6	9.0

Source : www.numbeo.com, Deutsche Bank.



Figure 35: Mortgage interest rate proxy (20 years fixed-rate mortgage yearly interest rate in %) *

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (pp)	7-yr cum change (pp)
1.	+0	+0	+0	+0	Argentina	Buenos Aires	22.6	25.6	51.4	48.3	54.7	884%	29.1	3.4
2.	+3	+2	+0	+0	Türkiye	Istanbul	12.7	12.8	20.5	43.1	41.1	663%	28.3	20.6
3.	+0	+0	+5	+0	Russia	Moscow	13.3	13.3	10.4	23.8	20.1	324%	6.8	9.6
4.	+5	+2	-1	+0	Egypt	Cairo	11.6	12.0	13.8	16.5	15.9	257%	3.9	2.1
5.	-3	-3	-1	+0	Colombia	Bogota	19.5	14.5	11.6	12.6	12.8	206%	-1.8	1.2
6.	+6	+1	+3	+3	Brazil	Rio de Janeiro	10.3	11.9	10.0	11.2	12.0	194%	0.1	2.0
8.	-2	-3	+3	-2	Brazil	Sao Paulo	12.6	12.4	9.9	12.3	11.5	186%	-0.9	1.6
7.	+3	+1	+2	+0	Mexico	Mexico City	11.4	11.4	11.4	11.6	11.5	186%	0.1	0.1
9.	+7	+2	-3	-1	South Africa	Cape Town	9.0	10.6	10.8	11.3	11.2	182%	0.7	0.4
10.	+5	+4	-3	+0	South Africa	Johannesburg	9.2	10.3	10.5	11.1	10.9	176%	0.6	0.4
12.	-4	-2	+1	-1	India	Delhi	11.8	10.7	9.4	9.4	8.9	144%	-1.8	-0.5
11.	+3	-2	-1	+1	Indonesia	Jakarta	9.4	11.2	9.9	9.3	9.9	160%	-1.2	0.0
14.	-3	-2	-2	+0	India	Bangalore	10.9	10.4	9.5	8.8	8.5	137%	-2.0	-1.0
13.	-6	+0	+1	+0	India	Mumbai	11.9	10.4	9.3	9.1	8.5	137%	-1.9	-0.8
15.	-11	+0	+0	+0	Philippines	Manila	13.0	8.6	7.9	7.5	7.4	120%	-1.1	-0.5
16.	+4	+20	+17	+0	Poland	Warsaw	6.8	3.9	4.1	7.4	6.9	111%	3.0	2.8
17.	-4	-1	+2	+0	Hungary	Budapest	10.1	5.5	5.0	6.9	6.9	111%	1.3	1.9
18.	+17	+14	+10	+1	United States	Chicago	4.9	4.0	4.3	6.4	6.4	103%	2.4	2.1
20.	+38	+14	+5	+1	United States	Boston	3.6	3.9	4.4	6.2	6.3	101%	2.4	1.9
19.	+21	+18	+13	+1	United States	Los Angeles	4.7	3.8	4.2	6.3	6.4	103%	2.6	2.2
21.	+24	+10	+8	+2	United States	New York	4.2	4.1	4.2	6.1	6.2	100%	2.1	2.0
22.	+16	+16	+7	+2	United States	San Francisco	4.9	3.8	4.2	6.1	6.1	98%	2.3	1.9
23.	+38	+2	-2	-1	Qatar	Doha	3.3	4.8	4.9	6.2	5.9	95%	1.1	1.0
24.	-3	+3	+11	+1	Australia	Sydney	6.4	4.6	3.9	5.8	5.9	95%	1.3	2.0
26.	-4	+2	+3	+2	Australia	Melbourne	6.4	4.5	4.2	5.7	5.6	90%	1.1	1.4
28.	+6	+2	-6	-2	Saudi Arabia	Riyadh	5.0	4.3	4.8	5.7	5.5	89%	1.3	0.8
27.	+1	-3	+0	+0	New Zealand	Auckland	5.8	4.8	4.3	5.7	5.6	90%	0.8	1.3
29.	-11	-12	-13	+2	Thailand	Bangkok	7.0	5.5	5.7	5.2	5.4	87%	-0.2	-0.3
30.	-5	-8	+8	-12	Chile	Santiago	6.3	4.8	3.6	6.6	5.3	85%	0.4	1.7
25.	-1	-3	-5	+4	New Zealand	Wellington	6.3	4.8	4.9	5.4	5.7	91%	0.8	0.7
32.	+18	+7	+5	+1	Israel	Tel Aviv-Yafo	4.0	3.4	3.7	5.1	5.0	81%	1.7	1.3
32.	+21	+21	+13	+2	Norway	Oslo	3.8	2.5	2.8	5.0	5.0	81%	2.6	2.2
31.	+5	+2	+13	+1	United Kingdom	Birmingham	4.9	4.0	3.3	5.1	5.1	82%	1.1	1.8
34.	+18	+7	+15	+1	United Kingdom	Edinburgh	3.9	3.3	2.6	5.0	4.8	78%	1.5	2.3
35.	+2	+5	+8	+1	United Kingdom	London	4.9	3.3	3.5	4.8	4.7	77%	1.4	1.3
37.	+4	+22	+9	+0	Czech Republic	Prague	4.4	2.1	2.8	4.7	4.7	75%	2.5	1.9
36.	-19	-16	-2	-6	United Arab Emirates	Abu Dhabi	8.5	5.1	3.9	5.3	4.7	76%	-0.4	0.8
41.	+18	+5	-2	-1	Canada	Montreal	3.5	2.8	3.6	4.5	4.5	72%	1.6	0.9
39.	-10	-18	-16	-1	Greece	Athens	5.6	4.9	4.7	4.6	4.5	73%	-0.4	-0.2
40.	-17	-11	-14	-1	United Arab Emirates	Dubai	6.4	4.3	4.3	4.5	4.5	73%	0.2	0.2
38.	+22	+7	+2	+3	Canada	Toronto	3.5	2.9	3.5	4.5	4.6	74%	1.7	1.1
42.	+6	+5	-6	+0	Canada	Vancouver	4.1	2.8	3.8	4.4	4.4	71%	1.6	0.6
43.	-17	-17	-19	+0	Malaysia	Kuala Lumpur	5.9	4.7	4.7	4.3	4.3	69%	-0.4	-0.4
45.	+17	+23	+24	+4	Finland	Helsinki	3.3	1.5	1.4	3.7	4.0	65%	2.6	2.7
44.	-11	-2	-3	+0	South Korea	Seoul	5.0	3.3	3.5	4.0	4.1	66%	0.8	0.6
46.	-2	+21	+16	-1	Germany	Munich	4.2	1.5	1.8	4.0	4.0	64%	2.4	2.2
49.	-10	-14	-7	-3	Ireland	Dublin	4.9	3.9	3.5	3.8	3.8	62%	0.0	0.4
47.	+10	+15	+13	+0	Germany	Berlin	3.6	2.0	1.9	3.7	3.9	63%	1.9	2.0
50.	-3	+2	+12	+6	Belgium	Brussels	4.1	2.5	1.8	3.5	3.8	61%	1.2	2.0
51.	+16	+11	+13	-3	Luxembourg	Luxembourg	2.2	2.0	1.7	3.7	3.8	61%	1.8	2.0
52.	+12	+3	-5	+1	Hong Kong	Hong Kong	2.9	2.3	2.6	3.5	3.7	60%	1.4	1.2
53.	-7	-2	+3	-1	Portugal	Lisbon	4.1	2.6	2.1	3.6	3.7	60%	1.1	1.6
47.	+9	+3	+20	+7	Denmark	Copenhagen	3.7	2.6	1.5	3.5	3.9	63%	1.3	2.4
54.	-11	+0	+4	-5	Germany	Frankfurt	4.3	2.4	2.0	3.7	3.7	59%	1.3	1.7
56.	-7	+3	-3	+1	Austria	Vienna	4.0	2.1	2.3	3.4	3.5	56%	1.4	1.2
55.	-24	-6	-4	-4	Netherlands	Amsterdam	5.2	2.7	2.3	3.6	3.6	57%	0.9	1.3
58.	-28	-14	-3	+2	Italy	Rome	5.3	2.9	2.2	3.3	3.4	55%	0.5	1.2
56.	-5	+8	-7	+6	Sweden	Stockholm	3.9	1.9	2.6	3.2	3.5	56%	1.6	0.9
59.	-17	-2	+6	-1	France	Paris	4.3	2.2	1.7	3.4	3.4	55%	1.2	1.7
62.	-36	-44	-45	-7	China	Beijing	5.8	5.2	5.3	3.5	3.1	51%	-2.1	-2.2
60.	-41	-41	-42	-2	China	Shanghai	7.0	5.2	5.1	3.4	3.2	52%	-2.0	-1.9
61.	-29	-18	-2	+2	Italy	Milan	5.2	3.1	2.0	3.1	3.2	51%	0.1	1.2
64.	-10	-3	-10	-2	Spain	Madrid	3.8	2.1	2.2	3.1	3.1	50%	1.1	0.9
62.	-7	-14	-11	+2	Spain	Barcelona	3.8	2.8	2.3	3.0	3.1	51%	0.4	0.9
65.	+0	-9	-18	+0	Singapore	Singapore	2.6	2.3	2.6	2.8	2.5	40%	0.2	-0.1
66.	+2	-8	-9	+0	Taiwan	Taipei	2.0	2.2	2.0	2.3	2.3	38%	0.2	0.4
67.	-4	-2	-6	+1	Switzerland	Geneva	3.0	1.8	1.8	2.0	2.0	32%	0.2	0.2
68.	+0	+1	+0	+1	Japan	Tokyo	2.0	1.4	1.4	1.6	2.0	32%	0.6	0.5
69.	-3	-3	-3	-2	Switzerland	Zurich	2.5	1.6	1.6	1.9	1.9	31%	0.3	0.4

Source : www.numbeo.com, Deutsche Bank. * Note: a rough proxy since some countries may not provide exactly 20-year, fixed-rate mortgage.



Figure 36: Mortgage as percentage of income *

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (pp)	7-yr cum change (pp)
1.	+9	+2	0	+1	Argentina	Buenos Aires	144	237	782	467	511	593%	274	-272
2.	+1	0	+6	-1	Russia	Moscow	227	247	193	474	374	435%	127	182
3.	+22	+10	+6	0	Turkiye	Istanbul	87	133	186	391	324	370%	191	138
4.	+3	+19	+8	0	Philippines	Manila	179	85	174	334	267	310%	182	92
5.	-4	-4	-3	0	India	Mumbai	398	344	411	285	246	286%	-98	-165
6.	-1	+2	+5	0	Brazil	Rio de Janeiro	188	187	182	260	245	284%	57	63
7.	+5	-1	0	0	Indonesia	Jakarta	130	217	136	244	241	280%	24	44
8.	0	+6	+2	+1	Thailand	Bangkok	164	128	185	205	241	280%	113	57
9.	+7	+9	+6	-1	Egypt	Cairo	110	99	143	227	207	241%	108	65
10.	+13	+7	+6	+4	South Korea	Seoul	88	101	141	160	186	216%	85	45
11.	+8	-1	-6	+1	Hong Kong	Hong Kong	102	172	227	173	185	215%	14	-42
12.	-10	-7	-8	-2	China	Shanghai	228	221	254	188	167	194%	-54	-67
13.	-9	-9	-10	-2	China	Beijing	191	229	289	174	161	187%	-68	-128
14.	-8	-7	-8	-1	Colombia	Bogota	183	204	208	160	152	176%	-52	-56
15.	-4	-4	+3	+3	India	Delhi	134	152	129	125	149	171%	-5	20
16.	+5	+3	+1	+1	Mexico	Mexico City	90	97	141	135	141	163%	43	0
17.	-3	+5	-4	-2	Taiwan	Taipei	127	88	160	155	139	161%	51	-22
18.	-5	-9	-4	-2	Brazil	Sao Paulo	130	180	159	148	122	141%	-58	-38
19.	-1	+2	+2	0	Israel	Tel Aviv-Yafo	103	89	100	114	113	131%	23	12
20.	-5	-4	-1	+1	Singapore	Singapore	112	107	118	104	110	127%	3	-8
21.	+1	-9	-1	+2	United Kingdom	London	89	152	112	103	109	127%	-43	-3
22.	+18	+5	+3	-2	Portugal	Lisbon	55	74	86	108	102	119%	29	16
23.	-6	-3	-1	+2	Italy	Milan	103	97	93	93	100	116%	3	7
24.	+2	+8	+4	-2	Czech Republic	Prague	86	67	84	104	96	111%	29	12
25.	+3	+3	-2	-1	Hungary	Budapest	78	73	88	95	95	111%	23	7
26.	+5	+7	+8	+4	Austria	Vienna	67	66	65	89	90	104%	24	24
27.	-18	-12	-1	+4	Italy	Rome	154	116	85	85	89	103%	-27	4
28.	+8	+2	+9	-2	United States	New York	57	69	64	93	86	100%	17	22
29.	+18	+8	+9	-1	Chile	Santiago	40	59	63	89	86	100%	27	23
30.	-1	-4	-6	+2	France	Paris	73	74	88	80	85	99%	11	-3
31.	+20	-7	+8	+9	Japan	Tokyo	35	81	61	69	80	92%	-2	18
32.	-8	+4	0	-5	Poland	Warsaw	88	60	68	91	79	92%	19	11
33.	+2	-2	+11	-4	Australia	Sydney	58	68	55	89	78	91%	10	23
34.	-4	+9	+7	0	Spain	Madrid	71	50	60	74	73	85%	23	13
35.	-8	-10	-8	+10	India	Bangalore	80	80	84	61	70	81%	-11	-15
36.	+2	+6	-5	+5	Germany	Munich	55	53	70	67	70	81%	17	-1
37.	-3	+9	+5	-4	Greece	Athens	59	46	57	75	69	80%	23	12
38.	+5	+11	+9	0	Finland	Helsinki	50	44	53	70	69	80%	24	16
39.	-6	-10	-6	-2	Sweden	Stockholm	62	69	67	71	67	78%	-2	0
40.	-9	+5	+5	-1	Norway	Oslo	39	47	55	69	65	76%	18	10
41.	-9	-2	+2	+1	Spain	Barcelona	64	55	56	64	64	74%	9	8
42.	+16	-2	-2	+1	New Zealand	Auckland	30	53	61	64	63	73%	9	2
43.	+3	+18	+17	-8	New Zealand	Wellington	43	35	35	72	62	72%	27	27
44.	-2	-10	-14	+2	Canada	Vancouver	51	62	71	61	60	70%	-2	-11
45.	+9	+12	-9	-1	Canada	Toronto	33	38	64	61	60	69%	22	-4
46.	-5	-5	-11	-10	Malaysia	Kuala Lumpur	54	53	65	71	58	67%	5	-7
47.	+1	-9	+5	+3	United States	Boston	40	58	45	54	58	67%	0	13
48.	-28	0	+13	+1	Switzerland	Zurich	94	45	35	55	58	67%	12	22
49.	+6	+7	+1	-1	Germany	Berlin	33	38	47	55	57	66%	19	10
50.	-6	+1	+3	-3	Switzerland	Geneva	45	43	43	57	55	64%	12	12
51.	+2	+8	+7	+3	Denmark	Copenhagen	33	36	38	46	55	64%	18	17
52.	-2	-2	-23	0	South Africa	Cape Town	35	44	79	53	53	62%	10	-25
53.	+15	-6	-4	-2	Luxembourg	Luxembourg	20	46	48	53	53	61%	7	5
54.	-15	-10	-8	-1	Netherlands	Amsterdam	55	47	54	47	50	58%	3	-4
55.	+7	+9	+11	+2	United Arab Emirates	Dubai	26	31	26	42	48	56%	17	22
56.	+7	-3	-5	0	Ireland	Dublin	24	42	45	44	46	53%	4	0
57.	-1	+3	-2	-2	United Kingdom	Edinburgh	32	36	41	45	45	52%	9	4
58.	-13	-6	-1	0	United States	Los Angeles	45	42	39	41	44	51%	2	5
59.	-2	-5	-5	0	Germany	Frankfurt	31	41	43	41	43	50%	2	-1
60.	-23	-5	-12	0	Australia	Melbourne	56	39	49	40	40	47%	1	-9
61.	+3	+4	+1	+1	Canada	Montreal	23	30	35	38	39	45%	9	4
62.	-3	+7	+7	+3	Saudi Arabia	Riyadh	29	13	18	31	36	42%	23	18
63.	-11	-28	-7	0	United States	San Francisco	34	61	40	36	36	42%	-25	-4
64.	-4	-1	+3	+2	Belgium	Brussels	26	31	26	30	34	40%	3	8
65.	0	+1	0	+2	United Arab Emirates	Abu Dhabi	23	30	27	29	34	39%	4	6
66.	0	-4	-3	-2	Qatar	Doha	22	33	33	35	33	39%	0	0
67.	0	-9	-8	-6	United Kingdom	Birmingham	21	36	37	40	32	37%	-5	-5
68.	-7	-1	-4	0	South Africa	Johannesburg	26	23	32	22	23	26%	0	-9
69.	0	-1	-1	0	United States	Chicago	16	19	23	19	22	25%	2	-1

Source : www.numbeo.com, Deutsche Bank. * Assuming purchasing a 90 sqm apartment in city centre with a 20% down payment, 2 people working.



Figure 37: Monthly basic utilities (electricity, heating, cooling, water, garbage)
for 85m2 apartment (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+39	+2	+2	+1	Germany	Berlin	141	278	266	391	410	162%	47.1	53.9
2.	+15	+14	+5	+10	Israel	Tel Aviv-Yafo	200	181	220	287	396	157%	119.0	80.0
3.	+0	+10	+13	+5	Norway	Oslo	263	193	192	342	384	152%	99.7	100.2
4.	-2	+2	+4	-1	United Kingdom	London	281	230	217	359	383	152%	66.9	76.7
5.	+4	+0	+16	-1	United Kingdom	Edinburgh	231	238	174	354	379	150%	59.0	117.8
6.	+2	-5	-4	-5	Germany	Munich	237	284	281	398	378	150%	33.0	34.8
7.	+5	-3	-6	+2	Germany	Frankfurt	221	254	305	341	376	149%	47.9	23.2
8.	+33	+12	+14	-2	Poland	Warsaw	140	172	173	347	342	135%	98.1	97.7
9.	+16	+8	+9	+1	Austria	Vienna	178	181	184	334	339	134%	87.1	84.2
10.	-6	-2	+16	-5	United Kingdom	Birmingham	251	203	165	353	334	132%	64.6	102.8
11.	+15	+11	+3	-4	Czech Republic	Prague	175	167	194	344	312	124%	87.2	60.4
12.	-5	+12	-1	-1	Netherlands	Amsterdam	238	163	200	306	311	123%	90.6	55.4
13.	-2	-11	-9	+0	Luxembourg	Luxembourg	225	284	262	287	297	118%	4.8	13.4
14.	+0	-4	-9	+0	Switzerland	Geneva	213	200	241	273	293	116%	46.6	21.3
15.	-14	+36	+36	+4	Sweden	Stockholm	298	83	90	243	281	111%	238.4	213.8
16.	+6	-7	+3	+2	Switzerland	Zurich	186	200	182	260	280	111%	40.1	54.0
17.	+34	+15	+22	+9	United States	Los Angeles	115	143	130	212	265	105%	86.0	104.9
18.	+5	-6	-1	-3	Ireland	Dublin	182	193	190	265	264	105%	36.8	38.8
19.	+5	-12	-10	-2	France	Paris	179	204	208	261	257	102%	26.1	23.6
20.	+12	+8	+10	+4	Belgium	Brussels	157	155	145	262	257	102%	65.9	76.4
21.	+22	+15	+16	+14	United States	New York	135	136	133	177	252	100%	85.2	90.0
22.	-3	+7	+11	-2	United States	Boston	195	152	138	243	246	97%	61.5	77.9
23.	+33	+15	+6	+0	United States	San Francisco	89	117	150	223	246	97%	109.1	64.0
24.	+21	-3	-4	-2	United Arab Emirates	Dubai	127	168	176	224	237	94%	41.1	35.7
25.	+6	+5	-12	-4	Italy	Milan	167	151	195	232	235	93%	55.6	20.6
26.	+3	-8	-1	-2	Greece	Athens	172	177	166	208	221	88%	24.8	33.0
27.	+6	+29	+25	+11	South Africa	Johannesburg	155	73	89	167	219	87%	199.0	144.5
28.	-18	+7	+10	+1	Australia	Sydney	230	137	131	202	218	86%	59.0	66.5
29.	+5	-18	-19	-5	Hong Kong	Hong Kong	154	196	204	222	217	86%	10.7	6.4
30.	-2	-5	-2	+1	Australia	Melbourne	173	160	151	196	210	83%	31.5	39.2
31.	+16	+13	+19	+5	Argentina	Buenos Aires	123	108	94	176	209	83%	94.5	121.7
32.	-12	-9	-20	-7	Italy	Rome	193	165	197	217	208	83%	26.0	5.8
33.	+4	-6	-1	-3	Spain	Madrid	149	159	142	197	206	82%	29.8	44.9
34.	+8	+9	+8	-7	New Zealand	Wellington	137	110	115	209	199	79%	81.1	72.2
35.	+1	+7	+0	+6	Russia	Moscow	149	111	134	157	193	76%	73.7	43.6
36.	-31	-3	-2	+1	United States	Chicago	243	141	135	176	186	74%	31.9	37.2
37.	-21	-18	-13	+8	Hungary	Budapest	202	175	167	149	183	73%	5.0	9.6
38.	-8	-24	-32	-4	Denmark	Copenhagen	172	189	225	182	181	72%	-4.2	-19.6
39.	-12	-5	+2	-7	New Zealand	Auckland	175	139	116	195	179	71%	28.9	54.2
40.	-22	-9	-13	-7	Spain	Barcelona	197	144	161	185	177	70%	23.2	10.2
41.	+9	+6	+2	+3	Portugal	Lisbon	116	101	113	154	173	69%	72.4	52.7
42.	-7	-16	-19	+1	South Korea	Seoul	150	159	168	155	165	65%	3.6	-1.8
43.	-22	-6	+1	-1	Singapore	Singapore	190	122	113	155	165	65%	35.2	45.3
44.	+16	+6	+8	+3	United Arab Emirates	Abu Dhabi	71	89	133	141	159	63%	79.8	19.7
45.	-7	-6	-14	-6	Chile	Santiago	146	117	143	161	157	62%	34.1	9.6
46.	-40	-31	-31	-6	Japan	Tokyo	239	189	193	159	154	61%	-18.5	-20.3
47.	-32	-2	+0	-1	Canada	Toronto	203	105	105	145	139	55%	32.1	32.7
48.	+10	+1	-3	+1	Philippines	Manila	84	91	110	128	123	49%	34.3	11.9
49.	+4	+8	+8	+1	South Africa	Cape Town	113	67	76	115	119	47%	78.6	58.0
50.	+5	-9	-1	-2	Finland	Helsinki	90	111	94	132	119	47%	7.4	26.5
51.	-12	-11	-11	+0	Brazil	Rio de Janeiro	146	115	125	104	117	46%	1.4	-6.5
52.	-8	+0	+2	+6	Colombia	Bogota	130	81	85	77	108	43%	32.1	27.2
53.	-5	-5	-7	+3	Indonesia	Jakarta	120	93	107	88	102	40%	10.0	-4.8
54.	+5	+0	-6	-2	Thailand	Bangkok	82	78	96	97	100	40%	28.6	3.9
55.	-6	+3	+7	+6	Brazil	Sao Paulo	117	67	62	72	98	39%	46.9	58.7
56.	-4	+11	-3	-2	Saudi Arabia	Riyadh	114	37	89	90	98	39%	163.2	9.5
57.	+0	-2	-2	+2	Qatar	Doha	87	75	78	90	94	37%	24.9	21.0
58.	-12	-12	+1	+1	Türkiye	Istanbul	124	105	71	77	90	36%	-14.5	27.0
59.	-46	+0	+1	-6	Canada	Vancouver	214	67	69	93	82	32%	22.7	18.1
60.	-6	-7	-2	-3	Canada	Montreal	96	78	74	77	81	32%	3.0	8.2
61.	+3	-1	-5	-1	Taiwan	Taipei	50	64	77	73	73	29%	13.6	-4.6
62.	+3	+1	+4	+2	Malaysia	Kuala Lumpur	49	49	54	58	66	26%	33.8	22.2
63.	+0	+3	+4	+3	Mexico	Mexico City	57	40	34	51	65	26%	61.5	88.7
64.	+2	-2	-1	+3	China	Shanghai	41	62	61	50	63	25%	2.6	4.3
65.	+2	-1	+0	-2	India	Mumbai	39	47	55	61	60	24%	27.6	10.4
66.	-5	-1	-5	-1	China	Beijing	62	47	67	55	55	22%	18.3	-17.5
67.	-5	-6	-3	-5	India	Delhi	59	62	61	66	43	17%	-29.9	-28.4
68.	+0	+1	+1	+0	India	Bangalore	29	22	27	33	28	11%	26.5	4.7
69.	+0	-1	-1	+0	Egypt	Cairo	26	31	28	25	28	11%	-9.9	-0.1

Source : www.numbeo.com, Deutsche Bank.



Figure 38: iPhone 17 Pro (256GB) (USD)

Rank	Country	Y2026	Relative to US
1.	Türkiye	2,592	219%
2.	Brazil	2,260	191%
3.	Egypt	1,872	158%
4.	Hungary	1,826	155%
5.	Norway	1,687	143%
6.	Mexico	1,657	140%
7.	Sweden	1,601	135%
8.	Poland	1,588	134%
9.	Czechia	1,585	134%
10.	Finland	1,578	134%
11.	Portugal	1,566	133%
11.	Greece	1,566	133%
13.	Italy	1,554	132%
13.	Ireland	1,554	132%
15.	Denmark	1,553	131%
16.	France	1,543	131%
16.	Netherlands	1,543	131%
16.	Belgium	1,543	131%
19.	Spain	1,531	130%
20.	Austria	1,508	128%
20.	Germany	1,508	128%
22.	Luxembourg	1,492	126%
23.	United Kingdom	1,476	125%
24.	India	1,427	121%
25.	Australia	1,413	120%
26.	Saudi Arabia	1,386	117%
27.	Switzerland	1,385	117%
28.	New Zealand	1,370	116%
29.	Singapore	1,364	115%
30.	Malaysia	1,352	114%
31.	Thailand	1,350	114%
32.	China	1,332	113%
33.	Vietnam	1,330	113%
34.	Philippines	1,326	112%
35.	Canada	1,291	109%
36.	UAE	1,279	108%
37.	Taiwan	1,264	107%
38.	Hong Kong	1,200	102%
39.	United States	1,181	100%
40.	South Korea	1,181	100%
41.	Japan	1,121	95%

Source : Official Apple Website, Online Retailers

Note: Prices are for iPhone 17 Pro (256GB). The prices are sourced from official Apple store websites. In the absence of official prices via websites, we have used prices from online retailers.



Figure 39: Groceries index * relative to New York

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+0	+0	+0	+2	Switzerland	Zurich	144	132	129	103	115	115%	-12.4	-10.2
2.	+1	+0	+0	-1	Switzerland	Geneva	139	124	116	105	114	114%	-7.8	-1.4
3.	+12	+0	+2	-1	United States	San Francisco	98	122	94	104	105	105%	-14.0	11.4
4.	+9	+0	-1	+0	United States	New York	100	100	100	100	100	100%	0.0	0.0
5.	+2	+4	+3	+0	United States	Boston	113	86	83	92	98	98%	13.5	18.2
6.	-4	+4	+0	+3	Norway	Oslo	139	86	93	78	91	91%	5.5	-2.5
7.	+28	+6	+6	+0	United States	Los Angeles	74	77	74	81	89	89%	15.6	19.5
7.	+30	+1	+7	-1	United States	Chicago	72	87	74	83	89	89%	1.7	19.8
9.	+23	+15	+10	+8	Israel	Tel Aviv-Yafo	77	70	67	68	82	82%	16.8	22.4
10.	+29	-5	-6	-2	South Korea	Seoul	71	92	97	81	81	81%	-11.4	-16.2
11.	-3	+11	+10	+4	Australia	Melbourne	113	71	66	69	79	79%	12.5	20.2
12.	-7	+3	+4	-1	Australia	Sydney	121	75	69	71	79	79%	5.3	14.8
13.	N/A	+7	-3	+5	Luxembourg	Luxembourg	N/A	73	80	66	78	78%	6.9	-2.4
14.	-4	+3	+9	+8	Denmark	Copenhagen	104	75	65	63	78	78%	4.7	19.8
15.	+2	-3	-3	-3	France	Paris	96	77	77	71	78	78%	0.8	1.7
16.	+10	-2	+20	-3	Singapore	Singapore	83	76	57	71	77	77%	1.9	34.4
17.	-5	+11	+5	-3	Canada	Vancouver	104	66	66	71	76	76%	14.5	15.5
18.	+7	-11	-7	-8	Hong Kong	Hong Kong	85	89	80	76	75	75%	-15.5	-6.5
19.	+8	+14	+6	+4	Austria	Vienna	82	62	64	63	74	74%	20.2	16.1
20.	+3	-2	-3	+11	Sweden	Stockholm	87	74	69	58	73	73%	-2.2	5.7
21.	+1	-2	+3	+8	Belgium	Brussels	91	74	65	61	72	72%	-2.6	11.1
22.	-13	+5	+9	+12	Ireland	Dublin	110	67	60	58	72	72%	7.6	19.5
23.	+17	+16	+3	+10	Netherlands	Amsterdam	70	59	64	58	71	71%	20.8	12.5
24.	-8	+7	-15	-5	Canada	Toronto	96	63	81	65	71	71%	12.3	-12.6
25.	+6	-2	+2	-2	Finland	Helsinki	81	70	63	63	71	71%	0.5	12.7
26.	+8	+0	+4	+0	Italy	Milan	74	67	60	62	70	70%	4.8	17.0
27.	-16	-16	-7	-7	New Zealand	Wellington	104	79	67	65	69	69%	-12.0	3.9
28.	-7	-7	+1	-3	United Kingdom	London	93	73	61	62	69	69%	-6.1	13.2
29.	-1	+1	-14	+1	Taiwan	Taipei	82	64	73	60	68	68%	5.0	-7.4
30.	-16	-14	-12	-9	New Zealand	Auckland	99	75	67	64	67	67%	-10.4	-0.3
31.	+12	+4	+7	+4	Germany	Frankfurt	66	61	55	58	67	67%	9.5	21.9
32.	-14	+0	+4	-16	Canada	Montreal	95	63	62	69	67	67%	6.1	7.1
33.	-13	+1	-1	-6	United Kingdom	Edinburgh	93	61	59	61	66	66%	7.3	11.5
34.	-10	+2	-1	+3	Italy	Rome	87	61	59	51	65	65%	6.6	10.7
35.	-16	-6	+6	+3	United Kingdom	Birmingham	94	64	49	51	65	65%	0.3	31.8
36.	-7	+4	-1	-4	Germany	Munich	82	59	58	58	64	64%	8.2	9.4
37.	-4	+4	+0	-1	Germany	Berlin	75	57	55	55	63	63%	11.2	13.7
38.	-32	-32	+3	-10	Japan	Tokyo	120	91	89	61	63	63%	-31.0	-29.2
39.	+10	+8	+3	+2	Spain	Madrid	61	50	48	46	56	56%	11.2	16.5
40.	+4	+5	+4	-1	Spain	Barcelona	65	54	47	49	56	56%	3.5	17.9
41.	-3	+8	+4	-1	Greece	Athens	72	46	45	46	55	55%	20.5	21.9
42.	+19	+14	+8	+6	Czech Republic	Prague	50	37	41	42	53	53%	41.9	30.2
43.	+10	+10	+3	+1	Portugal	Lisbon	59	43	43	44	50	50%	17.3	16.1
44.	+13	+13	+18	+6	Mexico	Mexico City	54	37	32	42	50	50%	36.6	55.6
45.	+18	+14	+9	+12	Hungary	Budapest	47	34	36	37	48	48%	41.5	33.2
46.	-4	-9	+7	-4	United Arab Emirates	Dubai	66	60	38	45	47	47%	-22.7	21.7
47.	+8	-3	+17	+4	Argentina	Buenos Aires	56	54	29	41	47	47%	-14.0	62.8
48.	+17	+13	+12	+8	Poland	Warsaw	42	33	33	37	46	46%	42.6	41.9
49.	-45	+2	+3	-6	Saudi Arabia	Riyadh	138	44	40	45	46	46%	4.9	14.9
50.	-14	-8	-10	-4	United Arab Emirates	Abu Dhabi	73	56	53	43	46	46%	-19.2	-14.1
51.	-5	-8	-17	-6	Thailand	Bangkok	63	55	58	44	45	45%	-17.5	-22.6
52.	+4	-2	-3	+0	Malaysia	Kuala Lumpur	55	45	42	40	45	45%	0.1	6.2
53.	+5	+1	-5	+0	Chile	Santiago	54	41	42	40	44	44%	6.1	3.6
54.	-24	-16	-11	-7	Qatar	Doha	82	59	48	43	42	42%	-29.0	-11.5
55.	-1	-3	+6	-1	Türkiye	Istanbul	56	44	32	39	42	42%	-4.1	29.8
56.	+6	-1	+3	-7	Philippines	Manila	48	40	33	42	41	41%	3.1	22.5
57.	+3	+1	+6	+8	Russia	Moscow	50	35	32	27	41	41%	14.8	27.3
58.	+1	-33	-19	-3	China	Shanghai	52	68	54	39	38	38%	-43.9	-29.4
59.	-12	+3	-2	+1	Brazil	Sao Paulo	63	32	34	31	37	37%	15.6	8.5
60.	-12	-14	-9	-1	China	Beijing	62	54	40	35	37	37%	-30.7	-6.4
61.	-20	-13	-14	-3	Indonesia	Jakarta	70	48	43	35	37	37%	-24.1	-14.5
62.	-10	+4	-6	-1	South Africa	Cape Town	60	29	36	29	35	35%	22.0	-0.2
63.	-12	+1	-8	-1	South Africa	Johannesburg	60	32	36	29	35	35%	8.8	-1.9
64.	-19	+3	+4	-1	Colombia	Bogota	65	28	25	28	34	34%	21.9	35.4
65.	-15	-2	-7	+1	Brazil	Rio de Janeiro	60	32	34	26	32	32%	-1.3	-5.5
66.	+0	-1	+0	+1	India	Mumbai	36	31	28	26	26	26%	-15.6	-8.4
67.	+0	+2	-2	-3	India	Bangalore	35	27	28	27	25	25%	-6.3	-10.1
68.	-4	-8	+1	+1	Egypt	Cairo	45	34	24	20	23	23%	-30.3	-2.3
69.	-1	-1	-2	-1	India	Delhi	35	27	26	23	23	23%	-16.0	-10.8

Source : www.numbeo.com, Deutsche Bank. * Note: Groceries index assesses the cost of grocery items in a city relative to New York City. It is calculated using the weighted prices of various food items commonly found in the "Markets" section of Numbeo's database.



Figure 40: Cigarettes 20 pack (Marlboro) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+1	+0	+0	+0	Australia	Sydney	16.5	19.3	24.0	33.6	44.5	234%	131.1	85.1
2.	-1	+0	+0	+0	Australia	Melbourne	16.6	19.0	21.4	31.7	42.4	223%	123.3	98.0
3.	+0	+1	+0	+0	New Zealand	Auckland	15.3	16.0	20.4	25.6	27.6	145%	73.3	35.7
4.	+1	-1	+0	+0	New Zealand	Wellington	13.7	16.1	19.9	25.3	25.6	135%	59.4	29.0
5.	+2	+0	+3	+0	United Kingdom	London	11.4	13.8	14.1	22.9	23.8	125%	72.0	68.2
6.	+3	+0	+0	+0	United Kingdom	Edinburgh	11.1	13.3	14.7	22.9	23.3	122%	75.0	58.7
7.	+3	+2	+2	+0	United Kingdom	Birmingham	11.0	12.8	13.8	22.8	22.2	117%	73.5	61.1
8.	+0	+3	-1	+0	Ireland	Dublin	11.3	12.0	14.4	20.9	21.8	114%	81.1	51.6
9.	-3	-1	-4	+0	United States	New York	12.9	13.1	15.0	18.4	19.0	100%	45.3	26.7
10.	-6	-3	+0	+0	Norway	Oslo	14.4	13.2	13.5	17.4	18.4	97%	39.2	36.4
11.	+1	-1	+0	+0	United States	Chicago	9.8	12.3	12.6	15.8	17.0	89%	38.7	35.1
12.	+2	+1	+3	+0	Canada	Vancouver	9.3	10.0	11.2	15.6	15.0	79%	49.2	33.7
13.	+13	+3	+4	+7	Israel	Tel Aviv-Yafo	6.6	8.7	10.0	12.7	15.0	79%	71.6	50.3
14.	+10	+9	+11	+1	Netherlands	Amsterdam	6.7	7.3	7.9	14.5	14.9	78%	104.1	88.1
14.	+5	+6	+5	+0	France	Paris	7.9	8.0	9.8	14.6	14.9	78%	85.5	52.3
16.	-5	-1	-3	-3	Canada	Toronto	10.4	9.4	11.8	15.4	14.6	77%	56.1	24.0
17.	-4	-3	-1	+2	Singapore	Singapore	9.4	10.0	10.7	13.2	13.9	73%	39.6	30.8
18.	+0	+8	+10	+0	Belgium	Brussels	6.4	6.8	7.4	13.4	13.7	72%	102.4	86.9
18.	+11	+7	+4	-1	Finland	Helsinki	6.4	7.0	8.8	13.7	13.7	72%	97.1	56.0
20.	+5	+2	+4	-4	Hong Kong	Hong Kong	6.6	7.4	8.0	14.1	13.4	70%	80.5	67.0
21.	+13	+0	-7	+1	United States	San Francisco	5.7	7.6	11.6	12.3	13.0	68%	70.8	12.3
22.	-5	-10	-10	-1	United States	Boston	8.0	10.4	11.8	12.6	12.5	66%	20.9	6.0
23.	-2	+8	-3	+0	United States	Los Angeles	7.2	6.6	9.7	11.8	11.9	62%	81.1	21.9
24.	-6	-6	-1	+1	Switzerland	Zurich	7.9	8.5	8.4	11.3	11.3	60%	33.8	35.0
25.	-10	-6	-7	-1	Canada	Montreal	8.6	8.4	9.8	11.7	11.3	59%	34.9	14.4
26.	-10	-9	-5	+0	Switzerland	Geneva	8.4	8.6	9.0	11.1	11.2	59%	30.6	24.5
27.	+5	+2	+2	+2	Germany	Berlin	6.3	6.7	7.2	10.1	10.8	57%	60.8	48.8
28.	+3	+0	-2	+0	Germany	Munich	6.3	6.7	7.7	10.1	10.3	54%	53.0	34.5
29.	-7	-3	-2	-2	Germany	Frankfurt	6.8	6.8	7.6	10.2	10.0	52%	46.7	31.7
30.	-7	-1	+1	+0	Denmark	Copenhagen	6.7	6.7	6.8	9.7	9.7	51%	44.2	42.8
31.	-11	-7	-1	+1	Sweden	Stockholm	7.3	7.1	6.9	8.2	8.6	45%	21.2	25.8
32.	+16	+11	+15	+8	Hungary	Budapest	3.2	4.1	4.6	6.8	8.3	44%	104.7	81.1
33.	+10	+11	+15	+1	Czech Republic	Prague	4.0	4.0	4.5	7.5	8.0	42%	99.8	80.0
34.	-1	-1	+3	-3	Luxembourg	Luxembourg	6.1	6.0	5.8	8.5	7.7	41%	29.2	32.5
35.	+28	+16	-3	+1	Saudi Arabia	Riyadh	2.1	3.2	6.6	7.2	7.5	39%	134.3	13.4
36.	+0	+0	-1	-3	Austria	Vienna	5.6	5.7	6.1	7.5	7.4	39%	31.7	21.6
37.	-10	-3	-4	+1	Italy	Milan	6.5	5.9	6.3	7.0	7.4	39%	25.7	18.1
38.	-3	-1	+0	-1	Spain	Barcelona	5.6	5.6	5.8	7.1	7.2	38%	27.9	22.4
39.	-9	-7	-5	+0	Italy	Rome	6.3	6.0	6.2	6.9	7.0	37%	15.9	12.4
40.	-1	-2	+3	+1	Portugal	Lisbon	5.2	5.5	5.6	6.4	6.9	36%	25.8	22.5
40.	-4	-5	+0	-5	Spain	Madrid	5.6	5.7	5.7	7.3	6.9	36%	21.0	20.3
42.	+14	+15	+0	+1	United Arab Emirates	Abu Dhabi	2.4	2.8	5.7	6.1	6.8	36%	142.3	20.3
43.	+2	+2	+7	+2	Poland	Warsaw	3.6	3.9	4.3	5.8	6.7	35%	69.7	55.8
44.	+8	+15	-8	-3	Qatar	Doha	2.5	2.8	5.9	6.4	6.4	34%	133.3	8.4
45.	+13	+13	-6	-1	United Arab Emirates	Dubai	2.3	2.8	5.8	6.0	6.1	32%	119.7	6.2
46.	-6	-6	-2	+0	Greece	Athens	4.7	4.6	5.1	5.5	5.7	30%	24.3	11.5
47.	-5	-8	-6	+0	Chile	Santiago	4.3	4.6	5.7	5.5	5.4	28%	16.9	-4.9
48.	+1	+14	+12	+0	Mexico	Mexico City	2.8	2.6	2.7	4.5	5.4	28%	105.8	95.3
49.	+15	+0	+0	+8	India	Delhi	2.0	3.3	4.4	3.9	5.0	26%	54.0	15.1
50.	+17	+1	+2	+5	India	Mumbai	1.8	3.2	4.2	4.0	4.6	24%	44.7	9.0
51.	+0	-3	-6	+3	Thailand	Bangkok	2.7	3.3	4.8	4.1	4.5	24%	35.8	-6.6
52.	-5	-11	-1	-3	Malaysia	Kuala Lumpur	3.2	4.3	4.3	4.3	4.4	23%	3.8	3.8
53.	+9	+1	+0	+0	India	Bangalore	2.2	3.1	4.0	4.1	4.2	22%	36.6	3.7
54.	+11	+7	+12	-3	Argentina	Buenos Aires	2.0	2.6	2.1	4.1	4.2	22%	58.8	100.0
55.	+0	+5	+3	-3	China	Beijing	2.5	2.7	3.0	4.1	4.1	22%	52.0	35.5
56.	+4	+12	+12	+5	Colombia	Bogota	2.3	1.5	1.9	3.2	4.1	21%	165.4	112.6
57.	-3	-2	-3	-1	Taiwan	Taipei	2.5	3.0	3.9	4.0	3.9	21%	29.6	0.8
58.	-20	-16	-12	-8	Japan	Tokyo	5.4	4.1	4.8	4.2	3.8	20%	-6.6	-19.9
59.	-15	-6	-2	+3	South Africa	Johannesburg	3.6	3.2	3.1	3.2	3.4	18%	7.9	9.6
60.	-14	+3	+1	+4	South Africa	Cape Town	3.5	2.5	2.7	3.1	3.4	18%	35.6	27.9
61.	+5	+5	+3	+4	Russia	Moscow	2.0	1.7	2.3	3.1	3.3	17%	98.2	47.1
62.	+7	+7	+3	-3	Philippines	Manila	1.3	1.3	2.2	3.5	3.1	16%	132.3	43.1
63.	-13	-13	-7	-5	China	Shanghai	2.7	3.3	3.5	3.6	2.9	15%	-9.5	-15.5
63.	-4	-16	-8	-3	South Korea	Seoul	2.3	3.9	3.8	3.2	2.9	15%	-25.0	-23.0
65.	-12	-1	-3	+1	Brazil	Sao Paulo	2.5	2.3	2.6	2.7	2.9	15%	25.3	13.6
66.	-9	-1	+1	+2	Brazil	Rio de Janeiro	2.4	2.3	2.1	2.4	2.9	15%	27.3	41.0
67.	+1	+0	+2	-4	Indonesia	Jakarta	1.4	1.6	1.8	3.1	2.8	15%	76.4	51.4
68.	-27	-23	-9	-1	Türkiye	Istanbul	4.5	3.9	2.8	2.4	2.4	12%	-40.2	-15.8
69.	-8	-13	-6	+0	Egypt	Cairo	2.3	3.0	2.5	2.0	2.1	11%	-30.0	-16.7

Source : www.numbeo.com, Deutsche Bank.



Figure 41: Domestic beer (0.5 liter bottle) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+1	+3	+3	+1	Australia	Melbourne	4.81	4.18	3.69	4.40	5.27	164%	26.1	42.8
2.	+5	+0	+3	-1	Singapore	Singapore	4.10	4.40	3.62	4.60	4.49	140%	2.0	24.0
3.	+1	+0	+0	+5	Australia	Sydney	4.71	4.34	3.80	3.61	4.10	128%	-5.5	7.9
4.	+19	+15	+10	+1	Israel	Tel Aviv-Yafo	2.27	2.25	2.53	3.69	3.83	119%	70.2	51.4
5.	+3	+6	+4	+3	Finland	Helsinki	3.97	2.68	2.85	3.61	3.74	117%	39.6	31.2
6.	+35	+11	+19	-2	United Kingdom	Birmingham	1.54	2.30	2.14	3.76	3.67	114%	59.6	71.9
7.	+5	+4	+3	-4	Ireland	Dublin	3.30	2.68	2.77	4.00	3.62	113%	35.1	30.7
8.	+8	+2	+4	+3	Malaysia	Kuala Lumpur	2.97	2.73	2.64	3.28	3.42	107%	25.3	29.5
9.	+16	+9	+7	+7	United States	San Francisco	2.08	2.26	2.50	2.95	3.40	106%	50.4	36.0
10.	-4	-4	-3	+2	Norway	Oslo	4.11	3.51	3.43	3.20	3.32	103%	-5.4	-3.2
11.	+11	+4	+9	-4	United Kingdom	London	2.29	2.42	2.27	3.67	3.27	102%	35.1	44.1
12.	+1	+17	+10	-7	Switzerland	Geneva	3.22	1.93	2.16	3.69	3.23	101%	67.4	49.5
13.	+7	+0	-2	-3	United States	New York	2.38	2.55	2.65	3.59	3.21	100%	25.9	21.4
14.	-5	-9	-1	-1	New Zealand	Wellington	3.91	4.01	2.62	3.16	3.16	98%	-21.2	20.6
15.	-0	+9	+7	+0	Canada	Vancouver	3.15	2.14	2.16	3.07	3.04	95%	42.1	40.7
16.	-11	+5	+10	-2	United States	Boston	4.17	2.20	2.07	3.12	3.03	94%	37.7	46.3
17.	+0	+6	+16	+4	United States	Los Angeles	2.67	2.17	1.95	2.78	2.94	92%	35.5	50.8
18.	+12	+12	+4	+0	Switzerland	Zurich	1.80	1.91	2.16	2.84	2.81	88%	47.1	30.1
19.	-5	-5	+8	-3	United Kingdom	Edinburgh	3.19	2.43	2.05	2.95	2.80	87%	15.2	36.6
20.	+18	+6	+13	+4	France	Paris	1.65	2.04	1.95	2.62	2.74	85%	34.3	40.5
21.	-20	-20	-20	+4	United Arab Emirates	Dubai	7.69	8.93	4.53	2.58	2.73	85%	-69.4	-39.7
21.	+22	+14	+23	+1	Belgium	Brussels	1.44	1.75	1.46	2.69	2.73	85%	56.0	87.0
23.	-4	-16	-21	-4	New Zealand	Auckland	2.47	3.28	3.87	2.83	2.67	83%	-18.6	-31.0
24.	+10	+13	-4	-4	United States	Chicago	1.75	1.68	2.27	2.82	2.64	82%	57.1	16.3
25.	-22	-16	-20	+1	United Arab Emirates	Abu Dhabi	4.77	2.86	3.62	2.49	2.48	77%	-13.3	-31.5
26.	-8	+2	-8	-1	Canada	Toronto	2.57	1.97	2.39	2.47	2.37	74%	20.3	-0.8
26.	-2	-6	+4	+2	Türkiye	Istanbul	2.20	2.24	2.01	2.41	2.37	74%	5.8	17.9
28.	+17	+3	+20	-5	Argentina	Buenos Aires	1.40	1.86	1.36	2.63	2.34	73%	25.8	72.1
29.	-18	+3	+2	+0	Canada	Montreal	3.48	1.83	2.00	2.30	2.33	73%	27.3	16.5
30.	-2	-8	+2	+8	Sweden	Stockholm	2.06	2.18	1.98	1.96	2.29	71%	5.0	15.7
31.	+6	+3	-4	+2	India	Mumbai	1.68	1.76	2.05	2.16	2.18	68%	23.9	6.3
32.	-7	+6	-13	-2	Luxembourg	Luxembourg	2.08	1.58	2.38	2.30	2.17	68%	37.3	-8.8
32.	-11	-16	-17	-1	Indonesia	Jakarta	2.31	2.36	2.51	2.21	2.17	68%	-8.1	-13.5
34.	N/A	N/A	N/A	N/A	Oman	Doha	N/A	N/A	N/A	N/A	2.08	65%	N/A	N/A
34.	-9	-7	-17	+0	South Korea	Seoul	2.08	2.03	2.40	2.14	2.08	65%	2.5	-13.3
36.	+0	+4	+5	+1	Greece	Athens	1.70	1.52	1.60	2.01	2.05	64%	34.9	28.1
37.	+24	+17	+19	+7	Mexico	Mexico City	0.92	1.13	1.04	1.73	2.02	63%	78.8	94.2
38.	+9	+6	-2	-7	India	Bangalore	1.36	1.45	1.89	2.21	1.95	61%	34.5	3.2
39.	-7	+2	+4	-4	Italy	Milan	1.76	1.51	1.51	2.13	1.92	60%	27.2	27.2
40.	-1	+10	+5	+6	South Africa	Johannesburg	1.63	1.24	1.41	1.65	1.90	59%	53.2	34.8
41.	-10	-5	-2	+5	Hong Kong	Hong Kong	1.78	1.70	1.68	1.65	1.85	58%	8.8	10.1
41.	-12	-16	-4	+7	Denmark	Copenhagen	1.92	2.10	1.81	1.63	1.85	58%	-11.9	2.2
43.	N/A	N/A	N/A	N/A	Saudi Arabia	Riyadh	N/A	N/A	N/A	N/A	1.83	57%	N/A	N/A
44.	-4	-11	-15	-3	Thailand	Bangkok	1.61	1.78	2.03	1.79	1.82	57%	2.2	-10.3
45.	-35	-37	-37	-9	Japan	Tokyo	3.89	3.17	3.04	2.06	1.79	56%	-43.5	-41.1
46.	+5	-3	-4	-7	India	Delhi	1.26	1.46	1.58	1.86	1.75	55%	19.9	10.8
47.	+1	+4	+2	-7	Chile	Santiago	1.33	1.19	1.34	1.82	1.74	54%	46.2	29.9
48.	+2	-1	-2	-5	Netherlands	Amsterdam	1.27	1.36	1.38	1.75	1.67	52%	22.8	21.0
49.	+8	-3	-11	-7	Taiwan	Taipei	1.01	1.39	1.72	1.76	1.64	51%	18.0	-4.7
50.	-7	-2	+2	-5	Italy	Rome	1.44	1.34	1.24	1.71	1.58	49%	17.9	27.4
51.	-19	-6	-5	+2	Brazil	Sao Paulo	1.76	1.42	1.38	1.32	1.57	49%	10.6	13.8
51.	-9	-12	-11	-2	Egypt	Cairo	1.50	1.56	1.61	1.59	1.57	49%	0.6	-2.5
53.	+0	+3	+5	+4	Spain	Barcelona	1.20	1.05	1.02	1.26	1.46	45%	39.0	43.1
54.	+9	+3	+0	-4	Austria	Vienna	0.91	1.04	1.17	1.54	1.45	45%	39.4	23.9
55.	-6	-6	-20	+2	South Africa	Cape Town	1.28	1.30	1.90	1.26	1.41	44%	8.5	-25.8
56.	-10	+8	+9	+0	Colombia	Bogota	1.37	0.91	0.87	1.27	1.39	43%	52.7	59.8
57.	-1	+2	-2	+4	Germany	Munich	1.04	1.01	1.07	1.16	1.36	42%	34.7	27.1
58.	+2	+7	+4	+5	Hungary	Budapest	0.93	0.88	0.91	1.14	1.33	41%	51.1	46.2
58.	+6	+5	+1	-7	Philippines	Manila	0.90	0.95	1.00	1.49	1.33	41%	40.0	33.0
58.	+7	+3	+2	+1	Germany	Berlin	0.87	0.96	0.99	1.17	1.33	41%	38.5	34.3
61.	-6	-6	-11	-7	Portugal	Lisbon	1.16	1.10	1.25	1.30	1.27	40%	15.5	1.6
62.	-1	+4	+1	-10	Poland	Warsaw	0.92	0.83	0.88	1.40	1.24	39%	49.4	40.9
63.	-11	-3	-10	-9	Spain	Madrid	1.24	0.98	1.19	1.30	1.23	38%	25.5	3.4
64.	-10	-3	-4	-5	Russia	Moscow	1.19	0.96	0.99	1.17	1.21	38%	26.0	22.2
65.	+2	+2	+1	-1	Czech Republic	Prague	0.71	0.72	0.74	1.12	1.19	37%	65.3	60.8
66.	-31	-24	-16	-2	Brazil	Rio de Janeiro	1.74	1.47	1.25	1.12	1.18	37%	-19.7	-5.6
67.	-8	-14	-11	-6	Germany	Frankfurt	0.96	1.16	1.04	1.16	1.16	36%	0.0	11.5
68.	-2	-11	-3	-2	China	Beijing	0.81	1.04	0.84	0.96	0.96	30%	-7.7	14.3
69.	-12	-17	-2	-2	China	Shanghai	1.01	1.17	0.70	0.83	0.90	28%	-23.1	28.6

Source : www.numbeo.com, Deutsche Bank.



Figure 42: Our "Oasis" index (USD) - 5 beers and 2 packs of cigarettes

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+0	+1	+1	+0	Australia	Melbourne	57.2	58.9	61.3	85.4	111.2	206%	88.8	81.4
2.	+0	-1	-1	+0	Australia	Sydney	56.6	60.2	67.1	85.2	109.5	203%	81.9	63.3
3.	+3	+2	+0	+1	New Zealand	Auckland	43.0	48.3	60.1	65.4	68.6	127%	42.1	14.2
4.	+0	-1	+0	-1	New Zealand	Wellington	47.0	52.2	52.9	66.3	67.1	124%	28.4	26.9
5.	+9	+3	+3	+1	United Kingdom	London	34.3	39.7	39.6	64.2	63.9	118%	60.8	61.3
6.	+13	+6	+5	-1	United Kingdom	Birmingham	29.7	37.1	38.3	64.5	62.8	116%	69.2	64.1
7.	+1	+4	+0	+0	Ireland	Dublin	39.1	37.4	42.6	61.8	61.6	114%	64.6	44.8
8.	+1	+2	+1	+0	United Kingdom	Edinburgh	38.1	38.7	39.6	60.5	60.5	112%	56.3	52.9
9.	+1	+0	-3	+0	United States	New York	37.7	38.9	43.2	54.8	54.1	100%	38.9	25.0
10.	-7	-4	-5	+0	Norway	Oslo	49.4	44.0	44.1	50.8	53.4	99%	21.4	21.0
11.	-4	-4	-1	+0	Singapore	Singapore	39.2	42.0	39.4	49.3	50.3	93%	19.9	27.7
12.	+13	+4	+6	+3	Israel	Tel Aviv-Yafo	24.5	28.7	32.6	43.9	49.1	91%	71.1	50.7
13.	+8	+0	-1	+0	United States	Chicago	28.3	32.9	36.5	45.6	47.2	87%	43.4	29.3
14.	+3	+4	+5	+0	Finland	Helsinki	32.6	27.3	31.9	45.5	46.2	85%	68.9	44.9
15.	-2	+0	+2	-3	Canada	Vancouver	34.3	30.8	33.2	46.6	45.1	83%	46.7	36.0
16.	+10	+6	+6	+1	France	Paris	24.1	26.2	29.3	42.3	43.5	80%	65.6	48.4
17.	+13	+3	-4	+4	United States	San Francisco	21.8	26.5	35.7	39.3	43.0	80%	62.1	20.6
18.	+16	+11	+14	+2	Belgium	Brussels	19.9	22.3	22.0	40.3	41.1	76%	84.2	87.0
19.	-4	-2	-5	-3	Canada	Toronto	33.6	28.6	35.6	43.2	41.1	76%	43.8	15.6
20.	-9	-6	-4	-2	United States	Boston	36.9	31.7	34.0	40.8	40.2	74%	26.7	18.3
21.	-5	-2	+3	-2	Switzerland	Geneva	32.8	26.8	28.8	40.6	38.5	71%	43.9	33.9
22.	+0	+4	+1	+1	United States	Los Angeles	27.7	24.0	29.2	37.4	38.4	71%	60.4	31.5
23.	+13	+8	+7	-1	Netherlands	Amsterdam	19.7	21.4	22.7	37.7	38.1	71%	78.3	67.7
24.	-1	-3	+1	+0	Switzerland	Zurich	24.8	26.5	27.6	36.7	36.7	68%	38.6	33.1
25.	+4	+3	+2	+0	Hong Kong	Hong Kong	22.2	23.3	24.4	36.4	36.0	67%	54.4	47.4
26.	-14	-3	-6	+0	Canada	Montreal	34.6	25.9	29.7	35.2	34.2	63%	32.2	15.1
27.	-4	-3	+1	+2	Sweden	Stockholm	24.8	25.1	23.6	26.3	28.7	53%	14.2	21.5
28.	-1	-1	+3	+0	Denmark	Copenhagen	23.0	23.9	22.6	27.5	28.6	53%	19.6	26.5
29.	+13	+10	+10	+3	Germany	Berlin	16.9	18.2	19.4	26.0	28.2	52%	55.0	45.1
30.	+9	+8	+4	+1	Germany	Munich	17.8	18.5	20.7	26.0	27.4	51%	48.0	32.6
31.	-3	+2	-2	-4	Luxembourg	Luxembourg	22.6	19.9	23.6	28.6	26.3	49%	32.4	11.6
32.	-12	+0	-11	+4	United Arab Emirates	Abu Dhabi	28.6	19.9	29.4	24.6	26.0	48%	30.6	-11.6
33.	-1	-3	+0	+0	Malaysia	Kuala Lumpur	21.2	22.2	21.7	24.9	25.9	48%	17.0	19.4
34.	-29	-30	-19	+0	United Arab Emirates	Dubai	43.1	50.2	34.2	24.9	25.9	48%	-48.4	-24.2
35.	+2	-1	+0	-5	Germany	Frankfurt	18.5	19.4	20.3	26.2	25.7	48%	32.7	26.6
36.	-5	-1	+0	-1	Italy	Milan	21.7	19.3	20.1	24.7	24.4	45%	26.3	21.5
37.	N/A	N/A	N/A	N/A	Saudi Arabia	Riyadh	N/A	N/A	N/A	24.1	24.1	44%	N/A	N/A
38.	+22	+17	+16	+8	Hungary	Budapest	11.0	12.5	13.8	19.2	23.3	43%	85.9	69.5
39.	N/A	N/A	N/A	N/A	Oman	Doha	N/A	N/A	N/A	N/A	23.3	43%	N/A	N/A
40.	+4	+2	+3	-3	Austria	Vienna	15.7	16.5	18.1	22.6	22.1	41%	34.1	22.3
41.	+15	+20	+17	+1	Czech Republic	Prague	11.5	11.6	12.6	20.5	22.0	41%	89.1	74.4
42.	-7	-5	-1	-4	Italy	Rome	19.8	18.7	18.6	22.3	21.9	40%	16.6	17.4
43.	-5	-2	-1	-3	Greece	Athens	18.0	16.8	18.3	21.1	21.7	40%	29.1	18.8
44.	-3	-1	+4	-1	Spain	Barcelona	17.2	16.4	16.8	20.4	21.6	40%	31.5	28.7
45.	+18	+18	+16	+6	Mexico	Mexico City	10.2	10.9	10.7	17.6	20.8	38%	91.7	94.8
46.	+7	+2	-6	+2	India	Mumbai	12.0	15.2	18.7	18.8	20.1	37%	32.6	7.5
47.	-4	-3	-1	-2	Portugal	Lisbon	16.2	16.4	17.5	19.3	20.1	37%	22.4	15.0
48.	+11	+2	+12	-9	Argentina	Buenos Aires	11.1	14.5	11.0	21.4	20.0	37%	37.7	82.7
49.	-9	-4	-2	-8	Spain	Madrid	17.4	16.3	17.4	21.0	19.9	37%	22.3	14.5
50.	+5	+8	+7	+0	Poland	Warsaw	11.7	12.0	13.0	18.6	19.5	36%	62.7	50.8
51.	-5	-4	-7	-7	Chile	Santiago	15.2	15.2	18.0	20.0	19.5	36%	28.4	8.0
52.	+10	-1	-3	+1	India	Delhi	10.4	13.8	16.6	17.2	18.8	35%	36.0	13.1
53.	-2	-7	-16	+2	Thailand	Bangkok	13.4	15.5	19.8	17.1	18.1	34%	16.6	-8.5
54.	+4	-1	-9	-7	India	Bangalore	11.3	13.4	17.5	19.2	18.1	34%	35.5	3.4
55.	-37	-30	-29	-6	Japan	Tokyo	30.3	24.1	24.8	18.7	16.6	31%	-30.9	-32.9
56.	-23	-20	-4	+1	Turkiye	Istanbul	20.0	19.1	15.6	16.8	16.6	31%	-13.2	5.9
57.	-9	-8	-6	-5	Indonesia	Jakarta	14.4	14.9	16.2	17.3	16.4	30%	9.7	1.1
58.	-13	-3	-3	+0	South Africa	Johannesburg	15.4	12.5	13.3	14.6	16.3	30%	30.3	22.9
59.	-12	-19	-21	-5	South Korea	Seoul	15.0	18.0	19.6	17.1	16.3	30%	-9.5	-17.1
60.	+4	-6	-10	-4	Taiwan	Taipei	10.1	13.0	16.3	16.8	16.0	30%	23.4	-2.1
61.	-4	+5	+6	-1	Colombia	Bogota	11.5	7.6	8.2	12.7	15.1	28%	98.0	84.5
62.	-12	+0	-9	-1	South Africa	Cape Town	13.4	11.5	14.8	12.5	13.8	26%	20.3	-6.6
63.	-14	-3	-4	+0	Brazil	Sao Paulo	13.8	11.8	12.0	12.1	13.7	25%	16.4	13.7
64.	+2	+0	+0	-2	China	Beijing	9.0	10.6	10.3	12.3	13.0	24%	22.8	26.8
65.	+2	+2	+1	-6	Philippines	Manila	7.1	7.4	9.3	14.4	12.8	24%	73.1	37.7
66.	-1	-1	-1	-2	Russia	Moscow	10.0	8.1	9.5	12.0	12.7	23%	55.7	34.1
67.	-13	-15	-11	-2	Egypt	Cairo	12.0	13.8	13.1	11.9	12.1	22%	-12.7	-7.9
68.	-16	-9	-5	-1	Brazil	Rio de Janeiro	13.4	11.9	10.4	10.3	11.7	22%	-1.8	12.9
69.	-8	-12	-7	-3	China	Shanghai	10.5	12.4	10.5	11.4	10.4	19%	-16.0	-0.8

Source : www.numbeo.com, Deutsche Bank.



Figure 43: Our "Cheap date" index (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+1	+1	+0	+1	Switzerland	Geneva	412	391	379	446	475	132%	21.5	25.2
2.	+1	-1	+0	-1	Switzerland	Zurich	398	406	373	454	467	130%	15.0	25.2
3.	+13	+12	+5	+2	Israel	Tel Aviv-Yafo	296	260	278	368	426	118%	63.8	53.5
4.	-3	-1	-1	+0	Norway	Oslo	416	336	334	377	377	105%	12.2	12.8
5.	-1	-1	-1	-2	Denmark	Copenhagen	372	333	331	381	377	105%	13.3	13.9
6.	+8	-1	+3	+0	United Kingdom	London	308	321	277	368	370	103%	15.5	33.6
7.	+0	-1	+0	+3	Sweden	Stockholm	354	320	281	341	362	101%	13.3	28.7
8.	+32	+13	+6	-1	United States	New York	234	250	262	356	359	100%	43.6	37.4
9.	+0	-2	-4	+0	Luxembourg	Luxembourg	331	296	285	346	345	98%	16.7	21.1
10.	+3	-2	-4	+4	Finland	Helsinki	317	293	283	320	345	96%	17.6	21.9
11.	+12	+0	+2	+1	Ireland	Dublin	279	266	265	321	343	96%	29.3	29.5
12.	+17	+4	+18	+1	United Kingdom	Birmingham	261	259	230	320	338	94%	30.6	46.8
13.	+38	+27	+20	+6	United States	Los Angeles	209	215	225	300	336	93%	56.0	49.4
14.	+23	+6	-2	-6	United States	San Francisco	247	252	267	353	328	91%	30.0	22.9
15.	+3	-1	+0	+1	France	Paris	292	261	261	319	325	90%	24.4	24.7
16.	+11	+10	+0	+1	Italy	Milan	269	239	259	318	324	90%	35.2	24.9
17.	+18	+11	+38	-2	Argentina	Buenos Aires	251	238	161	320	322	90%	35.6	99.9
18.	-1	-8	-7	+4	Belgium	Brussels	293	276	274	289	316	88%	14.4	15.6
19.	-14	-7	+0	+2	Australia	Sydney	368	265	243	289	315	88%	19.0	29.5
20.	+2	-7	+5	-2	United Kingdom	Edinburgh	283	264	235	308	311	86%	17.9	32.1
21.	-1	-12	-11	-10	Netherlands	Amsterdam	290	288	275	339	307	85%	6.6	11.7
22.	+30	+11	+6	+1	United States	Boston	208	225	231	288	305	85%	35.3	32.2
23.	-13	-1	-6	-3	Germany	Munich	327	248	256	296	303	84%	22.0	18.1
24.	-3	-1	-4	+3	Germany	Frankfurt	284	247	242	279	296	82%	19.9	22.3
25.	+6	+10	-2	+1	Germany	Berlin	255	224	237	283	296	82%	32.2	25.0
26.	+4	+1	-2	+2	Austria	Vienna	260	238	236	279	292	81%	22.6	23.7
27.	+16	-3	-6	-3	United Arab Emirates	Dubai	232	244	240	288	284	79%	16.5	18.3
28.	+33	+28	+24	+13	Hungary	Budapest	179	166	168	234	276	77%	65.8	64.6
29.	-18	-12	-3	+4	Australia	Melbourne	327	255	233	253	273	76%	7.1	16.9
30.	+25	+8	+1	-5	United States	Chicago	205	218	227	285	271	75%	24.5	19.3
31.	-19	+11	+8	+13	Russia	Moscow	318	207	207	223	268	75%	29.5	29.2
32.	+22	+21	+16	+4	Poland	Warsaw	205	188	180	252	263	73%	56.2	46.1
33.	+13	+12	+9	+1	Portugal	Lisbon	226	192	203	252	262	73%	36.2	29.1
34.	-2	+9	-7	-4	Greece	Athens	255	204	205	261	259	72%	26.7	26.1
35.	N/A	N/A	N/A	N/A	Qatar	Doha	N/A	N/A	N/A	N/A	259	72%	N/A	N/A
36.	-3	-5	+4	-7	Spain	Madrid	253	228	207	265	256	71%	12.3	23.8
37.	+1	-1	-2	-6	Singapore	Singapore	237	223	216	257	255	71%	13.9	18.0
38.	-10	+3	-2	-3	Canada	Vancouver	262	209	211	252	254	71%	21.6	20.2
39.	+14	+13	+6	+7	Czech Republic	Prague	207	170	188	216	250	69%	46.5	32.5
40.	-15	-21	-8	-8	New Zealand	Auckland	275	253	226	257	249	69%	-1.7	9.8
41.	+0	-4	-3	-1	Spain	Barcelona	233	220	208	234	249	69%	12.9	19.4
42.	+2	+2	-5	-4	Canada	Toronto	232	198	208	240	248	69%	25.0	19.0
43.	+15	-4	-16	+2	United Arab Emirates	Abu Dhabi	190	215	233	220	247	69%	14.4	6.0
44.	+4	-15	-10	-7	New Zealand	Wellington	221	235	220	247	236	66%	0.4	7.2
45.	N/A	N/A	N/A	N/A	Saudi Arabia	Riyadh	N/A	N/A	N/A	N/A	232	65%	N/A	N/A
46.	-10	-21	-17	-7	Hong Kong	Hong Kong	248	243	230	239	229	64%	-5.8	-0.8
47.	-5	+0	-1	-5	Canada	Montreal	233	190	187	233	227	63%	19.4	21.4
48.	-29	-18	-26	-5	Italy	Rome	292	232	237	232	227	63%	-2.2	-4.3
49.	+14	+14	+13	+1	Mexico	Mexico City	157	132	138	200	224	62%	69.3	62.1
50.	-1	+5	+10	+5	Colombia	Bogota	221	167	142	174	223	62%	33.8	56.7
51.	-25	-3	+2	+3	Brazil	Sao Paulo	272	186	167	175	216	60%	15.8	28.9
52.	+5	+6	-3	-3	Thailand	Bangkok	190	158	176	201	201	56%	27.2	14.3
53.	+7	+7	+13	-2	Türkiye	Istanbul	188	148	114	199	196	54%	32.5	71.1
54.	-15	-8	-4	-2	Taiwan	Taipei	234	191	172	194	190	53%	-0.3	10.6
55.	-21	-23	-12	-8	South Korea	Seoul	252	226	203	213	190	53%	-16.0	-6.2
56.	-6	+5	+3	-3	Malaysia	Kuala Lumpur	210	144	144	175	189	52%	30.8	31.1
57.	-51	-39	-39	-9	Japan	Tokyo	363	254	250	206	198	52%	-25.8	-24.8
58.	-13	+1	-2	-2	South Africa	Johannesburg	230	149	159	171	197	52%	25.9	17.6
59.	+0	+3	-2	+0	South Africa	Cape Town	190	137	156	162	181	50%	32.0	16.0
60.	-4	-6	-6	-2	Chile	Santiago	200	167	167	166	173	48%	3.3	3.5
61.	-37	-10	-14	-4	China	Beijing	277	175	186	168	170	47%	-2.7	-8.7
62.	-54	-12	-11	-1	Brazil	Rio de Janeiro	332	177	168	153	159	44%	-9.8	-5.3
63.	-48	-29	-19	-1	China	Shanghai	297	224	189	150	154	43%	-31.0	-18.1
64.	+1	+0	-1	-4	Philippines	Manila	143	131	128	156	146	41%	11.6	14.2
65.	-18	-16	-7	-1	Egypt	Cairo	222	180	151	120	141	39%	-21.7	-6.7
66.	-4	-9	-5	-3	Indonesia	Jakarta	175	158	141	143	135	38%	-14.4	-4.4
67.	-3	-1	-2	-1	India	Mumbai	144	117	116	114	115	32%	-2.2	-1.2
68.	-1	-1	-1	-1	India	Bangalore	112	110	111	108	106	30%	-3.0	-3.9
69.	-3	-4	-5	-4	India	Delhi	133	122	124	116	103	29%	-15.7	-16.5

Source : www.numbeo.com, Deutsche Bank. Note: We have defined a cheap date as follows: 1 bottle of wine, 1 pair of jeans, 1 dress, 2 coffees, 1 meal for 2 people in mid range restaurant, 2 cinema tickets, 2 transport tickets one way, 5km taxi trip.



Figure 44: Bottle of wine (mid-range) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+1	+1	+0	Singapore	Singapore	23.5	24.0	23.2	26.6	23.2	116%	-3.1	0.2
2.	-1	-1	-1	+1	Indonesia	Jakarta	30.7	31.0	29.6	19.9	21.1	105%	-31.9	-28.6
3.	+12	+7	+11	-1	United States	New York	15.5	16.2	15.3	20.0	20.0	100%	23.2	31.1
4.	+6	+9	+5	+5	Malaysia	Kuala Lumpur	17.2	15.0	16.4	18.0	19.0	95%	26.9	15.7
5.	-1	+4	+7	+6	Norway	Oslo	19.8	17.1	15.4	17.3	18.3	92%	6.9	18.7
6.	+0	-1	-3	+2	Thailand	Bangkok	18.9	19.2	20.8	18.1	18.0	90%	-6.0	-13.3
7.	+6	+1	-1	-1	South Korea	Seoul	15.7	18.5	17.7	18.2	17.6	88%	-4.6	-0.1
8.	+26	+39	+37	+20	Mexico	Mexico City	11.6	8.0	9.1	12.8	17.3	86%	114.7	89.9
9.	+24	+24	+11	+12	Israel	Tel Aviv-Yafo	11.7	11.0	13.4	13.5	16.7	83%	52.2	24.6
10.	+7	-3	-3	-6	Hong Kong	Hong Kong	15.0	18.8	17.6	19.4	16.6	83%	-12.0	-5.8
11.	-4	-5	-3	-6	United Arab Emirates	Dubai	18.5	19.1	17.6	18.3	16.3	82%	-14.5	-7.0
12.	+11	+15	-2	+3	Finland	Helsinki	13.6	12.1	16.4	16.0	16.0	80%	32.9	-2.1
13.	+16	-2	-8	-6	United States	Boston	12.2	16.1	18.2	18.1	16.0	80%	-0.3	-12.0
14.	+0	+17	+17	+27	Colombia	Bogota	15.7	11.4	11.1	10.8	15.0	75%	32.4	36.0
15.	+9	-3	-4	-5	United States	San Francisco	13.6	15.5	15.5	17.8	15.0	75%	-3.5	-3.2
15.	+24	+10	+7	+1	United States	Los Angeles	10.8	12.5	13.1	15.6	15.0	75%	20.3	14.4
15.	+9	+3	+3	-2	United States	Chicago	13.6	13.9	14.0	17.1	15.0	75%	8.1	7.1
18.	-10	-2	-5	-6	Canada	Vancouver	17.4	14.4	15.3	17.2	14.8	74%	2.9	-3.4
19.	-1	-5	-2	-2	China	Beijing	14.9	14.7	14.2	14.9	14.7	74%	0.0	3.7
20.	+8	+3	-4	-6	Taiwan	Taipei	12.3	12.7	14.6	16.5	14.0	70%	10.6	-3.6
21.	-9	-4	+3	-1	Australia	Sydney	16.2	14.3	12.9	14.2	13.9	70%	-2.5	7.7
21.	-5	+0	+7	+2	Australia	Melbourne	15.4	13.0	11.5	13.4	13.9	70%	6.8	21.1
23.	+8	-3	-2	-1	Ireland	Dublin	11.9	13.1	13.3	13.5	13.7	69%	4.6	3.6
24.	-4	-2	-1	+1	Canada	Toronto	14.6	12.7	13.1	13.1	13.7	69%	8.2	5.0
25.	+1	-6	-6	-8	Switzerland	Zurich	13.3	13.9	13.8	14.6	13.7	68%	-1.4	-0.8
26.	-24	-22	-22	-8	United Arab Emirates	Abu Dhabi	27.7	19.5	19.8	14.7	13.6	68%	-30.2	-31.4
27.	+11	+1	+5	+3	United Kingdom	London	10.9	11.9	11.0	12.6	13.4	67%	12.2	22.2
28.	+14	+11	+20	+15	Russia	Moscow	10.6	9.7	8.4	9.6	12.9	64%	33.0	53.1
29.	-7	+5	+7	-5	Türkiye	Istanbul	14.0	10.9	8.5	13.1	12.8	64%	17.8	50.9
30.	+7	+0	+4	+2	Sweden	Stockholm	11.1	11.4	10.5	12.2	12.5	62%	9.5	19.3
31.	-26	-16	+2	-4	Switzerland	Geneva	19.3	14.6	10.5	12.9	12.4	62%	-14.9	18.3
32.	-2	+6	+7	+1	United Kingdom	Birmingham	12.1	9.8	10.1	12.0	12.1	60%	23.6	19.9
33.	-22	-9	-18	+3	China	Shanghai	16.4	12.5	14.6	11.3	11.8	59%	-5.9	-19.3
34.	+6	+7	+10	+5	Denmark	Copenhagen	10.7	9.6	9.5	10.9	11.5	57%	19.5	21.1
35.	+19	+10	+0	+9	Luxembourg	Luxembourg	6.4	9.2	10.4	13.1	11.5	57%	24.2	9.9
36.	+7	-4	-7	+1	United Kingdom	Edinburgh	10.5	11.0	11.2	11.0	11.1	56%	1.0	-0.4
37.	+4	+7	+1	+3	New Zealand	Wellington	10.6	9.3	10.2	11.9	10.8	54%	16.6	6.0
38.	-17	-12	-12	-9	Canada	Montreal	14.4	12.5	12.4	12.7	10.6	53%	-15.2	-14.6
39.	+10	+1	+2	+4	India	Bangalore	7.6	9.7	9.9	11.7	10.5	52%	8.4	6.1
40.	+8	+8	+6	+2	Greece	Athens	8.7	7.9	8.5	10.1	10.3	52%	30.5	21.3
41.	N/A	N/A	N/A	N/A	Qatar	Doha	N/A	N/A	N/A	N/A	10.3	51%	N/A	N/A
42.	-7	-7	-2	-2	New Zealand	Auckland	11.5	10.8	10.0	10.9	10.3	51%	-5.3	2.7
43.	+16	+8	+8	+3	Philippines	Manila	6.1	7.3	7.8	9.2	9.8	49%	33.3	25.1
44.	-25	-2	-2	+7	Brazil	Sao Paulo	14.7	9.4	9.7	8.4	9.7	49%	3.8	0.7
45.	+5	+1	+5	+0	Belgium	Brussels	7.5	8.1	8.1	9.3	9.4	47%	16.5	17.3
46.	-10	-17	-19	-15	India	Delhi	11.2	11.7	12.0	12.5	9.4	47%	-19.1	-21.4
47.	-38	-44	-22	+5	Japan	Tokyo	17.4	20.2	12.6	8.4	9.3	46%	-54.2	-26.5
48.	+10	+6	+1	-1	Italy	Milan	6.2	6.4	8.2	8.9	9.2	46%	42.2	11.4
48.	-3	+1	-12	-4	France	Paris	9.7	7.6	10.3	9.6	9.2	46%	21.3	-11.3
50.	-2	+5	+2	+6	Austria	Vienna	6.8	6.4	7.4	7.5	8.6	43%	34.7	16.3
51.	-5	-14	-8	-13	India	Mumbai	9.6	10.0	9.5	11.0	8.4	42%	-16.5	-11.8
52.	-25	-16	-22	+2	Egypt	Cairo	12.8	10.6	11.1	7.8	8.2	41%	-22.5	-26.2
53.	-2	+0	+1	-5	Poland	Warsaw	7.2	6.6	7.1	8.7	8.0	40%	21.0	13.1
53.	+7	-1	+0	-4	Spain	Madrid	6.0	7.0	7.1	8.6	8.0	40%	14.9	13.0
53.	+9	+8	+4	+6	Spain	Barcelona	5.7	5.8	6.8	6.7	8.0	40%	39.3	17.3
53.	+0	-3	+3	-3	Netherlands	Amsterdam	6.6	7.5	7.0	8.4	8.0	40%	7.1	14.1
57.	N/A	N/A	N/A	N/A	Saudi Arabia	Riyadh	N/A	N/A	N/A	N/A	8.0	40%	N/A	N/A
58.	-26	-15	-21	-5	Brazil	Rio de Janeiro	11.8	9.3	10.3	8.3	7.8	39%	-16.4	-24.5
59.	+7	+4	+3	-4	Czech Republic	Prague	5.4	5.0	5.8	7.7	7.1	36%	41.1	23.2
60.	-13	-2	-2	-3	Germany	Munich	9.4	6.0	6.8	7.4	6.9	34%	13.7	1.5
61.	-5	-5	-2	-1	Chile	Santiago	6.2	6.3	6.2	6.5	6.4	32%	0.6	2.2
62.	+3	-4	-1	-4	Germany	Berlin	5.4	6.0	6.0	7.3	6.3	32%	4.3	5.9
62.	-7	-5	-2	-1	Germany	Frankfurt	6.3	6.3	6.2	6.5	6.3	32%	-0.3	2.4
64.	-3	+2	+1	-2	South Africa	Cape Town	5.8	4.2	5.2	6.3	6.2	31%	45.9	19.8
64.	-1	+3	+0	+1	South Africa	Johannesburg	5.6	3.9	5.3	5.8	6.2	31%	57.0	16.6
66.	+1	-1	+0	-3	Hungary	Budapest	4.8	4.5	4.7	6.1	6.0	30%	34.0	26.4
67.	-3	-5	+0	-1	Argentina	Buenos Aires	5.4	5.4	4.2	5.7	5.9	29%	9.1	41.1
68.	-12	-4	-5	-4	Portugal	Lisbon	6.2	4.6	5.4	6.0	5.7	29%	25.7	6.3
69.	-25	-9	-14	-2	Italy	Rome	10.0	5.9	7.1	4.9	4.6	23%	-22.0	-35.2

Source : www.numbeo.com, Deutsche Bank.



Figure 45: 1 pair of jeans (Levis 501 or similar) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+5	+1	+0	+1	Switzerland	Geneva	128	123	121	134	141	198%	14.6	16.6
2.	+4	-1	+0	-1	Switzerland	Zurich	128	125	120	135	137	193%	9.4	14.5
3.	+9	+16	+9	+2	Israel	Tel Aviv-Yafo	122	92	94	118	128	180%	38.7	36.6
4.	+9	+6	+2	-1	France	Paris	121	101	98	120	119	167%	18.0	20.5
5.	+26	+13	+15	+1	United Kingdom	London	99	94	86	117	117	165%	25.4	36.7
6.	-2	-2	+3	+4	Sweden	Stockholm	132	112	97	107	117	165%	4.4	20.7
7.	+34	+17	+11	+5	Ireland	Dublin	88	88	88	103	117	165%	32.9	32.7
8.	+39	+32	+29	+11	Hungary	Budapest	80	71	69	94	114	160%	60.2	66.0
9.	+23	+2	+37	+0	Argentina	Buenos Aires	98	98	61	108	114	160%	15.9	86.3
10.	-7	-7	-7	-6	Denmark	Copenhagen	137	114	117	120	111	156%	-2.9	-4.9
11.	+13	+10	-4	-3	Italy	Milan	106	90	98	109	110	154%	21.6	11.8
12.	+33	+27	+17	+3	Poland	Warsaw	83	72	73	100	107	151%	49.4	46.3
13.	+9	+9	+1	-2	Portugal	Lisbon	108	90	92	106	107	150%	18.7	16.1
14.	+9	+16	+26	+0	United Kingdom	Birmingham	107	83	66	102	107	150%	29.4	62.1
15.	+20	-6	-10	+12	Belgium	Brussels	97	102	104	84	105	148%	3.5	1.1
16.	+11	-8	-8	-9	Luxembourg	Luxembourg	103	102	98	114	105	148%	3.1	7.2
17.	+2	+12	+5	+0	Greece	Athens	112	83	83	97	98	138%	18.3	19.0
18.	-17	-13	-14	+3	Norway	Oslo	145	106	106	91	98	138%	-7.4	-7.8
19.	-8	-3	+5	+6	Germany	Frankfurt	122	94	81	84	96	134%	1.5	18.5
20.	-4	-13	-10	+3	Finland	Helsinki	113	103	95	90	95	134%	-7.4	-0.2
21.	-7	+6	-2	+8	Russia	Moscow	117	84	88	80	94	132%	10.8	6.4
22.	-1	-6	+1	-9	Spain	Madrid	109	94	82	102	94	132%	-0.4	14.1
23.	-8	+2	-10	-5	Germany	Berlin	114	86	93	94	92	130%	7.5	-0.2
24.	-8	-10	-8	-2	Austria	Vienna	113	96	91	90	90	127%	-5.9	-0.9
25.	+11	+12	+0	+7	Czech Republic	Prague	97	75	79	74	88	124%	18.2	11.3
26.	+3	-20	-15	-10	Netherlands	Amsterdam	100	104	94	99	88	124%	-15.2	-6.3
27.	-19	-4	-10	-7	Germany	Munich	127	90	89	93	87	123%	-2.9	-1.7
28.	+2	-15	-7	+2	Spain	Barcelona	100	96	84	77	83	117%	-13.4	-0.8
29.	+36	+34	+30	+2	United States	Los Angeles	46	49	50	75	82	115%	67.0	63.2
30.	+13	-2	-3	-6	United Kingdom	Edinburgh	87	84	78	86	82	115%	-2.2	4.9
31.	+6	+16	+22	+17	Colombia	Bogota	95	62	54	65	80	113%	29.6	49.9
32.	+7	+6	+3	+10	Australia	Melbourne	91	74	70	68	79	112%	8.0	13.4
33.	+25	+3	-5	-5	New Zealand	Wellington	60	76	76	82	79	111%	4.3	4.2
34.	+6	+0	-1	+4	Singapore	Singapore	90	77	71	70	78	109%	1.1	9.4
35.	+9	-3	+9	-1	United Arab Emirates	Dubai	87	80	63	72	76	107%	-4.7	21.6
36.	+28	+9	+6	-10	United States	San Francisco	47	64	65	84	76	107%	18.2	17.1
37.	+24	+27	+24	+0	United States	Boston	50	48	49	71	76	106%	58.3	54.4
38.	-18	-5	-4	+2	Australia	Sydney	111	79	70	68	75	106%	-4.9	7.3
39.	+29	+27	+24	+6	Mexico	Mexico City	44	42	45	66	74	105%	75.9	64.0
40.	+11	-5	-9	-4	New Zealand	Auckland	70	76	72	71	74	104%	-3.2	2.8
41.	+1	+9	+6	+3	Malaysia	Kuala Lumpur	87	59	56	67	73	103%	23.5	31.0
42.	+4	+4	+3	-9	Thailand	Bangkok	81	62	62	73	73	103%	17.8	18.1
43.	+17	+12	+7	-8	United States	New York	56	53	55	72	71	100%	33.2	28.5
44.	-39	-32	-29	-5	Italy	Rome	131	96	91	70	71	99%	-26.7	-22.6
45.	+21	-1	-7	+15	United Arab Emirates	Abu Dhabi	45	65	68	54	66	93%	2.1	-2.5
46.	-8	+2	+12	+13	Brazil	Sao Paulo	94	62	51	55	66	92%	6.3	29.6
47.	+15	+14	+7	+4	United States	Chicago	49	49	52	62	64	90%	29.7	22.3
48.	+6	+3	+7	-1	Canada	Vancouver	66	57	52	66	64	90%	11.6	22.5
49.	-40	+0	-17	-6	China	Beijing	126	60	71	67	63	88%	4.7	-11.9
50.	-2	-9	-7	-4	Qatar	Doha	77	70	64	66	62	88%	-11.0	-3.4
51.	+6	+14	+11	-10	Philippines	Manila	63	47	49	68	60	84%	27.0	22.1
52.	+1	+4	+12	-3	Saudi Arabia	Riyadh	69	53	45	63	59	83%	12.1	32.5
53.	-4	+9	-2	+9	South Africa	Cape Town	72	49	54	53	58	82%	18.7	7.9
54.	-21	-34	-28	-4	Hong Kong	Hong Kong	98	91	79	62	58	82%	-36.4	-26.5
55.	+1	+3	+5	+2	Canada	Montreal	64	52	49	55	57	81%	9.8	15.9
56.	+3	+4	-4	-1	Canada	Toronto	59	50	54	57	57	80%	14.6	6.2
57.	-30	+1	-7	+7	South Africa	Johannesburg	104	53	56	50	57	80%	8.5	2.7
58.	-30	-27	-19	-2	Taiwan	Taipei	101	81	66	57	56	79%	-31.2	-15.3
59.	-4	-6	+6	-7	Indonesia	Jakarta	66	54	44	59	54	76%	-0.3	22.8
60.	-8	-1	+6	-6	Türkiye	Istanbul	69	50	41	57	54	76%	6.8	32.6
61.	-11	-7	-13	+0	Chile	Santiago	71	53	56	53	53	74%	-0.9	-5.1
62.	-28	-20	-26	-9	Japan	Tokyo	98	69	69	57	51	71%	-26.9	-26.3
63.	-45	-37	-22	-5	South Korea	Seoul	112	85	65	55	49	69%	-41.7	-23.9
64.	-62	-49	-34	+0	China	Shanghai	141	94	72	45	48	67%	-49.3	-33.7
65.	-40	-22	-8	+1	Egypt	Cairo	105	66	51	31	41	58%	-37.8	-19.7
66.	-56	-14	-10	-1	Brazil	Rio de Janeiro	123	56	52	34	41	57%	-27.5	-21.7
67.	+2	+1	+2	+1	India	Bangalore	42	36	35	30	33	46%	-9.0	-6.4
68.	-5	+1	+0	-1	India	Mumbai	48	36	36	31	33	46%	-7.7	-8.4
69.	-2	-2	-2	+0	India	Delhi	44	39	36	29	24	34%	-36.9	-33.0

Source : www.numbeo.com, Deutsche Bank.



Figure 46: 1 summer dress in a chain store (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+7	+3	+9	+0	Argentina	Buenos Aires	74	62	47	101	94	183%	51.2	99.5
2.	+17	+5	+5	+1	Israel	Tel Aviv-Yafo	65	58	51	74	89	172%	52.6	72.4
3.	+6	+2	-2	-1	Switzerland	Geneva	72	62	66	80	88	170%	42.3	32.2
4.	-1	+5	+14	+0	Australia	Sydney	92	52	44	73	72	139%	37.8	63.6
5.	+1	+18	+6	+13	Russia	Moscow	78	46	47	51	61	119%	33.0	30.3
6.	+11	-4	-2	+8	Switzerland	Zurich	67	66	54	54	60	117%	-8.0	11.5
7.	+52	+40	+40	+14	United States	Los Angeles	40	39	36	50	60	117%	56.2	69.3
8.	+6	-5	-6	-2	Saudi Arabia	Riyadh	67	65	56	62	60	116%	-8.5	5.8
9.	+39	+12	+11	+2	Norway	Oslo	44	47	43	57	59	115%	26.9	37.2
10.	-3	+5	+11	+21	Brazil	Sao Paulo	77	49	43	47	59	114%	19.9	36.6
11.	+17	+16	+32	+17	Sweden	Stockholm	56	44	37	48	57	110%	30.5	52.9
12.	+8	+10	+5	-4	Denmark	Copenhagen	62	46	44	58	56	109%	20.7	27.7
13.	-3	-12	-8	-6	Qatar	Doha	71	66	52	59	55	108%	-15.9	6.1
14.	+29	-6	-11	-9	United Arab Emirates	Dubai	47	57	55	65	55	107%	-3.7	1.1
15.	+54	+33	+22	-2	Italy	Milan	34	38	38	55	54	105%	40.6	41.4
16.	+11	+4	+8	+0	United Kingdom	London	56	48	41	53	54	104%	12.2	29.4
17.	+13	+27	+33	+6	Mexico	Mexico City	55	39	35	49	53	104%	37.4	50.7
18.	+32	+10	+7	+2	France	Paris	43	43	41	50	53	102%	21.8	27.7
19.	+5	-2	-3	-4	Luxembourg	Luxembourg	59	49	44	54	53	102%	7.8	19.8
20.	+47	+18	+11	-11	United States	New York	35	40	39	57	52	100%	28.0	32.4
21.	+39	+16	+2	-9	United States	Chicago	39	40	42	56	52	100%	27.6	24.0
22.	+23	+33	+13	+16	Czech Republic	Prague	45	36	38	43	51	100%	43.8	33.3
23.	+8	+2	+3	+2	Ireland	Dublin	55	45	41	49	51	99%	14.0	23.7
24.	-12	+12	-10	+0	United States	San Francisco	70	41	44	49	51	98%	24.4	14.7
25.	+12	-9	+32	+15	Colombia	Bogota	51	49	34	43	51	98%	3.4	49.6
26.	-1	+5	+16	+4	United Kingdom	Edinburgh	59	42	37	47	50	98%	20.2	34.5
27.	+41	-13	+1	-8	United Kingdom	Birmingham	35	50	41	51	50	97%	0.6	22.5
28.	-5	+1	-15	+11	Belgium	Brussels	60	43	45	43	48	94%	11.8	7.1
29.	-16	-19	-7	+19	Australia	Melbourne	67	52	42	40	48	94%	-6.4	13.8
30.	+2	+0	+2	+5	Singapore	Singapore	54	43	39	45	48	93%	11.1	23.5
31.	+25	+18	+34	+1	United States	Boston	41	38	32	46	47	92%	23.0	47.6
32.	-10	-26	-26	+4	Egypt	Cairo	61	60	52	45	47	91%	-22.3	-9.5
33.	+24	+36	+20	+8	Portugal	Lisbon	41	32	35	42	47	90%	45.4	34.6
34.	+27	-17	-26	+13	United Arab Emirates	Abu Dhabi	39	49	50	40	46	89%	-5.6	-8.6
35.	-7	-9	-20	-25	Netherlands	Amsterdam	56	44	44	57	46	89%	3.3	4.0
36.	+30	+22	+27	+16	Hungary	Budapest	36	35	32	39	46	88%	29.6	41.8
37.	-32	-24	+2	-11	New Zealand	Auckland	78	50	38	48	46	88%	-8.1	20.0
38.	+3	+12	+18	-1	South Africa	Johannesburg	47	38	34	43	45	87%	16.7	31.6
39.	+8	+2	+9	-22	Finland	Helsinki	45	39	36	51	45	86%	13.6	25.0
40.	+13	+20	+11	+9	Canada	Toronto	42	35	35	39	44	85%	27.1	25.0
41.	-3	+21	-5	-19	Poland	Warsaw	51	35	38	49	44	85%	26.4	13.7
42.	-6	+21	-8	-15	Thailand	Bangkok	52	34	39	48	44	85%	27.9	12.7
43.	-42	-31	-34	-9	Brazil	Rio de Janeiro	110	50	49	46	43	83%	-14.9	-12.3
44.	+10	+16	+25	-11	Türkiye	Istanbul	42	35	28	46	43	82%	22.6	49.3
45.	-41	-3	-1	+12	Germany	Munich	85	39	37	36	42	82%	8.4	14.9
46.	+12	+8	-6	-4	Germany	Berlin	40	36	38	41	42	82%	17.1	11.6
47.	+3	+6	+19	-1	Spain	Madrid	43	36	31	40	42	81%	15.5	35.5
48.	-30	-14	-19	-5	Canada	Vancouver	67	41	40	41	42	81%	1.4	3.8
49.	-34	+16	+13	+2	Malaysia	Kuala Lumpur	67	34	32	39	41	79%	22.1	26.1
50.	+5	+7	+10	-21	Greece	Athens	42	35	33	48	40	78%	13.2	21.2
51.	-16	-12	+1	+2	Hong Kong	Hong Kong	53	40	35	38	40	77%	-0.2	13.6
52.	-36	-7	+6	-2	Taiwan	Taipei	67	39	33	39	40	77%	2.7	19.2
53.	+10	+13	+15	+7	Spain	Barcelona	37	33	30	36	40	77%	19.4	33.2
54.	+11	+10	-5	-11	Canada	Montreal	36	34	35	41	39	75%	13.4	8.7
55.	-5	+12	-14	+0	Austria	Vienna	43	33	38	37	38	75%	16.6	2.3
56.	-12	+12	-11	-2	Italy	Rome	46	32	36	38	37	72%	14.5	1.9
57.	-46	-14	-19	+2	China	Beijing	71	39	38	36	37	72%	-5.0	-3.0
58.	-32	-26	-12	+8	India	Mumbai	57	42	36	32	36	70%	-12.8	0.7
59.	-38	-35	-26	+2	China	Shanghai	61	45	39	35	36	70%	-19.7	-7.2
60.	-21	-25	-30	-4	Germany	Frankfurt	48	41	39	37	34	66%	-16.4	-12.8
61.	-20	-2	-7	+1	South Africa	Cape Town	47	35	34	33	34	66%	-2.2	-1.6
62.	-60	-22	-43	-4	Japan	Tokyo	106	39	44	36	33	63%	-17.0	-25.6
63.	-14	-11	-2	+0	Indonesia	Jakarta	44	37	33	33	30	59%	-17.9	-7.2
64.	-31	-13	+3	+1	Chile	Santiago	54	37	30	33	30	59%	-18.3	0.9
65.	-25	-32	-38	-1	India	Delhi	48	41	41	33	30	59%	-26.5	-26.2
66.	-32	-47	-54	-21	South Korea	Seoul	53	48	46	41	30	58%	-38.0	-35.7
67.	-3	-21	-12	+1	India	Bangalore	37	39	34	29	29	57%	-24.1	-14.4
68.	-22	-57	-9	-1	New Zealand	Wellington	45	51	33	32	29	56%	-43.4	-13.9
69.	-7	-13	-6	+0	Philippines	Manila	37	35	32	28	28	54%	-20.7	-12.7

Source : www.numbeo.com, Deutsche Bank.



Figure 47: Cappuccino (regular) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+3	+1	+3	+1	Switzerland	Zurich	4.90	5.20	5.13	6.64	7.13	119%	37.1	39.0
2.	-1	-1	-1	-1	Denmark	Copenhagen	5.66	5.41	5.84	6.78	6.96	116%	28.7	19.2
3.	+7	+6	+10	+1	Switzerland	Geneva	4.28	4.43	4.40	6.13	6.41	107%	44.7	45.7
4.	+9	+1	+1	+3	United Arab Emirates	Dubai	4.12	4.58	5.10	5.73	6.10	101%	33.2	19.6
5.	+14	+6	+2	+4	United States	New York	3.91	4.31	4.77	5.63	6.01	100%	39.4	26.0
6.	+2	-3	-4	-3	Qatar	Doha	4.41	4.70	5.35	6.18	5.85	97%	24.5	9.3
7.	-5	-4	-1	+9	Norway	Oslo	5.31	4.70	4.87	5.16	5.82	97%	23.8	19.5
8.	+3	+6	+2	-3	United States	Los Angeles	4.22	4.10	4.54	5.85	5.81	97%	41.7	28.0
9.	+15	+4	+0	-3	United States	San Francisco	3.68	4.23	4.55	5.83	5.77	96%	36.4	26.8
10.	-7	-2	-7	-2	United Arab Emirates	Abu Dhabi	5.18	4.48	5.21	5.70	5.74	96%	28.1	10.2
11.	+22	+21	+10	+16	Israel	Tel Aviv-Yafo	3.37	3.24	3.70	4.39	5.63	94%	73.8	52.2
12.	+7	+7	+3	-1	United States	Chicago	3.91	4.00	4.17	5.47	5.60	93%	40.0	34.3
13.	+12	+9	+13	+1	United Kingdom	London	3.65	3.85	3.61	5.24	5.53	92%	43.6	53.2
14.	+4	+2	-2	+1	Finland	Helsinki	4.03	4.06	4.44	5.21	5.44	91%	34.0	22.5
15.	+18	+11	+7	+3	Austria	Vienna	3.37	3.48	3.67	5.12	5.34	89%	53.4	45.5
16.	+1	-4	+0	-4	Sweden	Stockholm	4.08	4.30	4.10	5.37	5.26	88%	22.3	28.3
17.	+11	+9	+15	+5	United Kingdom	Birmingham	3.58	3.62	3.38	5.53	5.26	88%	45.3	55.6
18.	+11	+3	+16	-5	United Kingdom	Edinburgh	3.54	3.92	3.31	5.31	5.18	86%	32.1	56.5
19.	+3	-1	-1	+4	Singapore	Singapore	3.98	4.02	4.09	4.72	5.06	84%	25.9	23.7
20.	+12	+8	+8	+4	Ireland	Dublin	3.49	3.29	3.60	4.64	5.05	84%	53.5	40.3
21.	-9	-2	-5	-2	United States	Boston	4.17	4.00	4.10	4.98	5.04	84%	26.0	22.9
22.	-9	-7	-2	-1	France	Paris	4.12	4.09	3.94	4.80	5.00	83%	22.2	26.9
23.	-10	-16	-15	-7	Hong Kong	Hong Kong	4.12	4.50	4.59	5.16	4.96	83%	10.2	8.1
24.	+14	+19	+13	+2	Germany	Berlin	3.26	2.82	3.09	4.40	4.72	79%	67.4	52.8
25.	+17	+4	+14	+3	Germany	Frankfurt	3.11	3.28	3.08	4.75	4.70	78%	43.3	52.6
26.	+15	+7	+3	-6	Netherlands	Amsterdam	3.15	3.18	3.56	4.81	4.64	77%	45.9	30.3
27.	-4	-3	-1	+2	Luxembourg	Luxembourg	3.81	3.63	3.61	4.26	4.58	76%	26.2	26.9
28.	+21	+23	+15	+3	Poland	Warsaw	2.37	2.29	2.78	4.21	4.55	76%	98.7	63.7
29.	+6	+5	+2	-2	Germany	Munich	3.36	3.17	3.38	4.39	4.37	73%	37.9	29.3
29.	+8	+6	-4	-4	Belgium	Brussels	3.33	3.14	3.62	4.54	4.37	73%	39.2	20.7
31.	-22	-4	-1	+3	Greece	Athens	4.37	3.38	3.52	4.05	4.22	70%	24.9	19.9
31.	-5	+0	+2	+1	Canada	Vancouver	3.61	3.27	3.33	4.16	4.22	70%	29.1	26.7
33.	+15	+5	-11	+0	Saudi Arabia	Riyadh	2.45	2.95	3.67	4.12	4.14	69%	40.3	12.8
34.	+9	+14	+21	+3	Türkiye	Istanbul	3.05	2.62	2.04	3.90	4.06	68%	55.0	99.0
35.	-4	+9	+26	-5	Argentina	Buenos Aires	3.53	2.81	1.81	4.23	4.02	67%	43.1	122.1
36.	-1	+6	-2	-1	Canada	Toronto	3.36	2.83	3.31	4.04	3.94	66%	39.2	19.0
37.	-9	+2	+4	+5	Australia	Melbourne	3.56	2.93	2.99	3.59	3.92	65%	33.8	31.1
38.	+6	+2	-1	-2	Canada	Montreal	2.98	2.92	3.09	3.93	3.90	65%	33.6	26.2
39.	-10	+2	+4	+7	Australia	Sydney	3.54	2.86	2.78	3.49	3.87	64%	35.3	39.2
40.	-34	-4	+7	+7	Russia	Moscow	4.59	3.11	2.60	3.44	3.83	64%	23.2	47.3
40.	+14	+16	+13	+1	Mexico	Mexico City	2.22	1.97	2.20	3.65	3.83	64%	94.4	74.1
42.	+20	+15	+9	+1	Czech Republic	Prague	1.87	1.91	2.34	3.58	3.80	63%	99.0	62.4
43.	-24	-26	-25	-4	South Korea	Seoul	3.91	4.03	4.09	3.84	3.60	60%	-10.7	-12.0
44.	+21	+23	+23	+10	Hungary	Budapest	1.41	1.42	1.64	2.79	3.56	59%	150.7	117.1
45.	-40	-22	-21	-0	Japan	Tokyo	4.75	3.72	3.65	3.54	3.48	58%	-6.5	-4.7
46.	-7	-9	-6	-6	New Zealand	Wellington	3.22	3.06	3.05	3.70	3.46	58%	13.1	13.4
47.	-1	+0	-1	+1	Chile	Santiago	2.63	2.74	2.73	3.20	3.45	57%	25.9	26.4
48.	-8	-19	-12	-4	New Zealand	Auckland	3.21	3.28	3.11	3.55	3.43	57%	4.6	10.3
49.	+3	-4	-7	-11	Taiwan	Taipei	2.32	2.79	2.87	3.85	3.42	57%	22.6	19.2
50.	-34	-44	-39	+2	China	Beijing	4.11	4.56	4.45	2.92	3.27	54%	-28.3	-26.5
51.	-4	-5	-6	-2	Malaysia	Kuala Lumpur	2.60	2.77	2.75	3.16	3.21	53%	15.9	16.7
52.	-45	-42	-38	-1	China	Shanghai	4.46	4.37	4.30	3.07	3.16	53%	-27.7	-26.5
53.	-3	+1	+2	-3	Spain	Madrid	2.34	2.04	2.04	3.10	3.15	52%	54.4	54.4
54.	+2	+1	+0	-1	Spain	Barcelona	2.14	1.98	2.05	2.90	3.14	52%	58.6	53.2
55.	+9	+8	+9	+2	Portugal	Lisbon	1.58	1.61	1.70	2.62	2.94	49%	82.6	72.9
56.	+5	-4	-6	+1	Thailand	Bangkok	1.95	2.20	2.37	2.62	2.87	48%	30.5	21.1
57.	+11	+2	-5	-1	India	Mumbai	1.31	1.69	2.24	2.75	2.80	47%	65.7	25.0
58.	-3	-8	-9	-3	Philippines	Manila	2.17	2.50	2.49	2.76	2.51	42%	0.4	0.8
59.	-2	+5	-1	+3	South Africa	Cape Town	2.10	1.51	1.92	2.18	2.50	42%	65.6	30.2
60.	-8	-2	+6	+7	Brazil	Rio de Janeiro	2.32	1.82	1.68	1.99	2.49	41%	36.8	48.2
61.	-2	+0	+2	+4	Brazil	Sao Paulo	2.01	1.65	1.71	2.07	2.46	41%	49.1	43.9
62.	-2	+6	+7	+6	Colombia	Bogota	1.99	1.34	1.30	1.90	2.43	40%	81.3	86.9
63.	-5	+2	-3	+0	South Africa	Johannesburg	2.03	1.50	1.84	2.16	2.41	40%	60.7	31.0
64.	+3	-4	-7	-4	India	Delhi	1.32	1.68	1.93	2.34	2.38	40%	41.7	23.3
65.	+4	+1	-6	-6	India	Bangalore	1.07	1.43	1.85	2.42	2.37	39%	65.7	28.1
66.	+0	+3	+2	-2	Italy	Rome	1.39	1.31	1.44	2.11	2.24	37%	71.0	55.6
67.	-4	-5	-2	-6	Italy	Milan	1.78	1.62	1.69	2.20	2.18	36%	34.6	29.0
68.	-23	-19	-20	-2	Indonesia	Jakarta	2.76	2.54	2.56	2.03	1.96	33%	-22.8	-23.4
69.	-19	-17	-7	+0	Egypt	Cairo	2.34	2.20	1.78	1.70	1.89	31%	-14.1	6.2

Source : www.numbeo.com, Deutsche Bank.



Figure 48: Coke/Pepsi (0.33 liter bottle) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+1	+1	+0	+0	Switzerland	Zurich	4.07	4.38	4.24	5.88	6.03	172%	37.7	42.2
2.	+3	-1	+0	+0	Switzerland	Geneva	3.35	4.45	3.69	5.21	5.45	155%	22.5	47.7
3.	+0	+2	+1	+0	Denmark	Copenhagen	3.92	3.14	3.53	4.92	5.20	148%	65.6	47.3
4.	+0	+0	-1	+0	France	Paris	3.82	3.61	3.61	4.19	4.75	135%	31.6	31.6
5.	-4	-2	+0	+4	Norway	Oslo	4.74	3.63	3.34	3.84	4.14	118%	14.0	24.0
5.	+3	+8	+4	+1	Luxembourg	Luxembourg	3.02	2.51	3.07	3.99	4.14	118%	64.9	34.9
7.	+12	+10	+8	+9	Israel	Tel Aviv-Yafo	2.10	2.33	2.61	3.12	4.07	116%	74.7	55.9
8.	+8	-1	+5	-1	Netherlands	Amsterdam	2.35	2.78	2.66	3.97	3.96	113%	42.4	48.9
9.	+6	+3	+5	+3	Sweden	Stockholm	2.43	2.52	2.64	3.41	3.95	113%	56.7	49.6
10.	+13	-2	+2	-5	Germany	Frankfurt	2.00	2.73	2.67	4.08	3.94	112%	44.3	47.6
11.	+0	-2	-5	-3	Austria	Vienna	2.79	2.71	3.14	3.89	3.90	111%	43.9	24.2
12.	-5	-6	-5	-3	Germany	Munich	3.09	2.87	3.12	3.84	3.86	110%	34.5	23.7
13.	-1	-2	-2	+0	Italy	Milan	2.76	2.62	2.83	3.35	3.73	106%	42.4	31.8
14.	+4	+9	+4	-3	Germany	Berlin	2.17	2.03	2.31	3.59	3.69	105%	81.8	59.7
15.	+24	+11	+2	+4	United States	Los Angeles	1.53	1.93	2.40	3.05	3.55	101%	83.9	47.9
16.	+4	+6	+8	+4	United States	New York	2.04	2.07	2.10	3.03	3.51	100%	69.6	67.1
17.	-7	-7	-9	+0	Finland	Helsinki	2.82	2.65	3.09	3.09	3.50	100%	32.1	13.3
18.	-5	-3	-8	-3	Belgium	Brussels	2.75	2.41	2.84	3.22	3.34	96%	38.6	17.6
19.	-5	+0	+9	+1	Spain	Madrid	2.68	2.23	1.96	3.03	3.26	93%	46.2	66.3
20.	-11	-7	+1	+7	Australia	Melbourne	2.89	2.51	2.18	2.63	3.13	89%	24.7	43.6
21.	+11	+0	-5	-7	United States	San Francisco	1.81	2.09	2.58	3.26	3.10	88%	48.3	20.2
22.	+5	+6	+4	+2	Ireland	Dublin	1.92	1.85	1.99	2.78	2.98	85%	61.1	49.7
23.	-6	+2	+0	+2	Italy	Rome	2.19	1.97	2.15	2.73	2.93	83%	48.7	36.3
24.	+4	+6	+5	-2	United Kingdom	London	1.90	1.84	1.87	2.95	2.87	82%	56.0	53.5
25.	-3	-1	+4	+4	Spain	Barcelona	2.01	2.02	1.87	2.53	2.79	79%	38.1	49.2
26.	+9	+2	-1	-3	United States	Boston	1.73	1.85	2.04	2.85	2.78	79%	50.3	36.3
27.	+11	+0	+0	-9	United States	Chicago	1.54	1.88	1.97	3.08	2.71	77%	44.1	37.6
28.	+9	+5	+3	+5	Greece	Athens	1.60	1.59	1.83	2.36	2.68	76%	68.6	46.4
29.	-23	-13	-9	-1	Australia	Sydney	3.11	2.39	2.25	2.57	2.63	75%	10.0	16.9
30.	-4	+1	+5	+1	United Kingdom	Birmingham	1.96	1.77	1.60	2.43	2.54	72%	43.5	58.8
31.	-8	-11	-9	-5	New Zealand	Wellington	2.00	2.13	2.17	2.68	2.48	71%	16.4	14.3
32.	+20	+15	+16	+2	Poland	Warsaw	1.02	1.11	1.16	2.33	2.44	70%	119.8	110.3
33.	+3	-1	+0	-1	United Kingdom	Edinburgh	1.72	1.66	1.70	2.38	2.41	69%	45.2	41.8
33.	+8	+5	+3	-3	Portugal	Lisbon	1.49	1.43	1.56	2.45	2.41	69%	68.5	54.5
35.	+14	+13	+12	+7	Hungary	Budapest	1.03	0.98	1.17	1.69	2.17	62%	121.4	85.5
36.	-11	-2	+0	+0	Canada	Vancouver	1.98	1.54	1.56	2.19	2.16	62%	40.3	38.5
37.	-7	-1	-5	+2	Canada	Toronto	1.89	1.45	1.77	2.06	2.10	60%	44.8	18.6
38.	-7	-3	+6	-1	Argentina	Buenos Aires	1.82	1.46	1.27	2.08	2.08	59%	42.5	63.8
39.	-11	-21	-20	-4	New Zealand	Auckland	1.90	2.24	2.30	2.28	2.04	58%	-8.9	-11.3
40.	+2	+3	+1	+0	Czech Republic	Prague	1.41	1.22	1.35	1.91	2.02	58%	65.6	49.6
41.	-8	-4	-7	-4	Canada	Montreal	1.79	1.44	1.67	2.08	2.00	57%	38.9	19.8
42.	+4	-3	-3	+4	Singapore	Singapore	1.16	1.35	1.46	1.54	1.84	52%	36.3	26.0
43.	+0	+5	+12	+4	Türkiye	Istanbul	1.34	0.98	0.76	1.53	1.72	49%	75.5	126.3
44.	+0	+6	+9	+4	Russia	Moscow	1.32	0.97	0.90	1.43	1.67	48%	72.2	85.6
45.	+2	-3	+1	-4	Chile	Santiago	1.13	1.26	1.19	1.79	1.65	47%	31.0	38.7
46.	+10	+9	+13	-1	Mexico	Mexico City	0.82	0.67	0.69	1.55	1.62	46%	141.8	134.8
47.	-13	-7	-5	+4	Brazil	Rio de Janeiro	1.76	1.34	1.32	1.29	1.58	45%	17.9	19.7
48.	+14	+15	+1	+7	United Arab Emirates	Dubai	0.54	0.52	1.01	1.15	1.56	44%	200.0	54.5
49.	-28	-8	-9	+3	Japan	Tokyo	2.02	1.29	1.43	1.21	1.55	44%	20.2	8.4
50.	+12	+6	+10	-6	Philippines	Manila	0.54	0.65	0.68	1.57	1.46	42%	124.6	114.7
51.	-6	-8	-13	-8	South Korea	Seoul	1.18	1.22	1.53	1.65	1.42	40%	16.4	-7.2
52.	-13	-9	-9	+1	Brazil	Sao Paulo	1.53	1.22	1.29	1.16	1.38	39%	13.1	7.0
53.	-1	+0	-2	+0	South Africa	Cape Town	1.02	0.78	0.98	1.16	1.35	38%	73.1	37.8
54.	-5	-8	-9	-5	Hong Kong	Hong Kong	1.03	1.12	1.20	1.32	1.29	37%	15.2	7.5
55.	-6	-3	+0	+2	Colombia	Bogota	1.03	0.79	0.76	0.98	1.25	36%	58.2	64.5
56.	+8	+9	-4	-6	United Arab Emirates	Abu Dhabi	0.49	0.46	0.91	1.31	1.23	35%	167.4	35.2
57.	-10	-3	-8	-1	South Africa	Johannesburg	1.13	0.76	1.01	1.03	1.22	35%	60.5	20.8
58.	+0	+2	-1	-1	Qatar	Doha	0.74	0.57	0.75	0.98	1.03	29%	80.7	37.3
59.	-5	-8	-5	-2	Taiwan	Taipei	0.92	0.94	0.85	0.98	0.96	27%	2.1	12.9
60.	+6	+7	-2	+2	Saudi Arabia	Riyadh	0.42	0.43	0.71	0.72	0.85	24%	97.7	19.7
61.	-2	-2	+1	-1	Malaysia	Kuala Lumpur	0.68	0.60	0.66	0.92	0.82	23%	36.7	24.2
62.	-1	+0	+2	-1	Thailand	Bangkok	0.58	0.53	0.56	0.74	0.61	17%	15.1	8.9
63.	-3	-2	-2	+3	China	Shanghai	0.66	0.56	0.67	0.51	0.57	16%	1.8	-14.9
64.	-7	-6	+1	+4	China	Beijing	0.75	0.61	0.53	0.50	0.54	15%	-11.5	1.9
65.	-10	-8	-2	-1	Indonesia	Jakarta	0.84	0.63	0.64	0.59	0.46	13%	-27.0	-28.1
66.	+2	+2	+1	+0	India	Bangalore	0.37	0.42	0.51	0.51	0.42	12%	0.0	-17.6
67.	+0	-3	-1	-4	India	Mumbai	0.40	0.47	0.52	0.65	0.41	12%	-12.8	-21.2
67.	+2	-1	+1	-2	India	Delhi	0.36	0.45	0.50	0.52	0.41	12%	-8.9	-18.0
69.	-4	-1	+0	+0	Egypt	Cairo	0.45	0.42	0.32	0.36	0.38	11%	-9.5	18.8

Source : www.numbeo.com, Deutsche Bank.



Figure 49: Imported beer at a restaurant (0.33 liter bottle) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+0	+0	+0	+0 United Arab Emirates	Dubai	7.8	10.6	12.7	13.7	13.6	136%	29.0	7.0
2.	N/A	N/A	N/A	N/A	Qatar	Doha	N/A	N/A	N/A	N/A	13.4	134%	N/A	N/A
3.	-2	-1	+0	-1	Norway	Oslo	10.5	9.5	10.3	11.2	12.3	123%	29.0	19.1
4.	+7	-1	-2	-1	United Arab Emirates	Abu Dhabi	6.5	9.1	11.4	10.8	10.9	109%	19.1	-4.6
5.	+12	+7	+1	+7	Israel	Tel Aviv-Yafo	6.1	6.6	8.1	8.4	10.7	107%	61.9	31.4
6.	+8	-2	-1	+0	Singapore	Singapore	6.3	7.7	8.4	9.8	10.1	101%	30.8	20.3
7.	+2	-1	-3	-3	United States	New York	6.8	7.5	8.5	10.4	10.0	100%	33.2	18.3
8.	+21	+1	+2	-1	Switzerland	Geneva	5.3	6.8	7.4	9.2	9.3	93%	38.1	26.6
9.	+3	+2	+0	+0	United States	Los Angeles	6.5	6.6	7.6	8.6	9.3	93%	39.7	22.4
10.	+10	-2	-2	-2	United States	San Francisco	5.8	6.8	7.8	9.1	9.0	90%	32.5	16.1
10.	+12	+3	+3	+0	United States	Chicago	5.7	6.6	6.9	8.6	9.0	90%	37.4	29.9
10.	+20	-3	-3	-5	United States	Boston	5.2	7.0	7.8	9.8	9.0	90%	28.6	16.0
13.	+2	+1	+10	+0	Switzerland	Zurich	6.3	6.5	6.0	8.3	8.7	87%	34.7	44.3
13.	+10	+3	+12	+3	United Kingdom	London	5.6	6.3	5.8	7.7	8.7	87%	39.0	50.3
15.	-9	+4	+3	-1	Denmark	Copenhagen	7.1	5.9	6.4	8.0	8.4	84%	42.2	32.2
16.	-9	+1	+0	-1	Australia	Melbourne	7.0	6.2	6.5	7.8	8.4	84%	35.8	29.3
17.	+9	+6	+12	+10	United Kingdom	Edinburgh	5.4	5.7	5.4	6.5	8.0	80%	41.6	48.2
18.	-11	-3	-6	-1	Finland	Helsinki	7.0	6.3	7.2	7.6	8.0	80%	26.3	10.8
19.	-14	-14	-8	-8	Sweden	Stockholm	7.3	7.5	7.3	8.4	8.0	80%	6.4	10.0
20.	+4	+1	+1	+2	Ireland	Dublin	5.5	5.9	6.2	6.9	7.4	74%	26.3	20.4
21.	-8	+1	-4	-3	New Zealand	Auckland	6.4	5.8	6.4	7.5	7.1	71%	21.9	11.6
22.	-3	+3	-2	+1	Canada	Toronto	5.9	5.4	6.2	6.8	7.0	70%	31.6	13.2
23.	-19	-5	+1	-4	Australia	Sydney	7.3	6.1	5.9	7.4	7.0	70%	14.5	17.6
24.	+8	+22	+13	+6	Italy	Rome	4.9	3.9	4.6	5.9	6.9	69%	76.2	50.7
24.	+4	+4	+5	+6	Italy	Milan	5.3	4.8	5.4	6.3	6.9	69%	44.3	26.8
24.	+16	+7	+9	+7	Luxembourg	Luxembourg	3.9	4.7	4.9	5.9	6.9	69%	47.7	39.1
24.	-14	-14	-9	-4	France	Paris	6.7	6.7	6.8	7.3	6.9	69%	2.4	0.7
28.	-7	-1	-6	-1	Canada	Montreal	5.7	4.9	6.1	6.5	6.9	69%	39.1	13.0
29.	-4	-9	-10	-5	New Zealand	Wellington	5.5	5.9	6.2	6.7	6.8	68%	15.7	9.8
30.	+8	+2	+1	-1	United Kingdom	Birmingham	4.0	4.4	5.0	6.3	6.7	67%	51.0	33.5
31.	-14	-7	-5	-10	Canada	Vancouver	6.1	5.6	5.6	6.9	6.3	63%	13.4	12.8
32.	+5	+5	+3	+3	Malaysia	Kuala Lumpur	4.1	4.3	4.7	5.6	6.1	61%	44.0	21.0
33.	+21	-7	-19	-8	Hong Kong	Hong Kong	3.2	5.3	6.8	6.6	6.1	61%	14.6	-11.4
34.	+21	+21	+19	+7	Mexico	Mexico City	3.1	3.0	3.5	5.3	5.8	58%	94.9	66.2
35.	+0	+0	+3	-9	Netherlands	Amsterdam	4.3	4.4	4.5	6.6	5.7	57%	31.5	26.8
35.	+6	+7	+4	+2	Austria	Vienna	3.8	4.1	4.5	5.5	5.7	57%	39.2	28.5
35.	+1	+12	+5	+0	Belgium	Brussels	4.6	3.9	4.3	5.6	5.7	57%	48.6	31.8
35.	-8	-5	-3	-2	Greece	Athens	5.3	4.7	5.0	5.8	5.7	57%	20.9	14.6
35.	+7	+9	+10	+3	Germany	Munich	3.8	4.0	4.1	5.5	5.7	57%	44.1	40.9
35.	+18	+3	+9	+4	Germany	Frankfurt	3.3	4.2	4.1	5.4	5.7	57%	36.5	39.5
41.	+3	-2	+11	-7	Türkiye	Istanbul	3.8	4.1	3.7	5.6	5.5	55%	32.0	48.9
42.	+1	+7	+7	+1	Spain	Madrid	3.8	3.5	3.7	5.1	5.4	54%	54.5	47.4
43.	+20	+21	+17	+2	Poland	Warsaw	2.1	2.2	2.6	4.8	5.3	53%	148.4	103.0
44.	+6	+8	+13	+9	Russia	Moscow	3.5	3.3	3.0	3.9	5.2	52%	58.2	73.6
45.	-6	-11	-17	-3	South Korea	Seoul	4.0	4.4	5.6	5.2	5.1	51%	14.7	-8.8
46.	-13	+2	+15	+3	Argentina	Buenos Aires	4.8	3.6	2.5	4.5	5.0	50%	38.5	98.4
47.	-45	-18	-21	-7	Japan	Tokyo	8.2	4.7	5.6	5.4	4.9	49%	4.2	-12.1
48.	+8	+6	-6	-1	Chile	Santiago	3.0	3.0	4.2	4.6	4.9	49%	62.8	14.4
49.	-3	+2	+0	+5	Germany	Berlin	3.7	3.4	3.7	4.8	4.6	46%	35.9	24.1
49.	-2	+1	+5	-1	Spain	Barcelona	3.6	3.4	3.4	4.5	4.6	46%	35.5	34.3
51.	-1	-12	-15	+0	Thailand	Bangkok	3.5	4.1	4.7	4.3	4.5	45%	9.2	-3.0
52.	+5	+1	-18	-6	India	Mumbai	3.0	3.2	4.9	4.6	4.2	42%	29.0	-15.2
52.	+12	+4	-9	+0	India	Bangalore	2.0	2.7	4.2	4.2	4.2	42%	53.7	-0.9
54.	-23	-21	-8	+4	Brazil	Sao Paulo	5.2	4.4	3.9	3.1	3.9	39%	-12.0	0.0
55.	+10	+12	+12	+4	Hungary	Budapest	1.9	1.7	2.0	3.0	3.9	39%	131.0	96.0
55.	-19	-12	-14	-5	Indonesia	Jakarta	4.2	4.1	4.3	4.3	3.9	39%	5.4	-8.7
57.	+1	+0	+1	+2	South Africa	Cape Town	2.5	2.7	2.9	3.0	3.7	37%	37.0	29.4
58.	-6	+4	+7	+8	Colombia	Bogota	3.3	2.2	2.3	2.4	3.6	36%	67.1	58.3
59.	+3	+3	+4	-5	Taiwan	Taipei	2.2	2.2	2.5	3.9	3.6	36%	65.3	45.1
60.	+0	-1	-1	-4	Portugal	Lisbon	2.4	2.4	2.8	3.7	3.4	34%	42.3	22.9
61.	+5	+5	+5	-2	Czech Republic	Prague	1.8	1.7	2.0	3.0	3.3	33%	91.9	63.5
62.	-18	-26	-7	+1	Brazil	Rio de Janeiro	3.8	4.3	3.2	2.7	3.2	32%	-25.2	-0.3
63.	-4	-5	-16	-8	India	Delhi	2.5	2.7	3.9	3.7	3.1	31%	17.6	-18.4
64.	N/A	N/A	N/A	N/A	Saudi Arabia	Riyadh	N/A	N/A	N/A	N/A	2.8	28%	N/A	N/A
65.	-4	+0	-2	-1	South Africa	Johannesburg	2.3	2.1	2.5	2.6	2.8	28%	34.3	13.0
66.	+1	-5	-5	-9	Philippines	Manila	1.6	2.2	2.5	3.1	2.8	28%	26.5	9.9
67.	-18	-8	-11	-5	Egypt	Cairo	3.6	2.4	3.2	2.8	2.5	25%	2.1	-23.4
68.	-52	-27	-19	-3	China	Shanghai	6.1	4.1	3.7	2.5	2.2	22%	-46.4	-40.1
69.	-21	-24	-21	-2	China	Beijing	3.6	3.9	3.7	2.3	1.8	18%	-54.7	-52.4

Source : www.numbeo.com, Deutsche Bank.



Figure 50: McMeal at McDonalds (or equivalent combo meal) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+4	+3	+1	+2	Israel	Tel Aviv-Yafo	11.0	12.2	14.1	17.6	20.9	160%	71.0	48.4
2.	+0	+0	-1	+0	Switzerland	Zurich	14.3	14.4	14.2	19.0	18.6	143%	29.3	31.6
2.	+1	-1	+1	-1	Switzerland	Geneva	14.2	15.1	13.4	19.0	18.6	143%	23.6	38.9
4.	+21	+9	+6	+0	Netherlands	Amsterdam	8.1	8.7	9.4	15.6	16.0	123%	85.0	69.9
5.	-4	-2	-1	+1	Norway	Oslo	16.0	13.2	13.0	15.2	15.3	118%	16.3	18.3
6.	+37	+12	+3	+3	United States	San Francisco	6.3	8.1	9.5	13.6	15.0	115%	84.3	58.2
7.	+7	-1	-1	-2	Luxembourg	Luxembourg	8.6	10.4	10.8	15.4	14.9	114%	42.7	38.3
8.	+37	+24	+18	+10	United States	Los Angeles	6.1	7.4	8.0	12.8	14.0	108%	88.2	74.6
8.	+34	+23	+17	+12	United States	Chicago	6.6	7.5	8.0	12.7	14.0	108%	87.4	74.3
10.	-3	-2	-3	-2	Belgium	Brussels	9.5	9.3	9.9	14.4	13.9	107%	50.4	41.2
11.	-7	-6	-6	-4	Denmark	Copenhagen	12.2	10.9	11.1	15.0	13.8	106%	26.2	24.0
12.	-6	+4	+5	+2	Finland	Helsinki	9.7	8.3	8.9	13.1	13.7	106%	65.7	55.1
12.	+5	+0	-2	+4	Ireland	Dublin	8.6	8.7	9.4	12.9	13.7	106%	57.9	45.7
12.	+11	+13	+10	+10	Austria	Vienna	8.1	7.8	8.4	12.2	13.7	106%	76.4	63.8
12.	+19	+5	+2	+0	Germany	Munich	7.8	8.2	9.1	13.3	13.7	106%	67.2	50.5
12.	+10	+10	+4	+1	Germany	Frankfurt	8.2	8.0	9.0	13.3	13.7	106%	70.9	52.2
12.	+9	+9	+8	+3	Spain	Barcelona	8.2	8.1	8.5	13.1	13.7	106%	70.3	62.2
12.	-4	-2	+0	+7	Italy	Milan	9.4	8.8	9.2	12.7	13.7	106%	56.0	48.7
12.	-2	-5	-4	-2	France	Paris	9.0	9.3	9.8	13.5	13.7	106%	47.3	39.6
12.	+0	+11	+10	+5	Germany	Berlin	8.7	7.9	8.4	12.8	13.7	106%	73.3	63.8
21.	+16	-6	-7	-10	United States	New York	7.2	8.4	9.1	13.3	13.0	100%	54.9	42.4
22.	-13	-8	+24	-1	Argentina	Buenos Aires	9.2	8.5	6.1	12.3	13.0	100%	52.3	112.5
23.	+16	+4	-2	+7	United States	Boston	6.8	7.7	8.4	11.1	12.0	92%	55.0	42.2
24.	+6	+5	+4	+8	Australia	Melbourne	7.8	7.6	8.0	10.7	11.8	91%	55.9	47.9
25.	-11	-6	-6	-2	Spain	Madrid	8.6	8.1	8.7	12.0	11.7	90%	44.3	34.7
26.	-10	-15	-13	-2	Italy	Rome	8.6	8.8	9.1	11.8	11.5	88%	30.1	25.3
26.	+12	+17	+10	+7	Portugal	Lisbon	6.9	6.5	7.0	10.6	11.5	88%	76.7	62.9
28.	-11	-19	-10	-2	Sweden	Stockholm	8.6	8.9	8.8	11.6	11.4	88%	28.0	30.7
29.	+21	+20	+21	+17	Hungary	Budapest	5.7	5.4	5.6	8.4	11.3	87%	108.7	102.0
30.	-4	+7	-6	-2	Canada	Toronto	8.1	7.1	8.2	11.4	11.1	86%	56.5	35.9
30.	-4	+0	-1	+5	Australia	Sydney	8.1	7.6	7.9	10.2	11.1	86%	47.2	40.9
32.	+4	+7	+3	+4	Greece	Athens	7.3	7.0	7.1	10.2	11.0	85%	56.1	53.9
33.	-1	-13	-1	-4	United Kingdom	London	7.8	8.1	7.4	11.3	10.7	82%	32.2	44.3
34.	+23	+25	+18	+7	Poland	Warsaw	4.9	4.6	5.5	9.3	10.7	82%	133.7	94.9
35.	-2	-2	-4	-4	Canada	Vancouver	7.7	7.2	7.7	11.0	10.6	81%	46.1	37.5
35.	-11	-1	-8	-8	Canada	Montreal	8.1	7.2	8.0	11.6	10.6	81%	46.3	32.0
37.	+11	+21	+16	+8	Mexico	Mexico City	5.8	4.7	5.3	8.9	10.5	81%	126.2	98.9
38.	+10	+11	+7	+2	Czech Republic	Prague	5.8	5.4	6.1	9.5	10.4	80%	91.9	70.5
39.	+2	-11	+1	-5	United Kingdom	Edinburgh	6.7	7.6	6.9	10.6	10.2	79%	34.6	49.1
40.	+4	+8	+23	+3	Türkiye	Istanbul	6.2	5.5	4.0	9.0	9.6	74%	75.4	142.7
41.	-7	-15	-9	-2	New Zealand	Auckland	7.4	7.8	7.4	9.6	9.6	74%	23.5	29.0
42.	+5	-2	-8	-5	United Arab Emirates	Dubai	6.0	6.7	7.3	10.0	9.5	73%	43.1	31.1
42.	-2	-1	-7	+2	United Arab Emirates	Abu Dhabi	6.7	6.6	7.1	9.0	9.5	73%	44.2	33.5
44.	-9	-9	-14	-19	New Zealand	Wellington	7.3	7.1	7.9	11.6	9.4	73%	32.3	19.5
45.	-32	-21	-6	-7	United Kingdom	Birmingham	8.7	7.8	7.0	9.7	9.4	72%	19.7	33.7
46.	+0	-1	-2	-4	Chile	Santiago	6.1	5.9	6.3	9.3	9.2	70%	56.3	45.4
47.	-19	+5	+8	+2	Colombia	Bogota	7.9	5.3	5.0	7.4	9.0	69%	70.8	80.0
48.	-29	-10	-11	+3	Brazil	Sao Paulo	8.5	7.1	7.1	7.2	8.8	67%	23.2	23.4
49.	+2	-5	-7	+1	Qatar	Doha	5.5	6.0	6.8	7.3	8.2	63%	37.5	20.4
50.	+5	-3	-3	-3	Saudi Arabia	Riyadh	5.2	5.5	6.1	8.1	8.0	61%	44.8	31.0
51.	-31	+5	+6	+1	Russia	Moscow	8.3	5.0	4.8	6.6	7.8	60%	56.7	61.3
52.	-41	-16	-11	+4	Brazil	Rio de Janeiro	8.9	7.1	6.8	6.3	7.8	60%	9.3	13.7
53.	+0	-7	-5	-5	Singapore	Singapore	5.3	5.8	6.0	7.6	7.7	60%	33.0	28.1
54.	+10	+3	-3	+0	Thailand	Bangkok	3.9	4.8	5.6	6.5	7.5	58%	57.4	35.3
55.	+1	+7	+11	+3	South Africa	Johannesburg	5.0	3.7	3.8	5.5	7.4	57%	100.3	93.5
56.	-4	-5	-7	-3	South Korea	Seoul	5.5	5.4	5.7	6.6	6.5	50%	22.1	15.6
57.	+6	+3	+2	-2	Hong Kong	Hong Kong	4.0	4.4	4.6	6.3	6.4	49%	45.3	39.0
58.	+0	+5	+2	+3	South Africa	Cape Town	4.5	3.7	4.2	5.2	6.2	47%	67.2	45.9
59.	-5	-6	-1	+1	Egypt	Cairo	5.3	5.2	4.7	5.3	6.1	47%	19.0	29.8
60.	+2	+1	+0	-3	Taiwan	Taipei	4.0	3.9	4.2	5.6	5.7	44%	47.9	35.7
61.	-1	-6	-7	+1	China	Shanghai	4.3	5.0	5.3	5.2	5.2	40%	2.8	-2.1
61.	+0	-7	-5	+2	China	Beijing	4.2	5.1	4.9	5.0	5.2	40%	0.8	4.9
63.	-34	-21	-20	-4	Japan	Tokyo	7.8	6.5	6.5	5.4	4.9	38%	-24.5	-23.6
64.	+1	+4	+3	+3	Malaysia	Kuala Lumpur	3.7	3.2	3.6	4.6	4.9	38%	54.6	35.0
65.	+2	+2	+0	+1	India	Bangalore	3.4	3.3	3.9	4.7	4.3	33%	28.8	9.7
66.	+2	+0	-4	-2	India	Mumbai	3.1	3.4	4.2	4.7	4.2	32%	22.6	-0.9
67.	+2	+2	+2	-2	Philippines	Manila	2.9	2.7	2.9	4.7	4.1	31%	49.1	41.3
68.	-2	-5	-4	+0	India	Delhi	3.5	3.7	3.9	4.1	3.9	30%	4.6	-1.8
69.	-10	-4	-1	+0	Indonesia	Jakarta	4.4	3.5	3.4	3.8	3.3	26%	-3.8	-0.6

Source : www.numbeo.com, Deutsche Bank.



Figure 51: Three-course meal for 2 people at a mid-range restaurant (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+0	+0	+0	Switzerland	Zurich	105	118	101	146	149	100%	26.7	48.3
2.	+10	+6	+3	+1	United States	New York	77	83	91	133	140	100%	68.2	53.7
3.	-1	-1	+0	+2	Switzerland	Geneva	109	108	97	126	137	98%	26.1	40.5
4.	+21	+22	+4	+3	Israel	Tel Aviv-Yafo	60	59	76	113	133	95%	125.1	74.6
5.	-4	-2	-3	-1	Norway	Oslo	130	98	99	131	123	88%	25.7	23.9
6.	-1	-2	-2	+0	Denmark	Copenhagen	96	95	92	117	123	88%	29.6	33.7
7.	+13	+2	-1	-5	United States	San Francisco	67	78	83	135	120	86%	53.5	43.8
8.	+19	+12	+10	+5	United States	Los Angeles	59	64	66	97	115	82%	81.1	74.9
9.	+23	+4	+1	+5	United States	Boston	55	71	76	96	110	79%	55.0	44.8
10.	-1	+0	+1	+10	Finland	Helsinki	83	78	76	87	109	78%	39.5	43.5
11.	+5	-5	+3	-2	United Kingdom	London	74	84	70	105	107.11	77%	27.3	53.1
11.	+3	+3	+11	-1	United Kingdom	Edinburgh	75	71	63	104	107.11	77%	51.1	69.9
13.	-6	-8	-6	-5	Sweden	Stockholm	88	90	77	105	104	74%	14.9	34.9
14.	-1	+1	+1	-2	Ireland	Dublin	75	70	70	99	103	74%	46.4	47.8
14.	-10	-7	-5	+1	Luxembourg	Luxembourg	99	84	76	95	103	74%	22.7	35.0
16.	+14	+6	+15	+6	United Kingdom	Birmingham	58	61	58	85	102	73%	66.9	75.8
17.	-7	+4	-5	-1	Italy	Milan	81	63	72	93	94	67%	49.2	30.9
18.	-10	-2	+1	+8	Australia	Sydney	84	65	65	80	93	66%	42.0	42.5
19.	-4	-8	-6	-8	Netherlands	Amsterdam	74	75	71	101	92	65%	22.2	29.2
19.	+10	+12	+6	+2	Germany	Frankfurt	59	53	62	86	92	65%	72.1	48.0
19.	+15	+6	-3	+0	Germany	Munich	53	59	70	87	92	65%	54.2	31.5
19.	-8	-7	+1	-2	Belgium	Brussels	80	73	65	91	92	65%	25.9	41.5
19.	+20	+17	+17	+5	Austria	Vienna	50	56	50	85	92	65%	63.1	81.4
24.	+17	+17	+17	+10	Germany	Berlin	46	45	46	71	86	61%	88.7	85.8
25.	-6	+2	-1	+0	Canada	Toronto	67	59	62	83	84	60%	43.5	35.6
25.	-1	+7	+2	+3	Canada	Vancouver	62	53	60	79	84	60%	58.8	40.0
27.	+17	+9	-4	+0	United Arab Emirates	Dubai	44	49	63	79	82	58%	67.2	30.1
28.	+7	+2	-2	-5	Qatar	Doha	53	55	61	85	81	58%	47.4	31.5
29.	-8	-5	+0	+2	France	Paris	67	60	60	76	80	57%	32.7	33.7
30.	-2	-13	-13	-12	United States	Chicago	59	65	67	90	80	57%	22.9	19.7
31.	+14	+7	+3	+10	United Arab Emirates	Abu Dhabi	43	48	55	65	75	53%	55.5	37.3
32.	-15	+3	+6	+1	Canada	Montreal	71	51	50	74	74	53%	43.6	48.2
33.	+13	+12	+26	+9	Argentina	Buenos Aires	41	40	27	63	70	50%	76.0	159.6
34.	+21	+17	+15	+11	Hungary	Budapest	33	31	36	60	70	50%	121.6	91.9
35.	-29	-16	-5	-6	Australia	Melbourne	91	65	59	78	70	50%	7.6	17.9
36.	-5	-3	+3	+3	Spain	Madrid	55	52	49	68	69	49%	31.3	40.9
36.	-14	-13	-15	-4	Italy	Rome	64	61	63	75	69	49%	13.2	8.8
36.	+2	+1	+1	+2	Spain	Barcelona	51	48	50	68	69	49%	42.3	37.7
36.	+1	+6	+6	+4	Greece	Athens	52	43	45	65	69	49%	60.4	51.0
40.	-17	-22	-13	-10	New Zealand	Auckland	64	65	60	77	68	49%	5.5	13.4
40.	-7	-12	-7	-4	New Zealand	Wellington	53	57	56	71	68	49%	19.3	23.2
42.	-24	+2	+3	+5	Russia	Moscow	69	41	41	54	65	46%	58.1	59.6
43.	+7	+11	+10	+3	Poland	Warsaw	37	30	33	57	64	46%	116.8	95.0
44.	-2	-5	-9	-7	Hong Kong	Hong Kong	44	48	52	69	64	46%	33.3	23.3
45.	+3	+3	-1	-1	Portugal	Lisbon	40	37	41	60	63	45%	69.1	52.9
46.	+3	-6	-6	-11	Singapore	Singapore	38	47	49	71	62	44%	33.0	27.1
47.	+5	-1	+0	-4	South Korea	Seoul	35	37	39	61	59	42%	56.8	51.9
48.	+10	+9	+4	+1	Czech Republic	Prague	29	28	33	49	54	39%	97.9	62.8
49.	+2	-6	-6	+3	Chile	Santiago	36	43	45	47	54	38%	26.7	18.6
50.	+4	+8	+15	-2	Türkiye	Istanbul	34	27	20	50	53	38%	100.0	171.6
51.	+10	+4	+0	+0	Saudi Arabia	Riyadh	25	28	34	48	53	38%	87.6	56.8
52.	+4	+4	+2	-2	Mexico	Mexico City	30	28	33	49	52	37%	86.5	59.1
53.	-6	+0	-7	+3	South Africa	Cape Town	40	31	39	41	49	35%	58.0	24.3
54.	-14	-7	-4	+4	Brazil	Sao Paulo	48	37	35	38	49	35%	29.9	38.4
55.	-12	-3	-7	+0	South Africa	Johannesburg	44	31	37	43	48	34%	52.5	30.9
56.	+6	+3	+4	-2	Taiwan	Taipei	24	27	26	43	47	33%	75.4	80.4
57.	-31	-23	-25	-4	Japan	Tokyo	60	52	56	47	43	31%	-16.6	-22.6
58.	-1	+4	+3	+3	Colombia	Bogota	29	23	22	30	42	30%	81.4	87.8
59.	-23	-9	-2	-2	Brazil	Rio de Janeiro	52	34	29	39	40	28%	17.9	36.4
60.	+3	+3	-2	-1	Thailand	Bangkok	20	23	28	32	36	26%	56.6	29.5
61.	-2	-1	+1	+5	Egypt	Cairo	26	26	22	22	31	22%	17.2	41.3
62.	-2	-1	-7	+2	China	Beijing	25	25	32	25	29	21%	19.0	-8.4
62.	-9	-13	-6	-2	China	Shanghai	35	35	32	32	29	21%	-15.6	-7.7
64.	+1	+2	+0	-1	Malaysia	Kuala Lumpur	20	19	20	27	29	21%	55.4	49.2
65.	-1	-1	-2	-3	Philippines	Manila	20	22	20	29	26	19%	20.4	31.8
66.	+2	+2	+1	+1	India	Mumbai	15	16	19	22	21	15%	33.9	11.6
66.	+1	+1	+2	-1	India	Delhi	15	16	18	24	21	15%	31.3	16.1
68.	+1	+1	+1	+0	India	Bangalore	11	11	15	20	17	12%	55.7	12.3
69.	-3	-4	-3	+0	Indonesia	Jakarta	17	19	19	18	17	12%	-13.7	-13.3

Source : www.numbeo.com, Deutsche Bank.



Figure 52: 1 seat at a cinema, international release (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+1	+1	+1	+0	Switzerland	Zurich	20.0	19.5	18.8	25.2	24.9	110%	27.7	32.4
1.	+2	+0	+0	+1	Switzerland	Geneva	19.9	20.4	20.5	23.7	24.9	110%	21.8	21.3
3.	+10	+8	+2	+0	United States	New York	13.6	15.0	16.5	21.6	22.5	100%	50.1	36.8
4.	+7	+3	+2	+0	Denmark	Copenhagen	14.3	15.5	16.4	20.4	19.9	88%	28.8	21.2
5.	+4	+5	+4	+1	Finland	Helsinki	15.1	15.1	15.5	19.5	19.5	86%	29.3	25.3
5.	+27	+31	+20	+3	Luxembourg	Luxembourg	10.7	10.4	12.0	18.0	18.5	86%	87.3	61.6
7.	-1	-2	+6	+2	Sweden	Stockholm	17.1	15.5	14.3	17.7	19.2	85%	23.7	34.5
8.	-3	-2	+0	-3	Norway	Oslo	17.3	15.5	16.1	19.9	18.9	84%	21.9	16.9
9.	-1	-6	-3	-2	United Kingdom	London	16.1	18.3	16.4	19.0	18.7	83%	2.7	14.1
10.	+20	+4	+0	+0	United States	Los Angeles	11.0	13.9	15.3	17.7	18.0	80%	29.6	17.7
11.	+9	+4	+0	+0	United States	San Francisco	12.3	13.4	15.1	17.5	17.5	78%	30.2	16.0
12.	-8	+1	+2	+10	Australia	Melbourne	17.5	14.5	13.9	14.8	17.4	77%	20.2	24.7
12.	-5	-3	+3	+12	Australia	Sydney	16.8	15.1	13.5	14.7	17.4	77%	14.9	28.7
14.	+3	+2	+4	-1	Netherlands	Amsterdam	12.5	12.9	13.4	17.2	17.2	76%	32.9	28.3
14.	+9	+6	+7	+2	Germany	Munich	12.0	12.6	12.9	16.1	17.2	76%	36.8	33.5
14.	+14	+4	+2	+1	Germany	Frankfurt	11.3	12.8	13.5	16.4	17.2	76%	34.1	27.2
17.	+14	+7	+7	+1	Belgium	Brussels	11.0	12.0	12.2	15.6	17.1	76%	41.6	39.9
18.	+9	+1	+1	-4	United States	Chicago	11.4	12.6	13.1	17.1	17.0	76%	34.6	30.0
19.	+14	+15	+11	-2	United Arab Emirates	Dubai	10.6	10.5	11.6	16.1	16.3	73%	56.4	41.2
20.	+20	+11	+9	+6	Israel	Tel Aviv-Yafo	10.0	10.8	11.6	14.2	16.3	72%	50.4	40.2
21.	+8	+2	+5	-9	United Kingdom	Birmingham	11.1	12.1	12.0	17.2	16.1	71%	32.5	34.1
22.	-8	+3	+1	-3	France	Paris	13.0	12.0	12.3	15.6	16.0	71%	34.1	30.0
23.	+1	-1	-6	+0	Ireland	Dublin	12.0	12.3	13.4	14.8	15.5	69%	26.0	15.2
24.	-4	-7	-12	-3	United States	Boston	12.3	12.8	14.6	15.3	15.0	67%	17.1	2.7
25.	+11	+3	-3	-5	Germany	Berlin	10.4	11.7	12.5	15.3	14.9	66%	26.9	18.8
26.	-7	-14	+5	+1	United Kingdom	Edinburgh	12.4	14.5	11.3	14.0	14.7	65%	1.7	29.9
27.	N/A	-19	-24	+4	Saudi Arabia	Riyadh	N/A	15.3	18.1	13.0	14.6	65%	-4.6	-19.2
28.	-3	+2	-1	-3	Austria	Vienna	11.9	11.0	11.9	14.3	14.6	65%	32.2	22.8
29.	+15	+11	+7	+10	Italy	Milan	9.7	9.9	10.8	12.0	13.7	61%	38.2	27.7
30.	+15	-4	-10	+0	Hong Kong	Hong Kong	9.7	11.9	13.0	13.3	13.7	61%	14.8	5.5
31.	+15	+6	+1	-2	United Arab Emirates	Abu Dhabi	9.3	10.2	11.3	13.8	13.6	60%	33.2	20.9
32.	-16	+1	+2	+1	Canada	Toronto	12.8	10.5	10.9	12.7	13.4	59%	26.9	22.4
33.	-11	-12	-5	+3	New Zealand	Auckland	12.1	12.4	11.6	12.4	13.1	58%	5.4	12.6
34.	+18	+18	+15	+9	Czech Republic	Prague	7.8	7.2	8.5	11.1	13.0	58%	80.5	53.8
35.	-17	-3	-3	+0	Canada	Vancouver	12.5	10.7	11.3	12.5	12.6	56%	17.6	11.5
36.	-21	+7	+3	-4	New Zealand	Wellington	12.8	9.5	10.0	12.7	12.5	56%	32.4	25.9
37.	-36	-33	-33	-9	Japan	Tokyo	22.4	17.4	17.0	13.9	12.3	55%	-29.0	-27.5
37.	-5	+1	-2	-3	Qatar	Doha	9.8	10.2	10.8	12.6	12.3	55%	21.0	13.8
39.	+13	+0	+1	+1	Canada	Montreal	11.9	10.0	9.9	11.9	11.8	52%	18.1	18.5
40.	+10	+7	+3	-3	Singapore	Singapore	7.9	8.7	9.2	12.4	11.6	52%	33.6	26.7
41.	+0	+1	-3	+5	Italy	Rome	9.9	9.5	10.0	10.5	11.5	51%	20.4	14.5
41.	-6	-6	-4	-3	Spain	Madrid	10.4	10.4	10.1	12.1	11.5	51%	9.7	13.1
41.	-4	+0	+0	+0	Spain	Barcelona	10.3	9.7	9.7	11.4	11.5	51%	17.6	18.5
44.	+5	+1	+11	-2	Argentina	Buenos Aires	8.5	8.8	6.9	11.1	11.1	49%	26.5	62.2
45.	-2	+3	+2	+0	Greece	Athens	9.8	8.7	8.7	10.6	10.3	46%	18.9	18.7
46.	+11	+11	+11	+2	Hungary	Budapest	6.1	6.2	6.4	9.0	10.0	45%	63.1	57.7
47.	+1	+2	-3	+0	South Korea	Seoul	9.1	8.5	9.0	10.3	9.8	44%	15.6	8.8
48.	+3	+3	+3	+1	Portugal	Lisbon	7.8	7.3	7.8	9.0	9.7	43%	33.3	25.5
49.	-11	-3	-4	+3	Brazil	Sao Paulo	10.3	8.8	8.9	8.0	9.7	43%	10.8	9.2
50.	+5	+3	+1	+1	Poland	Warsaw	7.2	7.1	7.8	8.9	9.6	43%	35.7	24.0
51.	-4	-7	-9	-7	Taiwan	Taipei	9.3	9.4	9.3	10.7	9.4	42%	-0.3	0.6
52.	-18	+3	+4	+6	Russia	Moscow	10.6	6.7	6.6	7.5	9.1	40%	36.7	37.7
53.	+7	+0	+6	+4	South Africa	Cape Town	5.3	4.7	6.0	7.6	9.0	40%	92.1	49.2
54.	-44	-27	-8	+1	China	Shanghai	14.7	11.7	18.8	7.8	8.8	39%	-24.8	0.3
55.	+3	+6	+2	-1	South Africa	Johannesburg	5.9	4.9	6.4	7.9	8.6	38%	76.0	25.8
56.	-2	+0	+9	-6	Türkiye	Istanbul	7.4	6.5	4.3	8.9	8.5	38%	31.6	97.7
57.	-18	-7	-9	-1	Brazil	Rio de Janeiro	10.2	8.1	8.5	7.8	7.8	35%	-3.8	-8.6
58.	-5	-4	-4	+2	Chile	Santiago	7.7	7.0	7.2	7.1	7.8	34%	11.2	7.2
59.	+0	+0	-6	-7	Thailand	Bangkok	5.6	5.7	7.7	8.0	7.5	33%	31.8	-2.6
59.	-3	+6	+9	+6	Colombia	Bogota	6.5	4.4	4.1	5.0	7.5	33%	72.2	83.6
61.	-49	-32	-11	-3	China	Beijing	14.3	11.1	8.4	7.5	7.4	33%	-33.7	-12.3
62.	+5	-4	-2	-1	Philippines	Manila	4.2	5.9	5.9	6.6	7.3	33%	25.1	24.2
63.	+2	+4	+4	-1	Mexico	Mexico City	4.3	4.2	4.2	5.5	6.9	31%	65.9	63.9
64.	+4	+5	+2	-1	Malaysia	Kuala Lumpur	4.1	3.7	4.2	5.2	5.4	24%	44.1	27.1
65.	+0	-3	-4	-1	India	Delhi	4.3	4.8	4.9	5.2	5.2	23%	9.9	7.4
66.	-4	-2	-3	+0	India	Bangalore	4.6	4.5	4.8	4.8	4.5	20%	-2.0	-7.1
67.	-3	-1	-3	+0	India	Mumbai	4.5	4.3	4.6	4.6	4.2	19%	-3.5	-8.1
68.	-7	-8	-6	+0	Egypt	Cairo	5.2	5.3	4.8	3.9	4.1	18%	-23.0	-15.5
69.	-6	-1	+0	+0	Indonesia	Jakarta	4.5	4.1	3.8	3.4	3.2	14%	-22.4	-16.9

Source : www.numbeo.com, Deutsche Bank.



Figure 53: Fitness club, monthly fee for 1 adult (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+54	+13	+3	+1	United States	San Francisco	49	68	92	139	136	118%	101.2	48.1
2.	+21	+10	+11	+4	Israel	Tel Aviv-Yafo	74	71	75	89	136	118%	92.1	80.3
3.	+7	-2	-2	-2	Qatar	Doha	95	161	113	142	129	112%	-20.1	13.5
4.	+11	+6	+6	-1	United States	New York	87	75	81	126	115	100%	53.6	42.9
5.	+3	+1	-3	-1	Singapore	Singapore	100	94	103	111	104	90%	9.9	0.8
6.	+13	+9	+6	+1	Italy	Milan	83	66	76	84	89	78%	35.7	18.0
7.	+2	+1	-2	-2	Saudi Arabia	Riyadh	99	89	90	104	89	77%	0.7	-1.2
8.	-2	-1	+0	+3	Switzerland	Zurich	102	92	85	79	89	77%	-3.1	4.3
9.	+55	+21	+8	+0	United States	Boston	43	52	61	81	87	75%	67.1	41.6
10.	+15	-1	+1	+3	Hong Kong	Hong Kong	70	87	80	77	85	74%	-1.5	6.7
11.	-10	-7	-2	-3	United Arab Emirates	Dubai	117	97	83	81	85	73%	-13.2	1.4
12.	-10	-10	-5	+5	United Arab Emirates	Abu Dhabi	110	122	89	69	80	69%	-34.4	-10.3
13.	-8	-10	-10	-1	Switzerland	Geneva	105	109	99	79	79	69%	-27.3	-19.8
14.	-7	-1	+10	-4	Luxembourg	Luxembourg	102	70	51	80	77	67%	9.6	49.9
15.	+36	+32	+22	+7	Hungary	Budapest	51	43	43	64	74	64%	72.4	72.2
16.	-2	+7	+7	+4	Australia	Sydney	87	56	53	66	72	63%	29.9	35.7
17.	+24	+3	+9	-1	United States	Chicago	59	59	50	70	71	62%	20.3	43.1
18.	+21	+8	+4	+8	Germany	Frankfurt	61	54	54	61	71	61%	30.1	31.8
19.	+35	+6	-1	+4	China	Shanghai	50	55	55	63	70	61%	28.8	27.2
20.	+6	+16	+44	+1	Türkiye	Istanbul	70	49	29	64	70	61%	43.7	141.5
21.	+1	-4	-6	-7	Italy	Rome	75	63	66	72	69	60%	8.8	3.9
22.	+34	+34	+16	+5	Netherlands	Amsterdam	49	37	43	61	66	57%	78.5	54.2
23.	+26	+38	+29	-4	Czech Republic	Prague	51	35	38	66	66	57%	91.1	74.2
24.	+9	+8	+9	+11	Ireland	Dublin	64	51	44	53	66	57%	29.1	51.1
25.	-21	-20	-19	-1	Japan	Tokyo	106	97	89	71	63	55%	-34.4	-28.9
26.	+21	+22	+13	+10	United States	Los Angeles	54	43	43	52	63	55%	47.4	47.6
27.	+19	-11	-13	-3	Thailand	Bangkok	54	66	69	62	63	54%	-4.9	-9.9
28.	-10	-17	-7	-10	United Kingdom	London	84	73	54	67	62	54%	-14.4	16.0
29.	-2	+14	+14	+11	Germany	Munich	70	45	42	48	62	54%	39.1	48.7
30.	+22	+25	+12	-1	Denmark	Copenhagen	50	37	42	59	61	53%	63.7	46.3
31.	-7	-2	-11	-6	Mexico	Mexico City	71	54	54	62	57	50%	6.8	6.2
32.	+0	-4	-13	+10	Australia	Melbourne	65	54	54	48	56	49%	4.4	3.6
33.	+17	+30	+25	+1	Greece	Athens	51	32	32	54	56	49%	72.8	74.3
34.	-17	-13	-7	-1	Norway	Oslo	84	57	49	54	56	49%	-2.0	14.6
35.	-1	-4	-5	-3	Spain	Barcelona	64	52	47	55	54	47%	5.4	15.9
36.	+8	+10	+21	+11	Argentina	Buenos Aires	55	43	33	46	52	45%	19.2	58.0
37.	-17	-19	-21	-7	South Korea	Seoul	83	61	62	56	51	44%	-15.8	-18.4
38.	-2	+4	+7	-1	Spain	Madrid	63	45	42	51	51	44%	13.1	21.9
39.	+19	+20	+7	+14	Canada	Vancouver	48	36	40	42	50	44%	39.0	24.5
40.	+0	-5	-8	-1	United Kingdom	Edinburgh	60	50	45	49	50	44%	1.1	12.5
41.	+12	+12	+0	+5	Malaysia	Kuala Lumpur	50	40	42	46	50	43%	23.5	19.2
42.	-7	-8	-8	+1	Sweden	Stockholm	64	51	44	48	49	43%	-2.2	13.4
43.	+18	+14	+7	+8	South Africa	Cape Town	45	37	39	44	49	43%	34.2	26.1
44.	+21	+18	+18	+0	Poland	Warsaw	42	33	29	47	49	42%	47.1	65.9
45.	-24	-18	+3	-4	Finland	Helsinki	76	54	40	48	49	42%	-9.9	22.2
46.	-15	-6	+3	-1	Portugal	Lisbon	67	45	39	47	48	41%	6.3	21.9
47.	+13	-23	-22	-9	Philippines	Manila	46	55	50	50	44	38%	-20.0	-11.8
48.	-32	-29	-19	+0	France	Paris	85	60	48	45	44	38%	-26.7	-8.0
49.	-6	+1	-5	-21	Canada	Toronto	59	42	42	59	43	37%	1.2	2.2
50.	-37	-13	-19	+0	Russia	Moscow	88	47	46	44	42	36%	-11.5	-9.0
51.	+12	-29	-23	-20	China	Beijing	43	57	49	56	42	36%	-26.3	-13.7
52.	-22	+2	+2	+0	Austria	Vienna	67	38	33	42	41	36%	8.9	25.8
53.	-11	-20	-18	-4	Taiwan	Taipei	59	51	43	45	40	35%	-20.9	-7.2
54.	-25	-2	-7	+0	New Zealand	Wellington	67	41	40	41	40	35%	-3.1	-0.2
55.	+11	+13	+4	+5	Germany	Berlin	37	28	32	35	39	34%	41.2	24.1
56.	-8	+9	-5	+3	South Africa	Johannesburg	51	31	39	35	39	34%	25.3	1.3
57.	-19	-16	-21	+1	New Zealand	Auckland	62	45	43	36	38	33%	-15.2	-12.2
58.	-30	+0	-2	-2	United Kingdom	Birmingham	68	36	33	38	37	33%	3.3	14.1
59.	-47	-21	-4	+2	Brazil	Sao Paulo	93	47	33	33	37	32%	-21.1	12.6
60.	-49	+6	+1	-3	Belgium	Brussels	93	30	30	37	37	32%	20.4	21.3
61.	-16	-12	-8	+1	Chile	Santiago	55	42	35	31	35	30%	-16.9	1.8
62.	-25	-23	+4	+2	Colombia	Bogota	62	46	28	28	32	28%	-29.2	17.9
63.	-1	-3	+4	-8	Canada	Montreal	43	35	27	39	30	26%	-15.1	11.9
64.	-61	-20	-4	-1	Brazil	Rio de Janeiro	107	44	31	28	27	23%	-39.2	-14.2
65.	-6	-14	-2	+4	Egypt	Cairo	47	42	29	18	26	23%	-37.7	-9.8
66.	+2	+1	-1	+0	India	Mumbai	28	30	28	25	26	23%	-12.5	-5.6
67.	-10	-22	-27	-2	Indonesia	Jakarta	48	43	42	28	24	21%	-44.1	-42.5
68.	-1	-4	+0	-1	India	Delhi	28	31	26	22	19	17%	-38.3	-24.3
69.	+0	+0	+0	-1	India	Bangalore	23	24	24	18	17	15%	-30.3	-30.4

Source : www.numbeo.com, Deutsche Bank.



Figure 54: 1 liter of gasoline (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+3	+0	+0	+0	Hong Kong	Hong Kong	2.22	1.92	2.16	3.21	3.88	359%	102.1	79.6
2.	+15	+13	+3	+6	Israel	Tel Aviv-Yafo	2.03	1.55	1.76	2.08	2.65	245%	71.0	50.6
3.	+4	-1	-1	-1	Netherlands	Amsterdam	2.17	1.73	1.86	2.25	2.52	233%	45.7	35.5
4.	+6	+8	+8	-1	Denmark	Copenhagen	2.11	1.57	1.69	2.17	2.47	229%	57.3	46.2
5.	+27	+11	+8	+1	Singapore	Singapore	1.67	1.54	1.66	2.14	2.41	223%	56.5	45.2
6.	+16	+12	+7	-3	Switzerland	Zurich	1.97	1.49	1.66	2.17	2.35	218%	57.7	41.6
7.	+16	+9	+3	+0	Switzerland	Geneva	1.93	1.54	1.73	2.12	2.34	217%	51.9	35.3
8.	+3	+1	+2	+3	Finland	Helsinki	2.10	1.59	1.73	2.03	2.32	215%	45.9	34.1
9.	-3	+2	-4	+1	Greece	Athens	2.18	1.58	1.76	2.04	2.26	209%	43.0	28.4
10.	-8	-7	-7	-5	Norway	Oslo	2.39	1.72	1.85	2.15	2.22	206%	29.1	20.0
10.	+8	+8	-5	-2	France	Paris	2.02	1.49	1.76	2.08	2.22	206%	49.0	26.1
12.	+1	-6	-3	-1	Portugal	Lisbon	2.07	1.61	1.74	2.03	2.20	204%	36.6	26.4
13.	+7	+10	+5	+2	Germany	Munich	2.00	1.43	1.61	1.98	2.16	200%	51.0	34.2
14.	-1	+6	+4	+3	Germany	Berlin	2.07	1.47	1.61	1.95	2.14	198%	45.6	32.9
15.	+0	+6	+8	-4	Ireland	Dublin	2.04	1.45	1.59	2.03	2.12	196%	46.2	33.3
15.	-10	-11	-11	-4	Italy	Milan	2.21	1.63	1.78	2.03	2.12	196%	30.1	19.1
17.	+4	+6	+1	-1	Germany	Frankfurt	1.98	1.43	1.61	1.96	2.09	194%	46.2	29.8
17.	-8	-5	+0	+3	United Kingdom	Birmingham	2.14	1.57	1.62	1.89	2.09	194%	33.1	29.0
19.	-1	+3	-1	+2	Belgium	Brussels	2.02	1.44	1.61	1.82	2.04	189%	41.7	26.7
19.	-11	-15	-3	+0	United Kingdom	London	2.15	1.63	1.64	1.90	2.04	189%	25.2	24.4
21.	-10	-9	-3	+2	United Kingdom	Edinburgh	2.10	1.57	1.61	1.78	2.03	188%	29.3	26.1
22.	-19	-16	-17	-5	Italy	Rome	2.28	1.61	1.76	1.95	2.02	187%	25.5	14.8
23.	+5	+9	+7	+4	Hungary	Budapest	1.79	1.22	1.35	1.72	1.93	179%	58.2	43.0
24.	+2	+7	+5	+0	Austria	Vienna	1.80	1.23	1.40	1.77	1.92	178%	56.1	37.1
25.	+6	+8	+6	+3	Luxembourg	Luxembourg	1.70	1.21	1.33	1.70	1.90	176%	57.0	42.9
26.	+3	+4	+2	+4	Czech Republic	Prague	1.75	1.25	1.41	1.67	1.88	174%	50.4	33.3
27.	-12	-21	-12	-1	Sweden	Stockholm	2.04	1.61	1.65	1.73	1.87	173%	16.1	13.3
28.	+7	-3	-1	+3	New Zealand	Wellington	1.60	1.33	1.43	1.65	1.85	171%	39.1	29.4
29.	+5	-1	-3	+3	New Zealand	Auckland	1.63	1.27	1.45	1.63	1.80	167%	41.7	24.1
30.	-5	-3	-6	-8	Spain	Madrid	1.82	1.30	1.50	1.81	1.77	164%	36.2	18.0
31.	-5	-5	-6	-6	Spain	Barcelona	1.80	1.31	1.49	1.74	1.73	160%	32.1	16.1
32.	+1	+3	+0	-3	Poland	Warsaw	1.65	1.16	1.32	1.68	1.65	153%	42.2	25.0
33.	+2	+5	+4	+0	Chile	Santiago	1.60	1.08	1.15	1.32	1.61	149%	49.1	40.0
34.	+29	+23	+12	+0	Mexico	Mexico City	0.73	0.76	1.04	1.28	1.54	143%	102.6	48.1
35.	+6	+14	+6	+1	South Africa	Johannesburg	1.38	0.85	0.99	1.27	1.50	139%	76.5	37.6
36.	+20	+16	+9	-2	United States	San Francisco	1.12	0.81	0.95	1.28	1.49	138%	84.0	41.9
37.	+20	+18	+12	-1	United States	Los Angeles	1.11	0.77	0.92	1.27	1.47	136%	90.9	44.1
37.	+8	+13	+5	+5	South Africa	Cape Town	1.34	0.84	1.07	1.22	1.47	136%	75.0	37.4
39.	+12	+16	+15	+8	Thailand	Bangkok	1.27	0.77	0.97	1.18	1.41	131%	83.1	45.4
40.	+0	+7	+11	+5	Australia	Melbourne	1.41	0.92	1.00	1.20	1.36	126%	47.8	36.0
40.	+1	+4	-2	+5	Canada	Vancouver	1.38	0.94	1.13	1.20	1.36	126%	44.7	20.4
42.	-41	-33	-6	-6	Türkiye	Istanbul	2.46	1.59	1.16	1.27	1.35	125%	-15.1	16.4
43.	-5	+1	+6	-1	Australia	Sydney	1.44	0.94	1.02	1.22	1.33	123%	41.5	30.4
44.	-7	-10	+9	+5	Argentina	Buenos Aires	1.46	1.18	0.99	1.14	1.31	121%	11.0	32.3
45.	-15	-16	-12	-6	South Korea	Seoul	1.74	1.26	1.29	1.25	1.28	119%	1.6	-0.8
46.	-8	-10	-11	+8	Brazil	Rio de Janeiro	1.44	1.15	1.27	1.06	1.27	118%	10.4	0.0
47.	+3	-4	+1	+6	China	Shanghai	1.30	0.95	1.03	1.07	1.26	117%	32.6	22.3
47.	+0	+11	+13	+10	Colombia	Bogota	1.32	0.71	0.75	1.02	1.26	117%	77.5	68.0
47.	+7	+4	+4	+4	Philippines	Manila	1.22	0.83	1.00	0.99	1.26	117%	51.8	26.0
50.	-7	-2	+5	-1	Canada	Montreal	1.37	0.87	0.95	1.14	1.25	116%	43.7	31.6
50.	-3	-4	-4	+2	China	Beijing	1.32	0.93	1.04	1.08	1.25	116%	34.4	20.2
52.	-5	-11	-10	+3	Brazil	Sao Paulo	1.32	1.02	1.07	1.05	1.22	113%	19.6	14.0
52.	+1	+0	+5	+4	Canada	Toronto	1.25	0.81	0.90	1.03	1.22	113%	50.6	35.6
54.	+1	+5	+4	+5	United States	Chicago	1.13	0.67	0.83	1.00	1.15	106%	71.6	38.6
55.	-10	-16	-17	-15	India	Mumbai	1.34	1.04	1.13	1.24	1.11	103%	6.7	-1.8
55.	-12	-15	-15	-13	India	Bangalore	1.37	1.03	1.10	1.22	1.11	103%	7.8	0.9
57.	+2	+3	+2	+3	United States	New York	1.07	0.66	0.80	0.88	1.08	100%	63.6	25.0
58.	-34	-21	-25	-17	Japan	Tokyo	1.87	1.12	1.29	1.23	1.06	98%	-5.4	-17.8
59.	-7	-17	-17	-11	India	Delhi	1.26	0.99	1.07	1.16	1.05	97%	6.1	-1.9
60.	-2	-8	-4	-2	Taiwan	Taipei	1.10	0.81	0.93	1.01	1.02	94%	25.9	9.7
61.	-1	+2	+0	+0	United States	Boston	1.00	0.56	0.74	0.85	1.00	93%	78.6	35.1
62.	+3	+3	+3	+2	United Arab Emirates	Abu Dhabi	0.48	0.46	0.60	0.76	0.92	85%	100.0	53.3
63.	-2	-1	-1	-1	Russia	Moscow	0.90	0.58	0.71	0.82	0.91	84%	56.9	28.2
64.	+1	+0	+1	+0	United Arab Emirates	Dubai	0.48	0.47	0.61	0.76	0.89	82%	89.4	45.9
65.	-1	+0	+1	+2	Malaysia	Kuala Lumpur	0.60	0.46	0.51	0.56	0.79	73%	71.7	54.9
66.	-4	-5	-3	-3	Indonesia	Jakarta	0.77	0.59	0.65	0.80	0.76	70%	28.8	16.9
67.	+2	+2	+2	-1	Saudi Arabia	Riyadh	0.13	0.22	0.41	0.61	0.62	57%	181.8	51.2
68.	+0	-1	-2	-1	Qatar	Doha	0.27	0.36	0.51	0.56	0.56	52%	55.6	9.8
69.	-2	-1	-1	+0	Egypt	Cairo	0.31	0.33	0.45	0.37	0.45	42%	36.4	0.0

Source : www.numbeo.com, Deutsche Bank.



Figure 55: Public transport monthly pass (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+0	+0	+0	+0	United Kingdom	London	189	198	196	244	268	191%	35.2	36.6
2.	+1	+1	+0	+0	Australia	Sydney	131	124	150	142	151	108%	22.4	1.0
3.	+5	+1	+2	+2	United States	New York	105	117	126	134	140	100%	19.8	10.8
4.	-2	+4	+3	+2	Australia	Melbourne	142	103	110	126	138	98%	33.6	25.5
5.	+8	+0	-1	-2	New Zealand	Auckland	96	117	138	136	124	88%	6.3	-9.9
6.	+21	+22	+21	+1	Denmark	Copenhagen	74	67	70	121	119	85%	79.6	71.4
7.	+5	+8	+1	-3	Netherlands	Amsterdam	97	91	106	135	114	82%	25.8	8.0
8.	-2	+3	+6	+3	Sweden	Stockholm	108	97	94	111	110	79%	13.4	17.6
9.	-5	-7	-6	+3	Ireland	Dublin	123	139	144	103	110	78%	-21.0	-23.6
10.	-5	-4	-4	-2	Canada	Toronto	122	111	110	113	110	78%	-0.7	-0.5
11.	+6	+2	+7	-1	Switzerland	Zurich	86	95	85	111	109	78%	14.7	28.9
12.	+25	+20	+19	+10	Israel	Tel Aviv-Yafo	57	61	63	82	105	75%	73.2	65.9
13.	+21	-1	-2	+5	United States	Los Angeles	64	96	98	91	105	75%	9.3	7.4
14.	+7	+6	+5	-1	France	Paris	78	81	84	101	103	74%	27.5	22.6
15.	-8	-8	-3	-6	New Zealand	Wellington	105	107	96	112	103	74%	-3.6	6.9
16.	-1	+8	+6	-2	Singapore	Singapore	88	74	79	96	99	71%	33.2	25.4
17.	+9	+5	+9	-2	United Kingdom	Edinburgh	75	76	70	95	98	70%	28.0	40.1
18.	+0	-1	+3	-1	United Kingdom	Birmingham	85	86	82	93	94	67%	8.4	14.4
19.	+12	+18	+10	+12	South Africa	Johannesburg	67	55	67	75	93	66%	68.3	39.1
20.	+10	+1	-4	-1	United States	Boston	71	80	89	90	90	64%	12.7	1.6
20.	+12	+5	+0	-4	United States	San Francisco	66	74	83	95	90	64%	21.5	8.2
22.	+1	+4	+6	-1	Switzerland	Geneva	77	73	68	83	87	62%	19.8	28.3
23.	+13	+10	+7	+0	Finland	Helsinki	61	60	66	81	83	59%	38.4	25.6
24.	-14	-6	-7	+2	Norway	Oslo	100	85	86	77	82	58%	-3.4	-4.6
25.	+16	+9	+8	+5	Hong Kong	Hong Kong	56	58	63	77	82	58%	40.9	30.4
26.	-7	-7	-2	-6	Canada	Vancouver	84	81	77	89	81	58%	-0.5	4.0
27.	-18	-11	-14	-3	Japan	Tokyo	104	87	95	80	76	54%	-13.0	-20.8
28.	-12	-19	-19	+1	United States	Chicago	87	101	104	77	75	54%	-25.4	-28.1
29.	+11	-2	-4	-4	United Arab Emirates	Dubai	56	71	73	79	75	53%	5.6	3.2
30.	-1	-1	+2	-3	Canada	Montreal	73	65	63	77	74	53%	13.7	16.6
31.	-17	-17	-16	-3	Germany	Berlin	93	92	90	77	72	52%	-21.3	-20.0
31.	-20	-21	-21	+1	Germany	Frankfurt	97	98	103	73	72	52%	-26.4	-30.2
31.	-9	-8	-8	+2	Germany	Munich	78	75	78	66	72	52%	-3.5	-7.6
34.	+5	+2	+4	+0	Belgium	Brussels	56	56	55	63	63	45%	11.5	13.5
35.	-7	-4	+16	+0	Türkiye	Istanbul	73	62	35	57	62	45%	1.0	80.2
36.	+19	+22	+19	+3	South Africa	Cape Town	35	25	32	46	62	44%	151.3	94.4
37.	+1	+4	+3	+0	Austria	Vienna	57	51	48	47	58	42%	14.4	21.3
38.	-14	+2	+3	+5	Brazil	Rio de Janeiro	76	51	47	43	50	35%	-3.0	4.8
39.	+7	+15	+11	+9	Colombia	Bogota	49	30	37	39	48	34%	58.9	31.7
40.	-20	+2	-5	-2	Brazil	Sao Paulo	81	49	58	46	47	34%	-3.4	-19.0
41.	+6	+2	+4	+3	Portugal	Lisbon	47	42	40	42	46	33%	7.7	14.5
42.	+3	+14	+12	+5	Russia	Moscow	50	28	32	39	45	32%	60.1	40.6
43.	+8	+3	+0	-1	Italy	Milan	42	40	41	43	45	32%	12.7	7.9
44.	-1	-6	-8	-3	Chile	Santiago	52	52	57	44	43	31%	-17.2	-25.0
45.	+5	-6	-3	-5	South Korea	Seoul	43	51	45	46	41	30%	-19.0	-8.1
46.	-13	-16	-12	-10	Spain	Madrid	65	63	62	48	40	29%	-35.9	-34.9
46.	+2	-1	+2	+0	Italy	Rome	46	40	39	40	40	29%	1.1	2.2
48.	-4	-4	-4	-3	Taiwan	Taipei	50	40	41	40	37	27%	-6.8	-7.8
49.	+9	+13	-10	+0	Saudi Arabia	Riyadh	27	18	48	37	37	27%	105.0	-22.7
50.	+15	+7	+7	+0	China	Beijing	17	26	27	35	35	25%	35.5	29.2
51.	+5	-2	-2	+1	Thailand	Bangkok	32	36	39	33	35	25%	-2.4	-11.3
52.	-10	-1	+0	+2	Greece	Athens	53	34	34	33	31	22%	-9.9	-8.0
53.	+1	-1	+5	+4	China	Shanghai	36	33	27	29	29	21%	-9.8	8.3
54.	+5	+1	+2	+2	Poland	Warsaw	26	29	29	30	29	21%	3.0	2.5
55.	-3	-5	-2	+4	Hungary	Budapest	41	35	33	26	29	21%	-17.3	-11.3
56.	+7	+7	+7	-5	Argentina	Buenos Aires	18	18	17	35	28	20%	55.0	60.5
57.	-32	-9	-11	-4	Qatar	Doha	75	36	40	33	27	20%	-24.3	-31.0
58.	-23	-23	-21	-3	Spain	Barcelona	62	57	57	31	26	19%	-54.5	-54.3
59.	-2	+0	+3	+1	Czech Republic	Prague	27	23	20	22	26	19%	13.3	28.6
60.	+0	+0	+0	-2	United Arab Emirates	Abu Dhabi	19	22	24	26	26	18%	18.8	9.4
61.	+7	+4	+4	+2	Mexico	Mexico City	12	15	16	16	21	15%	36.6	29.8
62.	-1	-1	-1	+0	India	Bangalore	18	21	22	16	14	10%	-33.5	-36.0
63.	+3	+4	+3	+2	India	Delhi	16	12	15	14	13	9%	7.0	-11.1
64.	-2	+2	+0	-3	Philippines	Manila	18	12	16	17	13	9%	5.7	-18.9
65.	-12	-12	-6	+1	Malaysia	Kuala Lumpur	37	31	24	12	12	9%	-60.4	-49.4
66.	+1	-2	+1	-2	Indonesia	Jakarta	12	16	14	15	12	8%	-25.2	-18.2
67.	-3	+2	+1	+0	Egypt	Cairo	17	8	13	8	8	6%	5.0	-35.2
68.	+1	+0	+1	+0	India	Mumbai	10	8	5	4	5	4%	-31.0	8.9
69.	-20	-22	-22	+0	Luxembourg	Luxembourg	44	38	39	0	0	0%	-100.0	-100.0

Source : www.numbeo.com, Deutsche Bank.



Figure 56: One-way ticket on local transport (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+6	+0	+1	+0	Switzerland	Zurich	3.83	4.18	4.10	4.87	5.71	190%	36.6	39.3
2.	+13	+10	+6	+0	Germany	Munich	2.95	3.07	3.22	4.60	4.69	156%	52.8	45.7
3.	+15	+9	+9	+4	Germany	Berlin	2.90	3.07	3.09	4.17	4.58	153%	49.2	48.2
4.	-3	-1	-2	+2	Norway	Oslo	4.98	3.97	4.10	4.32	4.50	150%	13.4	9.8
5.	-3	-3	-4	-2	Sweden	Stockholm	4.85	4.14	4.20	4.49	4.47	149%	8.0	6.4
6.	+7	+4	+6	-2	Germany	Frankfurt	3.14	3.16	3.09	4.40	4.35	145%	37.7	40.8
7.	-4	-3	-1	-2	United Kingdom	London	4.70	3.96	3.39	4.35	4.02	134%	1.5	18.6
7.	+11	-1	+2	+1	United Kingdom	Birmingham	2.90	3.60	3.15	4.02	4.02	134%	11.7	27.6
9.	-4	+5	+5	+5	Australia	Sydney	4.03	3.05	3.06	3.20	3.90	130%	27.9	27.5
10.	+0	-2	-5	-1	Netherlands	Amsterdam	3.28	3.25	3.45	3.75	3.89	130%	19.7	12.8
11.	-5	+4	+4	+1	Australia	Melbourne	3.87	3.01	3.05	3.57	3.83	128%	27.2	25.6
12.	-1	-5	-2	-1	Finland	Helsinki	3.23	3.53	3.14	3.64	3.78	126%	7.1	20.4
13.	-4	-2	+3	-3	Switzerland	Geneva	3.54	3.10	3.00	3.67	3.73	124%	20.3	24.3
14.	-10	-9	-10	-1	Denmark	Copenhagen	4.51	3.63	3.51	3.52	3.68	123%	1.4	4.8
15.	+7	+7	+4	+8	Austria	Vienna	2.47	2.50	2.69	2.72	3.43	114%	37.2	27.5
16.	+6	+9	+10	+0	Spain	Barcelona	2.47	2.38	2.35	2.94	3.03	101%	27.3	28.9
17.	+16	+9	+0	+0	United States	San Francisco	2.00	2.33	2.75	2.93	3.00	100%	28.8	9.1
17.	+10	+0	+0	+1	United States	New York	2.30	2.75	2.75	2.92	3.00	100%	9.1	9.1
19.	+12	+8	+11	-4	United Kingdom	Edinburgh	2.15	2.31	2.15	2.97	2.95	98%	27.7	37.2
20.	+8	+10	+12	+1	France	Paris	2.27	2.08	2.10	2.82	2.86	95%	37.5	36.2
20.	+4	+2	+7	+0	Belgium	Brussels	2.42	2.50	2.31	2.83	2.86	95%	14.4	23.8
22.	-2	-4	-15	-3	New Zealand	Wellington	2.82	2.57	3.29	2.85	2.85	95%	10.9	-13.4
23.	+18	+15	+15	+9	Israel	Tel Aviv-Yafo	1.67	1.60	1.64	2.29	2.67	89%	66.9	62.8
24.	-7	-5	-1	-2	Canada	Montreal	2.91	2.54	2.49	2.74	2.64	88%	3.9	6.0
25.	-13	-9	-5	-1	New Zealand	Auckland	3.20	2.95	2.59	2.71	2.56	85%	-13.2	-1.2
26.	+9	+9	+5	-1	Italy	Milan	1.85	1.70	2.11	2.50	2.52	84%	48.2	19.4
27.	-1	-3	-3	-2	United States	Chicago	2.34	2.41	2.47	2.50	2.50	83%	3.7	1.2
28.	-7	+1	+1	-1	Canada	Vancouver	2.49	2.23	2.18	2.49	2.46	82%	10.3	12.8
29.	-1	-8	-7	-1	United States	Boston	2.27	2.51	2.52	2.43	2.40	80%	-4.4	-4.8
30.	-14	-10	-5	-2	Canada	Toronto	2.92	2.52	2.38	2.43	2.36	79%	-6.3	-0.8
31.	+6	+1	+2	+0	Portugal	Lisbon	1.84	1.81	1.93	2.32	2.29	78%	26.5	18.7
31.	-17	-22	-21	-3	Ireland	Dublin	3.03	3.17	3.14	2.43	2.29	76%	-27.8	-27.1
33.	-8	+8	-12	+0	South Africa	Johannesburg	2.35	1.49	2.58	1.94	1.85	62%	24.2	-28.3
33.	+10	+12	+11	+6	Czech Republic	Prague	1.47	1.14	1.15	1.53	1.85	62%	62.3	60.9
35.	+4	-1	+0	-1	United States	Los Angeles	1.80	1.76	1.76	1.88	1.75	58%	-0.6	-0.6
36.	+2	-1	+1	+0	Italy	Rome	1.82	1.70	1.68	1.72	1.72	57%	1.2	2.4
36.	-2	-3	+0	+0	Spain	Madrid	1.87	1.79	1.71	1.80	1.72	57%	-3.9	0.6
38.	+7	+2	+1	-1	United Arab Emirates	Dubai	1.43	1.50	1.61	1.56	1.63	54%	8.7	1.2
39.	+7	+5	+4	+4	Hungary	Budapest	1.34	1.28	1.20	1.33	1.62	54%	26.6	35.0
40.	+7	+3	+2	+0	Singapore	Singapore	1.29	1.31	1.27	1.49	1.55	52%	18.3	22.0
41.	+9	+11	+5	+7	South Africa	Cape Town	1.01	0.87	1.14	1.04	1.42	47%	63.2	24.6
42.	-2	-3	-2	+0	Greece	Athens	1.73	1.59	1.57	1.40	1.37	46%	-13.8	-12.7
43.	-13	-12	-9	-2	Japan	Tokyo	2.19	1.82	1.79	1.47	1.30	43%	-28.6	-27.4
44.	+7	-2	-3	-7	Hong Kong	Hong Kong	1.00	1.41	1.42	1.56	1.28	43%	-9.2	-9.9
45.	+12	+6	-1	-1	Thailand	Bangkok	0.67	0.97	1.15	1.10	1.20	40%	23.7	4.3
46.	+6	+3	+3	+0	Poland	Warsaw	0.98	1.07	1.07	1.09	1.17	39%	9.3	9.3
47.	+6	+9	+9	+8	Colombia	Bogota	0.95	0.68	0.71	0.78	1.07	36%	57.4	50.7
48.	-40	+12	+4	-1	Saudi Arabia	Riyadh	3.62	0.53	0.97	1.07	1.06	35%	100.0	9.3
49.	+6	+5	+5	+1	Russia	Moscow	0.84	0.73	0.75	0.89	1.04	35%	42.5	38.7
50.	-8	-4	-2	+0	Brazil	Sao Paulo	1.48	1.10	1.10	0.89	1.03	34%	-6.4	-6.4
51.	+2	-3	-1	-7	South Korea	Seoul	0.95	1.09	1.06	1.10	1.01	34%	-7.3	-4.7
52.	-8	-6	-2	-3	Brazil	Rio de Janeiro	1.46	1.10	1.06	0.95	0.97	32%	-11.8	-8.5
53.	-4	-1	+9	+0	Turkiye	Istanbul	1.10	0.87	0.51	0.80	0.90	30%	3.4	76.5
54.	-6	-4	-7	-2	Chile	Santiago	1.24	1.05	1.11	0.87	0.86	29%	-18.1	-22.5
55.	+3	+3	+0	-2	Taiwan	Taipei	0.60	0.61	0.74	0.80	0.78	28%	27.9	5.4
56.	+4	+0	+0	+2	United Arab Emirates	Abu Dhabi	0.54	0.68	0.71	0.69	0.75	25%	10.3	5.6
57.	-1	-2	+1	+0	Malaysia	Kuala Lumpur	0.72	0.69	0.65	0.70	0.73	24%	5.8	12.3
58.	+3	+4	+6	-2	Argentina	Buenos Aires	0.41	0.43	0.43	0.71	0.71	24%	65.1	65.1
59.	+0	+1	+2	+4	China	Shanghai	0.58	0.53	0.53	0.50	0.59	20%	11.3	11.3
59.	+7	-1	+0	+3	China	Beijing	0.31	0.61	0.57	0.53	0.59	20%	-3.3	3.5
61.	-29	-24	-8	+0	Qatar	Doha	2.15	1.62	0.91	0.55	0.55	18%	-66.0	-39.6
62.	-2	+2	-2	-3	India	Bangalore	0.32	0.38	0.56	0.59	0.52	17%	36.8	-7.1
63.	-1	+0	+2	-4	Philippines	Manila	0.32	0.40	0.34	0.59	0.49	16%	22.5	44.1
64.	-2	+1	-2	+0	India	Delhi	0.36	0.36	0.51	0.44	0.42	14%	16.7	-17.6
65.	+3	+4	+1	+0	Egypt	Cairo	0.24	0.21	0.31	0.33	0.37	12%	76.2	19.4
66.	+1	+0	+1	-1	Mexico	Mexico City	0.29	0.29	0.29	0.33	0.35	12%	20.7	20.7
67.	-5	+0	+1	+1	Indonesia	Jakarta	0.36	0.28	0.28	0.24	0.21	7%	-25.0	-25.0
67.	+2	+1	+2	+0	India	Mumbai	0.21	0.27	0.26	0.25	0.21	7%	-22.2	-19.2
69.	-34	-41	-41	+0	Luxembourg	Luxembourg	1.85	2.27	2.24	0.00	0.00	0%	-100.0	-100.0

Source : www.numbeo.com, Deutsche Bank.



Figure 57: Taxi trip (start + 5km, normal tariff, USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+3	+0	+0	Switzerland	Zurich	28.2	26.3	29.3	31.3	32.0	231%	21.9	9.1
2.	+0	+0	+0	+0	Switzerland	Geneva	28.7	27.7	28.9	27.0	27.6	199%	-0.2	-4.6
3.	-2	+3	+2	+0	Luxembourg	Luxembourg	30.9	19.7	20.8	26.0	25.2	182%	27.9	21.4
4.	+4	+7	+14	+5	Finland	Helsinki	18.3	15.9	13.5	20.2	23.0	166%	44.6	69.8
5.	+1	-4	-1	-1	United Kingdom	London	19.4	31.0	22.2	23.4	22.2	160%	-28.3	0.1
6.	+7	+2	+3	-1	Netherlands	Amsterdam	15.4	18.6	18.1	22.1	22.2	160%	19.2	22.9
7.	+5	+3	+6	-1	Germany	Munich	15.4	16.1	15.4	22.1	22.2	160%	37.8	44.3
8.	+3	+5	+2	+0	Germany	Berlin	15.6	15.2	16.9	21.3	20.9	151%	37.8	23.9
9.	-4	-4	-3	-2	Norway	Oslo	22.4	20.6	20.2	21.4	20.6	149%	-0.2	2.2
10.	+11	+5	+11	+1	Italy	Milan	13.9	14.5	13.3	18.5	19.5	141%	34.3	46.8
11.	-7	-8	-8	-1	Japan	Tokyo	23.9	27.7	24.6	20.1	18.5	133%	-33.2	-24.8
12.	+6	-3	-5	+0	United Kingdom	Birmingham	14.3	17.5	18.9	17.3	17.1	124%	-1.9	-9.3
13.	+11	+8	+4	+0	France	Paris	12.6	13.2	13.9	16.7	16.6	120%	25.5	19.2
14.	+12	+9	+8	+2	Austria	Vienna	12.2	12.8	13.1	15.8	16.3	118%	28.1	24.4
15.	+8	+3	-1	+4	Germany	Frankfurt	13.6	13.9	14.8	14.6	16.0	116%	15.3	8.2
16.	+3	+10	+3	-1	Italy	Rome	14.3	12.1	13.4	16.2	15.5	112%	28.2	15.1
17.	-8	-10	-9	+0	Denmark	Copenhagen	17.2	19.1	18.1	15.0	15.2	110%	-20.7	-16.2
18.	-4	-4	+10	+0	United Kingdom	Edinburgh	15.4	14.8	12.0	15.0	14.7	106%	-0.3	22.8
19.	-3	-2	-4	+2	United States	San Francisco	13.8	13.9	14.5	14.3	14.3	103%	2.7	-1.3
20.	+17	+17	+14	+8	Spain	Barcelona	9.9	8.7	9.9	12.5	13.9	101%	59.5	41.5
21.	-4	-5	-5	+2	Sweden	Stockholm	14.5	14.1	14.5	13.6	13.9	100%	-1.2	-3.9
22.	+13	+0	+1	-2	United States	New York	10.3	13.2	13.1	14.4	13.9	100%	5.2	6.0
23.	-8	-11	-11	-9	Belgium	Brussels	15.0	15.5	15.6	16.3	13.2	95%	-14.7	-15.3
24.	+10	+14	+11	+13	Colombia	Bogota	11.0	8.6	9.4	10.3	13.2	95%	52.2	39.6
25.	+0	+2	+1	-1	Ireland	Dublin	12.3	11.9	12.6	13.2	13.0	94%	9.9	3.4
26.	+14	+28	+24	-4	Qatar	Doha	9.2	4.9	5.5	14.2	12.9	93%	162.1	134.9
27.	+27	+20	+18	+2	Saudi Arabia	Riyadh	5.6	6.0	6.5	12.3	12.5	90%	107.5	91.0
28.	-21	-3	-17	-1	United States	Los Angeles	19.3	12.1	16.9	12.7	12.4	89%	2.2	-26.7
29.	-9	-10	+0	-4	New Zealand	Auckland	14.0	13.9	11.5	12.9	12.2	88%	-11.8	6.2
30.	+3	+4	+1	+0	Spain	Madrid	11.1	10.2	10.6	12.1	11.6	84%	13.9	9.3
31.	-21	-7	-6	+1	Australia	Sydney	16.1	12.2	12.8	11.3	11.4	83%	-6.3	-10.7
32.	-3	-3	+0	+4	Australia	Melbourne	12.0	11.1	10.5	10.4	11.4	82%	2.8	8.6
33.	-5	-13	-13	-2	United States	Boston	12.1	13.6	13.3	11.4	11.4	82%	-16.4	-14.7
34.	+10	+8	+7	+7	Czech Republic	Prague	8.0	7.0	7.5	10.0	11.2	81%	61.7	49.5
35.	-19	-7	-8	-2	New Zealand	Wellington	14.5	11.5	12.2	11.2	11.0	79%	N/A	-10.3
36.	-9	-5	-3	-2	Canada	Vancouver	12.2	10.8	10.0	10.9	10.8	78%	0.4	8.2
37.	+13	+7	+3	+6	Hungary	Budapest	7.4	6.7	7.7	9.3	10.7	77%	58.2	39.3
38.	+3	-2	-2	+0	Hong Kong	Hong Kong	8.8	9.0	9.1	10.2	10.4	75%	15.2	14.0
39.	-7	-4	-2	-13	Canada	Montreal	11.4	9.3	9.1	12.9	10.4	75%	11.0	14.0
40.	+5	-10	-16	-5	United States	Chicago	8.0	10.9	12.9	10.5	10.3	74%	-6.2	-20.8
41.	+7	-2	-3	+5	Israel	Tel Aviv-Yafo	7.7	8.3	8.7	8.5	10.2	73%	22.3	16.4
42.	+0	-2	-3	-3	Greece	Athens	8.5	8.1	8.4	10.1	10.0	72%	23.5	20.0
43.	+3	+0	+1	-3	Portugal	Lisbon	7.9	6.8	6.8	10.1	9.6	69%	41.5	40.9
44.	+11	+9	+8	+0	Poland	Warsaw	5.4	4.9	5.1	9.2	9.3	67%	89.0	81.3
45.	-15	-12	-15	-3	Canada	Toronto	11.7	10.5	10.7	9.8	9.1	66%	N/A	-14.4
46.	+3	+12	+7	+7	South Africa	Cape Town	7.6	4.2	5.1	6.3	8.2	59%	96.4	60.2
47.	+9	+5	+8	+1	Singapore	Singapore	5.2	5.0	5.0	7.4	7.4	54%	48.8	47.3
48.	+12	+13	+10	+6	Malaysia	Kuala Lumpur	3.7	3.0	4.0	6.3	7.3	53%	144.0	81.6
49.	-18	+7	+13	+7	Russia	Moscow	11.5	4.5	3.2	5.5	7.2	52%	57.8	121.4
50.	+18	+15	+7	+0	Thailand	Bangkok	2.0	1.9	4.2	6.9	7.1	51%	263.4	69.5
51.	+8	-3	-3	+1	United Arab Emirates	Dubai	4.0	5.8	6.1	6.4	6.7	49%	15.9	10.9
52.	-5	-11	+8	-7	Argentina	Buenos Aires	7.9	7.8	3.6	9.2	6.7	48%	-14.9	86.0
53.	+0	-4	-7	-4	Taiwan	Taipei	5.7	5.8	6.2	7.3	6.6	47%	13.1	5.3
54.	-18	-4	-3	+1	Brazil	Sao Paulo	10.1	5.5	5.3	6.0	6.2	45%	13.9	17.8
55.	-3	-4	+6	+3	Türkiye	Istanbul	5.8	5.0	3.3	5.0	6.0	43%	19.4	80.7
56.	+1	+4	-7	-5	United Arab Emirates	Abu Dhabi	4.5	3.8	6.0	6.4	6.0	43%	59.2	0.3
57.	-14	-25	-14	-10	South Korea	Seoul	8.1	10.6	6.9	7.4	5.8	42%	-44.7	-15.7
58.	-19	-13	-11	-1	South Africa	Johannesburg	9.3	6.6	6.2	5.0	5.5	40%	-17.0	-10.4
59.	-8	-4	-5	+0	Brazil	Rio de Janeiro	6.9	4.9	5.1	4.7	5.5	39%	12.1	7.5
60.	-22	-14	-18	+0	Chile	Santiago	9.5	6.2	7.2	4.6	5.0	36%	-18.9	-30.4
61.	+3	+2	+3	+1	Mexico	Mexico City	2.5	2.4	2.2	3.8	4.8	34%	100.8	112.5
62.	-4	-5	-6	-1	China	Shanghai	4.3	4.3	4.2	4.2	4.4	32%	2.3	5.8
63.	-2	-4	-4	+0	China	Beijing	3.3	3.8	3.7	3.6	3.6	26%	-5.5	-2.4
64.	-1	+2	+1	+0	India	Bangalore	2.8	1.9	2.1	2.8	2.4	17%	23.0	9.8
65.	+1	+3	+2	+1	India	Delhi	2.3	1.7	1.9	2.3	2.4	17%	39.9	24.3
66.	-1	-4	-3	+1	Indonesia	Jakarta	2.4	2.4	2.5	2.3	2.2	16%	-11.5	-12.2
67.	-5	-3	-1	-2	Philippines	Manila	3.0	2.1	2.0	2.3	1.9	14%	-7.2	-2.5
68.	+1	-1	+0	+0	India	Mumbai	1.8	1.8	1.8	2.0	1.9	13%	1.1	1.6
69.	-2	+0	+0	+0	Egypt	Cairo	2.1	1.3	1.4	1.7	1.4	10%	11.0	1.4

Source : www.numbeo.com, Deutsche Bank.



Figure 58: Volkswagen Golf 1.5 (or equivalent new car) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+0	+0	+0	+0	Singapore	Singapore	104,749	92,816	83,772	146,860	156,223	441%	68.3	86.5
2.	+4	+3	+3	+2	Israel	Tel Aviv-Yafo	32,867	34,051	38,148	49,343	57,859	163%	69.9	51.7
3.	+1	-1	+1	-1	Denmark	Copenhagen	43,520	44,484	39,543	51,424	51,297	145%	15.3	29.7
4.	+33	+17	+12	-1	Türkiye	Istanbul	23,652	25,996	27,125	50,091	49,121	139%	89.0	81.1
5.	-2	-1	-2	+1	Norway	Oslo	44,189	34,373	40,115	42,403	47,014	133%	36.8	17.2
6.	+15	+7	+1	-1	Hong Kong	Hong Kong	27,062	28,309	33,556	43,573	44,627	128%	57.6	33.0
7.	+0	+13	+2	+1	Switzerland	Geneva	32,831	26,893	30,802	39,597	44,105	124%	64.0	43.2
8.	+4	+1	+4	-1	Netherlands	Amsterdam	29,737	29,891	28,697	42,172	42,339	119%	41.6	47.5
9.	-7	-6	-3	+2	Malaysia	Kuala Lumpur	44,611	34,957	35,617	38,982	41,757	118%	19.5	17.2
10.	+1	+6	+22	+2	United Kingdom	Birmingham	30,256	27,661	23,509	38,976	40,257	114%	45.5	71.2
11.	+12	+8	+4	-1	Ireland	Dublin	26,531	27,014	27,316	39,515	39,866	112%	47.6	45.9
12.	+39	+51	+33	+8	Russia	Moscow	20,408	16,673	21,102	36,021	38,959	110%	133.7	84.6
13.	+16	+4	+8	+1	United Kingdom	London	25,611	27,570	26,496	38,124	38,688	109%	40.3	46.0
13.	+23	-3	+10	-1	United Kingdom	Edinburgh	23,846	29,284	25,938	38,976	38,688	109%	32.1	49.2
15.	-6	-4	+2	-7	Switzerland	Zurich	32,302	29,041	27,074	39,597	38,514	109%	32.6	42.3
16.	-8	-4	+8	-1	Finland	Helsinki	32,407	28,395	25,343	37,797	37,435	106%	31.8	47.7
17.	+14	+17	+5	+10	Italy	Milan	24,877	22,991	26,239	35,508	37,124	105%	61.5	41.5
18.	+30	+22	+8	-2	United States	San Francisco	21,333	21,886	25,004	37,215	36,998	104%	69.0	48.0
18.	+42	+9	+9	+8	United States	Los Angeles	17,000	24,167	24,898	35,613	36,998	104%	53.1	48.6
20.	+41	+18	+15	-3	United States	Boston	16,672	22,405	23,086	36,605	36,584	103%	63.3	58.5
21.	+20	+12	+21	-3	United States	Chicago	22,663	23,133	22,302	36,249	36,531	103%	57.9	63.8
22.	+35	+31	+34	+15	South Africa	Johannesburg	18,286	19,135	19,491	31,768	36,348	103%	90.0	86.5
22.	+2	+37	+31	+21	South Africa	Cape Town	26,253	17,716	20,026	30,762	36,348	103%	105.2	81.5
24.	+15	+7	+7	+5	Italy	Rome	23,045	23,619	23,879	34,758	36,341	102%	53.9	52.2
25.	-7	+1	+14	-4	Belgium	Brussels	28,807	24,984	22,508	35,995	36,249	102%	45.1	61.1
26.	+8	+17	+14	+4	Spain	Madrid	23,990	21,655	22,451	34,496	36,055	102%	66.5	60.6
26.	+16	+18	-1	-1	Spain	Barcelona	22,473	21,435	25,103	35,802	36,055	102%	68.2	43.6
28.	+19	+1	+15	-6	United States	New York	21,667	23,769	21,894	35,964	35,458	100%	49.2	62.0
29.	-11	+1	-1	-1	France	Paris	28,807	23,688	24,425	35,120	35,368	100%	49.3	44.8
30.	-15	-24	-10	-11	Egypt	Cairo	29,596	32,032	26,520	36,142	34,602	98%	8.0	30.5
31.	+35	+27	+20	+1	Saudi Arabia	Riyadh	14,333	17,963	20,130	33,734	34,589	98%	93.6	71.8
32.	-15	-17	-19	-8	Portugal	Lisbon	28,858	27,799	28,553	35,890	34,387	97%	23.7	20.4
33.	-6	-10	-3	-2	Sweden	Stockholm	25,676	25,819	23,907	34,442	34,293	97%	32.8	43.4
34.	-18	-9	-16	-1	Germany	Munich	29,012	25,116	26,740	33,508	33,645	96%	34.0	25.8
34.	+10	+3	-23	+0	Germany	Frankfurt	22,222	22,420	19,163	33,409	33,645	96%	50.1	15.4
34.	+6	+11	+3	+1	Germany	Berlin	22,983	21,336	22,674	32,921	33,645	96%	57.7	48.4
37.	+15	-30	-27	-14	Taiwan	Taipei	20,114	31,868	29,226	35,933	33,300	94%	4.5	13.9
38.	-3	+12	+12	+16	Hungary	Budapest	23,849	19,899	20,147	26,831	32,362	91%	62.6	60.6
39.	+20	+21	+18	+2	Czech Republic	Prague	17,708	17,708	19,464	31,207	32,116	91%	81.4	65.0
40.	+15	+17	+12	-2	Poland	Warsaw	19,166	18,037	20,101	31,760	31,973	90%	77.3	59.1
41.	-31	-23	-39	-1	Thailand	Bangkok	30,945	27,291	44,096	31,351	31,555	89%	15.6	-28.4
42.	-10	-18	-13	-3	Luxembourg	Luxembourg	24,691	25,587	23,946	31,591	31,454	89%	22.9	31.4
43.	-18	-15	-24	+1	Austria	Vienna	26,189	24,103	26,629	29,116	30,320	86%	25.8	13.9
44.	-11	-5	+14	+2	Argentina	Buenos Aires	24,349	22,169	19,411	28,656	30,000	85%	35.3	54.6
45.	+8	-3	-1	-3	United Arab Emirates	Abu Dhabi	20,000	21,690	21,387	30,825	29,949	84%	38.1	40.0
46.	+0	+2	+0	+2	Greece	Athens	22,078	21,008	21,057	28,370	28,592	81%	36.1	35.8
47.	-34	+8	+16	+8	Australia	Melbourne	29,730	18,449	18,022	26,144	28,166	79%	52.7	56.3
48.	+2	+14	+11	-3	Canada	Vancouver	20,481	16,763	19,312	29,080	27,865	79%	66.2	44.3
48.	+1	+13	+14	+1	Canada	Montreal	20,631	17,022	18,050	28,034	27,865	79%	63.7	54.4
50.	-28	+4	+10	+7	Australia	Sydney	26,989	18,469	18,317	25,504	27,822	78%	50.6	51.9
51.	+11	+0	-15	+1	United Arab Emirates	Dubai	16,432	19,815	22,787	27,078	27,226	77%	37.4	19.5
52.	-9	+4	+2	+4	Canada	Toronto	22,423	18,388	19,716	26,126	26,805	75%	44.7	34.9
53.	+5	-4	-5	-17	Qatar	Doha	18,201	20,567	20,448	32,353	26,044	73%	26.6	27.4
54.	-40	-40	-40	-7	South Korea	Seoul	29,705	27,971	27,771	28,814	25,719	73%	-8.1	-7.4
55.	+12	-19	-21	-5	New Zealand	Wellington	14,286	22,479	23,090	27,401	25,639	72%	14.1	11.0
55.	-17	-20	-14	-4	New Zealand	Auckland	23,496	22,509	22,413	27,308	25,639	72%	13.9	14.4
57.	-27	-16	-19	+3	Brazil	Sao Paulo	24,930	21,818	22,577	22,171	23,328	66%	6.9	3.3
58.	-38	-6	-3	+3	Brazil	Rio de Janeiro	27,113	19,770	19,506	20,716	23,112	65%	16.9	18.5
59.	-54	-51	-51	-6	Indonesia	Jakarta	32,886	30,379	32,475	26,834	22,724	64%	-25.2	-30.0
60.	+5	+6	+5	+2	Mexico	Mexico City	14,518	13,720	16,880	19,270	22,264	63%	62.3	31.9
61.	-5	-29	+0	-3	Philippines	Manila	18,525	23,266	18,107	24,342	21,825	62%	-6.2	20.5
62.	-36	-15	-29	-3	Japan	Tokyo	25,872	21,054	23,284	23,924	21,142	60%	0.4	-9.2
63.	-18	+1	+1	+2	Colombia	Bogota	22,125	16,638	17,101	18,066	20,740	58%	24.7	21.3
64.	-36	-18	-15	+0	China	Beijing	25,664	21,136	20,249	18,511	19,117	54%	-9.6	-5.6
64.	-10	-42	-17	+2	China	Shanghai	19,407	25,874	20,764	17,796	19,117	54%	-26.1	-7.9
66.	-3	-1	+0	-3	Chile	Santiago	15,389	15,271	16,664	18,589	18,685	53%	22.4	12.1
67.	+2	+1	+1	+0	India	Mumbai	13,914	11,970	12,461	17,339	15,954	45%	33.3	28.0
68.	-4	+1	+1	+0	India	Delhi	14,522	11,113	12,381	17,096	12,031	34%	8.3	-2.8
68.	+0	-1	-1	+1	India	Bangalore	14,002	12,523	13,433	14,397	12,031	34%	-3.9	-10.4

Source : www.numbeo.com, Deutsche Bank.



Figure 59: Monthly internet cost (60 Mbps or higher, unlimited data) (USD)

Rank	vs 2012	vs 2016	vs 2019	vs 2025	Economy	City	2012	2016	2019	2025	2026	Relative to NY	10-yr cum change (%)	7-yr cum change (%)
1.	+2	+1	+1	+0	United Arab Emirates	Dubai	74.0	89.7	102.0	98.1	94.7	123%	5.6	-7.1
2.	+2	-1	-1	+0	United Arab Emirates	Abu Dhabi	73.0	89.7	105.5	96.6	89.7	117%	-0.1	-15.0
3.	-2	+0	+0	+0	Qatar	Doha	100.7	71.9	91.1	89.1	88.6	115%	23.1	-2.7
4.	+9	+5	+0	+0	United States	Boston	52.8	57.2	69.8	74.7	84.5	110%	47.6	21.1
5.	+25	+10	+5	+1	United States	Los Angeles	41.5	53.0	64.1	72.8	78.9	103%	48.8	23.1
6.	+22	+4	+5	+3	United States	New York	42.3	56.6	63.7	64.4	76.8	100%	35.6	20.5
7.	+11	+5	-2	+0	United States	San Francisco	46.8	54.8	69.4	70.7	71.3	93%	30.2	2.8
8.	-2	+19	-1	-3	Saudi Arabia	Riyadh	67.5	42.1	66.5	72.9	69.5	91%	65.3	4.6
9.	+23	+4	+0	-1	United States	Chicago	40.3	54.7	64.9	66.7	64.3	84%	17.5	-0.9
10.	+12	+6	+8	+4	Switzerland	Zurich	45.9	51.9	53.7	56.1	59.2	77%	13.9	10.2
11.	+3	+11	+6	+1	Luxembourg	Luxembourg	52.5	46.1	54.0	58.1	59.1	77%	28.2	9.4
12.	+11	+5	+1	+3	Ireland	Dublin	44.7	50.0	60.8	55.1	57.5	75%	15.0	-5.5
13.	-5	-7	-1	-3	Switzerland	Geneva	64.7	60.1	61.7	58.2	56.6	74%	-5.8	-8.3
14.	+1	+0	+8	+7	Australia	Melbourne	50.9	53.6	49.8	50.4	56.4	73%	5.2	13.3
15.	-4	+3	+10	+2	Australia	Sydney	56.2	49.6	49.6	53.0	55.9	73%	12.8	12.8
16.	+8	+7	+5	-5	Belgium	Brussels	44.7	46.0	49.8	58.2	55.1	72%	19.8	10.6
17.	+20	+16	+10	-1	Netherlands	Amsterdam	35.8	36.3	47.0	53.7	53.4	70%	47.3	13.7
18.	-6	-14	-3	-5	New Zealand	Wellington	54.0	60.4	55.1	56.4	52.9	69%	-12.5	-4.0
19.	+12	+1	-3	-1	Canada	Vancouver	41.4	47.8	54.1	53.0	52.5	68%	10.0	-2.9
20.	-10	+6	-1	+2	Norway	Oslo	56.2	43.0	51.8	50.1	52.5	68%	22.1	1.4
21.	+45	+27	+18	+2	Germany	Berlin	20.5	27.0	34.8	49.7	50.8	66%	88.2	46.1
22.	-15	-17	-8	-2	New Zealand	Auckland	66.0	60.3	55.4	52.1	50.4	66%	-16.5	-9.0
23.	+19	+21	+9	+1	United Kingdom	Birmingham	31.8	28.6	38.9	47.3	48.8	64%	70.8	25.3
24.	-8	+0	-1	-5	Canada	Toronto	48.9	44.5	49.7	52.5	47.3	62%	6.3	-4.7
25.	+20	+21	+15	+2	Germany	Frankfurt	30.9	27.9	34.7	45.1	46.7	61%	67.1	34.4
26.	+15	+19	+5	+2	Germany	Munich	33.8	28.1	39.0	44.8	43.3	56%	54.4	11.0
27.	+17	+8	+7	-1	United Kingdom	Edinburgh	31.4	33.2	38.0	45.1	43.3	56%	30.3	14.0
28.	+12	+13	+19	-3	Sweden	Stockholm	34.5	29.8	32.4	46.1	42.3	55%	42.0	30.5
29.	-27	-22	-21	+6	South Africa	Johannesburg	91.1	57.9	66.4	36.5	41.8	55%	-27.8	-37.0
30.	-4	-5	-2	+0	Canada	Montreal	43.1	44.2	46.6	42.5	41.6	54%	-5.9	-10.7
31.	-26	-23	-25	+1	South Africa	Cape Town	68.9	57.8	68.1	39.2	41.2	54%	-28.6	-39.5
32.	+14	-1	-2	-3	United Kingdom	London	30.7	37.6	40.5	42.5	40.7	53%	8.2	0.4
33.	+16	+7	+0	-2	Denmark	Copenhagen	29.6	30.3	38.7	39.8	40.3	52%	32.8	4.2
34.	-7	-6	-10	-1	Spain	Barcelona	42.9	41.8	49.7	37.5	36.8	48%	-12.1	-26.0
35.	+17	+24	+18	+9	Israel	Tel Aviv-Yafo	28.4	21.9	27.0	31.3	36.0	47%	64.7	33.3
36.	-4	-6	-10	+2	Spain	Madrid	40.3	40.3	48.2	34.5	35.3	46%	-12.3	-26.7
37.	+11	+5	-1	-1	Portugal	Lisbon	29.8	29.8	36.8	35.9	35.0	46%	17.5	-5.0
38.	+19	+16	+14	+5	Mexico	Mexico City	24.4	23.4	27.8	31.7	34.5	45%	47.3	24.1
39.	+21	+10	-2	-5	Austria	Vienna	22.9	26.6	35.6	37.0	34.1	44%	28.4	-4.2
40.	+7	-1	+1	+5	Italy	Rome	30.3	30.8	33.6	31.2	32.3	42%	5.1	-3.8
41.	+13	+12	-3	-2	Greece	Athens	25.9	24.2	35.2	33.3	31.8	41%	31.5	-9.6
42.	-7	-4	+2	-5	France	Paris	37.0	30.8	33.0	35.3	31.8	41%	3.3	-3.6
43.	-22	+4	+3	+6	Colombia	Bogota	45.9	27.5	32.7	26.9	30.9	40%	12.4	-5.5
44.	-1	-7	-1	+3	Italy	Milan	31.5	31.5	33.1	29.1	30.1	39%	-4.6	-9.2
45.	-6	-13	-16	-5	Japan	Tokyo	34.9	37.1	44.1	32.8	30.0	39%	-19.0	-31.9
46.	-9	-27	-1	-4	Argentina	Buenos Aires	35.8	48.5	32.7	32.1	29.1	38%	-39.9	-10.9
47.	+6	+4	+10	-1	Taiwan	Taipei	26.8	25.3	22.5	29.7	27.3	36%	7.8	21.3
48.	-31	-19	+0	+3	Malaysia	Kuala Lumpur	48.0	41.0	29.2	25.0	26.4	34%	-35.7	-9.7
49.	-13	-13	-7	-8	Singapore	Singapore	35.9	32.7	33.4	32.5	26.3	34%	-19.6	-21.4
50.	+6	+7	+4	-2	Finland	Helsinki	24.7	22.6	25.5	27.7	25.9	34%	14.5	1.5
51.	+16	+16	+13	+4	Hungary	Budapest	20.3	15.6	14.7	20.8	25.3	33%	62.5	72.7
52.	-27	-41	-32	-2	Philippines	Manila	43.5	55.3	50.9	25.5	23.8	31%	-57.0	-63.3
53.	+6	+11	+6	-1	Czech Republic	Prague	23.0	17.7	22.0	22.8	22.8	30%	28.4	3.7
54.	-34	-2	-4	+4	Brazil	Rio de Janeiro	46.0	25.3	28.6	19.0	21.7	28%	-14.0	-24.0
55.	-21	-21	-20	-1	Chile	Santiago	40.1	35.3	36.9	21.3	21.3	28%	-39.7	-42.2
56.	-27	+2	-7	+3	Brazil	Sao Paulo	41.6	22.3	28.7	18.5	21.2	28%	-5.0	-26.0
57.	+5	-14	-2	-1	Hong Kong	Hong Kong	22.6	28.9	24.4	20.6	21.0	27%	-27.2	-13.9
58.	-39	-8	-7	-5	Indonesia	Jakarta	46.0	26.5	28.5	21.3	18.6	24%	-29.7	-34.6
59.	+4	+6	-3	+1	Thailand	Bangkok	22.4	17.5	22.6	18.5	18.0	23%	2.5	-20.6
60.	+8	+8	+5	+1	Poland	Warsaw	17.7	11.9	14.6	17.7	17.8	23%	50.5	22.5
61.	+4	-5	-3	+4	South Korea	Seoul	20.6	22.8	22.2	20.3	17.6	23%	-22.8	-20.9
62.	-12	-2	+0	+0	Turkiye	Istanbul	29.6	20.2	16.2	15.4	16.5	22%	-18.1	1.7
63.	-12	-1	+3	+0	China	Beijing	29.2	18.9	14.1	13.3	14.5	19%	-23.7	2.6
64.	-55	-43	-4	+0	Egypt	Cairo	57.5	47.1	17.9	12.6	12.6	16%	-73.3	-29.8
65.	-10	+1	-4	+0	China	Shanghai	25.2	16.9	16.3	11.5	10.7	14%	-36.7	-34.6
66.	-5	-3	-3	+0	India	Bangalore	22.9	18.3	15.4	10.4	10.5	14%	-42.8	-32.0
67.	+2	+2	+2	+2	Russia	Moscow	12.2	5.9	7.4	8.5	10.0	13%	70.4	35.9
68.	-4	-7	+0	-1	India	Mumbai	21.8	19.5	11.5	9.4	9.5	12%	-51.1	-17.2
69.	-11	-14	-2	-1	India	Delhi	24.4	23.2	11.7	9.2	7.3	9%	-68.7	-37.8

Source : www.numbeo.com, Deutsche Bank



Appendix

Figure 60: Foreign Exchange Rates

Rank based on change from 2016	Economy	Y2012	Y2016	Y2019	Y2025	Y2026	cum change since 2012 (%)	cum change since 2016 (%)	cum change since 2019 (%)	cum change since 2025 (%)
1	Israel	0.2557	0.2591	0.2801	0.2969	0.3359	31.4	29.7	19.9	13.2
2	Switzerland	1.0542	1.0232	1.0243	1.2591	1.2366	17.3	20.8	20.7	-1.8
3	Czech Republic	0.0496	0.0409	0.0447	0.0476	0.0470	-5.2	14.9	5.1	-1.1
4	Mexico	0.0748	0.0541	0.0520	0.0531	0.0572	-23.5	5.7	10.0	7.7
5	Thailand	0.0317	0.0285	0.0326	0.0308	0.0301	-5.1	5.7	-7.7	-2.3
6	Poland	0.2987	0.2534	0.2679	0.2770	0.2655	-11.1	4.8	-0.9	-4.2
7	Singapore	0.7905	0.7412	0.7391	0.7859	0.7725	-2.3	4.2	4.5	-1.7
8	Austria	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Belgium	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Finland	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	France	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Germany	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Greece	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Ireland	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Italy	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Netherlands	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Portugal	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Spain	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
8	Luxembourg	1.2660	1.1074	1.1373	1.1763	1.1414	-9.8	3.1	0.4	-3.0
20	Denmark	0.1704	0.1489	0.1524	0.1577	0.1527	-10.4	2.6	0.2	-3.1
21	Taiwan	0.0335	0.0310	0.0323	0.0342	0.0314	-6.2	1.3	-2.7	-8.2
22	United States	1.0000	1.0000	1.0000	1.0000	1.0000	0.0	0.0	0.0	0.0
23	United Arab Emirates	0.2723	0.2723	0.2723	0.2723	0.2723	0.0	0.0	0.0	0.0
24	United Kingdom	1.5706	1.3264	1.2698	1.3714	1.3250	-15.6	-0.1	4.3	-3.4
25	Qatar	0.2746	0.2746	0.2733	0.2744	0.2743	-0.1	-0.1	0.4	0.0
26	Saudi Arabia	0.2666	0.2666	0.2667	0.2666	0.2662	-0.2	-0.2	-0.2	-0.2
27	Hong Kong	0.1289	0.1289	0.1280	0.1274	0.1275	-1.1	-1.1	-0.4	0.1
28	Malaysia	0.3147	0.2482	0.2420	0.2376	0.2448	-22.2	-1.4	1.2	3.1
29	China	0.1574	0.1504	0.1456	0.1396	0.1473	-6.4	-2.0	1.2	5.6
30	Australia	1.0238	0.7447	0.7021	0.6572	0.6912	-32.5	-7.2	-1.6	5.2
31	Canada	0.9837	0.7705	0.7637	0.7338	0.7038	-28.4	-8.7	-7.8	-4.1
32	Hungary	0.0044	0.0035	0.0035	0.0029	0.0032	-27.6	-8.8	-8.9	8.9
33	South Africa	0.1225	0.0681	0.0710	0.0564	0.0610	-50.2	-10.4	-14.1	8.1
34	Sweden	0.1445	0.1179	0.1077	0.1055	0.1030	-28.7	-12.7	-4.4	-2.4
35	Colombia	0.0006	0.0003	0.0003	0.0002	0.0003	-48.0	-14.8	-6.4	19.2
36	Norway	0.1678	0.1193	0.1172	0.0991	0.1009	-39.9	-15.4	-13.9	1.8
37	Russia	0.0308	0.0156	0.0158	0.0128	0.0127	-58.8	-18.7	-19.6	-0.4
38	New Zealand	0.8013	0.7123	0.6717	0.6090	0.5673	-29.2	-20.4	-15.5	-6.8
39	Philippines	0.0237	0.0212	0.0195	0.0178	0.0163	-31.3	-23.1	-16.4	-8.2
40	South Korea	0.0009	0.0009	0.0009	0.0007	0.0006	-26.0	-25.6	-25.4	-12.6
41	Indonesia	0.0001	0.0001	0.0001	0.0001	0.0001	-47.3	-26.2	-21.1	-9.3
42	Chile	0.0020	0.0015	0.0015	0.0011	0.0011	-45.7	-28.1	-26.4	1.0
43	India	0.0180	0.0148	0.0145	0.0117	0.0106	-41.2	-28.7	-27.1	-9.4
44	Japan	0.0125	0.0097	0.0093	0.0069	0.0061	-50.9	-36.6	-33.7	-11.3
45	Brazil	0.4977	0.3112	0.2598	0.1841	0.1937	-61.1	-37.8	-25.4	5.2
46	Egypt	0.1650	0.1125	0.0599	0.0202	0.0203	-87.7	-81.9	-66.1	0.8
47	Türkiye	0.5530	0.3480	0.1726	0.0251	0.0214	-96.1	-93.8	-87.6	-14.6
48	Argentina	0.2209	0.0665	0.0235	0.0008	0.0007	-99.7	-99.0	-97.1	-18.9

Source : Bloomberg Finance LP, Deutsche Bank.
Note: All rates are USD in terms of unit of local currency as on 30 June each year.



Figure 61: Changes in Consumer Price Indices

Rank based on change from 2016	Economy	cum change since 2012 (%)	cum change since 2016 (%)	cum change since 2019 (%)
1.	Argentina	31,708.3	9,946.3	3,752.3
2.	Türkiye	1,439.0	1,033.4	661.1
3.	Egypt	503.7	299.0	146.5
4.	Russia	148.3	74.5	56.5
5.	Hungary	77.1	73.2	58.9
6.	Colombia	93.4	63.2	46.4
7.	Mexico	77.8	55.9	35.3
8.	Brazil	108.5	55.8	40.0
9.	Czech Republic	58.6	54.4	44.1
10.	Poland	54.0	54.0	46.6
11.	South Africa	89.1	51.2	32.1
12.	India	95.7	51.0	35.5
13.	Chile	72.5	49.5	39.8
14.	Philippines	55.1	40.1	26.1
15.	Austria	45.2	37.7	29.9
16.	United Kingdom	44.1	37.5	28.3
17.	Netherlands	40.1	35.7	28.4
18.	United States	40.3	34.2	26.0
19.	Norway	47.6	34.0	24.8
20.	New Zealand	37.8	33.3	26.8
21.	Belgium	38.7	33.1	25.7
22.	Germany	36.0	31.5	25.1
23.	Australia	41.3	30.9	24.0
24.	Sweden	34.2	30.9	23.8
25.	Indonesia	60.4	28.6	16.7
26.	Canada	34.9	27.9	20.8
27.	Luxembourg	31.1	27.9	20.8
28.	Spain	27.6	27.1	21.5
29.	Italy	26.3	24.3	20.4
30.	Portugal	25.8	24.1	20.4
31.	France	26.5	24.0	18.6
32.	Greece	18.7	22.7	19.8
33.	Ireland	22.8	22.1	19.9
34.	South Korea	27.0	21.7	17.2
35.	Finland	26.0	21.5	17.8
36.	Denmark	22.7	21.3	18.3
37.	Singapore	22.1	19.3	17.4
38.	Israel	19.7	18.8	16.5
39.	Malaysia	28.3	16.9	10.8
40.	Hong Kong	34.3	16.8	9.3
41.	Saudi Arabia	26.0	15.3	15.8
42.	Taiwan	18.2	15.0	11.5
43.	Japan	18.5	14.1	11.9
44.	China	22.5	13.1	6.0
45.	United Arab Emirates	21.3	10.9	7.6
46.	Thailand	14.5	10.7	8.1
47.	Qatar	22.5	9.2	9.7
48.	Switzerland	6.2	8.1	6.2

Source : Haver Analytics, Deutsche Bank.



Figure 62: Relative Price Levels as Implied by PPP (US = 100)

Rank 2025	Economy	Y1980	Y1985	Y1990	Y1995	Y2000	Y2005	Y2010	Y2015	Y2020	Y2025
1.	Switzerland	95.6	79.2	128.2	141.0	94.2	106.4	145.4	123.2	124.8	118.6
2.	Israel	N/A	49.0	92.2	100.9	98.5	82.7	107.5	98.9	116.6	115.7
3.	Luxembourg	N/A	N/A	N/A	N/A	68.6	87.0	109.2	93.8	102.5	100.7
4.	United States	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
5.	Australia	110.8	74.3	100.9	93.1	70.6	98.2	153.8	106.0	110.2	95.2
6.	Denmark	104.3	78.4	124.2	123.3	87.8	110.3	128.6	101.3	104.6	93.1
7.	Netherlands	N/A	N/A	N/A	N/A	76.8	98.7	110.0	85.6	91.5	90.6
8.	Sweden	151.5	96.5	155.9	139.5	96.1	109.3	131.9	103.5	102.7	90.1
9.	United Kingdom	117.3	78.2	122.8	103.5	99.3	114.5	106.2	100.3	89.2	89.7
10.	Ireland	N/A	N/A	N/A	N/A	75.0	101.9	101.1	87.3	94.8	89.0
11.	Finland	N/A	N/A	N/A	N/A	85.4	101.6	114.2	95.6	98.1	88.5
12.	Norway	111.9	88.4	119.9	105.0	91.0	126.0	165.4	110.6	111.7	87.3
13.	Austria	N/A	N/A	N/A	N/A	74.3	90.8	102.0	84.2	89.2	86.6
14.	New Zealand	73.2	50.5	81.7	86.0	57.5	88.1	108.1	95.0	102.3	86.1
15.	Germany	N/A	N/A	N/A	N/A	76.7	91.4	99.7	81.0	86.5	84.8
16.	Canada	93.1	83.7	105.6	87.5	80.0	104.1	122.1	87.2	94.1	84.8
17.	Belgium	N/A	N/A	N/A	N/A	74.6	91.9	103.1	83.3	87.6	83.2
18.	France	N/A	N/A	N/A	N/A	79.4	98.1	109.1	85.1	84.7	76.8
19.	Hong Kong SAR	83.5	62.5	79.6	102.7	98.4	75.8	71.4	77.0	79.3	71.0
20.	Italy	N/A	N/A	N/A	N/A	65.0	84.0	94.9	75.5	77.3	70.8
21.	Spain	N/A	N/A	N/A	N/A	58.9	80.8	92.4	69.3	74.1	69.7
22.	Portugal	N/A	N/A	N/A	N/A	52.9	70.3	79.6	62.2	66.4	66.5
23.	Czech Republic	N/A	N/A	N/A	37.5	33.6	51.9	67.5	49.7	56.6	62.9
24.	Greece	N/A	N/A	N/A	N/A	57.9	75.3	89.1	64.4	64.8	62.4
25.	Singapore	N/A	50.3	59.5	76.1	57.9	54.5	71.1	61.7	63.2	61.5
26.	Japan	123.3	105.1	141.9	172.6	139.8	113.2	141.9	89.3	97.5	60.4
27.	United Arab Emirates	N/A	N/A	45.8	42.0	46.6	56.1	72.7	63.3	56.9	60.3
28.	Qatar	N/A	N/A	N/A	32.1	47.9	70.6	72.8	66.7	62.4	57.9
29.	Hungary	N/A	N/A	N/A	36.4	32.7	52.6	60.2	45.6	47.8	57.2
30.	Mexico	56.0	25.3	37.0	32.6	52.8	57.3	58.9	47.0	49.2	57.1
31.	South Korea	N/A	59.2	86.6	103.5	65.2	84.0	76.8	73.3	76.3	54.5
32.	Poland	N/A	N/A	N/A	42.6	39.7	51.9	60.0	44.8	45.4	53.2
33.	Chile	N/A	N/A	38.5	56.1	42.7	54.8	73.2	52.6	57.7	52.5
34.	China	N/A	48.8	36.5	36.6	36.6	39.5	56.0	60.8	61.5	48.6
35.	Saudi Arabia	N/A	N/A	31.1	29.2	31.4	38.9	48.8	44.5	51.2	46.8
36.	Brazil	N/A	N/A	N/A	60.3	42.3	49.4	90.8	51.2	43.6	46.5
37.	South Africa	66.4	28.7	51.7	58.9	38.5	59.2	73.6	37.7	49.4	44.8
38.	Taiwan	N/A	59.4	84.8	86.0	68.5	58.7	56.6	48.8	52.2	43.5
39.	Colombia	N/A	N/A	N/A	44.4	35.4	42.3	59.7	38.9	37.1	41.1
40.	Argentina	N/A	N/A	N/A	71.5	64.0	31.3	48.7	46.0	32.0	39.1
41.	Türkiye	N/A	0.0	34.5	33.2	36.3	49.7	58.1	40.6	28.4	38.8
42.	Russia	N/A	N/A	N/A	24.9	17.9	34.3	52.8	31.4	31.2	37.3
43.	Malaysia	62.0	49.8	41.4	47.1	36.8	38.5	50.8	36.4	37.9	33.4
44.	Philippines	N/A	N/A	N/A	43.4	29.7	31.8	43.7	41.6	40.5	32.3
45.	Thailand	N/A	32.1	36.7	40.5	25.1	27.3	40.0	33.8	38.0	32.0
46.	Indonesia	N/A	N/A	N/A	31.8	18.4	25.5	43.0	33.2	34.1	28.3
47.	India	51.3	39.3	34.1	25.4	23.3	26.7	36.7	29.8	27.8	22.3
48.	Egypt	N/A	N/A	N/A	22.2	22.2	17.8	26.5	31.9	23.5	15.9

Source : IMF, WEO, Bloomberg Finance LP, Deutsche Bank calculations

Note: The PPP conversion rates as published by the IMF have been adjusted with actual foreign exchange rates (as on December 31 each year) to derive the implied price levels. Also note that some of the above numbers may not match those published in previous editions exactly due to data revisions.



Figure 63: Relative Price Levels as Implied by PPP (US = 100), ranks from 1980

Economy	Y1980	Y1985	Y1990	Y1995	Y2000	Y2005	Y2010	Y2015	Y2020	Y2025
Switzerland	8	6	3	2	7	6	3	1	1	1
Israel	N/A	16	10	9	4	21	13	8	2	2
Luxembourg	N/A	N/A	N/A	N/A	19	18	10	11	7	3
United States	7	2	9	10	2	10	18	7	9	4
Australia	5	9	8	11	18	12	2	3	4	5
Denmark	6	7	4	4	9	4	6	5	5	6
Netherlands	N/A	N/A	N/A	N/A	13	11	9	15	14	7
Sweden	1	3	1	3	6	5	5	4	6	8
United Kingdom	3	8	5	7	3	2	14	6	15	9
Ireland	N/A	N/A	N/A	N/A	15	8	17	13	12	10
Finland	N/A	N/A	N/A	N/A	10	9	8	9	10	11
Norway	4	4	6	5	8	1	1	2	3	12
Austria	N/A	N/A	N/A	N/A	17	16	16	17	16	13
New Zealand	11	13	13	13	27	17	12	10	8	14
Germany	N/A	N/A	N/A	N/A	14	15	19	19	18	15
Canada	9	5	7	12	11	7	7	14	13	16
Belgium	N/A	N/A	N/A	N/A	16	14	15	18	17	17
France	N/A	N/A	N/A	N/A	12	13	11	16	19	18
Hong Kong SAR	10	10	14	8	5	23	30	20	20	19
Italy	N/A	N/A	N/A	N/A	22	19	20	21	21	20
Spain	N/A	N/A	N/A	N/A	24	22	21	23	23	21
Portugal	N/A	N/A	N/A	N/A	28	26	24	27	24	22
Czech Republic	N/A	N/A	N/A	26	40	34	32	32	31	23
Greece	N/A	N/A	N/A	N/A	26	24	23	25	25	24
Singapore	N/A	14	15	15	25	32	31	28	26	25
Japan	2	1	2	1	1	3	4	12	11	26
United Arab Emirates	N/A	N/A	17	24	31	30	29	26	30	27
Qatar	N/A	N/A	N/A	31	30	25	28	24	27	28
Hungary	N/A	N/A	N/A	28	41	33	33	36	36	29
Mexico	14	21	20	30	29	29	36	34	35	30
South Korea	N/A	12	11	6	21	20	25	22	22	31
Poland	N/A	N/A	N/A	23	34	35	34	37	37	32
Chile	N/A	N/A	19	19	32	31	27	30	29	33
China	N/A	17	22	27	37	39	39	29	28	34
Saudi Arabia	N/A	N/A	25	33	42	40	42	38	33	35
Brazil	N/A	N/A	N/A	17	33	37	22	31	38	36
South Africa	12	20	16	18	35	27	26	42	34	37
Taiwan	N/A	11	12	14	20	28	38	33	32	38
Colombia	N/A	N/A	N/A	21	39	38	35	41	42	39
Argentina	N/A	N/A	N/A	16	23	44	43	35	44	40
Türkiye	N/A	22	23	29	38	36	37	40	46	41
Russia	N/A	N/A	N/A	35	48	42	40	47	45	42
Malaysia	13	15	18	20	36	41	41	43	41	43
Philippines	N/A	N/A	N/A	22	43	43	44	39	39	44
Thailand	N/A	19	21	25	44	45	46	44	40	45
Indonesia	N/A	N/A	N/A	32	47	47	45	45	43	46
India	15	18	24	34	45	46	47	48	47	47
Egypt	N/A	N/A	N/A	36	46	48	48	46	48	48

Source : IMF, WEO, Bloomberg Finance LP, Deutsche Bank calculations

Note: The PPP conversion rates as published by the IMF have been adjusted with actual foreign exchange rates (as on December 31 each year) to derive the implied price levels. Also note that some of the above numbers may not match those published in previous editions exactly due to data revisions.



Appendix 1

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Deutsche Bank
Research

Q3 2026 Corporate Bank Quarterly

A guide to Macro, Markets, Geopolitics & Tech for CFOs and Corporates

July 13, 2026

**Marion Laboure | Camilla Siazon | Luke Templeman |
Adrian Cox | Helen Belopolsky | Miha Hribernik | Jim Reid**

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
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Introduction: Macro, Markets, Geopolitics & Tech

A guide for CFOs & Corporates



We're pleased to share our July 2026 Guide for CFOs and Corporates, providing essential insights into macroeconomics, markets, geopolitics and technology. This report offers critical analysis and perspectives to help corporate clients navigate today's complex economic environment.

Key highlights:

- **Macroeconomic outlook:** Resumption of US-Iran attacks in the Strait has reignited fears of an oil-driven energy shock. So far, the global economy has proved to be resilient.
- **Market analysis:** Corporate earnings growth is expected to be strong and credit spreads have proved resilient to several shocks. Brent crude averages \$80/bbl in 2026 (\$75 in Q3).
- **Geopolitics:** Corporates continue to cite geopolitics as a key risk, although other 'normal' risks are high priorities. Q3/Q4 events to watch: expiry of US global 10% Section 122 tariffs in Jul, expiration of original US-Iran ceasefire in Aug, US midterms in Nov.
- **Artificial Intelligence:** AI adoption is increasing, but corporates cite challenges in scaling across business functions. Disruption worries are relatively latent. The costs around AI capex demand are shifting into the spotlight.
- **Monetary policy:** The energy shock is influencing central bank trajectories, with both the Fed and ECB expected to hike rates. Momentum around policymakers considering tokenization of assets is growing.

Upcoming key macro events for rest of 2026

07 \ July			08 \ August			09 \ September		
01	IE	Ireland holds EU Council presidency	Early	CH	China Politburo meeting to set H2 economic policy direction	06-20	DE	German regional elections
07-08		NATO summit in Ankara				08	US	UNGA convenes in New York, starts next UN Secretary-General selection process
17	UK	Andy Burnham expected to become PM	Mid	US	60-day US-Iran ceasefire extension expires	09-10	EZ	ECB decision
22-23	EZ	ECB decision				12-13	IN	India hosts BRICS Summit
24	US	US global 10% Section 122 tariffs expire	29	IS	Iceland referendum on EU accession talks	13	SE	Swedish general election
28-29	US	Federal Reserve meeting				15-16*	US	Federal Reserve meeting
30	UK	Bank of England decision				17	UK	Bank of England decision
30-31	JN	Bank of Japan decision				17-18	JN	Bank of Japan decision
Late	CH	China Politburo meeting to set H2 economic policy direction				18-20	RU	Russian legislative election
						24	CH	President Xi expected to visit US after May China-US Summit
						27	FR	France Senate election
						30	US	Fiscal year ends- multiple agencies' funding expires

10 \ October			11 \ November			12 \ December		
04 & 25	BR	Brazilian general election	03	US	US midterm elections (House & Senate)	-	CH	China Central Economic Work Conference
27-28	US	Federal Reserve meeting	05	UK	Bank of England decision	08-09*	US	Federal Reserve meeting
28-29	EZ	ECB decision	09-20	COP31	COP31 in Türkiye	14-15	G20	G20 summit in US
29-30	JN	Bank of Japan decision	18-19	CH	APEC Economic Leaders' meeting in China	16-17	EZ	ECB decision
						17	UK	Bank of England decision
						17-18	JN	Bank of Japan decision

DB forecasts



GDP Growth (%)				
	2024	2025	2026F	2027F
Global	3.5	3.5	3.1	3.3
US	2.8	2.1	2.3	2.2
Eurozone	0.9	1.5	0.5	1.1
Germany	-0.5	0.2	0.5	1.3
France	1.4	0.9	0.5	0.9
Italy	0.6	0.7	0.4	0.7
Spain	3.5	2.8	1.8	1.8
Japan	-0.2	1.1	0.7	0.7
UK	1.0	1.3	1.0	1.2
China	5.0	5.0	4.7	4.5
India	7.2	7.5	6.7	7.2
Russia	4.9	1.0	0.6	1.3
Asia (ex Japan)	5.4	5.5	5.2	5.1
EEMEA	3.5	3.1	2.4	3.4
LatAm	2.6	1.8	1.9	1.9
EM	4.6	4.5	4.2	4.4

- *CPI (%) forecasts are period averages. **Germany CPI local definition taken.
EEMEA: Czech Rep., Israel, Egypt, Hungary, Poland, Russia, Saudi Arabia, South Africa, Turkey, UAE
- LATAM (ex Venezuela): Argentina, Brazil, Chile, Colombia, Mexico, Peru
- ASIA (ex Japan): China, HK, India, Indonesia, Korea, Malaysia, Philippines, Singapore, Sri Lanka, Taiwan, Thailand, Vietnam
- Current prices as of July 10, 2026.
- *China rates are 7d OMO rates.
- **Gold, Brent and WTI forecasts are for quarterly periods.

Source: Deutsche Bank Research, Bloomberg Finance LP.

CPI Inflation, YoY* (%)				
	2025	2026F	2027F	2028F
US	2.7	3.2	2.2	2.6
Eurozone	2.1	2.8	2.3	2.2
Germany**	2.3	2.9	2.7	2.4
Japan	3.2	2.1	2.6	2.1
UK	3.4	3.2	2.7	2.1
China	0.1	1.3	1.6	-
India	2.2	4.6	4.2	
Russia	8.7	5.5	5.0	

Central Bank policy rate (%)				
	Current	Dec 26F	Jun 27F	Dec 27F
US	3.625	4.125	4.125	4.125
Eurozone	2.25	2.50	2.50	2.50
UK	3.75	3.75	3.50	3.25
Japan	1.00	1.25	1.75	1.75
China*	1.40	1.40	1.40	1.40
India	5.25	5.75	6.25	6.25

Key market metrics				
	Current	Dec 26F	Jun 27F	Dec 27F
US 10Y Yield (%)	4.56	4.80	4.75	-
EUR 10Y Yield (%)	3.06	3.20	3.20	3.20
EUR/USD	1.14	1.20	-	1.25
USD/JPY	162	150	-	140
USD/CNY	6.78	6.55	6.50	6.30
S&P 500	7,575	8,000	-	-
**Oil WTI (USD/bbl)	71	67	67	67
**Oil Brent (USD/bbl)	76	70	70	70
**Gold	4,119	4,800	5,200	5,400



01. Macroeconomics



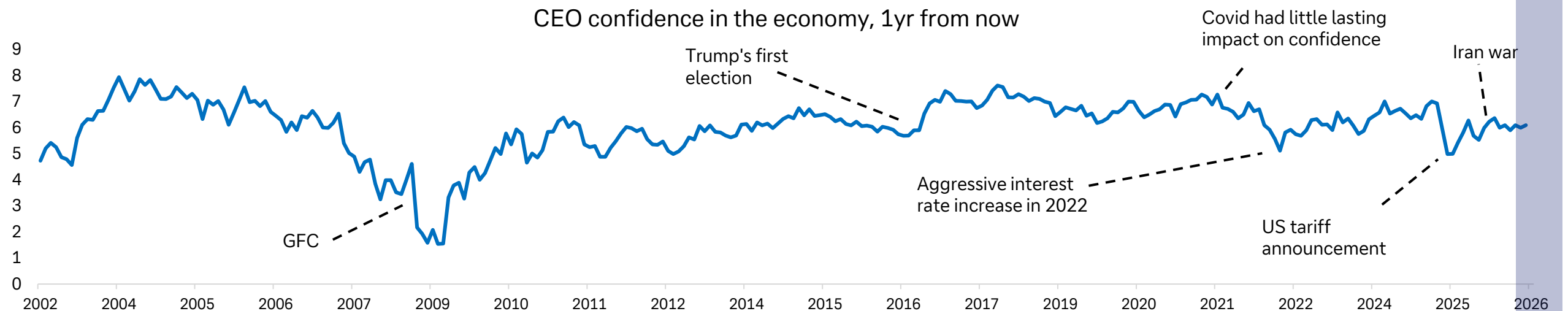
2026 Global growth

The situation in the Middle East conflict adds uncertainty, but growth remains far more resilient than pre-war forecasts assumed for a Strait of Hormuz closure



View	Rationale	View	Rationale
 Growth at 3.1% in 2026 and 3.3% in 2027.	— Main risk is if the war continues/worsens, oil market impact from prolonged SoH closure.	 Growth at 4.7% in 2026, 4.5% in 2027.	— Softer domestic demand and policy adjustments offset by strong trade and anticipated reflation.
 2.3% GDP in 2026 and 2.2% in 2027.	— Growth to stay resilient in 2026, supported by easy credit conditions, fiscal policy, and AI-driven capex. A more substantial oil shock can weaken real disposable income and tighten financial conditions.	EM 4.2% growth for the EM region in 2026, 4.4% in 2027.	— In Asia, growth momentum is resilient, with fiscal balance sheets actively working to dampen the transmission of the energy-price shock.
 Euro area growth 0.5% in 2026 and 1.1% in 2027. German GDP at 0.5% in 2026 and 1.3% in 2027.	— Europe remains materially impacted by the Middle East crisis and energy shock.		— CEEMEA's economic growth proved relatively durable through the conflict, though 2026 growth was cut to 2.4%. Sharp reversal in oil prices during MOU has lifted inflation and allowed financial conditions to normalize.

CEO confidence edged back up in Q2, as a ceasefire was announced – Q3 might be a different story



Source: Deutsche Bank Research, Bloomberg Finance LP.

Three key risks for the global economy

Collapse of the Iran ceasefire is renewing stagflation concerns, while cybersecurity risks have become a key concern for corporates

1. Collapse of Iran ceasefire

- Renewed flare-ups in the Middle East are driving oil prices higher again, compounded by energy infrastructure attacks between Russia and Ukraine. Corporates remain concerned with economic conditions [vs six months ago](#).
- Long term, however, the global economy seems more resilient to geopolitical shocks.

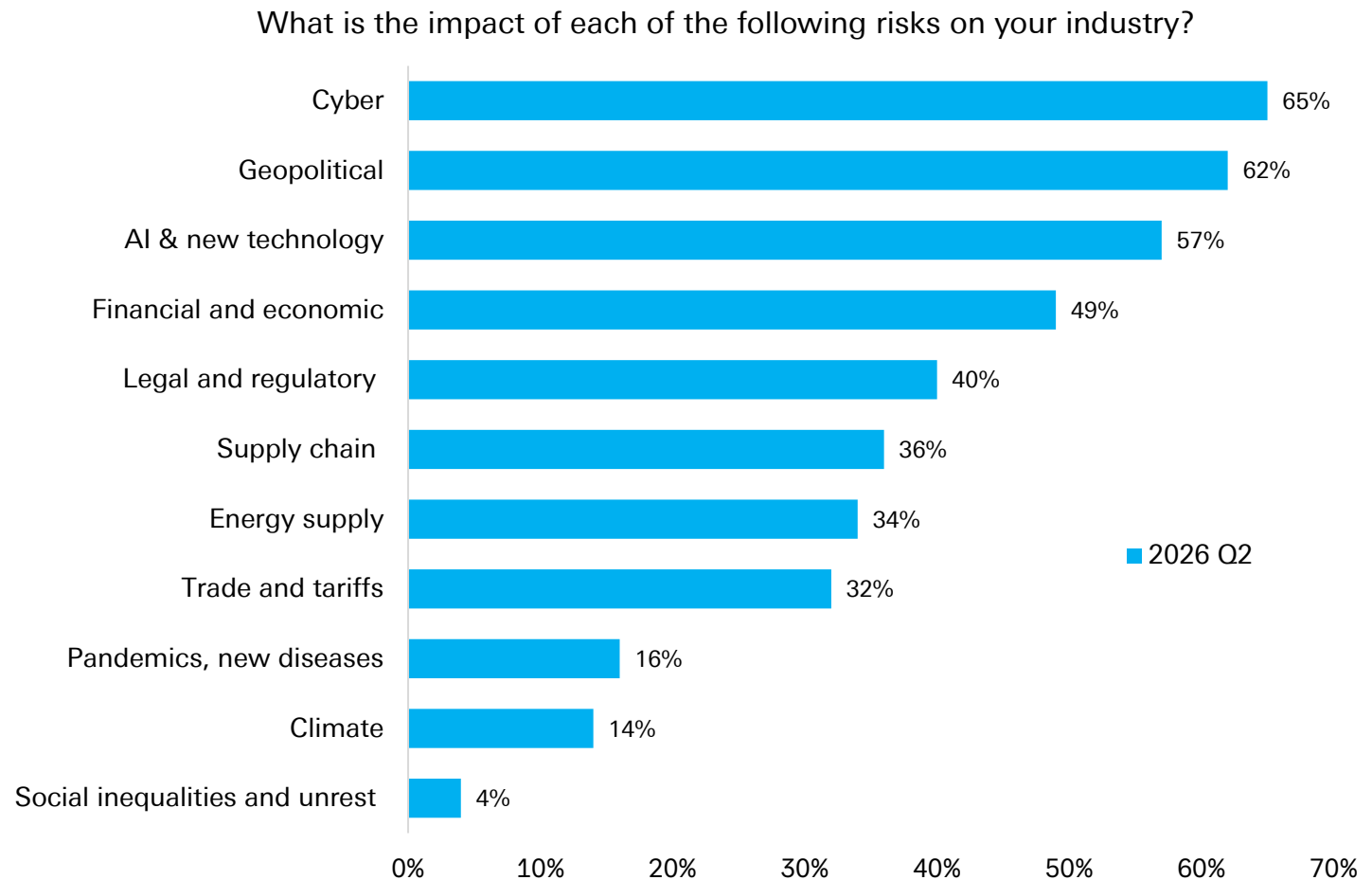
2. AI disruption

- Concerns that AI will disrupt the labor force have yet to be borne out by the data. CEOs see little impact on jobs in the near term, while companies struggle to diffuse adoption.
- Central banks are becoming more attuned to AI as a source of upwards price pressure.

3. Hawkish central bank trajectories

- Expectations for higher US rates will grow as markets absorb the apparent demise of the ceasefire. Rates expectations had already decoupled from oil prices even during the ceasefire.
- Expectations for ECB rate hikes have also shifted due to the energy shock.

Cyber risk has emerged as the top concern among CEOs



Source: Deutsche Bank Research, The Conference Board Measure of CEO Confidence in collaboration with The Business Council

Central banks

The inflation shock has altered central bank trajectories



Fed

- +25bps hike in Sep and Dec, bringing Fed funds rate to 4.1% YE, on hold in 2027. 50bps of rate cuts pencilled in for H1 2028.
- Core PCE sticky at 3.2% in 2026 and 2.5% in 2027.



BoE

- On hold at 3.75% through 2026, with rate cuts to resume from spring next year.



BoJ

- Quarterly +25bp rate hikes starting in Oct, 1.75% policy rate by Apr 2027.



PBoC

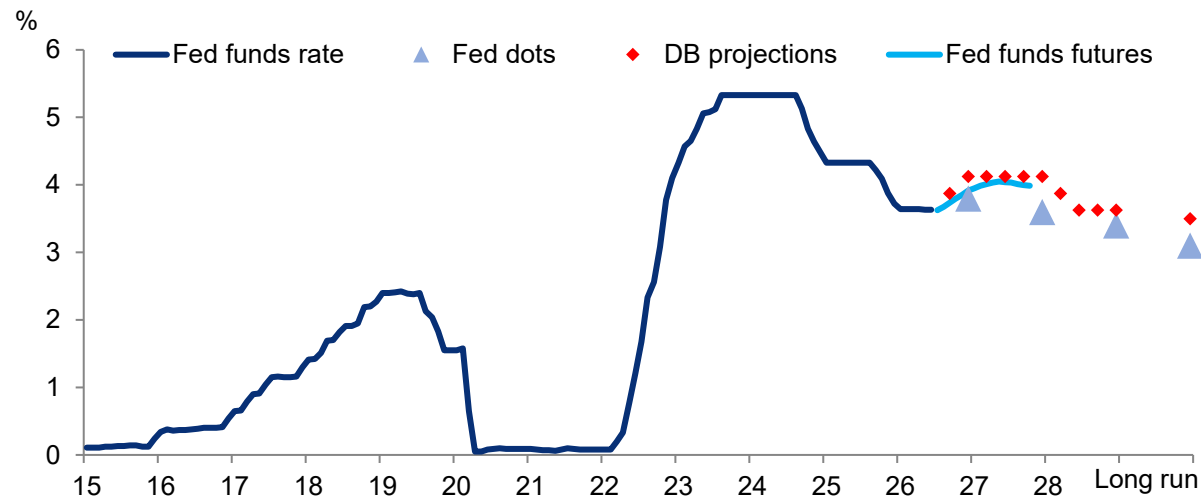
- Hold the 7-day reverse repo at 1.40% through 2026.



ECB

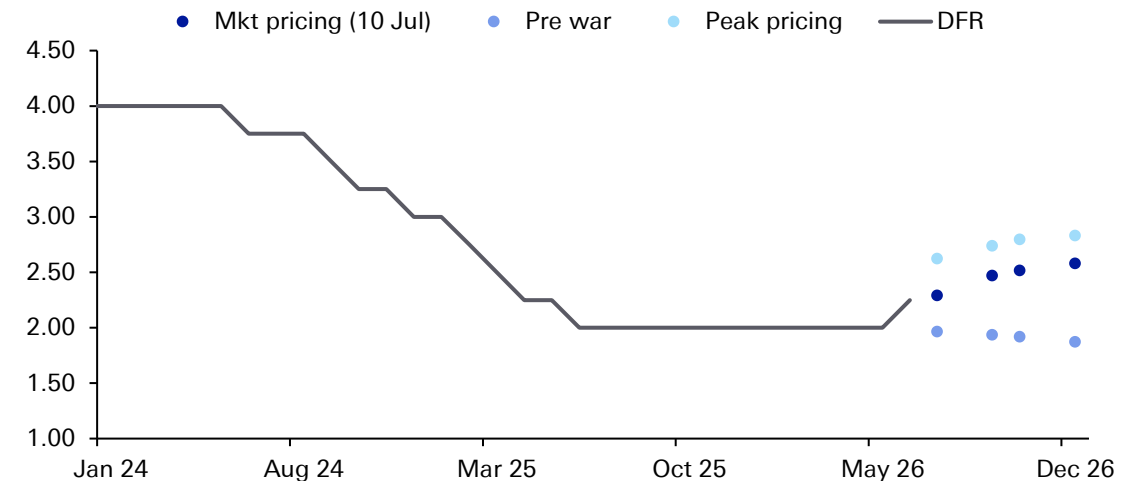
- A further +25bp hike in Sep to 2.50% year-end.
- ECB staff “baseline scenario” consistent with 2.75% rates, ‘milder’ scenario consistent with 2.50% rates.

Fed: a December hike if fully priced



Source: FRB, Bloomberg Finance LP, Haver Analytics, Deutsche Bank Research. Updated July 2 2026.

ECB: 2.50% YE



Source: Deutsche Bank Research, ECB. Updated July 10 2026.



02. Geopolitics



Midterm Moderation: Will Trump Turn Dove?



Trump has a strong incentive not to shock markets or stoke inflation ahead of the vote in November. But markets should expect more policy volatility, not less, as we move into the second half of Trump's term. An administration looking to demonstrate strength through quick wins may pursue other issues more aggressively. Investors should watch rising US pressure on Cuba and the political relationship with Europe as possible sources of nearer-term movement.

The **July expiry of Section 122 tariffs** and their likely replacement by Section 301 levies on dozens of trading partners should keep pressure on countries to strike bilateral deals favourable to the US, spanning trade, investment, industrial and technology concessions. In addition, Section 232 tariffs could expand into new categories.

While **EU-US trade relations are more stable**, there remains a tail risk that a transatlantic security or sovereignty dispute (e.g. renewed American claims on Greenland) spills over into the economic relationship. At the 7-8 July NATO summit, Trump reinforced his expectations that European allies carry more of the security burden and his disappointment in their lack of support for US efforts in Iran. Trump also reiterated his continued interest in acquiring Greenland.

Trump will likely renew his push for a ceasefire in the **Russia-Ukraine war**. In early July, Trump suggested “we’re getting much closer than people realize.” While a ceasefire this year is far from guaranteed, any deal brokered by the Trump administration could offer Russia sanctions relief, access to frozen assets and an expectation of an accelerated Ukrainian accession to the EU.

What happens after the midterms actually matters more. **Trump may become a “foreign policy president”**, focused on cementing his legacy. This will include: embedding existing tariff and investment deals with key US partners; deepening US leadership in the Americas; creating the conditions for a pull-back of American resources from Europe; and broadening US efforts to reduce rare-earths and critical-minerals dependence on China.



Source: [Library of Congress](#) on [Unsplash](#)

Other geopolitical flashpoints



Key topics to monitor during 2026-Q3

- **Iran:** The 60-day limit for US-Iranian MOU (focusing on Iran's nuclear programme and deal implementation) will expire on 20 AUG but negotiations will almost certainly run past the initial 60 days. A significant escalation on over the past week saw the US launch strikes on hundreds of targets across Iran and reimpose sanctions on Iranian oil sales in retaliation for Iran striking ships in the Strait of Hormuz. Iran has fired a number of waves of strikes against Gulf nations. While Trump has suggested that the ceasefire is over, we retain our assessment that both parties are incentivized to find a near-term resolution as kinetic attacks will not drive a resolution. That said, we should prepare for more volatility in the coming months and a persistent threat to the Strait of Hormuz. The Middle East will likely remain a focus for the rest of Trump's term, even if markets move on.
- **China-US Relations:** Post-summit strategic stability could deliver tactical trade and investment gains and slow the pace of escalation into late 2026. But corporates and investors should still prepare for a start-stop cycle of engagement and restriction as technology, tariffs and economic security remain central to national security agendas in Beijing and Washington. A key watchpoint will be Xi's expected visit to the US on 24 September, just over a month before the US midterm election set to be dominated by voter cost-of-living concerns. This could incentivize the Trump administration to seek an economic agreement at the summit. An extension of last **October's Busan agreement** and an operationalization of the potentially commerce-boosting **US-China Board of Trade** – which China hopes could include a ten-fold increase in tariff-free trade – are possible deliverables.
- **What next for Europe?** An unresolved war in Ukraine, greater geopolitical fragmentation, and a less predictable transatlantic relationship. At the same time, the geopolitical landscape is presenting myriad opportunities – increased defence investment and a chance to build more strategic autonomy. There is a growing recognition that the EU needs to overcome its often slow, consensus-driven decision-making processes to respond more effectively to evolving geopolitical dynamics. The key question will be whether Europe can convert these pressures into coordinated action, or whether internal divisions stymie real progress.

For more on geopolitical flashpoints to watch out for, read our recent publications here: [Geopolitical Outlook 2026: At the Midpoint](#), [This Month in Geopolitics: July 2026](#), [Geopolitics beyond Iran](#), [The Dragon and the Eagle: Decoding China-US Ties After Beijing](#)

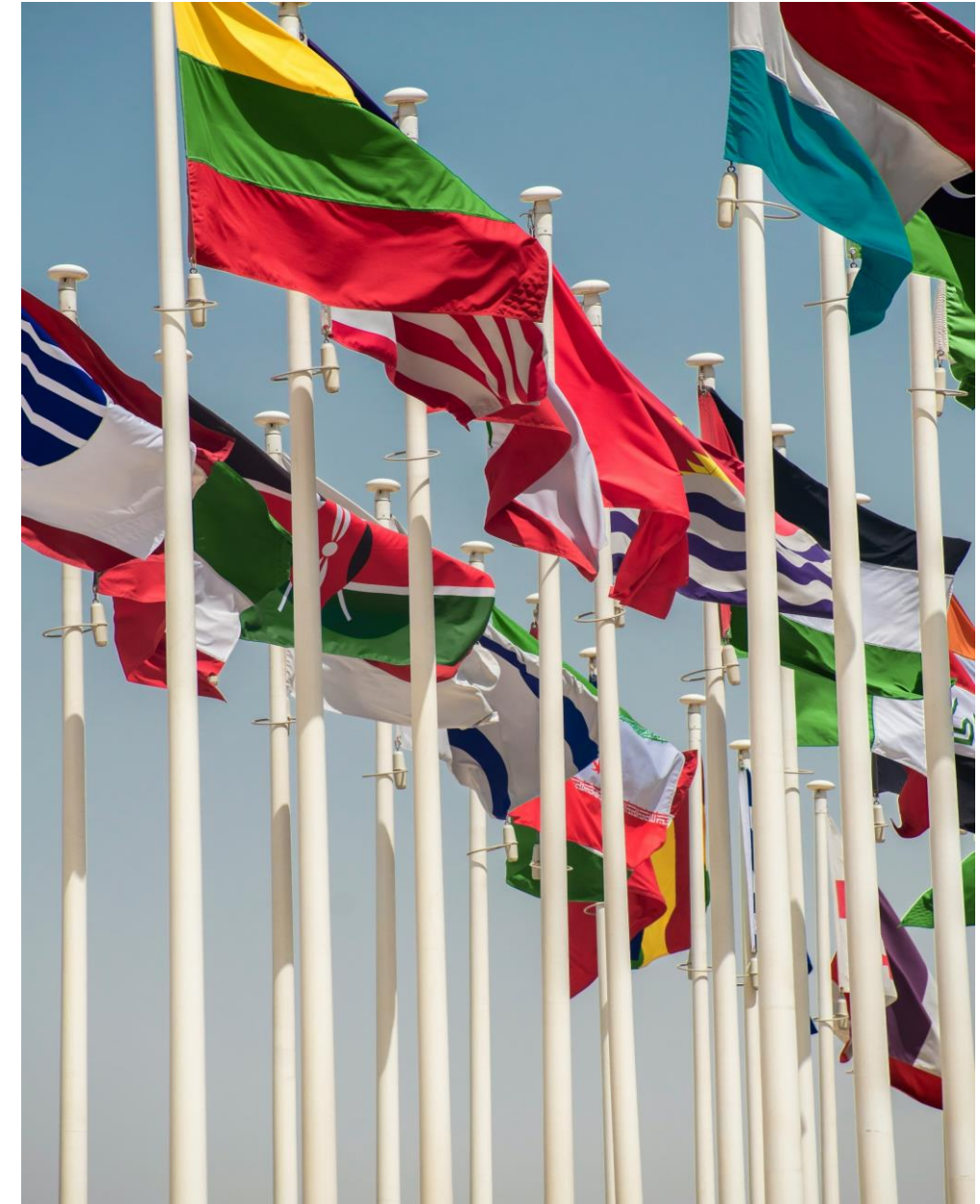


Source: IStock ugurhan

Selected geopolitical events in July - September



- **Mid-July:** **UK:** Andy Burnham expected to formally become Prime Minister.
- **24 July:** The **US** global 10% Section 122 tariffs are set to expire.
- **Late July/early Aug:** The **Chinese** Politburo meeting will likely set the economic policy direction for the rest of the year.
- **Mid-August:** **US/Iran:** The 60-day ceasefire extension between the US and Iran is set to expire.
- **29 Aug:** **Iceland** referendum on EU accession talks.
- **8 Sep:** The UN General Assembly in **New York** will convene and initiate the process to choose the next UN Secretary-General.
- **12-13 Sep:** **India** will host the BRICS Summit in New Delhi.
- **13 Sep:** **German** state and local elections.
- **13 Sep:** **Swedish** general election.
- **18-20 Sep:** **Russian** legislative election.
- **24 Sep:** **China's** President Xi is expected to visit the United States following the China-US Summit in Beijing in May.
- **30 Sep:** Key **US** government funding deadline. Failure to pass appropriations or a stopgap measure could trigger a government shutdown.



Source: [aboodi vesakaran](#) on [Unsplash](#)



03. Markets



Commodity markets

Oil was trading around \$73/bbl, but prices are now rising back towards \$80/bbl. Several transport- and materials-exposed companies have pointed to 2027 for normalization (see [here](#)).

Oil: Under the ceasefire, we had placed Brent at \$75/bbl (Q3'26) and \$70/bbl (Q4'26).

- Oil prices fell faster than expected during the MOU-interim, although outlook is uncertain for now with the reignition of Strait conflict.
- Crisis has revealed need for broader reserves beyond crude oil and gas, where disruption can rapidly cascade through supply chains and consumer prices (i.e. refined fuels, petrochemicals, fertilizers, food, metals).
- Companies should now be expected to maintain larger inventories, build in more redundancy, spend more on insurance, hedging, and transport, and accept duplicated routes and suppliers.

Gold: revised base case to reach \$4,800/oz in Q4

- Tighter Fed monetary policy and resilient US macro data has pushed gold prices lower.

Latest oil price forecasts

USD	2025	Q3 2026	Q4 2026	2026	2027	2028
WTI (bbl)	64.9	72.0	67.0	76.1	67.0	73.3
% change		-23.4%	-17.3%	-10.5%	-11.8%	-2.8%
Brent (bbl)	68.20	75.0	70.0	80.0	70.0	76.4
% change		-24.2%	-18.6%	-11.1%	-12.5%	-2.7%

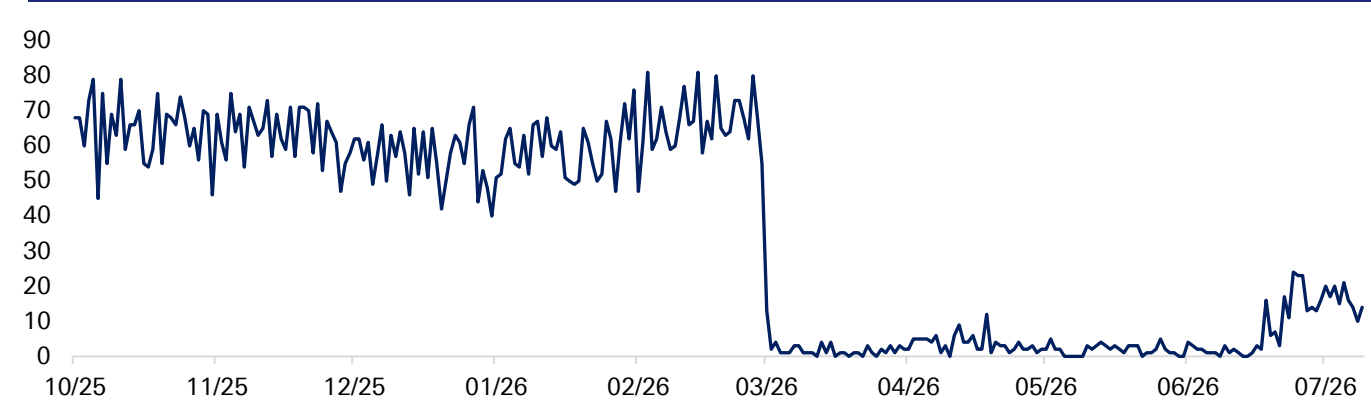
Reports: [Hsueh On Oil: Swinging to surplus - Deutsche Bank Research](#) and [Precious Special Report: Hawks drive out bulls - Deutsche Bank Research](#)

Brent crude oil prices (\$/bbl) have gone from around \$100/bbl to just \$73/bbl during MOU



Source: Deutsche Bank Research, Bloomberg Finance LP

The number of vessels going through the Strait of Hormuz is still at a fraction of its pre-conflict level



Source: Deutsche Bank Research, Bloomberg Finance LP.

Rates & Credit markets

Despite higher rates, financial conditions are relatively accommodative, with credit spreads remaining tighter than expected. Potential area for concern lies in private credit



Private credit is pricing in more stress now that full Q1 data is in. There is potential for greater AI-disruption stress on software and other adjacent subsectors.

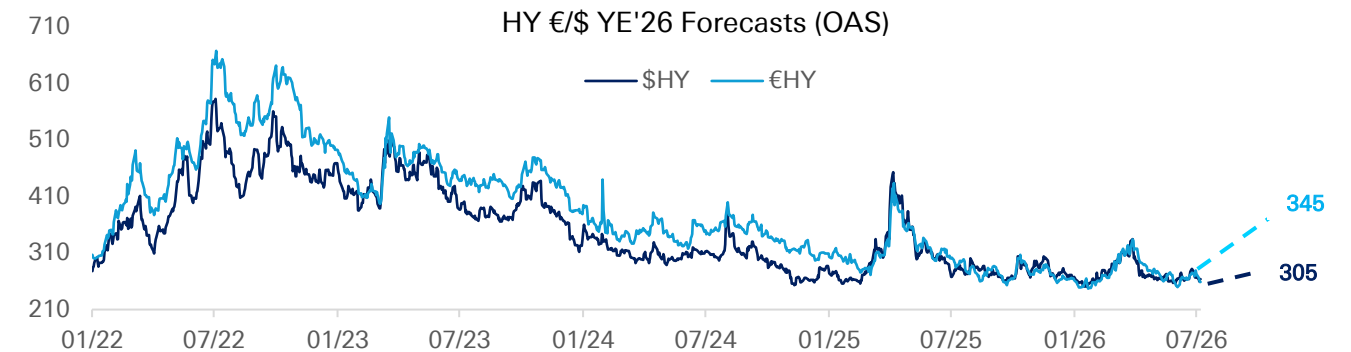
- Software spreads are near Covid (March 2020) levels, but ex-software spreads are trading much better. Market is pricing software as an isolated issue, although there is a risk the stress will spread more broadly over the next 6-12 months.
- Credit-spread forecasts slightly lower throughout 2026, with \$IG 82bps; \$HY 305bps. €IG 95bps; €HY 345bps.

Raised our YE-2026 targets for the 10yr US Treasury (4.80%) and 10yr Bund (3.20%)

- Rationale: Expectations of more Fed hikes are now baked into our forecast, while front loading of ECB hikes in response to the energy shock has pushed our yield forecasts higher across the curve.
- Downside risks: AI tails from either a bubble or significant impact on the market, or a concrete end to the Iran War leading to materially lower oil prices.
- Upside risks: Demand for additional fiscal and capex globally or a waning in foreign demand for US Treasuries that further lifts term premia.

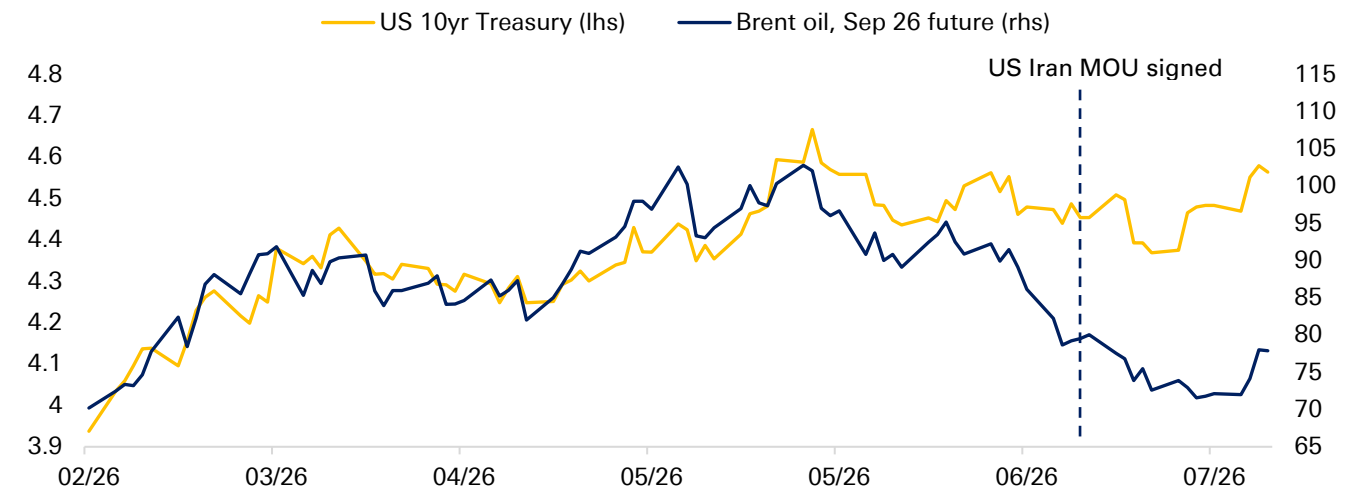
Reports: [Global Fixed Income Weekly: Strategy Update - Deutsche Bank Research](#) and [Fixed Income Blog: EUR Rates Forecast revision - Bund to 3.2% - Deutsche Bank Research](#) and [IG & HY Strategy: Global Default Monitor: Software woes, BDC selling putting pressure on Loans - Deutsche Bank Research](#)

Outlook until year-end calls for more dispersion and some spread widening



Source: Deutsche Bank Research, Bloomberg Finance LP.

The ten year treasury yield initially traded in a tight correlation with oil prices, but that's completely opened up in the last month. Yields have stayed high even during the ceasefire



Source: Deutsche Bank Research, Bloomberg Finance LP.

Foreign exchange markets

Hawkish FOMC is supporting the dollar

Hard to argue for significant dollar weakness

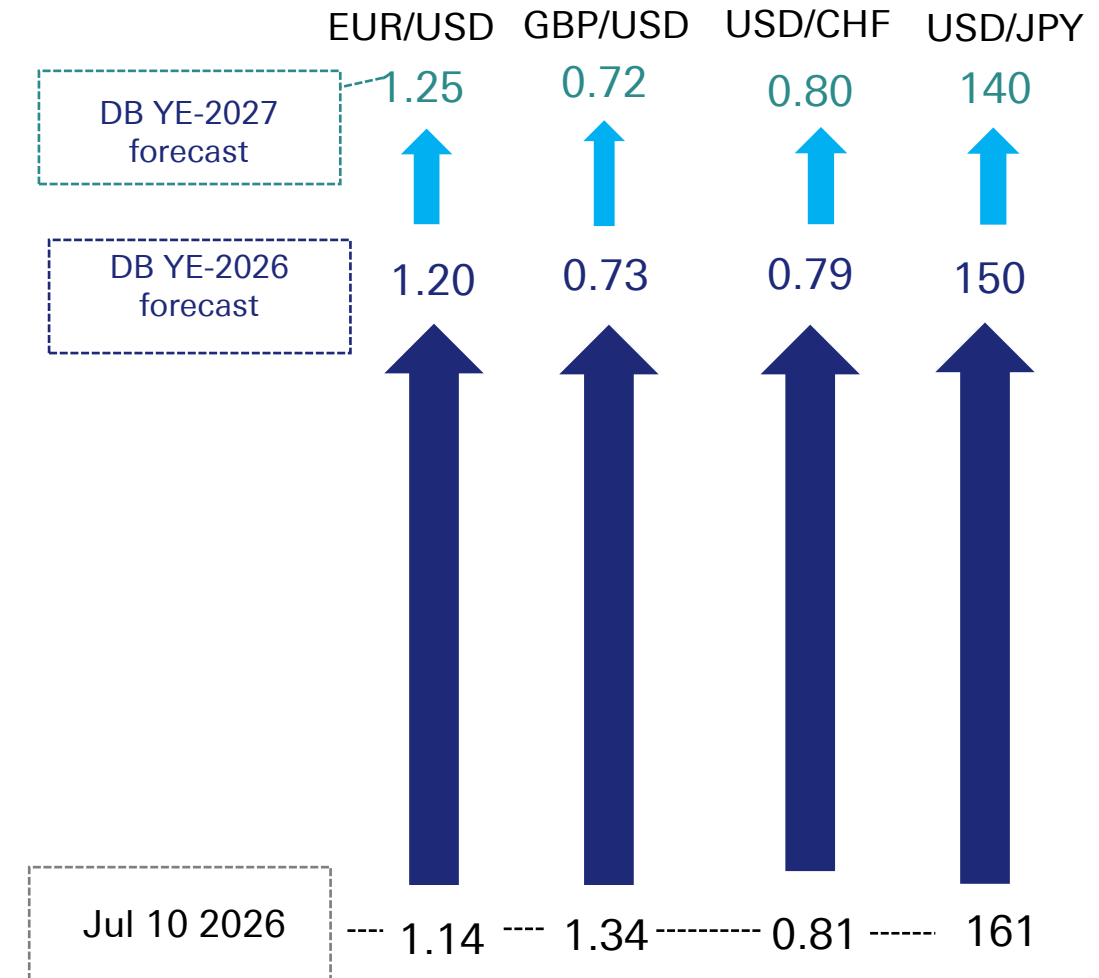
- The dollar has modestly benefited from rising energy prices, while equity-led inflows into the US have also started to recover.
- Near-term, the second-order effects of the Iran War is dollar supportive.
- However, the level of front-end US Treasury yields, the interest-rate differential with the rest of the world and relative shape of the US yield curve are not sending dollar bullish signals.

Currency outlook

- **EUR/USD:** 1.20 by YE (prevs 1.25). Escalation in Iran and more hawkish repricing of the Fed could push EUR/USD below 1.15.
- **EUR/JPY:** Hawkish signals from BoJ officials, firm recent data, and potential for repatriation sees the yen rising against the Euro. Despite yen relatively flat against the Euro since our [May blueprint](#), we still expect the yen to strengthen in H2, with two potential avenues: either the BoJ hikes rates to 2% more quickly than expected, or the government undertakes a more concerted effort to shift domestic investment back onshore.
- **USD/CNH:** Renminbi's appreciation will continue. Domestic reflation, strong real-GDP growth, surpluses, and a greater use of RMB in cross-border transactions/internationalisation should all act as significant tailwinds.

Read: [FX Blog: one thing - Deutsche Bank Research](#)

Currency movements and DB forecasts for YE-2026 and 2027



Source: Deutsche Bank Research, Bloomberg Finance LP.



04. Technology



Artificial intelligence

Companies are committing growing sums of capital to strengthen their AI capabilities



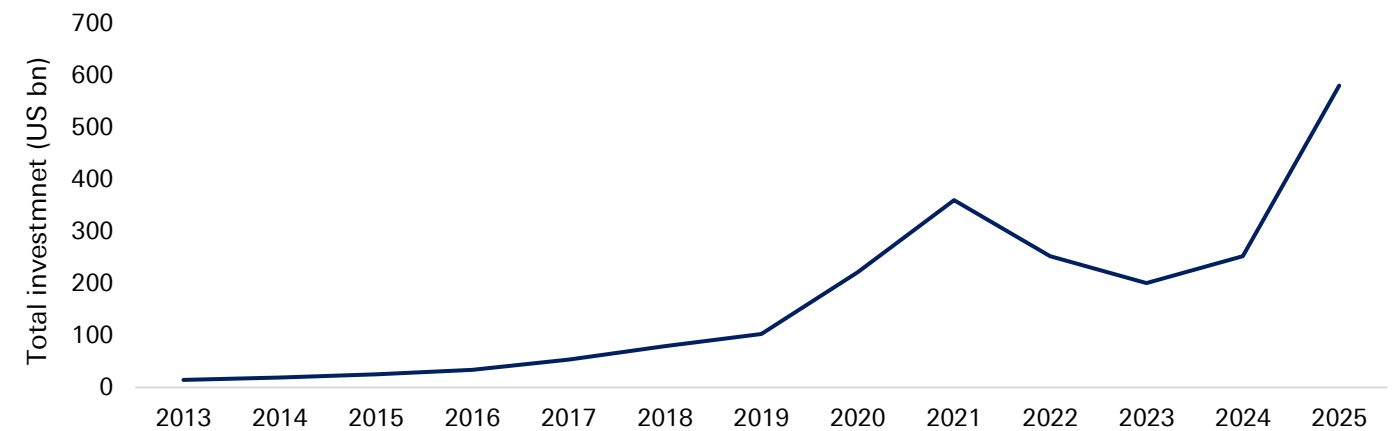
2025 was a breakout year for corporate AI investment

- Global AI corporate investment hit \$582bn in 2025, +130% from 2024, mostly driven by Gen AI. (Source: [McKinsey/Stanford](#)).
- Private financing (VC/PE) was the largest share of activity.
- The US continues to lead private AI investment; China's true total AI spending is understated, as government guidance funds deploy a significant amount.

The costs for AI are growing

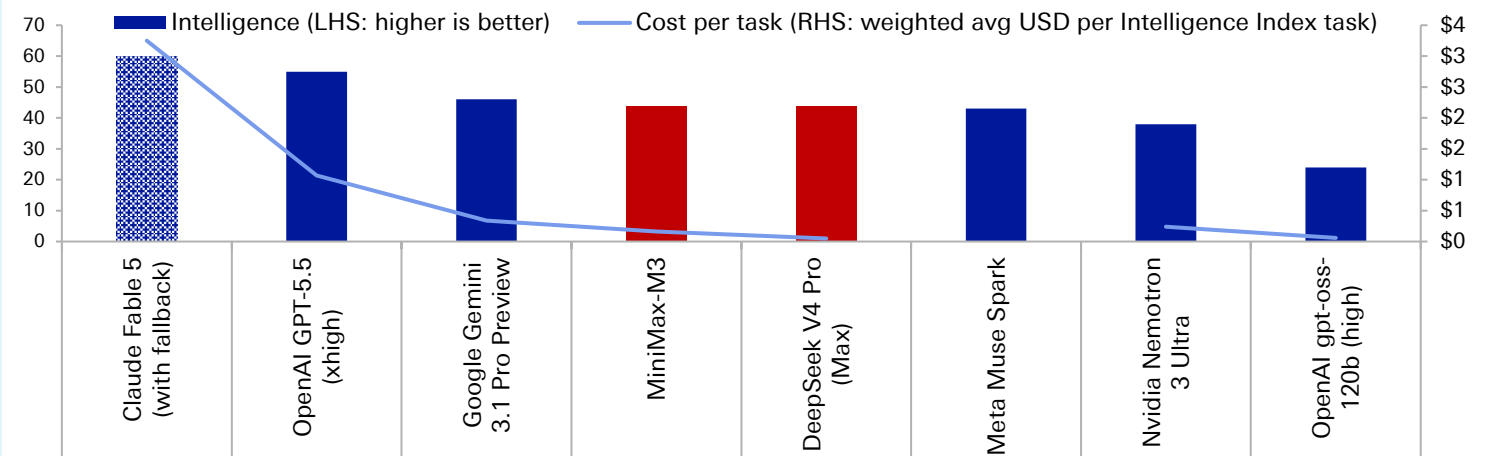
- AI company revenues are rising at record pace – but so are compute and infrastructure costs.
- Frontier-model token costs specifically have become exponentially higher than still very intelligent mid-tier models. This is one of the biggest challenges for wide enterprise adoption.
- There is also evidence the AI-capex demand shock is feeding into inflation, most visibly in computer and semiconductor prices, specifically memory.

Global corporate investment in AI



Source: Deutsche Bank Research, McKinsey/Stanford Artificial Intelligence Index.

The cost of tokens for frontier models is exponentially higher than still very intelligent mid-tier models. This is one of the biggest challenges for widespread company adoption.



Source: Deutsche Bank Research, Artificial Analysis Intelligence Index. "Fallback" mechanism routes safety-flagged messages to Opus 4.8 model. Meta Muse Spark price not available. US models in blue; Chinese models in red

Artificial intelligence

Job displacement from AI does not seem to be a near-term (~2 years) concern



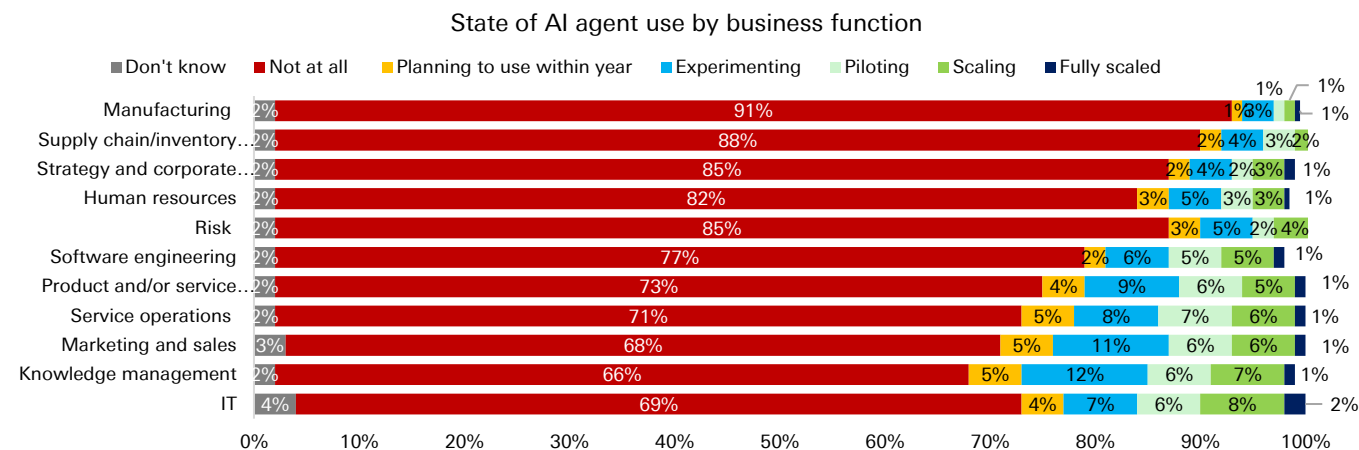
Scaling AI adoption remains a challenge

- 88% of firms now use AI in at least one business function (78% in 2024). Source: [McKinsey/Stanford AI index](#)
- Biggest y/y gains: Europe (+11pp to 91%) and China (+13pp to 88%) (McKinsey/Stanford).
- Usage is highest in knowledge management for business, legal, and professional services (58%), software engineering (58%), and IT (56%).
- But scaling adoption lags: <10% of firms are scaling AI agents across their firm.

AI job displacement not a near-term concern

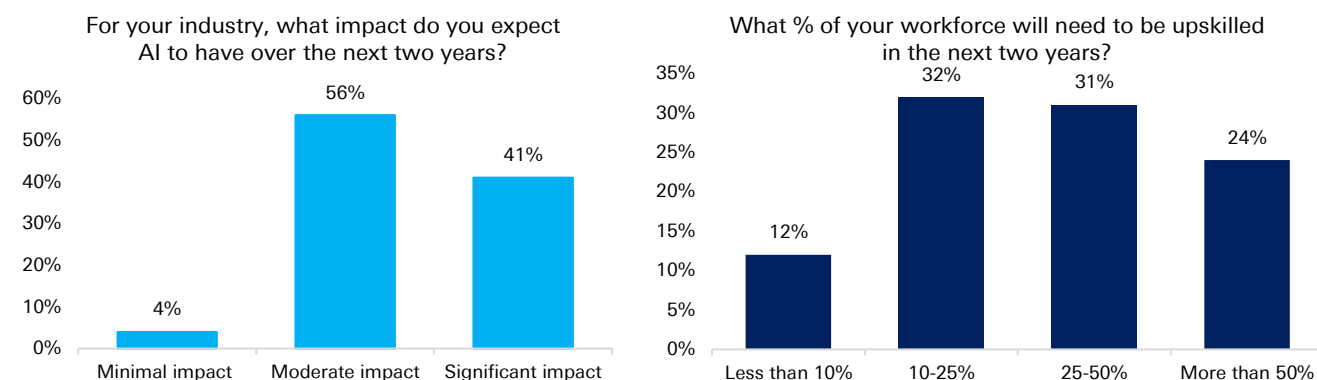
- 56% of CEOs expect AI to have only a moderate impact on their industries (Business Council and Conference Board).
- Evidence so far shows little relationship between US labour-market churn and AI adoption rate (see [here](#)).
- Yet, a third of firms expect AI to cut headcount in the coming year – larger firms more so. The largest share (43%) still expect little or no change. McKinsey/Stanford

Organizations are beginning to experiment with agentic AI, but diffusion is not yet widespread



Source: Deutsche Bank Research, McKinsey/Stanford Artificial Intelligence Index. Number of respondents = 1,933

Majority of CEOs don't think AI will fundamentally transform their firms, but most also anticipate upskilling their workforce in the short term



Source: Deutsche Bank Research, The Conference Board Measure of CEO Confidence in collaboration with the Business Council, Q2 2026. Left hand chart: Minimal impact (very few business processes will change; the industry remains largely unchanged); moderate impact (many business processes will change, but the industry will not fundamentally transform); Significant impact (AI will affect nearly every business process the industry will be fundamentally transformed.)



Digital assets

The push toward 24/7, digital, and programmable financial markets has brought tokenization into focus for policymakers and market infrastructure providers

Asset Tokenization

- Transferable tokenized assets on-chain total \$33bn (rwa.xyz).
- [The SEC is reportedly](#) moving to permit trading of tokenized stocks; ongoing DTCC progress on tokenized collateral infrastructure.
- Several US banks have announced a tokenized-deposits clearing network operated by The Clearing House, live from 2027 onwards, with settlement in central bank reserves.

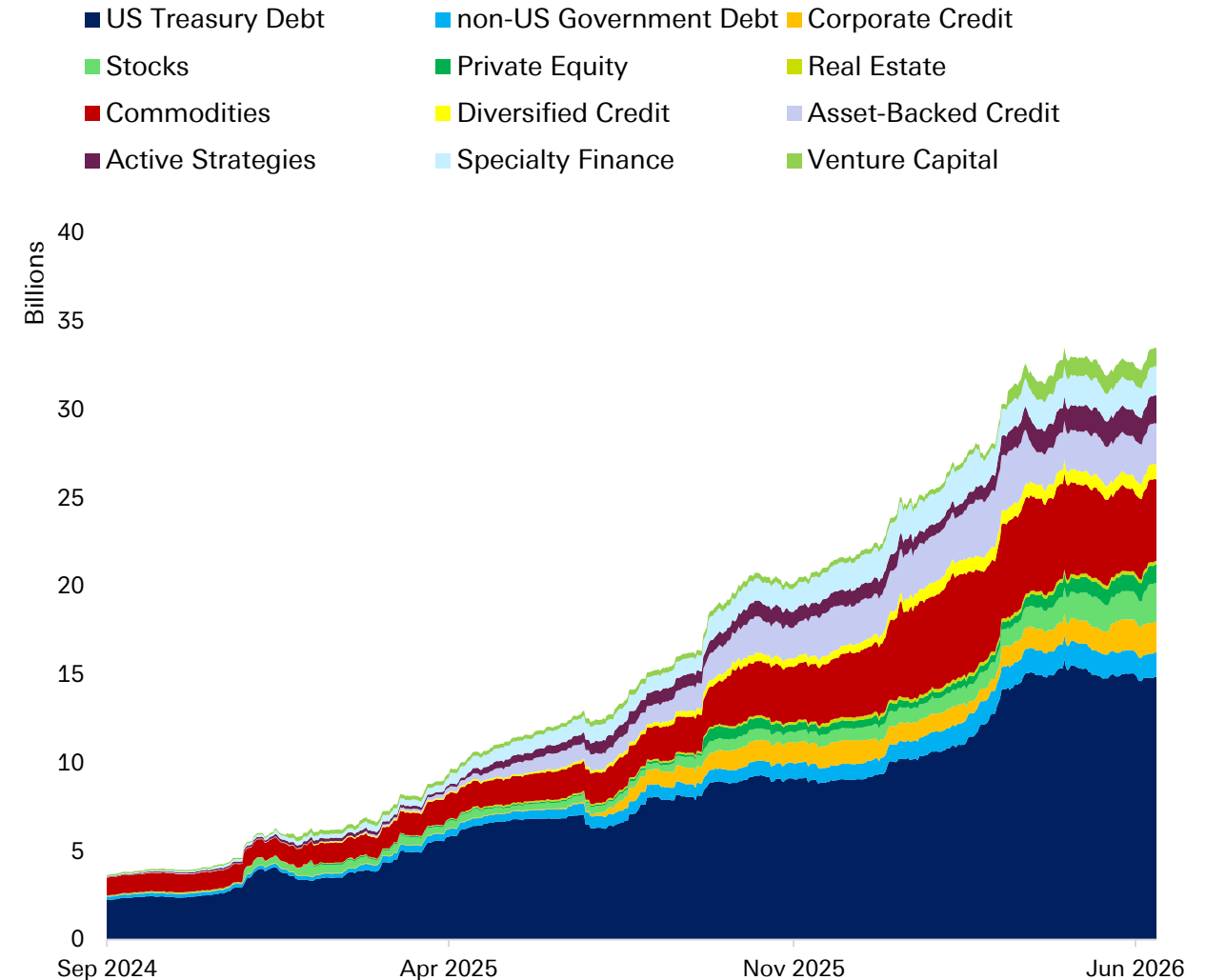
Stablecoins

- Adjusted stablecoin transaction volume is projected to reach [\\$719trn](#) by 2035 (Chainalysis).
- Stablecoin payment volumes could match Visa and Mastercard's off-chain transactions sometime between 2029-31.

CBDCs

- Wholesale: ECB's Project Pontes – a settlement solution for DLT-based transactions in central bank money – tracking for Sep 2026 launch.
- Retail: EP committee approved the digital euro framework June 23, final law targeted end-2026. Pilot mid-2027; first issuance possible 2029.

Tokenization is growing fast



Source: Deutsche Bank Research, RWA database. Data up to July 10.



Appendix 1

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Transatlantic Tensions? The NATO Summit Verdicts

July 9, 2026

Transatlantic tensions are brewing! Our latest analysis dissects the NATO Summit's verdict, revealing critical shifts in defence spending, new industrial initiatives, and unresolved strains between Europe and the US. Discover what these developments mean for markets, from potential trade wars to a looming Ukraine ceasefire.

Context

- NATO leaders met in Ankara on 7-8 JUL for a summit, which reaffirmed the alliance's unity on key issues and embedded the 'NATO 3.0' plan to shift more of the burden of Europe's conventional defence onto European members.
- At the same time, the summit was fractious given high-profile criticism by President Trump of NATO's willingness to assist the US in Iran, renewed claims on Greenland, and political differences with several European countries. The Europe-US relationship is under stress.
- While the summit delivered some concrete industrial initiatives, the underlying transatlantic strains remain unresolved, and there is almost undoubtedly going to be more strains ahead relating to burden-shifting, spending and issues like Greenland.

What did the summit deliver?

- The summit's headline outcome is that NATO allies are moving quickly to **increase defence spending** and production, and to develop new warfare capabilities. Following the 2025 declaration that allies' "core" defence spending would rise to 3.5% of GDP by 2035, leaders highlighted a \$139 Bn increase in 2025 by European NATO countries and Canada. The NATO Summit Defence Industry Forum saw allies announce over \$50 Bn in new defence procurement, including a \$40 Bn commitment over five years to build drone and counter-drone systems. They also pledged to provide Ukraine with €70 Bn EUR in weapons deliveries and defence assistance in 2026 and an equivalent amount in 2027.
- Countries have launched **NATO-adjacent coalitions** to address financing and procurement bottlenecks. These include: the Canada-led Defence Security and Resilience Bank (DSRB), which reportedly aims to raise up to \$135 Bn in lending capacity; and the UK-championed Multilateral Defence Mechanism (MDM), meant to coordinate joint procurement to improve efficiency and reduce unit costs. Discussions are ongoing on aligning or consolidating the DSRB and MDM, and participation in these and similar initiatives is likely to gather pace.
- **More efficient engagement with industry:** Secretary General Rutte launched the NATO Front Door for Industry (a new platform to simplify how companies engage with NATO by giving companies a single point of access to NATO procurement opportunities, innovation events, and other

Authors

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engagement channels). This is paired with NATO's first public unclassified demand signal, developed by NATO's defence planners laying out capability priorities for the coming years and distributed through the Front Door. Rutte also launched the NATO Engine, a framework to expand industrial production capacity, which connects defence and civilian manufacturing lines across the Alliance.

What should markets be watching next?

- **Europe-US political and security relations:** Tension in the transatlantic relationship has grown since the start of the year and may increase further, particularly as ties between Trump and traditional centre-right European allies have come under strain. European leaders fear rapid US withdrawal of defence capabilities that cannot be quickly replaced (e.g. intelligence, reconnaissance and surveillance; air-to-air refuelling; and long-range fires). US Defence Secretary Pete Hegseth's announcement on 18 June of a six-month Pentagon review of US force posture in Europe will heighten these concerns, as well as fears of political retaliation against individual countries.
- European capitals are pushing the US for a clear roadmap on force drawdowns that would allow them to plan their corresponding build-up, but it is unclear whether they will receive any concrete commitments. As the US begins to face constraints on resources and weapons stockpiles owing to the Iran war, the Trump administration will likely intensify its demands that Europe urgently take on more responsibility for its own conventional defence.
- **Europe-US economic ties:** A key consideration for investors is whether worsening political ties will spill over into the transatlantic trade and investment environment. The 2025 EU-US Turnberry agreement took effect in the EU on 1 July, but US adherence to the 15% all-inclusive rate on covered EU exports has been thrown into doubt by the shifting basis of US tariff authority this year. The next watchpoint will be the findings of the US Section 301 investigation into structural overcapacity, in which the EU is one of 16 jurisdictions in scope.
- This may lead to renewed EU-US negotiations over implementing Turnberry; the European Commission could propose to suspend certain concessions if it deems the US in breach of the agreement's terms. Political tail risks to the economic relationship will also remain: Trump's restated claims on Greenland, or his demand to cut economic ties with Spain, could prompt the EU to threaten retaliatory measures similar to those it considered applying in January of this year, and could trigger renewed market volatility.
- **Russia-Ukraine ceasefire negotiations:** In line with our expectations, Trump is renewing his diplomatic push for a ceasefire, meeting President Zelensky on 8 July and speaking to President Putin over the 4-5 July weekend. This re-engagement has notably coincided with closer Europe-US alignment on Ukraine, the expiry of the US waiver for Russian oil sanctions on 17 June, and Trump's announcement on 8 July that the US would grant Ukraine a licence to produce Patriot air defence missiles domestically. Taken together, these measures may form part of a US strategy to increase pressure on Moscow to re-enter mediated negotiations with Ukraine.
- Growing war fatigue, a shift in public attitudes in both Russia and Ukraine, and greater openness among some European leaders to direct talks with Moscow all point to possible progress towards a deal. A ceasefire this year is by no means assured, but any agreement could have wide-ranging implications for markets, including sanctions relief, increased agricultural



exports, and substantial investment in infrastructure, energy, transport, minerals, defence and other sectors associated with post-conflict reconstruction.

- The author of this report wishes to acknowledge the contribution made by **Scott R Martin**, an independent expert engaged by Deutsche Bank, in the preparation of this report.



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